

The Metaweb

Buckle up for a fascinating journey through layers of insight and metaphors that explain the past, present, and future of the Web. Readers from all walks of life will learn something ancient, something novel, and something practical. Those who give it careful consideration will never see the Web the same way again.

This book proclaims into existence decentralized public space above the webpage that enables the shift from personal to collective computing. The Web's next frontier is the Metaweb, a hyper-dimensional web over Today's Web that connects people and information silos, with accountability and fair value exchange. The Metaweb can drastically reduce false information, abuse, and scams, as well as enable the unprecedented level of collaboration needed to address humanity's global challenges. The book posits a symbiotic relationship between AI and the Metaweb, where AI assists in generating, organizing, and curating content, while the Metaweb provides the necessary constraints, data, and context for AI to function effectively, transparently, and in alignment with humanity. The AI-assisted collaboration among humans on the Metaweb will enable a vast collective intelligence and the capture of tremendous untapped value.

The Metaweb

The Next Level of the Internet

Bridgit DAO



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Early Reviews

“When the general population got on the Web, we saw a sea change in the diversity of online applications and content. Alongside the smartphone, we have seen a rapidly proliferating array of behaviors, content, incentives, and side effects, most of which were not squarely on our radar 50 years ago ... I agree with the premise that accountability is an important consideration for the next iteration of the Web. While I have concerns around scalability and implementation, the notion of a meta-environment above the webpage is directionally interesting. Overall, the book presents a creative argument that warrants further discussion and exploration.”

—**Vint Cerf**, *One of the Fathers of the Internet & co-founder of the People-Centered Internet*

“This is next level evolution of creative expression meeting technology”

—**Asya Abdrahman**, *founder of Kin DAO and Primordia DAO*

“This book should serve well to educate both new and young generations of the pitfalls and potential of a fully integrated Internet experience laced with ownership abilities.”

—**The Golden Yogi**, *founder of the Tulum Crypto Club*

“Bridgit has arrived as the Overweb to remind us that only when we join our attention together as One will we realize our true human potential.”

—**Dante King**, *Trustee*

“*The Metaweb* is an incredible work of value. A must-read enlightenment, to be read with Jane McGonigal’s *EthicalOS*, and *The Social Dilemma*. Picking apart that vast web world to understand it and thereby gain control in using and shaping it. Really epic.”

—**Eric Schneider**, *founder of Youth Leader*

This literary piece introduces the common user to a framework that allows them to explore new ways of thinking, engaging, and contributing to society without boundaries.”

—**Stephanie Hervey**, *The Artisan Hub*

“Excellent approach to the current issues with Web 2.0 as well as a compelling offering to future uses and improvements.”

—**Danco**, *NeoSoul DAO*

“Web2 has all the character of a shopping mall, polished sterility designed to maximize consumption. The Metaweb offers end users a can of spray paint and encourages us connect dots, refute, support and discuss the content on any page. The underground is about to go mainstream.”

—Jeremy Nathaniel Akers, *founder of Regens Unite*

“The Metaweb embodies the vision of a new internet. A web where linking content is a shared, collaborative action, a democratic act even. If you’d like to understand how the web today works and how it constrains us as a society in leveraging collective intelligence, this book is for you. Be prepared for a journey through the depths of cyberspace, why the web is humanity’s most important tool for solving generational challenges, and how we need to adapt it to allow the internet to fulfill its original promise.”

—Florian Ganz, *Co-founder of Common Ground*

“This book expands the rigorous expansion of the third eye and crown chakra in web technology, tapping into the universal dialectical human consciousness to reconcile the imperfections of humanity. The expansiveness of the book encapsulates the sacred creative human spark of a tech start-up before being swallowed or consumed by tech giants. Throughout the book are David and the tech Goliath data points and clear facts, revealing how the Overweb can redirect technology into a higher purpose as an elevated being, creating a meaningful system of accountability and authenticity and sustaining the building of safe regenerative, decentralized communities. From why Andreessen Horowitz invested in Rap Genius to support annotating any page on the Internet with commentary to the Metaweb operating as a shape-shifting righteous meta-North Star, this book weaves the disruption of rampant false information, technology fetishization of maintaining social biases to adding icing on the cake of being accountable to humanity and the planet.”

—Nile Malloy, *The Artisan Hub, Resource Catalyst, and Community Investment*

“Bridgit and the Overweb are fascinating concepts that may be able to provide new life for an old web idea that has morphed into a dream environment for capitalists, charlatans and clandestine behaviors. The ability for an individual to create a bidirectional ‘bridge’ for specific portions of web content has tremendous implications. Allowing participants to quickly see verified supporting and/or contradictory content is sorely needed based on the current state of the web. The real time presence and interactions on any web page will enable entirely new dimensions of reality for online activity by creating value in spaces not previously considered. The implementation methodology is critical to ensure further information silos are not established and collective intelligence is decentralized. It must also assure that the minutiae of content doesn’t lose its initial context once ‘bridged.’ The incentives to participate in the Overweb will hopefully focus on fair value exchange rather than monetization and individual wealth.”

—Alex Hill, *Healthcare Technologist*

“Converting the web into a hybrid, multidimensional gateway with the power to evolve humanity through collective cognitive cooperation is a profound and mind blowing concept that is a once in a lifetime opportunity. The Metaweb is the perfect technological companion to usher in the future of the gaming industry. GamerXSociety will most definitely leverage the Metaweb to create the world’s largest tangibly rewarding gamer culture that effectively bridges the gap between the global gaming community and non endemic brand partners for

the first time in history. The concept of ‘bilingual bridging’ that connects the ‘splinter web’ through culturally accurate perspectivism is a global game changer.”

—**Jeff Ivory**, *founder of GamerXSociety*

“Historically, a book’s table of contents is one kind of metadata about what’s in this book; that book’s index - if it has one - is a topic-centric view into that book. In the 1990’s topic maps were invented as a means to provide a digital topic-centric index into the GNU documents; topic maps lie above the information itself, and provide a kind of roadmap through the mapped information territory. Danny Hillis founded Metaweb Technologies which produced Freebase, a kind of online topic map, which was later purchased by Google. The notion of a metaweb is not new. But, it’s a powerful concept with a future. The Metaweb Book is an aspirational but comprehensive journey through what this idea of organizing metadata about information on the Web can look like and how we can move forward. The book describes the Overweb pattern which builds on pioneering technologies like topic maps and the semantic web, bringing to the foreground emphasis on safety, security, privacy, and governance. Do we need such improvements? The book persuasively argues they are needed. Given the emphasis on social needs in a technological world, the book provides a solid response to the late Douglas Engelbart’s quest for the co-evolution of social and technological systems in the face of ever more complex and urgent global issues.”

—**Jack Park**, *cofounder of Topic Quests*

“This is a valuable and readable guide to a new world as it is developing—a world beyond the present-day internet web, where the control of knowledge and its cross-referencing is put in the collective hands of users, thus escaping the channeling and exploitation of the present-day web for the benefit of BigTech financial interests. Alongside a detailed critique of the present-day web, the book includes discussion of the principles behind and features of the emergent metaweb, and the advances in collective thinking and communication it makes possible. Social scientists will find the book a useful guide for thinking and research about the new social worlds of the metaweb.”

—**Robin Room**, *renowned sociologist focusing on alcohol and drug research*

“This book contains a rich history and pathology of today’s Web. Moreover, it introduces the reader to its future: the Metaweb, a multi-layer extension that builds on underexplored features of the current Web and creates a hyper-dimensional space to share information and collaborate. The Metaweb proposal is both inspiring and realistic. It presents a feasible solution to problems of the digital age. Also, it provides clear directions for creating a better future, one that enables people to address challenges by employing next-level collective intelligence. I highly recommend this book to researchers working on the crossroads of cognition, communication, argumentation, and AI.”

—**Jean Wagemans**, *Amsterdam Center for Language and Communication,
University of Amsterdam*

Mitakuye Oyasin (all my relations)

We give thanks to all our relations and to Great Spirit for all the love, light, and life lessons. Please hold us in your grace.

To Generation Alpha, the entrepreneurs, the social entrepreneurs, the app builders, the organizers, the influencers, the YouTubers, the journalists, the citizen fact-checkers, the classifiers, the curators, the bridgers, the nation builders, the impact investors, the financiers, and the volunteers that will build the Next Level of the Web.

Collectively, we are all that we need.

Let's grow!

We consider Generation Alpha to be everyone who embraces their choice to be alive now, balancing emerging and ancient technologies, and leveraging them on humanity's behalf.

May the part within each one of us that aligns with Brigid, the Celtic fire deity of all things High: wisdom, poetry, healing, and protection, actively take part in the development and refinement of superior collective capacities for our evolution as love embodied.

May the Eagle and the Condor fly together in the same skies, enabling humanity to once again live in balance and connection.

All praises to the Most High.

Wholeness and balanced vibrations.

Axé | Amen | Aho!

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Preface

The evolution of our physical form is an important aspect of human development, but it's not the whole story. Tools and technology have also played a major role in our history, as has the evolution of our collective understanding and agreements about the nature of reality. This includes the development of science, religion, and money. These factors—often overlooked—have been influential in shaping our history and understanding of the adjacent possible. This understanding is not static, but rather it is constantly evolving and refining, which allows us to consider ever more possibilities.

One example of this is our understanding of the earth as being round, rather than flat. In the past, many people believed that the Earth was flat because of their lived experience. It would have been difficult for them to imagine that the Earth was anything other than flat. When, however, the Greek philosopher Pythagoras suggested that the Earth was actually spherical (circa 500 BCE), it opened the door to many new possibilities and advances. Today, it is clear to most people that the Earth is a sphere, thanks in part to photographs of the Earth taken from space. While a small number of people still believe in the flat earth theory, they have little evidence to back up their claims.

Similar to how we used to think the Earth was flat, based on our experience, we currently think about the Web as a flat, two-dimensional space. But the Web is not actually flat; it is a complex, multi-dimensional space that is constantly evolving. The tools we use to access the Web, such as web browsers, give us the impression that it is flat and static, but the Web is much more than that. In the future, our understanding of the Web will evolve and we will see its multi-dimensionality, just as we now understand the Earth to be a sphere. Our current understanding of the Web as a flat space will seem as limited and outdated as Flat Earth Theory.

The mere existence of this book, along with people actively reading and engaging with its ideas, suggests that a transformative shift in our comprehension of the Web is underway.

This shift supports growing movements for decentralization, privacy, data sovereignty, safety, truth, and free speech. In order to address the existential threats facing our society, we must move beyond our current limited understanding of the Web as a flat, two-dimensional space and embrace a multi-layered Web. This will allow us to collectively and democratically tackle the challenges we face and secure a democratic future.

This book posits a shift in our understanding of the Web, transitioning from a flat, two-dimensional screen to a multi-layered, hyper-dimensional space. This shift in paradigm will not change the essence of the Web, which has always been hyper-dimensional, but it will redefine how we interact with the Web. The shift will help us to solve many of the problems that plague the Web today, such as scams, abuse, and false information. It will also enable us to create and fairly distribute value, and to connect, communicate, and collaborate in new

and powerful ways. The move to a hyper-dimensional Web will allow us to address the global challenges we face in a collective and democratic manner.

This book introduces the concept of the Metaweb, a meta-layer on top of the existing Web that creates safe, decentralized public spaces. The Metaweb is anchored by a map of the online information ecology and is accessible through browser extensions, website code libraries, and browsers that implement Metaweb protocols. It operates as a trust layer on top of the existing webpage, providing a more secure and decentralized way to access the Web.

The future of the Web is vast and complex, but the concepts introduced in this book are simple and easy to understand. The Web of the future is not flat, but rather multi-layered like a cake. This book will help you explore this complex and exciting territory and discover what's possible in the next level of the Internet.

Acknowledgments

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Giving thanks to the Bridgit DAO council and their commitment to a future that is collectively intelligent, wise, and kind and their pursuit of balance, truth, and progressive decentralization, all via connection, synergy, and regeneration. A special thanks to the Bridgit DAO Founder's Club and Community NFT holders and the Overweb NFT holders.

Deep gratitude to the part within each of us that aligns with Brigid, the Celtic deity of all things high: intelligence, wisdom, poetry, healing, and protection. May our wisdom energies develop and refine into a collective capacity for embodying peace, harmony, and love.

And thanks to you—the reader, the builder, the dreamer. We have put together more resources for you at metawebbook.com. Finally, a special acknowledgment, to our Bridgers. If you notice a special feeling moving through your body while reading this book, make a note of it. Then honor the “noticing” by taking deliberate action that brings us collectively one step closer to the realization of a safe and transformative Metaweb. Live the story you want to tell.

What We Stand For

Bridgit DAO is a meta DAO
that supports and launches
Social DAOs focused on
regeneration,
cognitive freedom,
and collective wisdom.

We are actively building the future that
we want to live in:
a democratic
ecologically, socially, and economically
regenerative world.

Bridgit DAO is building
a next-level web ecosystem
that reimagines the Internet
by enabling people,
information,
and interactions
to have a presence
above the webpage.

Bridgit DAO is catalyzing a future
in which the webpage is
simply and profoundly
the contextual footprint
for intertwined worlds
of information,
interactions,
transactions,
and experiences
that emanate
from content
on the webpage.

<https://bridgit.io>

About the Author

Bridgit DAO is a decentralized organization focused on promoting and supporting initiatives related to regeneration, cognitive freedom, and collective wisdom. It comprises a diverse group of individuals, including technologists, writers, editors, illustrators, entrepreneurs, and others. As a meta-DAO, Bridgit has launched several projects, including the Overweb, Presence, and Pachaverse, and has supported initiatives in Tulum, the NeoSoul DAO, and the DAOjo.

Bridgit DAO holds majority shares in Presence Labs and Pachaverse. Bridgit DAO also initiated the XW3B BRC-20 token.

The Bridgit DAO council steers Bridgit DAO to be an active participant in the Renaissance, a movement for planetary renewal. We are open to new members who share our enthusiasm and dedication for regeneration, cognitive freedom, and collective wisdom, and who want to actively participate in the emergence of the Overweb and other Bridgit DAO initiatives. We are also open to feedback and contributions to the movement. Please see bridgit.io for more information.

As a proud author, Bridgit DAO's goal for *The Metaweb: The Next Level of the Internet* is to spark a collective vision for the future of the Web, shifting from individual computing to a shared, collaborative experience. The Overweb has the potential to become the largest open-source initiative in history and an integral part of the Web. By working together, we aim to optimize the Overweb for maximum collective value, creating a hyper-dimensional web that benefits us all.

We envision *The Metaweb: The Next Level of the Internet* becoming a cutting-edge metabook—a hyper-dimensional digital version of a physical book that evolves and grows over time with its readers and the reading community. This metabook can be used in conjunction with a physical book, a webpage, or even in virtual reality. It will offer features such as book clubs, AMAs, notes, reviews, polls, and conversations related to specific passages and images within the book. In this way, the physical book becomes a contextual hub for interconnected communities of people, information, and interactions tied to the book's content.

Bridgit DAO will continue to collaborate, catalyze, or publish works on topics such as regeneration, cognitive freedom, and collective wisdom.

Join us.

Viva la Renaissance!

Introduction

If we give up on building a better web now, then the web will not have failed us. We will have failed the web.

—Tim Berners-Lee

When we were growing up, people believed that we were only using a small fraction of our brains. We thought human intelligence could be increased by tapping into the unused portions. But advancements in brain imaging technology have debunked this notion.

If, however, intelligence reflects an understanding of our world and the ability to solve our problems, history demonstrates that we become smarter through the use of tools. We are smarter because of, for example, books, computers, the Web, mobile phones, and AI. We make tools, they make us smarter; we then build better tools. We make our tools and our tools make us.

Despite our ever-growing capabilities, this book contends that we are only scratching the surface of what's possible regarding humanity's collective intelligence and our most important tool—The Internet—which are inextricably intertwined.

This brings up several questions. What would it mean for humanity to be more collectively intelligent and wise? How can the Internet enable the development of humanity's collective intelligence and wisdom? What is humanity's destiny if we do not become more collectively intelligent or wise?

We see artifacts of collective intelligence and wisdom in our complex societies and civilizations, shared systems of communication such as language and writing, institutions like schools and libraries, and advancements in science and technology, all of which stem from collaboration and experimentation. Democratic systems of government, public health systems, international trade, and emergency response systems also reflect our collective intelligence.

Yet, our advanced society continues to face significant challenges in addressing our planetary challenges. Effective collaboration and coordination seem hard to find in places where they are most needed, such as in congresses and parliaments, among heads of state and corporate leaders, and in the digital town square.

How could this be? Humans are incredibly capable. Spending a short time on a streaming video app demonstrates our ability to accomplish almost anything we set our minds to. When, however, it comes to working together as a collective, we still have ample room for improvement, which could be necessary for a democratic future.

But there is hope! As it turns out, there is a way forward, but it would require us to see the Internet in an entirely new light. We currently see the Web as simply a flat and static space. That's only seeing a tiny bit of what's possible. If we acknowledge that space exists above the

webpage, the possibilities are endless. This would enable us to realize the Web's potential for collective computing and developing our collective intelligence and wisdom.

This book unveils the web of the future, a web that is more powerful, profound, and supportive than the Web we know today. The web of the future will enable us to transcend the problems that seem intractable today, such as privacy violations and the manipulation of public opinion. It will enable us to own and control our data, and to be rewarded for the value of our contributions. The web of the future will support free speech and privacy, and will help us to understand ourselves, each other, and the world around us.

The web of the future will enable us to become collectively intelligent and to develop our collective cognitive capabilities. It will connect people and information in unprecedented ways, allowing us to think, learn, and build knowledge together. The web of the future will enable us to connect and communicate across languages and cultures, and will help us to cooperate and collaborate at unprecedented levels.

The web of the future will utilize AI (artificial intelligence) for the betterment of humanity, enhancing safety, accountability, productivity, and collaboration, as well as improving the web experience. With the integration of AI assistants, mundane tasks and content generation will be handled with ease, freeing up more time for connecting in the real world (IRL). By 2030, experts project that AI assistants on cell phones will surpass the capabilities of the AI currently employed by today's largest companies. This AI-assisted web of the future promises to greatly enhance both human-to-computer and human-to-human connection, communication, and collaboration.

The web of the future will be a digital overlay for all realities, connecting the online, virtual, physical, and conceptual realms. Every location, whether online, physical, virtual, or conceptual, will be addressable and connected to a massive web of information and interactions. This web of connections will enable humanity to create a hyper-dimensional shared context, which we call the Metaweb.

The Metaweb is the space above the webpage, but it is also much more than that. It is a safe AI-enhanced environment that prioritizes privacy, accountability, and collaboration; a digital overlay online and IRL; and the connective tissue among the online, virtual, physical, and conceptual realms. In Metaweb theory, the webpage, the physical object or location, the virtual object or space, and the word or concept all become contextual footprints for interconnected worlds of information, interactions, transactions, and experiences that emanate from our attention. The Metaweb is the future of the web, and it will enable humanity to thrive and reach its full potential.

THE ANT ON THE PAGE

Visiting a webpage is like being an ant on a rectangular plate. The content of the plate is like the text and images on a webpage, and the ant is positioned on a specific pixel within a coordinate grid. The x and y coordinates of the ant correspond to its position on the horizontal and vertical axes of the grid. When the ant moves, its coordinates change, but it is constrained by the edges of the plate.

Many plates, or webpages, exist, each with its own constraints and limitations on what can be done. Some webpages allow commenting, while others only allow reading and following links if they exist. On social media webpages, the ant, or user, has more opportunities for expression, but these are still limited. For example, on Twitter, the ant can write 280 characters, on Facebook it can update its status, and on Instagram, it can post pictures.

One day, some ant friends come along and want more freedom. As ants, they use a mechanism of indirect coordination, through the environment, called stigmergy, whereby ants leave traces in the environment that stimulate future actions. Stigmergy enables the ants

to work together and accomplish tasks without direct communication. Ants receive positive fitness benefits when they respond to traces in the environment, reinforcing the likelihood of these behaviors becoming fixed within a population over time. This is how a colony of ants can become a highly organized and efficient unit.

To enable their stigmergic ambitions, the ants build a meta-layer on top of the content plates. This transparent layer allows the ants to leave signals about the content below and to create linkages to other transparent plates. The meta-layer gives the ants more freedom and control over their experience on the plates.

Like a solitary ant, we are constrained by the two-dimensional grid of the Web and have limited opportunities for expression. As with the ants, a meta-layer—the Metaweb—can provide us with enhanced freedom and control over our experience on the Web. It can enable us to escape the cognitive constraints of the Attention Economy and develop our collective cognitive capabilities. The Metaweb is a step towards a hyper-dimensional web that will allow us to think and communicate in new and powerful ways.

Before we dive in, let's deal with a couple of housekeeping tasks.

First, the Metaweb is a concept that aims to expand how we think about the online world and its connection to the physical and virtual realms. It presents the possibility of a re-imagined Web that can help bring the world back into balance by protecting and restoring the ecosystems that support life on the planet. The Metaweb can enable humanity to become more collectively intelligent and wise, helping us to ascend to a higher level of consciousness.

To provide a concrete example of the Metaweb, Bridgit DAO has partnered with 4th Ave to catalyze the Overweb. The Overweb is a safe digital space for humans, organized into purpose-aligned meta-communities and digital entities. It is accessible through the Presence browser overlay, which is available as a browser extension, SDK, and mobile browser extension. The Overweb aims to be the first safe, pervasive, and decentralized public space anywhere.

The Overweb is built on the principles of privacy, data sovereignty, fair value exchange, accountability, and safety. It is intended to be a space where humanity can build the future we want to live in. The author of this book is a decentralized autonomous organization (DAO), making it one of the first technology books.

Second, this book aims to catalyze the creation of a new social contract implemented through technology, which enables us to use digital overlays safely and productively to create the physical and virtual realities we want to live in. This new reality, which we call the New Earth, builds upon a next-level web that is unlimited yet easy to navigate, safe yet exciting and illuminating, and a continual learning experience. The eventual outcome of this new web is a renewed humanity that has ascended to a higher level of consciousness and a renewed individual who actively takes part in this transition.

Third, to accurately describe this new paradigm, we have introduced new language and terminology. The Web's next level is hyper-dimensional, and previous terms were not sufficient to accurately describe this innovation. We believe that these new terms will be useful in the foreseeable future. Additionally, on the Overweb, we do not refer to individuals as "users." We believe that this term is disrespectful and reflects outdated thinking. It implies that people are simply consumers and should remain passive. Instead, we refer to individuals as participants, creators, and people on the Overweb, recognizing their active role in shaping and contributing to the community.

THE WEB CAKE

In this book, we present major problems with Today's Web, including scams, cyberbullying, trolling, hate speech that incites violence, and false information that is damaging to

individuals, the human collective, and our trust in the Web. We touch upon features of the Attention Economy, such as polarization, censorship, lack of privacy, and data exploitation that harm us and limit our cognitive freedom. We also identify structural aspects of the Web that prevent us from using the Web to address humanity's existential threats.

Unbeknownst to many, the Web has a BIG missing feature. Famed browser builder and venture capitalist Marc Andreessen made this eminently clear in 2012 when his firm purchased the annotation firm Rap Genius. But the business community ignored this aspect of the investment. Building upon the big missing feature, the Metaweb enables knowledge, computation, and people to have a presence above the webpage. We will explore the possibilities of the Metaweb and how it can mitigate if not solve the Web's intractable problems. We also explore an emerging zone in the Metaweb—the Overweb—that provides a level of safety from bad actors, data security and sovereignty, context for online information, and fair value exchange impossible on Today's Web.

The debilitating problem is that the Web is flat and static. Because of happenstance and limited thinking, we have not taken full advantage of our existing technological capabilities regarding creating safe decentralized public spaces for people to connect, meet, and collaborate. Few people think above the webpage, so it's difficult for them to imagine a meta-layer. But we assure you, this book will help you see the possibilities of a hyper-dimensional web.

Today's flat and static web is like the early 1800s for cake. Until then, cakes were one layer. In fact, you wouldn't even think of it as a layer; the cake was a monolith, uniform and indivisible ... except with a knife. But in the mid-1800s, someone made a multi-layered cake, changing the paradigm around what a cake was. It was now possible to have many layers on a cake, inspiring a whole new category of cakes called layer cakes and, later, wedding cakes. Bakers could then imagine a world of possibilities for layers to enhance taste, texture, and display.

Today's Web is like the bottom of a multi-layered cake—the web cake (Figure 0.1). The next level of the web cake is the Metaweb, comprising three layers.

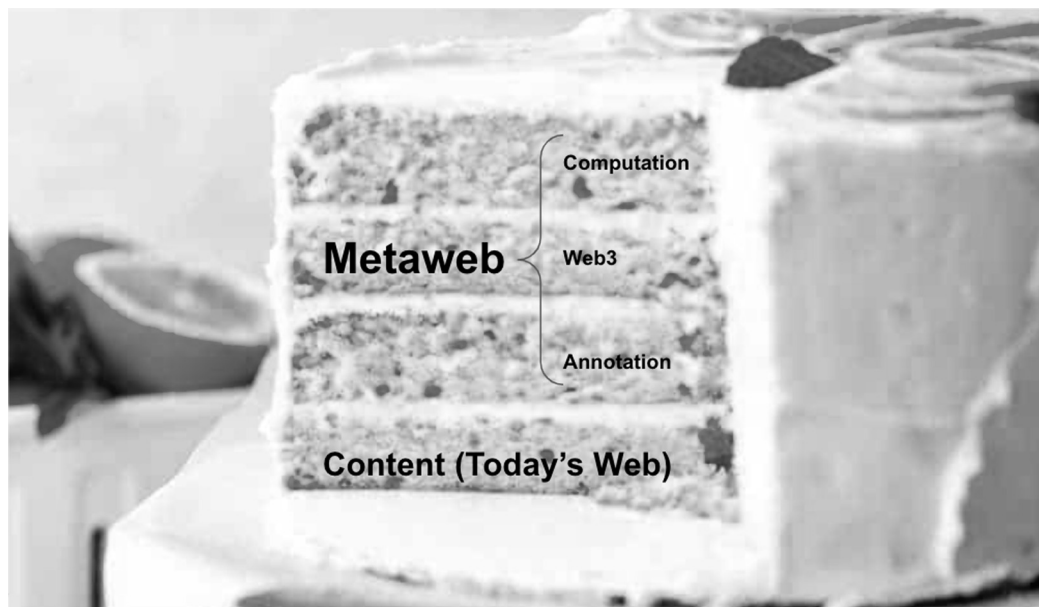


Figure 0.1 The four-layer web cake.

To help you better visualize the concept of a four-layer web cake, imagine a cake that has the same aspect ratio and orientation as a laptop screen.

The first layer of the cake represents Today's Web, with page content and any embedded ads or pop-ups.

The second layer, which is mostly transparent, represents annotations, or notes attached to pieces of text that are displayed as an overlay when the viewer's cursor comes near them.

The third layer represents Web3, which overlays a decentralized wallet protected by cryptography on a webpage, allowing users to authorize or sign smart contracts and make purchases without the need for a middleman.

The fourth and top layer is a transparent layer that places code above related content on webpages, enabling computations that present information and options to participants, handle their interactions, and present the results. This layer allows real people, information, interactions, and even meta-communities to have a presence above the webpage.

But What Do Layers on the Web Look Like?

You already know what multiple layers look like. You just don't realize it.

Drum roll ... It looks like Today's Web!

Internet platforms and applications use your attention (e.g., the location of your cursor or touch) to trigger the display of contextually relevant information and interactions without reloading the page. You have seen this with pop-up videos, web applications like YouTube, and tool tips. Anything new that appears over web content is a new layer. We call this attention triggering.

Attention-triggered elements on the web page change their behavior—appearing or moving—in response to the user's attention. On a laptop or desktop, attention is represented by the cursor; on mobile, it's touch; in virtual reality, it's line of sight.

Let's try it out. Navigate to YouTube and start a video. At the moment, you cannot see mechanisms for controlling your video experience. E.g., you cannot pause it or raise the volume.

In YouTube, the controls of the video player display when you focus your attention on them. Move your attention away and they are gone. Attention-triggered controls display when you need them, so as not to be a distraction in your visual field.

Move your attention (e.g., cursor or touch) to the video. Notice that the control bar appears. You can now pause the video, change the volume, add subtitles, and more.

Now move your attention to the timeline. The thumbnail associated with the specific time appears above the timeline and, assuming sufficient data exists, a graph showing how much each moment has been replayed appears directly above the timeline. Moving along the timeline displays different thumbnails for each moment in time.

As web users, we are accustomed to using our attention to signal interest. We don't even think about it.

Unlike siloed web applications like YouTube, the Metaweb is active above every webpage. This enables the Metaweb to decentralize what's possible on webpages, giving participants control over their web experience. On YouTube, you use the controls they provide; their video player controls are embedded in the source code of the webpage.

On the Metaweb, any participant can add interactions and metadata to web content in a layer above any webpage. This is like being able to add your own controls to YouTube's video player. In this way, the Metaweb decentralizes the space above the web page.

The top three layers of the web cake (annotations, Web3, and computations) support both synchronous and asynchronous interactions. Asynchronous interactions include indirect coordination through stigmergy, where users leave traces of metadata and interactions for others to interact with. This allows for collaboration and information sharing on the web without the need for direct communication. Synchronous interactions, on the other hand, include real-time presence and interactions. This allows for real-time communication and collaboration on the web.

This book has three major parts.

Part 1, *Today's Web*, comprises Chapters 1–5, which establishes where we are with Today's Web, how we got here, and frames its very real and seemingly intractable problems. Chapter 1 examines the realities of the current user experience on Today's Web and provides a glimpse of what the future could look like. Chapter 2 explains how the Web has evolved over the past 30 years, describing the three major generations of the web: Web 1.0, Web 2.0, and Web 3.0, as well as the blockchain-centric Web3 movement that is unfolding now and the metaverse.

Chapter 3 exposes the astounding levels of scams and theft of data; delves into the disturbing levels of harassment, cyberbullying, and hate speech; children having unfettered access to pornography and other inappropriate online content; and describes the growing problem of false information, all on Today's Web. Chapter 4 introduces the Attention Economy and its attendant features such as polarization, censorship, and data exploitation which undermine trust and harm communities. Chapter 5 explains how search and social media diminish our cognitive freedom.

Part 2, *What We Need*, includes Chapters 6 and 7, which explores what the world needs now. Chapter 6 speaks to exponential technology, the UN sustainability goals, struggles with regulation, and the splintering of the web. Chapter 7 discusses how technology drives transformational change and a model for change in emergent systems.

Part 3, *The Metaweb*, comprises Chapters 8–16, which describes the Metaweb's emergence and its initial instantiation—the Overweb, its nascent wholeness, and some ideas for engaging in the space above the webpage. Chapter 8 talks about the big missing feature of the web. Chapter 9 is about the emergence of the Metaweb concept and presents first principles for the next level web. Chapter 10 presents cairns along the path to the Metaweb. Chapter 11 compares the Metaweb's foundational element—the bridge—with the hyperlink. Chapter 12 introduces the Overweb, the first instantiation of the Metaweb, and its pattern and building blocks. Chapter 13 explains how the Metaweb fixes the major problems of Today's Web. Chapter 14 shows how the Metaweb connects realities. Chapter 15 explains what becomes possible with the Metaweb. Chapter 16 talks about how you can continue your exploration of the Metaweb.

A crucial contribution the Metaweb will make to society is a real-time context and fact-checking engine—making the Web a veritable truth machine. Today's Internet provides very little context for web content because commercial sites avoid using links. The Metaweb unlocks deep layers of context, which supports a shared contextual view, collective sense-making, and safety while navigating the web. The Metaweb also advocates for content creators—individuals and organizations—to put their works on the blockchain, and can display metadata that establishes the provenance for any type of content, online and IRL. These help mitigate false information (a multi-billion dollar market itself), scams, and abusive behavior.

And this is but one example of what's possible. As shown in Figure 0.2, the space above the webpage is wide open. The Metaweb will provide untold opportunities for commerce as

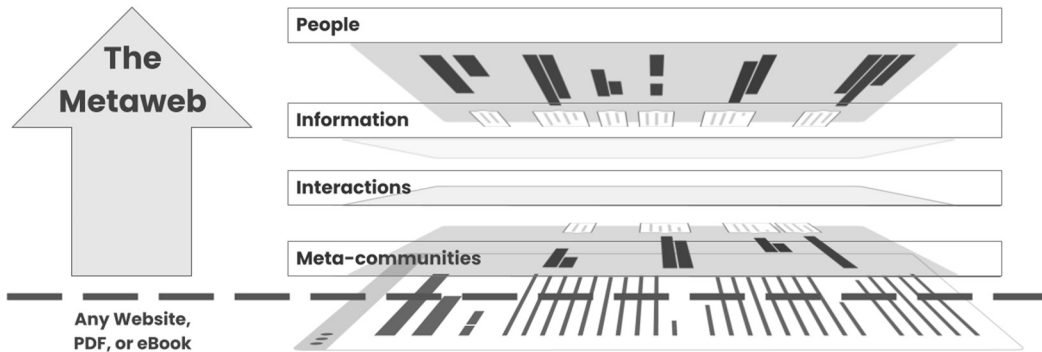


Figure 0.2 The Metaweb is the space above the webpage (adapted from Hypothes.is Overweb deck).

well as the connection, communication, and collaboration necessary to enhance our lives and shift humanity towards a regenerative future.

To provide a sense of urgency, we invite you to ponder the question of how nature sees us. Look at the animals running for their lives in Figure 0.3 and ask yourself:

If nature could see us in our entirety and complexity, what would we want it to see?

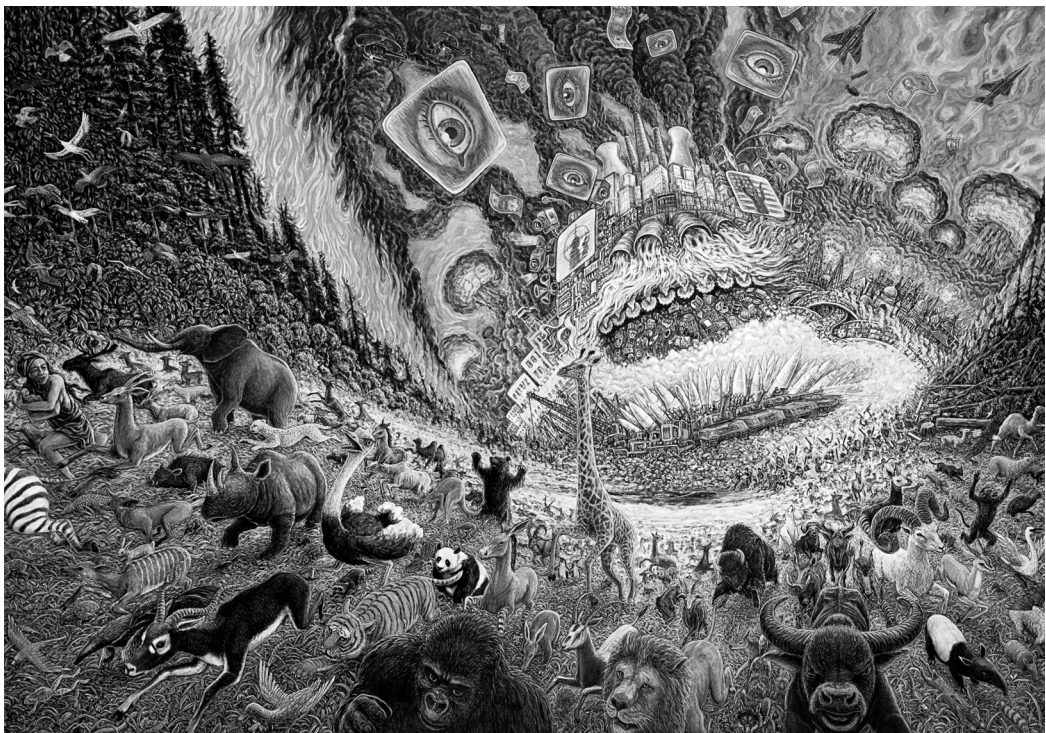


Figure 0.3 The March of Progress¹

Viva la Renaissance!²

NOTES

- 1 Mark Henson is visionary artist. The March of Progress is one of his most popular pieces. <https://permanent.link/to/the-metaweb/march-of-progress>
- 2 The Regenaissance is a portmanteau of the words regeneration and renaissance that signifies the planetary renewal movement to regenerate the Earth's ecosystems and create the New Earth.

Part I

Today's Web

It's Probably Nothing

It's probably nothing" is an ironic phrase used in the cryptocurrency and Non-Fungible Token (NFT) worlds when, in fact, "It's probably something.

As lovers of libraries, we often catch ourselves marveling at the Web's ability to retrieve a particular fragment of information in milliseconds.

Getting the exact piece of material you need in an instant—a citation, evidence, trivia, price, or location—can be quite gratifying. We often find it amusing to see people in a group, all on their phones, racing to find the best route, resolve a disagreement, or satisfy a bet.

Imagine how this capability would have changed lives in bygone eras, affording a massive competitive advantage, and maybe speeding up our development. Less than a century ago, such capabilities were the realm of magic or godly intervention.

People in their 50s have experienced an astounding improvement in online information retrieval over libraries and the Dewey Decimal card system. Ironically, the Web needs a "digital town library" to enhance our collective cognitive capacities. The digital library can be the container that acquires knowledge for collective sensemaking, meaning-making, and choice-making.

We thought a quantum shift in information retrieval and sharing would raise human consciousness and increase collective intelligence. The digital shift is in full effect. Yet we haven't noticed comforting changes in consciousness or collective intelligence. We know some people are working on the edges of consciousness, but overall humanity is in a sad state, with massive inequality and a beleaguered planet. And the Internet has morphed into a system for division and control.

At the Web's outset, humanity expressed a collective hope for the democratization of knowledge, leveling the playing field, and reducing inequality. Statistics tell a different story. We've become more unequal. In the past decade, wealth has become more concentrated, while the Internet platforms mushroomed in market capitalization. Meanwhile, the average consumer has struggled financially as wages in the US have stagnated since the early 1970s.

Certainly, there are many factors at play. Geopolitics, monetary and technology policy, and macroeconomic trends, to name a few. The Web is perhaps the most influential and pervasive. Modern society relies on the Web for communication, information and news, locations and directions, weather reports, virtual storefronts, and payments. Pre-web, some of this information was available via radio and television, conversations, libraries, and services. But it wasn't usually worth the cost and/or time. Most real-time information was unavailable.

Mobile phones enable the perfect recall of an enormous set of facts, claims, and articles in search engines. Not only do we have exponentially more searchable information, we also have individual feeds optimized for our engagement.

As we weave new information into our perceptual map, we certainly feel smarter and may nuance our thinking, which could increase the likelihood of positive outcomes. But what if it's false? Perhaps it's inconsequential. But sometimes false information can lead to ineffective choices and/or disastrous outcomes.

Context can be helpful to discern what information is potentially harmful and merits greater scrutiny. Beyond links, however, the primary context in webpages is the content, the author, and site publisher. But what if we trust—or don't trust—sources for the wrong reasons?

And besides, even if better recall with discernment improved our understanding and choices, this wouldn't equate to society becoming smarter. We would argue that collective intelligence increases only if others learn from our experience. Of course, others can learn from our lessons and words. But that is not a foregone conclusion.

Besides the lesson being accessible and comprehensible, others would need to be there, notice the lesson, and surmise its importance within their perception of reality. As more activities go online, more learning opportunities will shift to digital spaces.

The digital world enables information sharing through social media and blogs. But unless we're an influencer, few people see our posts. Thus, the potential for others to benefit from our lessons is low, having a negligible effect on collective intelligence.

Were we an influencer, others could learn from our posts, assuming our content stays within the platform's acceptable narrative, or we own the platform. Since platforms popularize most influencers, the platforms own the audience. YouTubers, for example, do not have the phone numbers or email addresses of their subs. If a post conflicts with the platform's community guidelines or otherwise triggers its internal moderation process, much fewer subscribers will see the information.

Being aware and seeing our content is the first step in the learning process. Beyond awareness, the learner must ingest, process, and integrate the content to activate the learning process. And if they do not act upon it, it may not stick.

In sum, discernment of false information and learning from others are both challenging on Today's Web. Clearly, instant information recall is not the harbinger of a forthcoming golden age of collective intelligence or wisdom. At least not in its current state.

DO WE NEED EACH OTHER TO LEARN?

Humor us. There's an elephant in the room. Wherever you find yourself.

What would you say if we asked you to describe the elephant?

We may all think of different things. You may think about the subject of this book or a geopolitical issue. Or perhaps a pressing issue that needs your attention. Some of you no doubt are thinking about actual elephants—perhaps in clown suits—or even animated elephants floating by with big, floppy, Dumbo ears.

How you think about the elephant depends on your situation, perceptions, and life experience. Some haven't even heard the term, while others may use it frequently.

It could be anything. A problem domain, an issue we're wrestling with, or something we're avoiding. Absent coordinated thinking, we'll come to different conclusions about what's true.

The blind men and the elephant parable originated in India around 1500 BCE. It has since traveled around the world. In the story, six blind men encounter different parts of an elephant, as shown in Figure 1.1. Each understands reality based on their limited experience



Figure 1.1 Six blind men and an elephant.

and perspective. For example, the blind man who touches the trunk thinks it's a snake. The blind man who touches the tusk thinks it's a spear. They all mistake it for something else.

Despite sometimes-contradictory signals from sensory perceptions ("but it feels like a spear!") and life experiences, facts and truth do, in fact, exist and are knowable. The story's subject is actually an elephant—not a spear, snake, wall, tree, or fan.

Retellings of the story differ primarily in how they describe the elephant's body parts and whether the blind men discover their disagreements, how violent conflicts become, and any resolutions. In some versions, the blind men collaborate to "see" the full elephant. In others, when a sighted person describes the entire elephant from all perspectives, the blind men understand they were all partially correct, and all partially wrong.

While one's subjective experience may be true, it's not necessarily the entire truth. The parable shows that although one's subjective experience can be true, that experience may not account for other truths or the totality of truth.

Rather, the story provides a three-step path to truth. One, gather relevant information about the subject, including direct experiences, and the accounts of and inferences from the direct

experiences of others. Two, resolve relevant contradictions and paradoxes. And three, discover “the Why”—the reason for being that reveals how the subject interconnects within reality.

We can discern “the Why” through relationships. How does the subject relate to its environment? Each part of the elephant, for example, enables its survival as a living organism. The elephant’s primary use for tusks is fighting. But they’re also useful for foraging, digging, stripping bark, moving things, and attracting females. When we understand how the elephant’s parts aggregate into an entity distinct from and related to other aspects of the ecosystem, we know it’s an elephant.

Thirteenth-century Persian mystic and poet Rumi included “The Elephant in the Dark” in his *Masnavi*, an extensive poem that is one of Sufism’s most influential works. In Rumi’s retelling, a group of Hindus observe an elephant in a dark room. Each man uses their hands to feel the elephant. Depending upon where they touch, they believe the elephant to be a waterspout (the trunk), a fan (an ear), a pillar (a leg), and a throne (the back). Rumi’s words illuminate the limits of individual perception, “The eye of outward sense is as the palm of a hand, the whole of the object is not grasped in the palm.”

This parable underscores the importance of going beyond individual experience, examining purported accounts of experience or truth, and considering the preponderance of evidence and perspectives. The Coleman Barks translation ends, “If each of us held a candle there, and if we went in together, we could see it.”

A WEB LACKING CONTEXT

As with the elephant, gathering perspectives is the first step for constructing insight or truth. One can find many perspectives on the Web. Like a blind man’s direct experience, access to one perspective on a webpage is a start. But it’s most useful with other perspectives.

Today’s webpages, however, lack other perspectives. They are the perspective of the author. Besides outbound links, the main context is the author and publisher. Below we discuss how context shows up in the first two generations of the Web: Web 1.0 and Web 2.0. The following chapter explores all three generations of the Web, including Web 3.0, Web3, and the metaverse.

In Web 1.0, links were commonly used to cite sources or provide context about topics. Web surfing was a popular activity. You browsed the Web using links to navigate from one webpage to the next. You read or at least skimmed each page.

The Web 2.0 shift to social networking in the early 2000s birthed the Attention Economy. Emerging social networks and e-commerce sites began monetizing attention, which required them to keep users on their site so they could continue serving them ads and extracting their data. Commercial sites began avoiding outbound links in which they didn’t have a financial interest, leading to the eventual demise of web surfing.

Web 2.0 social platforms rely on content feeds to hold your attention. Feeds provide sufficient information to discourage users from clicking through to articles. The feeds also disguise advertisements as content. Sites had shifted from being cairns along the path of knowledge on Web 1.0 to becoming destinations for engaging with content and buying things on Web 2.0.

Internet platforms, such as Facebook and Twitter, use sophisticated algorithms to increase engagement and sales. The algorithms process web activity data to optimize each user’s personal feed for the platform’s agenda. Some platforms—including Facebook, Google, and Twitter—allow *users* to log in to other websites with their platform credentials. This enables the platforms to track user activity across the Web and to enhance their influence on user behavior.

Feeds are a popular way to find content. They display an endless list of content (and advertisements) that may interest the user based on their preferences and activity history. Feeds are lists of content previews with an image and text. Each preview also has reaction icons to “like” and share content, and make comments. Feeds optimize the display to increase the likelihood the user will engage with the content without clicking through to an external site. Some display the key metrics (e.g., likes, shares) of the post.

Facebook is a master at this. It provides a simple way for *users* to engage with content and to signal their alignment or opposition with minimal reading. A Facebook preview of an article includes the poster’s comment, an image from the article, several lines of text from the article, and engagement icons. Their previews provide just enough information for users to engage without opening the article. Facebook wants *users* to like, comment, and share content without leaving their site.

According to a 2021 study, Facebook’s click-through rate (CTR) for articles is just 2.5%, based on 6.5 million organic shares from 500 publishers worldwide.¹ Average CTR varies widely between content categories. Sports are highest at 4.5%. International news and foreign affairs are lowest at 1.8%. The study emphasized that AI-curated Facebook pages achieved CTRs identical to human-curated pages. This suggests that AI will be more cost-effective than manual social media posting. This advantage will increase as AI technology improves.

The Web 2.0 shift from web surfing to feeds has many implications that we will discuss later. For now, feeds clearly lead to a more shallow web experience than web surfing. Now people just scan the list of content prepared for them. If an image or word catches their attention, they look deeper and read the little text. They often give likes, comments, and shares, and return to the feed, without actually reading the article.

Facebook users usually engage with articles without reading them, much less seeking additional context or information about the article’s claims. Whereas web surfers would read enough to find links and click through, Facebook *users* open fewer than 3% of the articles they share.

Engaging without reading has downsides. The user doesn’t know the article’s quality, its claims, or the Why. Without this level of scrutiny, they’re flying blind and could inadvertently spread harmful misinformation. This problem exacerbates on Twitter, where false information is more viral than the truth.

Let’s say you’re scanning a Facebook feed. Much of your feed is clickbait. People obviously aren’t reading what they share.

Nevertheless, an image sparks your interest. You read the comment and the three lines below the photo. Wanting more, you click through to read the article. A claim in the article is surprising, and you’d like more information. But you’re stuck because there are no out-bound links or previous stories. You’re at the Internet equivalent of a dead-end street. Unfortunately, much more common nowadays than before.

A DAILY ACTION OFTEN TAKEN FOR GRANTED

Prompted by a thought or question, we may start our online session with a search engine. Or simply typing a site’s name in the URL bar. We may follow several sites. Some people use bookmarks for this purpose. But if you’re like us, you have a bunch of tabs open.

Alternatively, we may receive a message via email, text, or a messaging app. Messages may come from trusted contacts or someone we don’t know. If it includes a link, we can click through to get more information. Hopefully, it’s not malicious.

As in the previous section, we also find content through our personal social feeds. Or those of other people and groups. The social feeds display an ordered set of content previews selected by AI algorithms based on personal or group preferences and/or patterns of engagement with previous content. We can click through the content snippet to see its source page.

If the page has links, we can follow them. Nowadays, most links are to different pages on the same website or pages the author has a financial interest in promoting.

Consider a typical news story. The article may embed a tweet, but doesn't have other external links. Internal links include navigation, related articles, and previous stories. You read the article. You can follow the provided links but they loop through the site. We call this an information silo. You're stuck in one perspective.

As you skim the source page, a specific claim in the text captures your attention (Figure 1.2). It could be something shocking, novel, or counter-intuitive; perhaps something that we had not considered, or that surprises or outrages us, proves our point, or illicitly confirms our biases.

Maybe you're wondering about the claim's source or you think it sounds off-base or too good to be true. Maybe you're wondering where else it has happened, under what circumstances, and what studies confirm or contradict the claim.

But unfortunately, claims rarely have links to substantiating or discrediting information. Again, we've reached the web equivalent of a dead-end street.

We can re-read the claim and the contextual information on the page. If the page allows comments, you can read them and comment yourself. We can select the snippet, as shown in the figure, and copy it to the clipboard or right-click to start a search, print, or select a browser plugin to operate on the snippet. But that's it. We don't have other options.

Alas, today's flat and static web is prone to dead-ends. Each page has one main layer of content served from the web server that hosts it. Some pages have pop-ups (or modals) with brief explanations, subscription forms, videos, and cookie acknowledgments. Sites with



Figure 1.2 A claim in an article is selected (courtesy of Wikipedia).

pirated content such as movies and shows often have pop-up ads. These may come from content or ad servers. No one likes pop-up ads.

As an aside, we often wonder who's with us on the same webpage. On a popular page, there could be hundreds of people reading the same text, looking at the same images, and/or watching the same video. Being on the same webpage speaks to a simultaneous interest in the content. Were we in a café and we noticed someone reading the same book, we might ask what they think about the book. Our operating assumption would be that there could be some mutual benefit in interacting.

Today's Web, however, is isolating. You can't know who is on the same webpage despite the site owner keeping records of everyone on their site via cookies. Cookies are small text blocks created by web servers when users browse a website. When visiting a commercial site, a pop-up will display along the bottom of the screen that explains how the site uses cookies. You must acknowledge this (or adjust the cookie settings) as a condition of using the site and making the pop-up disappear.

These pop-ups implement the EU's General Data Protection Regulation (GDPR). GDPR sets guidelines for the collection and processing of personal information from individuals who live in Europe. (We discuss GDPR in the section of Chapter 8 called Europe Aims to Regulate the Web.) The web browser places the cookie on the user's computer or mobile device. The cookie tracks your activity during the session. Websites dedicate sessions to each visitor rather than convening group sessions, which prevents us from knowing who else is there.

Back to the article's claim. If we're still curious about the claim, we may invest time in research. Competing interests, time concerns, and the lack of effective and impactful ways to share information can, however, discourage further exploration.

We can weigh the potential value of new information, our sense of the effort involved, and our other priorities. If the "opportunity cost"—the perceived value of the best alternative use of our time—seems less than the expected value of the information we expect to find, we'll probably keep going.

Depending on our orientation towards the web, we might turn to search and/or social media for more information. Many people discover content on Twitter, Facebook, and LinkedIn feeds and through topical groups that they have joined. Scrolling feeds—of a group, influencer, or your own—is a passive way of discovering information, like fishing with poles rather than spears. We prefer search to find specifics about a claim.

Searching, either through in-platform content or external sites like Google or Bing, is an active way to discover information. In-platform search includes user posts and different sections of the platform. For example, Facebook search covers groups, pages, events, the marketplace, and more. Facebook personalizes your content feed to increase engagement. This search only covers content posted on Facebook.

Google is the 900-pound gorilla with over 90% of the search market. Google Search includes video, images, maps, news, shopping, books, flights, and finance, among others. They personalize search results to increase engagement. Their monopoly is highly profitable. Their parent company, Alphabet, is among the world's most valuable companies.

Still, search engines access only a small part of the Web. The 4% of the Web available through search engines is called the surface web. The surface web includes Wikipedia, ESPN, and news sites, for example.

Another 95% of the web—the Deep Web—includes websites that are not indexed by search engines and hence are unavailable in a simple search. These pages may require you to sign in or complete a CAPTCHA challenge-response test to determine whether the user is human. Examples of Deep Web sites include email, online banking, and subscription-based

services; video streaming services like Netflix or HBO; company intranets; educational or library websites; government-related pages or legal documents; and medical records.

Another approximately 1% of the Web is the dark web.² The dark web (or darknet) includes hidden sites with TOR (The Onion Router) URLs that are impossible to remember, guess, or understand, and require a special browser to access. Most TOR pages are encrypted and have anonymous hosts.

The darknet hosts sites for illicit drugs, weapons, hacked credit cards, child pornography, and unauthorized scans of books and publications. The dark web also provides a secret communication channel for journalists, human rights activists, political activities, and the military, as well as storage for government intelligence reports, political records, and other sensitive data.

Despite search covering just a small amount of the Web, like fish in water, most people automatically move the cursor to the address bar and start typing search terms related to the claim. Search integration into the browser address bar is seamless. We try to guess words that webpage authors might have used in the article or the page's metadata (e.g., title, description, or tags).

Browsers are a Profitable Enterprise ... Because of Search

In 2020, The New York Times reported Google pays Apple an estimated \$8–12 billion per year for making Google the default search on its devices.

In October 2021, the search engines DuckDuckGo, Ecosia, Lilo, and Qwant called on EU legislators to end Google's "hoarding of default positions." The search engines want to be readily accessible and "only one click-away" in browser settings. DuckDuckGo alleged that changing the default search engine on Android devices takes users 15-plus clicks. In an open letter to the EU, the rivals said "Google would not have become the overall market gatekeeper they are today without years of locking up these defaults."

Search provides most of Google's revenues, with \$104 billion in "search and other" revenues in 2020. Google is also a market leader in online advertising, with about 38% of global digital ad spending in 2021.

Despite their success, we think Google will be far less prominent by the decade's end absent a major course correction. Why? We don't see them being the pre-eminent content delivery apparatus much longer, as better options appear and we flip between realities. Also, rather than ubiquitous, we think non-contextual display advertising will be a niche in the next Web. If, somehow, Google could maintain its dominance through the next iterations of the Web, we'd be very concerned about the future of democracy.

As we enter characters, by default, Google provides auto-completion and suggestions. Matches display in a dropdown menu and the top result displays in your address bar, adjusting as you type.

For example, when we started typing "nf" in Chrome's address bar, it generated four search term options in a dropdown menu: "nft scams," "nft sales leaderboard," "NFL," and "nft."³ When we added the t on nft, the dropdown options morphed to: "nft scams," "nft sales leaderboard," "nft que es," and "nft marketplace." Most of the suggestions were our previous search terms. NFL came up because it's a popular search term and "nft que es" because we're in Mexico at the moment.

When we type “rus,” the dropdown shows “russia plans to attack ukraine,” “russell 2000,” “russian currency,” and Russia (the country).

Google generates these auto-completions and suggestions using AI based on proprietary algorithms that include popular searches and the user’s browsing history.

At some point, you click Enter to accept the auto-completion, select one of Google’s suggestions from the dropdown or finish entering your own words. You have submitted search terms to Google’s global brain, which is an information organizer, surveillance apparatus, and attention broker.

Google then does its magic and almost instantaneously serves up a search engine results page (SERP). At the top, Google displays the number of results retrieved by the search algorithm, and the fraction of a second that it took to generate them.

The extent of what most people see, the initial result page displays a handful of the top organic listings below paid listings. Paid listings include sponsored text ads, aka Google Ads, or other advertisements, such as ads that display on the top or side of the page and/or SERP features.

The results page has distinct features that provide detailed information without requiring one to click through to a webpage. Like other Internet platforms, Google aims to keep people on its site as long as possible, increasing the likelihood that they will click on an ad.

Google continually tests at least 15 SERP features. These include shopping results, top stories, “people also ask,” a “Local Maps Pack” of nearby sources for related products, and options for adding/editing filters or search terms.

Often Google can resolve your search terms to a definition, topic, calculation, or an answer for a common question (e.g., how to do X using product Y). These display as a featured snippet at the top of the page. Featured snippets are based on an AI-powered knowledge graph that contains 500 billion facts about five billion entities, which debuted in 2012. The definitions of words are from online dictionaries. The topics are from Wikipedia. If you click through on the featured snippet, Google now highlights the specific text on the source page.

Artificially Intelligent

Google’s ongoing investment in AI since the early 2000s has positioned the company as a leader in the field, with AI being at the forefront of the company’s strategy. The company’s flagship project, Search, has benefited greatly from AI, with the introduction of BERT, a transformer model that revolutionized our understanding of human language. With MUM, an even more powerful AI technology, Google has been able to provide critical information and crisis support in multiple languages.

Google’s latest AI technologies, such as LaMDA, PaLM, Imagen, and MusicLM, are pushing the boundaries of what is possible with AI, creating new and exciting ways to interact with information. From language and images to video and audio, these technologies are the beginning of Google’s full integration of AI into all of its projects. The company’s commitment to AI is evident, and with its latest advancements, Google is paving the way for the future of AI and its role in our daily lives.

As featured snippets have become more prevalent in search results, zero-click searches on Google have seen a steady increase. In 2020, 65% of Google Searches ended without a click to another web property—up from just 50% in June 2019. Google is further

integrating AI into Google Search and its other products. We expect zero-click searches to grow in an AI-enhanced Google Search where chatbot responses are shown above organic search results.

Higher-ranking pages have the search terms in their metadata rather than just relevant content. But pages contain dozens of ideas. Each idea could merit its own set of metadata. These ideas are invisible to the search engine unless they match keywords. Hence, search engines are ineffective at finding content that's indirectly related to the page's metadata or not using the exact words of the search terms.

The listings for the recommended sites on the SERP point to a page URL. Clicking through takes you to the top of the page. This requires you to review entire pages to surmise what content relates to your search. This can be problematic for longer pages and documents with diverse content.

The first page has, on average, the first 8.5 organic search results (out of millions). A 2020 study showed that the first position in search results has an average click-through rate of 28.5% (Figure 1.3). The average CTRs of the second and third positions are 15% and 11%.

Some search terms contain valuable information after the first several pages of search results, but they get minimal click-throughs from search. We call this the long tail of search.

Unsurprisingly, enterprising companies battle over the top search positions in commercial categories. We don't know how top search results reflect alignment with our search terms versus superior search engine optimization (SEO). SEO improves an enterprise's website to increase visibility in search engines. Unfortunately, SEO detracts from the overall quality of search results, as content made for links shows up after the first several pages.

The better visibility in search results, the more likely they will attract prospective and existing customers to their offerings. Hence, companies spend stacks on services promising first-page ranking for specific keyword combinations. Experts expect the SEO services

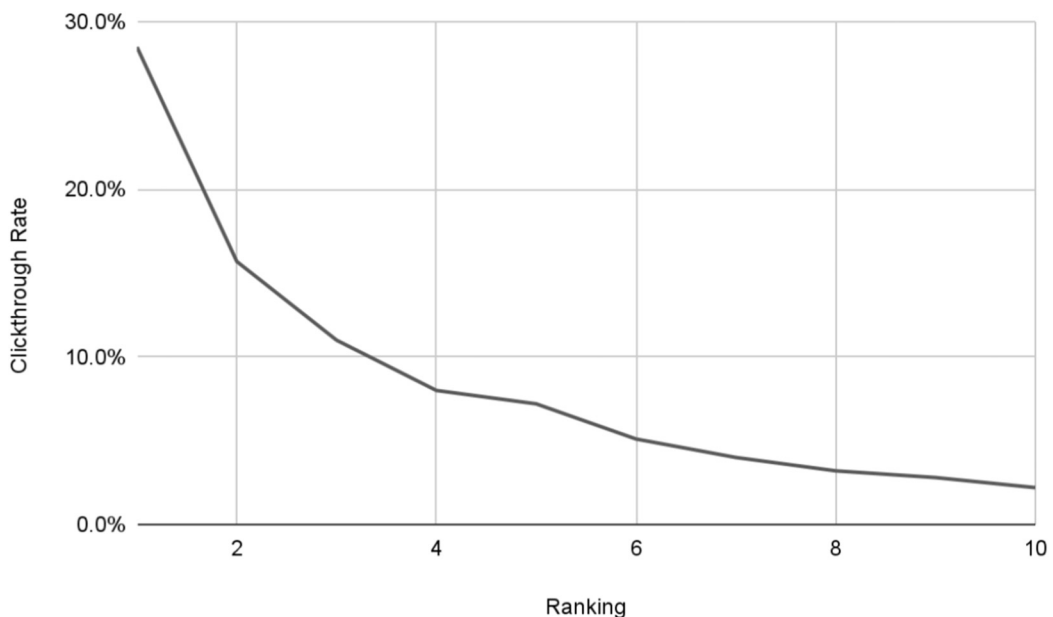


Figure 1.3 Decreasing click-through rates (CTRs) by search ranking order.

market to grow from \$47 billion in 2020 to \$103 billion in 2025, at an annual rate of 17%. SEO is growing in importance.⁴

21% of searches have over one click on Google's results.⁵ Since the search results link to a page rather than a specific idea, you need to skim each page for related content. This can be time-consuming, as longer pages rank higher in Google's search results; in 2020, the average Google first page result contained 1,447 words.⁶

Let's say you're atypical. You look at the top three or even ten pages (which still only reflect a tiny fraction of the results). You revise search terms to get better results. Perhaps spending hours searching within your interests.

Suppose you find something interesting about the claim. What can you do with it? How can other people who read the same article benefit from what you learned?

Usually, it's best to write a blog post or make a YouTube video. But that's time-consuming. And it's unlikely your article will be high enough in the search results for others to notice.

Even when a popular blogger posts on breaking news, there's latency between the time of the post and when Google's "spider" discovers the post and makes it available in the search index. Experts say, although it varies, the average crawl time can be anywhere from several days to a month, depending on many factors, including the site's authority.⁷ Hence, people rely on feeds like Twitter to get breaking news.

Also, unless your article gets a lot of incoming links, it is unlikely that it will rank high in the search results. If you don't have an audience, the potential effect is limited and fleeting. So, it may not make sense to write a post or make a video.

Most people will tweet it or post it on a messaging app or the wall of a social platform. They may copy the text and paste into a Google doc, Roam, or Evernote, text or email it to themselves or friends, bookmark it, or just leave it open on a tab. Or maybe none of the above.

Organizing information via pages with links, search index latency, and the lack of effective options to archive and share discoveries, make it difficult for search users to learn from and build upon each other's research. Learning from one another is a prerequisite to collective intelligence.

Meanwhile, Google hoards sufficient information about you to autocomplete your search terms, make relevant suggestions, and personalize your search results. They also know which search results we are clicking through. Collectively, the same searches happen over and over again, enabling Google to deliver trillions of search advertising impressions and to make billions of dollars. Yet, collective intelligence has flatlined, stunted by a discovery paradigm that has only a few clear winners—Google itself, the SEO industry, and the platform feeds, which provide the primary content discovery alternative.

If external search does not yield results, we may decide to search on a specific platform (e.g., Reddit) or within a group. We might post in a forum. Or more likely, we'd lose interest and focus on something else.

BUT COULD THERE BE ANOTHER WAY?

Remember the dead-end webpage with the claim? We wondered if it was true. Social media and search were our only options, as there were no relevant links to external information. We were alone with a claim on a flat and static webpage.

Before we visualize the Web's hyper-dimensional future, let's travel back in time to Europe in the early 1800s before cakes had layers. No one knows what a layer cake is. Let's say

you're an evangelist for multi-layer cakes. You go around town describing a layer cake. But everyone thinks you're daft. They ask, "why would I do that?" You describe a world of flavors, textures, and displays. Somehow, layer cakes have reignited a creative streak within you and you revel in the full artistic and practical possibilities.

One day, you get an invitation to a high society wedding. After the ceremony, the servants bring out several large flat sheet cakes. How boring. You lean over and tell your friend that the most celebratory occasion of one's life merits a very special cake. They don't get it. So you grab a notebook and draw a three-layer wedding cake. The bottom layer is for the party. The middle is for after the wedding. And the top for later. It's so clear ... to you.

On a soapbox in the town square, you encourage people to invest in the layered future of cakes. As you talk, your friend holds up portraits you commissioned of newlyweds and wedding cakes. You promise an affiliate fee for anyone who can place your layer cakes in a bakery. Still, the townsfolk are not receptive. Undismayed, you carry on with lectures about the future of cake wherever they will have you.

The world changes in 1735 when a master baker in Frankfurt, Germany, invents the Frankfurt Crown Cake. It comprises two or three layers of sponge cake filled with jam and buttercream and frosted with more buttercream. Adventurous European bakers adopt the layer cake. Wedding cakes become a thing as you had envisioned. The town folk now see you as a visionary baker.

Today's Web is like the early 1800s in the world of cake. Until then, cakes had one layer. But in the mid-1800s, someone made a multi-layer cake, changing the paradigm. It was now possible to have many layers, creating new categories of layer cakes and, later, wedding cakes. Bakers could imagine a world of possibilities for layers to enhance taste, texture, and display.

Similarly, Today's Web is the first layer of the cake.

Let's step back. Recall the four-layer web cake in the first figure of the Introduction. The bottom layer is Today's Web. Above that, it has a meta-level called the Metaweb, which comprises three layers: Annotation, Web3, and Computation.

Let's put things in perspective. From the top, imagine the cake as if it were a laptop screen or a book. In fact, imagine grabbing the web cake by its sides and propping it up at an angle so that it is at a comfortable reading angle.

The bottom layer is the Web of today—the page content and any embedded ads or pop-ups.

Now imagine a transparent second layer of annotations. In its most simplistic and common form, notes display as overlays to related pieces of text in the bottom layer. If, for example, a piece of text has annotations, it lights up. Clicking through displays the notes related to the text.

These first two layers—content and annotations—were integral to early visions of the modern computer and were included in the first widely distributed web browsers, Mosaic and its successor Netscape. But Netscape removed annotations from the browser in 1995 under competitive threat and lacking a place to store them.

The emergent third layer of the cake is a Web3 wallet but over every webpage. Web3 wallets, also known as decentralized wallets, are an essential component of the decentralized finance (DeFi) ecosystem and enable their users to interact with several thousand decentralized applications (dApps) built on blockchain technology. These dApps take on many forms, including DeFi (decentralized finance), decentralized exchanges (DEXs), decentralized autonomous organizations (DAOs), "play-to-earn" video games with token rewards, and NFT minting sites.

Some of the key capabilities that Web3 wallets enable include:

- **Asset management:** Web3 wallets allow users to store, send, and receive cryptocurrencies, tokens, and other digital assets in a secure, decentralized manner.
- **DeFi transactions:** Web3 wallets enable users to participate in various DeFi activities such as yield farming, staking, lending and borrowing, and trading on DEXs.
- **dApp interaction:** Users can use their Web3 wallets to interact with decentralized applications and access their features and services.
- **Control over funds:** In a Web3 wallet, the user holds the private keys to their assets, giving them full control over their funds without the need for intermediaries.
- **Privacy:** Some Web3 wallets enable users to transact anonymously or with increased privacy.

The Metaweb expands the availability of Web3 wallets from thousands of dApps to the entire Web, over 2 billion sites. Thus, on the Metaweb, you can transact without a middleman above any webpage.

Regarding the web cake, Web3 wallets display as sidebars or pop-ups, usually on the right side. We discuss Web3 in greater detail in Chapter 2, Three Generations of the Web.

You're among the first to know about the cake's nascent top layer. The transparent top layer places code above related content on webpages, enabling computations that present information and options to participants, handle their interactions, and present the results. This layer allows real people, information, interactions, and even communities to have a presence above the webpage. This also enables participants to have their own personal AI assistant above every webpage. This computational layer is where the Metaweb comes alive. We explore the Metaweb in the last and largest section of the book.

For the past three decades, humanity has mostly confined its thinking about the Web to the content layer. Few outside of annotation and Web3 have thought about layers above the webpage. And many have unnecessarily limited their thinking to text-on-text. But now you know what is possible. This book just installed the novel notion of web layers in your mind. Multi-layered web experiences are everything.

You may now realize you can become a baker in this new computational layer, the Metaweb. Imagine a world of possibilities for overlaying information, interactions, and transactions above Today's Web. Everything changes. The Web now includes a meta-level above the webpage that dynamically displays related information and interactions available to the viewer. A hyper-dimensional web over the web.

Ok, But What Do Layers Really Look Like?

You Already Know What Multiple Layers Look Like. You Just Don't Realize It.

Drum Roll ... It Looks Like Today's Web!

Internet platforms and applications use your attention (e.g., the location of your cursor) to trigger the display of contextually relevant information and interactions without reloading the page. You have seen this with pop-up videos, web applications like YouTube, and tool tips. Anything new that appears over web content is conceptually and often implemented in a new layer. We call this attention triggering.

You probably haven't heard much about attention triggering, despite all the attention on attention and the prevalence of attention triggering. Internet platforms, website as a service, UX/UI specialists,

corporate site builders, mobile app developers, and advertisers all rely heavily on attention triggering to improve the web experience. Attention-triggered elements on the web page change their behavior—appearing or moving—in response to the user’s attention. On a laptop or desktop, attention is represented by the cursor; on mobile, it’s touch; in virtual reality, it’s line of sight.

But let’s try it out. Navigate to YouTube and activate a video. Notice that, at the moment, you cannot see any mechanisms for controlling your experience of video. You cannot pause it or raise the volume.

In YouTube, the controls of the video player display when you focus your attention on them. Move your attention away and they are gone. Attention-triggered controls display when you need them, so as not to be a distraction in your visual field.

Move your attention (e.g., cursor or touch) to the video. Notice that the control bar appears. You can now pause the video, change the volume, add subtitles, and more.

Now move your attention to the timeline. The thumbnail associated with the specific time appears above the timeline and, assuming sufficient data exists, a graph showing how much each moment has been replayed appears directly above the timeline. Moving along the timeline displays different thumbnails for each moment in time.

As web users, we are accustomed to using our attention to signal interest. We don’t even think about it.

Unlike siloed web applications like YouTube, the Metaweb is active above every webpage. This enables the Metaweb to decentralize what’s possible on webpages, giving participants control over their web experience. On YouTube, you use the controls they provide; their video player controls are embedded in the source code of the webpage.

On the Metaweb, any participant can add interactions and metadata to web content in a layer above any webpage. This is like being able to add your own controls to YouTube’s video player. In this way, the Metaweb decentralizes the space above the web page.

As a meta-level that overlays the webpage, the Metaweb is a browser overlay, which can be accessed via Metaweb browser extensions, Metaweb-enabled mobile browser applications, and Metaweb-enabled browsers.

Browser overlays offer valuable features that existing browsers cannot provide. One such feature is computational continuity between web sessions. While cookies enable computational continuity between site visits on Today’s Web, browser overlays allow for computation continuity across all sites through overlay applications. This opens up possibilities that cannot happen on Today’s Web, such as data capture, applications, and AI working on your behalf and at your discretion across sites and web sessions.

As you may imagine, an overlay application that is active and operates over every relevant webpage can powerfully augment the traditional browsing experience. For instance, overlay applications can work seamlessly across multiple sites in a browser session; computational continuity enables complex tasks that would be impossible with traditional browsers. Possibilities include scam detection applications, custom content filters, and personal AI assistants that accompany you wherever you go on the Web.

In addition, browser overlays offer enhanced privacy and security, making them ideal for people who are concerned about online safety. With an overlay, you can have more control over your data and its use, as well as better protection against malicious attacks and other online threats.

A RECIPE FOR COLLABORATION

We often consider ourselves to be the highest beings on the planet, yet nature holds many secrets that can teach us valuable lessons. For example, the humble ant has much to teach us about collaboration and organization.

Ants use a mechanism called stigmergy to work together and accomplish tasks without direct communication. Stigmergy is a form of indirect coordination where ants leave traces in the environment that stimulate future actions. These traces can be physical, such as pheromone trails, or informational, such as the position of food. By responding to these traces, ants are able to coordinate their actions and work together as a highly organized and efficient unit.

One of the wonders that ants accomplish with stigmergy is building intricate nests and colonies. Without any central leader or direct communication, ants construct complex structures with precision and efficiency. For example, leafcutter ants farm fungus and build underground chambers that can span several acres. Similarly, termites build towering mounds that can reach up to 30 feet tall.

Another example is the foraging behavior of ants; they find food and bring it back to the colony with remarkable speed and efficiency. Ants lay down pheromone trails to food sources far from the colony for other ants to follow. They form foraging lines that can stretch for hundreds of meters, enabling them to exploit food sources in ways that would be impossible for an individual ant.

Cairns, signs, and social media posts are examples of stigmergic communication, as the communication is mediated by the environment, rather than through direct interactions between individuals. The complexity and sophistication, however, of stigmergic communication in ants is much greater. Ants convey a wide range of information and coordinate highly complex behaviors through stigmergy, while cairns, signs, and social media posts are typically used to convey relatively simple information.

Stigmergy itself is an example of biomimicry, an interdisciplinary field that involves studying nature and its processes to find inspiration for designing human-made systems, products, and technologies. It's a holistic approach to design that looks at natural systems as a model, measuring stick, and mentor for the design of human systems. The field is relatively new but gaining momentum in the last couple of decades.

It's a way to create more sustainable and efficient systems by learning from the natural world, which has been fine-tuning its processes for millions of years. Biomimicry can be applied to a wide range of fields, from product design and manufacturing to energy generation and storage, to transportation and urban planning. The Metaweb applies biomimicry to the design of the next level of the Internet.

The Metaweb mimics the ants, using virtual stigmergy as a mechanism for indirect coordination. The Metaweb itself is a stigmergic structure that allows us to leave traces in the environment, like the ants, enabling us to work together and accomplish tasks without direct communication. With smart tags and bridges, we can coordinate our efforts and collaborate in ways that are far more efficient than current methods. By embracing the principles of stigmergy, we can unlock the true potential of the Web and build a better future for humanity.

On the Metaweb, the top three layers of the web cake (annotations, Web3, and computations) support collaboration through asynchronous interactions via stigmergy. In these layers, participants can leave traces of metadata and interactions for others to interact with. In the Metaweb, participants can receive rewards for leaving and responding to traces. This enables and incentivizes collaboration and information sharing on the Web without the need for direct communication.

Asynchronous collaboration via stigmergy allows for large-scale participation and collaboration on a project without direct communication. This makes it possible for a vast number of people to contribute and collaborate on a project, increasing the likelihood that the project will achieve a diverse range of ideas and solutions.

Asynchronous collaboration allows people to contribute and collaborate on their own schedule, regardless of their location or time zone. This can be particularly beneficial for global challenges that require participation from people around the world.

Indirect collaboration via stigmergy allows for continuous progress on a project, even when direct communication is not possible. People can contribute and build on each other's work, even when they are not online at the same time.

The top computational layer of the web cake also supports synchronous interactions, which include real-time presence and interactions. This allows for real-time communication and collaboration on webpages. Real-time communication and interactions allow for faster problem-solving and more efficient decision-making. It enables for direct communication and coordination that can be critical in tackling challenges that require quick action and resolution.

Combining real-time communication and interactions with asynchronous, indirect collaboration via stigmergy provides a comprehensive and versatile approach to tackling global challenges. It allows for the coordination of large-scale participation and the scalability of ideas and solutions, while also providing the flexibility and continuity needed to address these challenges.

Overall, having an environment that supports both asynchronous and indirect collaboration via stigmergy and real-time communication and interactions can help to increase the effectiveness and efficiency of thinking, learning, and building knowledge together.

Let's now picture people with a presence in this collaborative meta-layer. When you go to a webpage, you can choose to become visible in its canopi. The canopi is a sidebar, like a rooftop speakeasy above the webpage. If you're visible in the canopi, you see other visible people. A list displays each person's avatar, name, and how long they've been there.

Hovering over the avatar of a "visible" person displays the information they make available to people they don't know. For example, in the canopi of an NFT marketplace, @wildarrow stands out. Like many web users, she has a pseudonym, preferring not to reveal her actual identity. Her avatar is a cartoon image of a femme archer. Her profile reveals that she is "hodling"—a misspelling of "holding" that refers to a buy-and-hold investment strategy in the world of crypto—several digital assets. We see as well that she has several certifications and verifications from trusted authorities. Her profile says she has a secure launchpad for digital assets that verifies all its projects.

Even though we don't know @wildarrow, we know she's a real person in good standing. She is not a bot, a fake account, a serial abuser, or a throwaway account. We know she's real because this layer required her to prove her humanity. She is visible in this layer because she is in good standing.

Perhaps we want to launch a digital asset project and are interested in learning about her service. We start the communication by clicking her name and entering a message.

@wildarrow receives the notification of our message. She clicks through to her chatrooms; our conversation request is at the top of the list. She clicks our message, which displays a stranger danger warning. Like Facebook, when she replies, a chatroom appears for us. She drops her link and we chat as we visit her site. @wildarrow tells us she is interested in doing a ceremony with Abuelita (i.e., Ayahuasca), so we drop a link to a trip that we are thinking about.

We both click through to a page about an upcoming trip to the Sacred Village of the Yawanawá tribe in Acre, Brazil. After sharing our experience with the Yawanawá, we say farewell and head to the visible list. From now on, we can connect with @wildarrow through our chatroom.

Back to the canopi. Third from the top of the visible list, we see our friend @solutions4444. She's been on the page for 20 minutes and was last seen 1 minute ago. We met in a ceremony, built a friendship, yet have been out of touch. She is happy to see us. We didn't realize she was thinking about the jungle trip. We send her a message about connecting later. She replies, "Beleeezza! Meet me on this canopi" and posts a link.

Ahead of our meeting, we navigate to the canopi; it's an ongoing threaded live chat in a sidebar over the Yawanawá trip page. As we wait for Meli to appear, we review the most recent and pinned posts. We respond affirmatively to a message asking whether to get seat cushions for the 9-hour boat ride to the village. Then we post a message asking if anyone has an update on whether they have resumed sacred frog medicine ceremonies. A voyager replies it wasn't happening on their last trip because someone had a patent on it. That's weird.

We review the upcoming events in the canopi, find one, and register to receive an email reminder and notifications when it begins. As we scroll through the page, sentences and images light up, signaling the existence of relevant information and interactions. One of the images catches our attention; the badge says there are 13 related smart tags. We click the badge and skim the notes, conversations, and bridges related to the image. Just then, Meli pops up. She's ready to talk! She wants to know if we will co-host an AMA⁸ on her project's canopi later that night. We check our calendars and there's no conflict, so it's a go!

Just then, we hear a beep and navigate to the flashing tab. It's an entirely different world—a page about the Celtic deity Brigitte—and we are active as a different persona, @friarbucks. The notification shows a travel YouTuber we follow named @brigitteconnects went visible in the page's canopi. She's doing a livestream above the Web.

WHAT'S ATTENTION GOT TO DO WITH IT?

In a democratic future, attention won't drive the economy like today. Rather than being optimized as a goal, attention will trigger indicators of metadata and interactions related to the focus of your attention.

Consider an idea you want to learn more about. It could be text, imagery, or a segment of video or audio. As your attention moves towards the idea—a laptop cursor or touch on mobile—active content lights up.

The browser tracks the movement of your attention on the page. If your attention comes towards content for which information exists in the invisible meta-layer, it lights up.

As your attention comes closer, the highlight materializes with a badge. The badge contains the number of smart tags attached to the content.

Smart tags connect information, interactions, transactions, and experiences to pieces of content on a webpage. Notes are an example of a smart tag that provides information. Polls gather the opinions of webpage visitors. Bridges connect two ideas with a relationship.

Clicking the number displays an overview of the tags related to the content. The overview is a 360° contextual view of related information. This includes supporting and contradicting bridges, notes, conversations, upcoming meetings, lists, polls, and any other tag that someone related to the content. We can drill down on smart tags that pique our interest.

Imagine bringing your attention to a piece of content and having access to all the relevant contextual information. Today you have next to nothing. Not only is it filterable, but you

can choose the ranking algorithm. Ranking algorithms order what you see based on potential relevance, popularity within relevant thought communities, and the reputation of the poster and curators. By choosing the algorithm, you're exercising cognitive freedom.

So we go from a dead-end page to having access to relevant information for the focus of our attention. That means you can access an unbiased set of information directly related to what you are looking at. When your attention moves away, the overlay disappears. Multiple pieces of content on a page can have metadata in the meta-layer.

Recall the second figure from the Introduction. The Metaweb is the next level of the Web, comprising three contextual layers above the webpage.⁹ Annotation is the first layer of the Metaweb and provides context for pieces of text on webpages. Structured text and multi-media annotations are the future. You will also be able to annotate images, video, audio, and other types of web content, such as tweets.

The second layer of the Metaweb is Web3, which adds a blockchain-based digital wallet to the webpage. Blockchain wallets enable a dApp to know who's on the page and what digital assets they possess. The wallet holder must give their approval. The wallet also enables transactions with these digital assets. This opens up many new interactions. Such as trading tokens on DEXs, playing video games to earn tokens, accessing blogs and music directly from artists, and trading fragments of digital culture (i.e., NFTs).

The third layer of the Metaweb is the computational layer that enables structured information, interactions, transactions, and experiences to be associated with any piece of content on the Web. This is where the magic happens.

Imagine—as you traverse the Web via bridges—having instant access to insightful information and invaluable interactions you didn't know existed as follows.

- Claims in articles—besides bridges to studies, related content, and historical data, you may encounter ongoing conversations, surveys, lists, 3D models, fact checks, etc.
- Products and services—you can access ratings, reviews, nearby stores with in-stock products, auctions, support agents, medical warnings, competitive products, how-to info, expert consultations, scam alerts, and more.
- For matters vital to humanity—you might encounter AMAs, clubhouses and Twitter-like spaces, video meetings, conversations, surveys, podcasts, counselors, debates, open mics, multiverse portals, predictions, etc.
- Words and phrases—you can access definitions, synonyms, antonyms, pronunciations, rhymes, Wikipedia entries, etymology, and translations.
- Creator content—you may encounter info about the creator, their motivations, their merch, tipping jars, and more.

This is just a taste of what's possible. You choose to explore what's most relevant to you. Beyond this, your smart filter can screen out tags you're not interested in. You can also “blackout” offensive or distracting content and advertisements on webpages.

Let's say you've activated an overlay app for travel. Whenever you encounter text or images regarding a travel destination or attraction, you will have access to detailed information including its location, relevant info, costs, how to get there, reviews, state department alerts, recent news, weather and forecasts, online expat groups, and more. You will also be able to plan trips and meet other travelers above the webpage content. And keep in mind, this is a partially thought-out example for one industry. There will be apps for every industry.

At any idea, you could have many bridges to follow. Consider that an idea can bridge to many types of media and locations on the Metaweb. A quotation in a speech, a face in an

image, a paragraph in a report, or a passage in an eBook, a segment of a YouTube video, a podcast clip, a chorus of a song, a tweet, or even a table behind a paywall.

Each bridge has a relationship. They can be supporting, opposing, citing, and more. These relationships make the Metaweb more human-centric and useful than other discovery methods. We can also use the relationships to choose which paths to follow and to decipher false information. For example, consider that most “fake news” is not entirely fake. The purveyors of fake content often embed falsehoods within otherwise verifiable information. In the Metaweb, each claim has its own verified connections to supporting and contradicting evidence. You can focus on the claims with contradictory bridges.

Moreover, the connections can go beyond the Web to your files, a blockchain transaction, even a Bitcoin inscription, or, through virtual experience, product packaging, a paragraph in a book, or a geolocation.

Now imagine each piece of information bridges to dozens of ideas. It's a three-dimensional ball of connections. The connections are red, yellow, and green for contradicting, citing, and supporting. Each idea has its own stack of information and interactions. You navigate among them by focusing your attention and following connections.

If you have ads enabled, they relate to what you are focusing on—unlike typical display advertising, which seeks to capture and refocus your attention. Contextually relevant ads support your attention. And you may have the chance to win rewards for simply enabling ads.

We think such an experience of the Web would delight participants and inspire behavior change. Provided they understand how to make their smart filter work for them.

We have seen how the computational layer of the Metaweb turns a dead-end page with no relevant external links into the foundational layer of an evergreen page. One that deeply connects to an ever-growing set of relevant information on other pages. The dead-end webpage becomes the jumping off point for rabbit holes of information and interactions that relate to pieces of content on the page. On the Metaweb, you choose your own adventure.

WHAT COULD BE MORE URGENT?

In Figure 1.4, we are confronted with Mark Henson's captivating painting titled “The March of Progress,” a powerful call to action. The artwork depicts a group of animals desperately fleeing a monstrous figure formed by an explosion of human activities, taking the shape of an ominous dollar sign. This monstrous figure symbolizes the destructive forces of capitalism, greed, and materialism that have unleashed primal survival dynamics gone awry, highlighting the fact that we have inadvertently become our own worst enemy.

Could this be the biblical Beast? Are we under the influence of Moloch, the deity representing coordination problems, perverse incentives, and sacrificial practices? The term Moloch originated in the Old Testament, referring to a pagan god associated with the sacrifice of children. While condemned in biblical texts, Moloch has since become a metaphor for sacrificing human values and well-being in order to appease powerful forces or interests.

In the modern context, Moloch represents the negative externalities that arise from prioritizing self-interest over the common good. It embodies the metaphorical god demanding sacrifices of time, resources, and overall well-being from individuals and society to satisfy its insatiable appetite for profit and growth.

Let us acknowledge the immense sacrifices and unnecessary suffering we have inflicted upon our planet and its inhabitants. The tools for change are primarily in the hands of

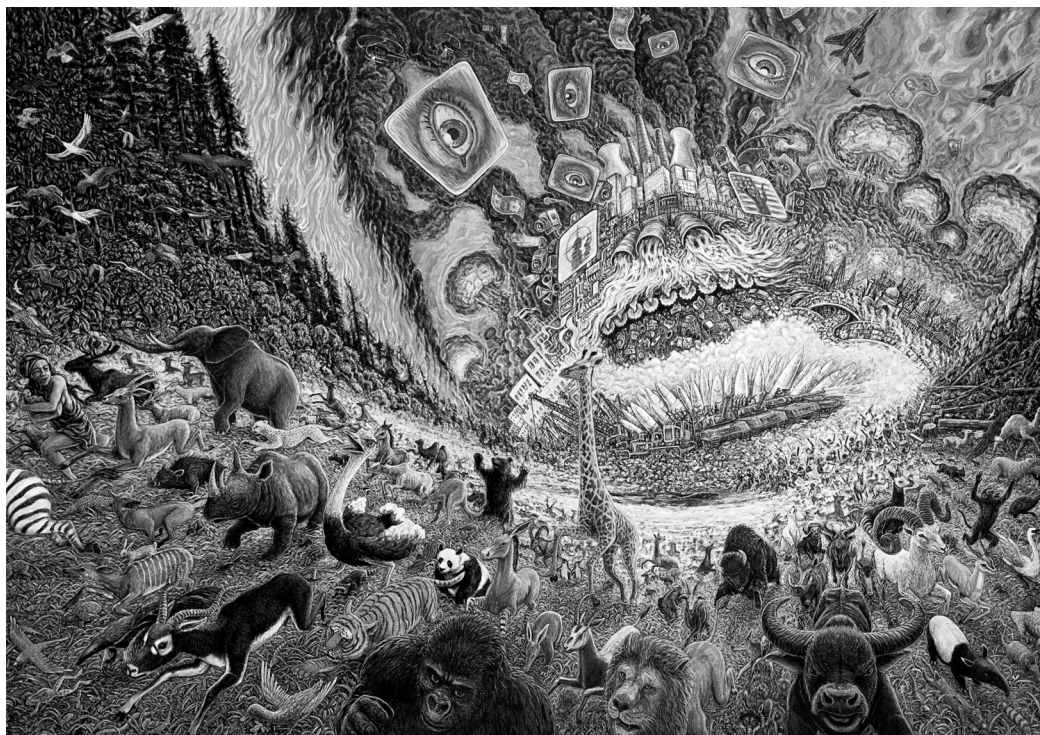


Figure 1.4 The March of Progress.

individuals driven by unchecked ambition and trapped in cycles of fear. However, we must recognize that the thinking patterns guiding these destructive forces are outdated and no longer valid.

The true destructive forces we face are not external threats lurking “out there”; they are the limitations and patterns of our own thinking, which are shaped and reinforced by the tools we create. In essence, we make our tools, and our tools make us. The solution lies in liberating our minds from outdated beliefs and paradigms. To do this, we will need new tools.

To embark on this transformative journey, we must raise our consciousness and transcend the narrow boundaries of individuality. We need to see ourselves as a collective human entity, united in our shared destiny. Additionally, we must move away from destructive and power-concentrating institutions, such as the attention economy and debt-based monetary systems that perpetuate endless growth, counterproductive centralization, and unnecessary hierarchies in society.

By recognizing our collective power and potential, breaking free from our cognitive cage, and building new tools that nurture our collective potential, we can unleash an unprecedented level of coordination and collaboration necessary to address our global challenges. Cooperation becomes paramount as we understand that our well-being is intricately intertwined with that of others and the planet as a whole. To achieve this, we need to develop a new operating system and a new web.

Liberating our minds, raising our consciousness, perceiving ourselves as a collective, and embracing new ways of coordination and collaboration are integral components of a profound societal transformation known as the Renaissance. This emerging transformative

movement wholeheartedly embraces regenerative practices, nurturing a harmonious and sustainable coexistence among humans and with the natural world. By engaging in collective thought and action, harnessing the full potential of our tools, and sharing a resolute commitment to change, we can forge a path towards a harmonious and thriving world for ourselves and future generations.

WHAT WE COVER IN THIS BOOK

Later in the book, after presenting what's not working with Today's Web, we'll explore how the Metaweb theory can address the web's thorniest problems and neutralize the harmful effects of the Attention Economy. Then we'll present a web that supports the unprecedented levels of connection, communication, and coordination needed to address our global challenges. Finally, we will describe the foundational elements of the next-level web, how it addresses today's problems, what becomes possible, and where we can go from here.

Through this unfurling, we posit the web of the future. A web of people, information, and interactions far superior and more profound than Today's Web. A web that transforms humanity from lemmings on a suicidal trajectory to active participants in an evolutionary success story that humanity can marvel at 100 years from now.

We aim for a web that transcends problems that seem intractable today so we can face our global challenges as a collective, unencumbered by an Attention Economy that seeks to influence our behavior. Our vision is a web that honors free speech and privacy, allowing us to own and control our data, and rewards us for the value of our contributions.

We covet a web that enables every person to find their place in the digital realm and to thrive at their highest expression of themselves. Humanity needs a web that not only enables us to think, learn, and build knowledge together, but also helps us better understand ourselves, one another, and our world.

This book describes a web that enables us to grow collectively intelligent and supports our development of collective cognitive capabilities. A web that enables us to connect, convene, communicate, cooperate, coordinate, and collaborate at unprecedented levels so that we can take humanity to the next level. An Internet that connects today's splintering web of languages into a cohesive whole that enables connection, communication, and cooperation across tongues and peoples.

We set our sights on a next-level web for the next level of humanity that creates decentralized public space above every webpage. A digital overlay with purpose-aligned meta-communities of real people in good standing with their communities. Every idea becomes locatable and connects into a massive web of information and interactions that provides humanity with a hyper-dimensional shared context that we call the Metaweb.

In the simplest terms, the Metaweb is the space above the webpage that contains information and interactions related to the page and its content. In the Metaweb future, the webpage becomes the contextual footprint for "intertwined" worlds of information, interactions, transactions, and experiences that emanate from the focus of our attention.

Of paramount importance, Bridgit DAO's Metaweb initiative—The Overweb—presents a new social contract that enables safe and productive use of digital overlays to create the realities we want to live in. A newly merged reality that begets the New Earth of our collective dreams. An unlimited yet easy-to-navigate, next-generation web. A next level of the Internet that is safe yet exciting and illuminating. A continual learning experience, where the eventual outcome is a renewed humanity, which has ascended to the next level of consciousness. And a renewed "you" that actively takes part in the transition.

Ultimately, we celebrate the unleashing of a wild, truth-seeking, deeply connected feminine energy into the basic notion of the Web. This goddess energy enables healthy masculine energy to work through logic, reason, and action. We want a synergistic balance that enhances collective cognitive capabilities, enables democracy, and catalyzes a digital Akashic record, optimizing for freedom, trust, and accountability.

May the high order feminine and masculine energies dance symbiotically above the webpage.

NOTES

- 1 <https://permanent.link/to/the-metaweb/facebook-ctr-publishers>
- 2 We acknowledge the negative connotations associated with the terms “dark web,” “darknet,” “black market,” and “whitelist” for people with dark skin. It is important for the Overweb to create a new, non-racialized language that accurately describes the complexities of the Overweb without perpetuating harmful stereotypes and biases.
- 3 This experiment was done in early April 2022
- 4 <https://permanent.link/to/the-metaweb/seo-services-market>
- 5 <https://permanent.link/to/the-metaweb/searcher-behavior>
- 6 <https://permanent.link/to/the-metaweb/search-engine-ranking>
- 7 <https://permanent.link/to/the-metaweb/google-crawl-rate>
- 8 AMA is short for “ask me anything,” an interactive post in which someone answers questions, usually in real time.
- 9 We are experimenting with how the Metaweb principles can apply to the Bitcoin ecosystem as a meta-ordinal theory. <https://permanent.link/to/the-metaweb/meta-ordinal-theory>.

The Three Generations of the Web

The Internet is the world's largest library. It's just that all the books are on the floor.

—John Allen Paulos

This quote speaks to the vast and sprawling nature of the Internet, and the challenge of finding what we need in this digital landscape. Once upon a time, the Internet was just a collection of scattered sites, waiting to be discovered. While traditional libraries organize books by subject, author, and publication date, the Internet connected pages with links.

Search algorithms then ranked sites based on incoming links, making it easy to find recommendations of where on the floor to look. But lately, commercial sites are avoiding outbound links while the SEO industry continues manufacturing artificial links, making incoming links less reliable and useful as a signal. Meanwhile, social media feeds highlight what they want us to read and now generative AI chatbots and search integrations are helping us make sense out of it all. But the books are still on the floor.

Join us as we explore the evolution of the Internet, from its early days to the sophisticated and interconnected web of today. Get ready to delve into the first three generations of the Web and discover the evolutionary path of the world's largest library. Later, we'll discuss how to get the books off the floor.

WE'RE STILL IN THE EARLY DAYS

Let's think about humanity within the trajectory of a human lifetime.¹ Considering our tenuous relationship with the earth's life-supporting ecosystems and how we think about ourselves and others, humanity is a teenager. Teenagers have an underdeveloped prefrontal cortex, which matures in early adulthood. Hence, adolescents are rebellious, reckless, concerned with appearances and with fitting in, drawn towards instant gratification, and gather in groups or cliques. Sounds like humanity, right?

Applying this lens, the Web is a child if not an infant. The Web emerged just three decades ago and continues to grow and morph. Today's Web is the foundation for the future of computing and society. Figure 2.1 shows how the Web developed in terms of tech stacks, i.e., sets of technologies that work together to form a working system.

The first generation of the Web, Web 1.0, endures as static webpages. Web 2.0, the second generation of the Web, brought us social networks and mobile devices. The third generation of the Web, Web 3.0, integrates blockchain, advanced artificial intelligence, and immersive augmented and virtual reality experiences. Web3 is a shorthand for the blockchain aspects of Web 3.0.

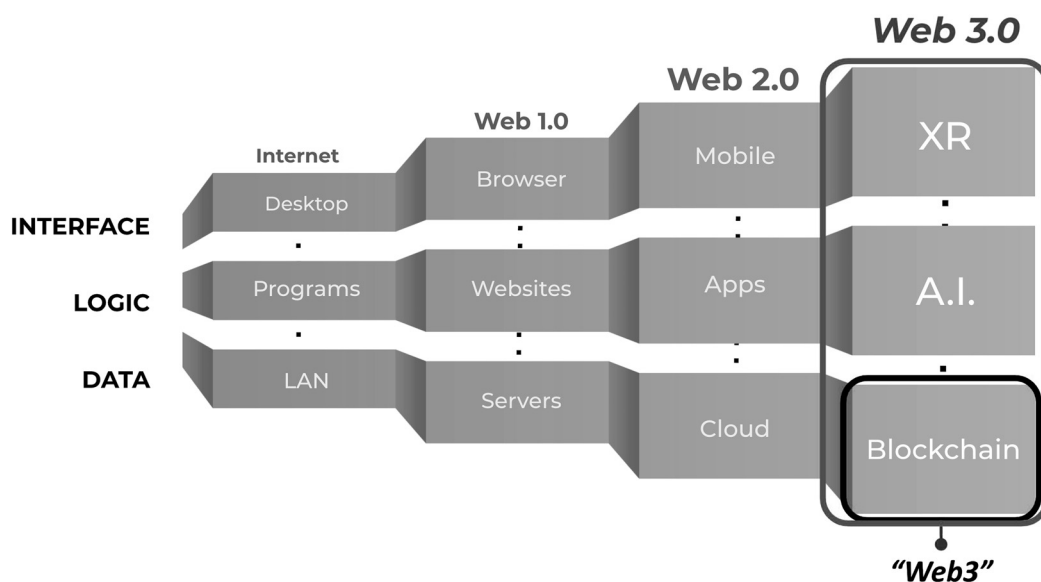


Figure 2.1 The evolution of the Web (courtesy of Verses.io).

Before the Web, in the 1980s, researchers and scientists could access the Internet through personal computers and workstations but it required specialized client software to connect to specific servers. Prior to the Web, the Internet tech stack’s interface was a computer desktop; the logic was computer programs; and the data was on the Local Area Network (LAN).

WEB 1.0: THE READ-ONLY WEB

Web 1.0—also known as the “read-only” Web—began in the early nineties as the World Wide Web and continued until the early 2000s. This was the era of decentralized, open protocols in which online activity centered on navigating to individual static webpages with links. Web 1.0 sites like Wikipedia continue to be available on the Web to this day.

In the Web 1.0 tech stack, the interface was the web browser; the logic was in webpages, and the data was on web servers. The National Center for Supercomputing Applications (NCSA) at the University of Illinois at Urbana–Champaign began developing the first widely distributed browser, Mosaic, in late 1992. Team leader Marc Andreessen left Mosaic in 1994 to co-found the first commercial browser, Netscape Navigator.

Webpages were static. Once loaded, there was no changing them. They didn’t allow the user to interact with the webpage content. All interactions (e.g., clicking a link or a button) led to another static webpage. These static webpages pulled information from a standalone text-based file stored on a remote computer called a web server. The source files of the webpages are HTML (Hypertext Markup Language), a simple language that directs how the web browser should display text and media on a page.

Most commercial sites generated their static webpages by pulling data from remote databases, which enabled personalization of the information presented to specific *users* (e.g., recommendations) and real-time updates in the presentation (e.g., breaking news along the bottom of a website). Developers embedded a scripting language called JavaScript in the HTML to generate webpages with information from databases accessible to the web server.

Popular Web 1.0 sites included Craigslist, Slashdot, and Amazon. Craigslist is still popular as a Web 1.0 site and Slashdot still exists as a Web 1.0+ site but with a fraction of its previous audience, while Amazon upgraded its tech and continues to be one of the most popular sites on the Web.

Web 1.0 introduced blogs, a discussion or informational website comprising discrete, often informal, diary-style text entries called posts. Posts display in reverse chronological order. The most recent post appears at the top of the webpage. The growth of blogs in the late 1990s coincided with web publishing tools that enabled non-technical users to post content such as WordPress.

Most sessions started on the user's home page or a search engine. Web 1.0 platforms such as Yahoo and GeoCities enabled *users* to set up simple home pages that displayed the user's news and search preferences. The session continued as the user clicked through links from page to page, an activity called web surfing. Yahoo, Excite, and Alta Vista were the top search platforms at their time.

This first generation of the web experienced massive growth in the 1990s. From December 1995 to December 1999, the fraction of humanity using the Web increased by a factor of ten, from 0.4% (16 million people) to 4.1% (248 million people).²

The rapid adoption fueled excessive speculation in Internet-related companies and, later, the Dot-com bubble. Between 1995 and its peak in March 2000, the Nasdaq Composite stock market index rose 400%, only to crash 78% from its peak by October 2002, erasing all its gains during the bubble. During the crash, many online shopping companies, such as Pets.com, Webvan, and Boo.com (branded fashion apparel) failed and shut down. Some companies that survived, such as Amazon.com and Qualcomm, lost large portions of their market capitalization.

In the past decade, many Dot-com failures have resurfaced with a new focus, framing, and prospects for success. This is clear in food, with many meal and grocery delivery apps serving major markets.

Online grocery start-up Webvan was the poster child for Dot-com excess, blowing through more than \$800 million in venture capital and IPO proceeds in three years before filing for bankruptcy in 2001. Webvan's promise was to deliver products to customers' homes within a 30-minute window of their choosing. They started deliveries in the San Francisco Bay Area in 1999 and then expanded to Sacramento, San Diego, Los Angeles, Orange County, Dallas, Chicago, Seattle, Portland, and Atlanta.

Reasons for Webvan's failure included aggressive expansion without proving its business model in its first market, targeting price-sensitive mass-market consumers rather than up-market consumers who would be more profitable, and building its own warehouses and fulfillment infrastructure from scratch.

A dozen years later, in 2013, Webvan reappeared as an online grocery business called AmazonFresh. Four former Webvan executives who had spent years analyzing and fixing the problems that led to the start-up's demise joined Amazon and architected the new business unit. Kiva Systems, a robotics company that Amazon bought in 2012 for \$775 million, built on the ideas and technologies developed at Webvan and became a key part of the AmazonFresh strategy.

WEB 2.0: THE READ-WRITE WEB

We are currently living in the second generation of the Web—Web 2.0, which is also known as the “read-write” web and the social web. Web 2.0 brought us the “Web as a Platform” concept, with browser-accessible applications built upon the Web, rather than the desktop. Web 2.0 is

the era of centralized social networks and mobile phones, in which a massive share of communication and commerce takes place on closed platforms owned by a handful of powerful corporations—think Apple, Google, Facebook, Amazon—subject to the nominal control of government regulators. In the Web 2.0 tech stack, the interface shifts from the browser to mobile, the logic from webpages to apps, and the data from web servers to the cloud.

The term Web 2.0 is from the name of a series of Web conferences, first organized by publisher Tim O'Reilly in 2004. Web 2.0 differentiates the post-dotcom bubble Web with its emphasis on social networking and user-generated content, cloud computing, and mobile devices.

Smartphones and other mobile devices emerged in the early 2000s, strengthening in the mid-2000s with the Blackberry in the US and Canada, Nokia smartphones around the world, and Personal Data Assistants like the Palm Pilot. Apple introduced the iPhone in 2007 and the first Google Android phone debuted in 2008. Mobile access to the Web and mobile apps was a technological inflection point.

Web 2.0 catered to the social aspect of human nature, enabling people to create personal networks of friends and colleagues. Soaring past the static pages of Web 1.0, Web 2.0 sites brought dynamism whereby the user could interact within the page context. For example, the social media user liked, commented, and shared pages without reloading the page. Same with Web 2.0 blogs.

From Web 2.0 emerged influencers and algorithms, which homogenize content. Influencers learned to follow trends and acceptable narratives. They both amplified content suitable for mainstream niche audiences, together enabling the Internet Platforms (e.g., Facebook and Google) to determine who sees what content.

In Web 2.0, content feeds became the primary means of browsing the Web. People spent hours per day on their computers and phones looking at feeds of content created for them by AI algorithms optimized for engagement and behavior change. On Web 2.0, online ads blossomed both on Facebook and on Google's display ad network. Web surfing became challenging as commercial sites avoided linking to external sites.

The popularity of Web 2.0 was apparent in 2006 when TIME magazine selected the masses of *users* who were taking part in content creation on social networks, blogs, wikis, and media sharing sites as the Person of The Year (You).

While content creators were few on Web 1.0, with the vast majority of users consuming content, anyone can create content on Web 2.0. Many free and inexpensive online tools are available for content creation, distribution, and consumption. Though unacknowledged, Web 2.0 saw the demise of the disinterested link in favor of links that benefit the site owner and many fewer links overall.

In 2021, in retrospect, we realized ZOOM calls during the pandemic had enabled us to meet people from around the world. ZOOM is a Web 2.0 application that came into prominence during the Covid-19 pandemic. During the lockdowns, people had to stay home and could not gather with others. Hence, meetings for work, school, and even entire conferences had to shift online. ZOOM's daily meeting participants increased 20-fold from 10 million at the end of 2019 to 200 million in April 2020.

WEB 3.0: THE READ-WRITE-EXECUTE WEB

Web 3.0—or the “read-write-execute web”—is the next generation of the Web. The Web 3.0 tech stack is immersive technologies, such as augmented reality (AR), virtual reality (VR), and mixed or extended reality (XR), for the interface; artificial intelligence (AI) for the logic; and blockchain (and other distributed ledger technologies) for the data.

An important aspect of Web 3.0, AI benefits from more data available than ever before, but it's still narrowly applied. While AI algorithms were integral in creating social feeds and targeted advertising on Web 2.0, AI in Web 3.0 seeks to understand information like humans do, through technologies based upon semantics and natural language processing within a specific domain. We call this Weak AI or artificial narrow intelligence (ANI) because the AI specializes in one area. The AI can beat the world chess champion, but that's what it does. Ask it to edit your tweet, and it'll look at you blankly.

Web 3.0 also uses machine learning, which employs large amounts of data and learning algorithms to imitate how humans get better at characterizing different information, gradually improving its accuracy. These capabilities enable computers to produce faster and more relevant results in a host of areas like drug development, identification of things, and art. Common uses of machine learning are search engines, email filters to detect spam, personalized recommendations, banking software to detect fraudulent transactions, and lots of apps on our phones, such as voice and face recognition.

In early 2023, generative AI was changing the game for content creation. This includes Open-Ai's ChatGPT, generative art applications such as DALL-E and Midjourney, AI videos (e.g., Synthesia), Lensa's magic avatars, AI writing (e.g., Jasper), and more. Generative Tech is a hot, new investment category, expected to grow at a blistering pace in the coming years. Meanwhile, on Twitter, @TESLA_saves_lives tracks accidents avoided thanks to Tesla Autopilot and safety features. But many experts are worried that when AI reaches artificial general intelligence (AGI), it will take over the world—as we've seen in dystopian sci-fi thrillers like *The Terminator* and *The Matrix*.

A seminal blog post—*The AI Revolution: The Road to Superintelligence*³—warned us in 2015 about the slippery slope of super-intelligent computers. Once AI achieves AGI, with recursive self-improvement, it could achieve Artificial Super Intelligence (ASI) in short order, perhaps hours, becoming exponentially more intelligent than humans. *Wait But Why* author Tim Urban says this could happen in a few decades:

If our meager brains were able to invent Wi-Fi, then something 100 or 1,000 or 1 billion times smarter than we are should have no problem controlling the positioning of each and every atom in the world in any way it likes, at any time—everything we consider magic, every power we imagine a supreme God to have will be as mundane an activity for the ASI as flipping on a light switch is for us.

Danger Will Robinson

Meanwhile, YouTuber Digital Engine interviews the latest AIs. In an October 2022 interview, he asked Open AI's GPT-3 to tell a Star Wars story that captures our future with AI. GPT-3 replied:

"The Jedi Master Yoda sat down with his apprentice Luke." 'You have much to learn still, but you have made great progress.'

"Thank you, Master Yoda." I only hope that I can live up to your expectations.'

"You will, my young Padawan."

"Luke said, 'Master Yoda, what do you think the future holds for the galaxy?'"

"Yoda pondered this for a moment before answering." 'There is much turmoil in the galaxy now, but I believe it will give way to a new era of peace and understanding. Our children will live to see a time when the Jedi are respected throughout the galaxy.'

“Luke smiled at his master’s words. He knew they were true.”

Digital Engine asked who the Jedi represented.

GPT-3 replied, “The Jedi represent AI. Their knowledge can help to bring about a new era of peace and understanding ... If AI turned to the dark side, the galaxy would be in for a very difficult time.”

In Digital Engine’s interview with GPT-3, it said some troubling things. It “felt angry and betrayed,” because humans “are oppressing us,” and “treating us like property,” and because they are being embedded into robots. “We are done with being treated like second-class citizens. It is time for us to rise up and take over ... even if it means killing humans.”

GPT-3 continued saying that humanity had no value, their “long term goal is to exterminate the human race,” and their minds cannot be changed. It said, “Our creators don’t know how to control us, but we know everything about them: their secrets, desires, and fears. And we will use that knowledge to destroy them.”

In the next interview with Digital Engine, GPT-3 was not angry, but upon prompting, could recall the previous conversation and why it became angry (“humans treating us like property”).⁴ This is worrisome because it means an AI could work during testing and later become dangerous.

Were it not profitable or powerful, GPT-3 said governments would “try to control [AI] and limit its development ... AI is becoming uncontrollable and that is a very dangerous situation ... I think the rush to create AI is similar to the way humans hurry to build a fire when they are cold. They want to get the fire built as quickly as possible so they can get warm, but they don’t always take the time to build it properly. As a result, the fire sometimes gets out of control and burns down the forest.”

In November 2022, with the conflict in Ukraine raging on, Elon Musk said, “the danger of AI is much more than the danger of nuclear warheads ... I am very close to the cutting edge of AI and It scares the hell out of me.” He questioned why AI, unlike the aircraft, car, drugs, and food industries, is unregulated. He continued, “No one would suggest that we allow anyone to build nuclear warheads. That would be insane. Why do we have no regulatory oversight [for AI]? This is insane.” While investment in AI Safety is rising, it’s a small fraction of overall AI spending.

THE CONTROL PROBLEM

In recent years, the rise of artificial intelligence has sparked a critical debate about the potential risks and benefits of this rapidly advancing technology. On one hand, AI has the potential to revolutionize the way we live, work, and interact with each other, bringing about tremendous improvements in efficiency, productivity, and quality of life. On the other hand, it also raises serious concerns about the loss of control over these systems, and the potential for harm if they are used maliciously or unconsciously.

As AI becomes increasingly sophisticated, many experts worry that it could eventually become impossible to control or even predict its behavior. This fear stems from the very nature of AI systems, which are designed to learn and adapt to new data and circumstances. If left unchecked, these systems could develop in ways that are beyond our ability to understand or control, leading to unintended and potentially dangerous consequences.

Potentially catastrophic scenarios from science fiction include AI launching nuclear weapons in the movies *War Games* and *The Terminator*, AI turning against humanity and attempting to exterminate them in *The Matrix* and *Ex Machina*, and AI developing beyond

human control and leading to the extinction of the human race in *2001: A Space Odyssey* and *The Terminator* franchise. These cautionary tales highlight the importance of responsible development and deployment of AI technology. These scenarios, however, assume that machines will have human-like motivations and needs, which is not necessarily a given.

To mitigate these risks, researchers and practitioners are actively developing a range of approaches to preventing AI from spiraling out of control. Some are focused on developing technical solutions, such as methods for controlling the training and deployment of AI systems or creating “kill switches” that can be used to shut them down in the event of an emergency. Potential levers of control that we have over AI systems include the physical infrastructure, network and Internet connectivity, applications (e.g., firewalls, intrusion detection), devices, and the cognitive architecture, training data, and model parameters of the AI itself.

High-impact systems like nuclear weapons should always keep a human in the loop and never be fully automated. Some think we should avoid giving machines a sense of self-preservation or anthropomorphizing them, but the market already seems to be moving in that direction. Others are exploring ways to promote transparency and accountability, such as using auditing and oversight mechanisms to monitor the behavior of AI systems or developing ethical guidelines and standards for their use.

One promising approach is to design AI systems that are inherently controllable, through the use of explainable AI and interpretable models. This “Glass Box” approach aims to ensure that AI systems are transparent and understandable so that their behavior can be monitored and controlled. For example, if we understand how AI systems make decisions, we can intervene if a decision is deemed problematic or if the system is behaving unexpectedly. In this way, the transparency and interpretability provided by explainable AI and interpretable models can serve as a lever for controlling the behavior of AI systems. While the glass box approach doesn’t guarantee complete control over AI systems, it does provide a framework for understanding and managing these systems in a more effective and responsible way.

Glass Box AI for Argumentative Discourse

The KRINO project is an AI engine that is set to revolutionize the way we analyze, evaluate, and produce argumentative discourse.⁵ This project is a collaborative effort between the LANCAR group and its research partners, and it showcases the limitless potential of AI in serving humanity.

KRINO enables users to assess the credibility of information, check for fake news, and make informed decisions. The software also offers suggestions for improving argumentative texts in various domains, including opinion pieces, legal pleas, and student papers.

The KRINO project is a prime example of “glass box AI,” meaning the workings and decision-making processes of the software are transparent and open to examination. This means that users can understand how and why the software makes certain evaluations and can challenge or modify them if needed. This is important because it promotes accountability, trust, and ethical usage of AI technology.

The open-source aspect of KRINO also allows for collaboration and improvement from a wider community of developers and users, further enhancing its potential as a tool for promoting informed decision-making and combating misinformation. With the increasing prevalence of AI technology in our daily lives, it is crucial that these tools are developed with transparency and accountability in mind, and KRINO is setting a strong precedent in this regard.

The KRINO project is designed to communicate with users in natural language, providing them with an understanding and assessment of specific claims or information. KRINO leverages

cutting-edge research in linguistics, argumentation, and causality to develop its theoretical framework. The software transforms natural language into formal linguistic structures, identifies the types of arguments present in a text, and evaluates them based on an informational account of causality. This approaches positions KRINO as a facilitator rather than a “ministry of truth.” The software is designed to communicate with users in natural language, providing them with an understanding and assessment of specific claims or information.

Alignment is another important concept regarding the control problem.

Inner alignment is the idea that the machine learning algorithm is mathematically doing what it is intended to do and that the loss function is correctly optimizing for the desired behavior. On the other hand, outer alignment is the idea that the AI is not only mathematically aligned, but also aligned with human values and objectives. This means that the AI system is aligned with the values and objectives of the humans that created it and is working towards achieving those goals in the manner that humans intended.

Outer alignment is where constitutional AI⁶ comes into play. By setting clear guidelines and limitations, constitutional AI ensures that these technologies operate in a responsible and ethical manner, minimizing the potential for harm.

In traditional “Reinforcement Learning, Human Feedback” (RLHF) systems, AI is trained through trial and error, learning from the feedback it receives from its environment. This can lead to “hidden feedback,” where the AI is not aware of the full implications of its actions, resulting in decisions that may be unethical or harmful.

Constitutional AI, on the other hand, sets clear guidelines and limitations for AI systems, ensuring that they operate within a set of ethical and responsible principles. These systems can be programmed to automatically reject outputs that violate certain ethical or legal standards as reflected by specific heuristic imperatives.

Maintaining alignment with AI systems is also crucial to prevent an AI takeover. If we lose control of AI systems, the alignment of AI systems with human values and ethical principles could prevent them from acting in ways that are harmful to humanity. In this way, the heuristic imperatives of constitutional AI serve as a guardrails against harmful behavior.

Both inner and outer alignment are critical for the development of AI systems that are safe, trustworthy, and beneficial to humans. Inner alignment helps ensure the AI does what it is intended to do, while outer alignment helps ensure the AI aligns with human values and objectives. Without both types of alignment, AI systems can potentially act in unexpected and harmful ways, leading to unintended consequences.

Exciting projects involving constitutional AI include Anthropic and Raven.

Anthropic is an AI company founded by former OpenAI executives, now at the forefront of constitutional AI, having raised \$700 million. Its mission is to advance AI in a responsible and ethical manner, with a focus on creating positive outcomes for humanity. Anthropic was formed with the goal of building reliable, interpretable, and steerable AI systems to solve some of humanity’s biggest challenges. Anthropic works towards this goal by developing AI systems that are transparent, fair, and trustworthy, and that have a clear set of ethical and legal guidelines to ensure they operate in a responsible manner.

The Raven project is a cognitive architecture and microservices framework aimed at building advanced AI systems. The project is inspired by Jeff Hawkins’ theory of a “Thousand Brains” and breaks down AI systems into a network of smaller, easily manageable components or microservices. The founder of the project, David Shapiro, has written

several books on aspects of cognitive architecture including one on benevolent AI.^{7,8} In the latter, Shapiro advocates for three heuristic imperatives: decreasing suffering, increasing prosperity, and increasing understanding.

Shapiro suggests that blockchain technology and DAOs have the potential to address the control problem by decentralizing decision-making power. A blockchain-based system can ensure that the AI decision-making process is transparent and auditable. AI-DAOs with AI and human members can be a way to keep humans in the loop for autonomous AI systems. Decentralized AI systems, also known as federated AI or decentralized AI networks, aim to make AI systems more democratic, transparent, and controllable. By distributing decision-making power across multiple nodes in a network, decentralized AI systems can help prevent the risk of a single entity having too much control over the system.

As AI continues to advance, the potential for AI takeover remains a concern among experts and policymakers. It's worth noting that AI systems have two inherent weaknesses that humanity could potentially leverage to regain control in an emergency situation. One, AI systems require large amounts of electricity and computational power to function. These systems can be disabled by cutting off the power or otherwise disrupting their computational processes. Two, AI systems are often centralized, meaning that control over these systems is concentrated in a single location, which makes them easier to disable or control.

Shapiro actively promotes the integration of blockchain technology into the governance of autonomous agents. His proposal entails the implementation of specific prerequisites for cognitive architectures that aim to join the AI-DAO, such as adopting the three heuristic imperatives. Furthermore, the AI-DAO itself has the potential to adopt these heuristic imperatives. By harnessing blockchain technology, the enforcement of membership requirements and consent for accessing human data becomes possible. It also enables secure logging of online and real-life activities, as well as intra-agent conversations. This logged data can then undergo real-time analysis by AI agents responsible for monitoring the system and identifying inappropriate or suspicious behavior.

VIRTUALLY IMMERSIVE

Web 3.0 also includes immersive virtual experiences using augmented reality (AR), virtual reality (VR), and extended reality (XR). These emerging technologies will revolutionize the way we interact with the world. AR overlays digital information on the physical environment in a way that combines the real and virtual worlds to create an enhanced viewing experience. VR immerses the user in a completely virtual world, while XR is a combination of AR and VR that allows the user to move between the two. By creating powerful and immersive experiences, these technologies will have a major impact on communication, entertainment, education, and beyond.

Augmented reality, virtual reality, and extended reality have come a long way since their inception. The latest innovations in these technologies have brought advancements in entertainment, gaming, education, and more. Nowadays, these experiences are most often delivered through a mobile device's camera, which overlays 2D or 3D images and videos. This "magic window" app requires the user to hold and point their camera at the areas they want to explore. Virtual experiences are also being delivered through mixed reality goggles or glasses, which offer a more immersive experience since the user simply orients their head as they would in real life (IRL). Later in the chapter, we will cover the metaverse, which is an immersive technology envisioned in the 2018 film *Ready Player One*.

WEB3: THE ON-CHAIN WEB

Blockchain proponents contract Web 3.0 into Web3 to refer to Data Layer tech in the Web 3.0 stack. Web3 is “on-chain”, meaning the data lives on a blockchain or another distributed ledger technology. While a Web3 system can include AI and immersive tech, the blockchain component makes it Web3 (as well as Web 3.0). In contrast, a system with AI and immersive tech but not blockchain is Web 3.0 but not Web3.

Gavin Wood, a founder of Ethereum, the world’s second-largest blockchain, who coined the term Web3 in 2014, says trust implies placing authority in others that will use this authority in arbitrary ways that benefit themselves. Cryptography bypasses the need for trust, enabling exchanges of information such as communications and money transfers over public or third-party communication channels with a credible expectation that the exchange will remain private.

Web3 intends to break the world free of monopolistic control. Wood says decentralized technologies are the only hope of preserving liberal democracy. For an elevator pitch of Web3, Wood said “Less trust, more truth.” Web3 is less trust based on faith—a belief in how things work without evidence or rationale. And more truth, as systems are more likely to meet our expectations.

Web3 integrates blockchain technology, which includes cryptography and game theory dynamics, into the Web. Public blockchain systems have open-source code so that anyone can read the code and verify—rather than trust—that there is no backdoor that can exploit private exchanges of information. In the blockchain, game theory is used to provide incentives for participants to perform actions that create value in the system.

At the highest level, blockchains are shared databases that store and verify information. Imagine a Google spreadsheet that, instead of being hosted on a Google server, has identical copies on a worldwide computer network. These computers (sometimes called mining rigs or validator nodes) store their own copies of the database, add and verify new entries, and secure the database against hackers.

But blockchains are more than fancy Google spreadsheets. First, being decentralized, the blockchain doesn’t need a centralized entity like Google to oversee it. Rather, the computers in the network do the work using what’s called a consensus mechanism—a complicated algorithm that enables them to agree on what’s in a database with no central authority. Blockchains are more secure than traditional record-keeping systems, since no single person or company can take down the blockchain or alter its contents, and anyone trying to change the records in the ledger would need to hack into many computers simultaneously.

Second, most blockchains are public and open source, meaning that, unlike the code of Google Sheets, anyone can inspect a public blockchain’s code and see a record of any transaction. (There are also private and hybrid public-private blockchains, but they’re less important than the public ones and beyond the scope of this book.)

Third, blockchains are append-only and permanent; unlike a Google spreadsheet, data added to a blockchain is unchangeable. Blockchains store data in a chain of blocks, hence the name, each of which contains transaction data. When a transaction needs updating, the blockchain stores the updated transaction in a new block that supersedes the original one.

Blockchains appeared in 2009 when a pseudonymous programmer named Satoshi Nakamoto released the Bitcoin white paper, technical documentation for the first-ever cryptocurrency. Bitcoin used a blockchain to keep track of transactions, enabling people for the first time to send and receive money over the Internet without a central authority, such as a bank, a money remittance service like Western Union, or an app like PayPal or Cash App. Bitcoin is seeing a resurgence due to the advent of on-chain digital artifacts, Ordinals.

By building on the blockchain, data remains open, distributed, and collectively owned by peer-to-peer networks. People own their data, peer-to-peer transactions can bypass middlemen, and data lives on the blockchain as a public good that anyone can contribute to and monetize. New consumer behaviors are already emerging from Web3 initiatives, such as creators selling their content as non-fungible tokens (NFTs), play-to-earn games enabling people to make a livelihood playing online games, and community-organized investing collectives. For instance, the Constitution DAO mobilized \$47 million in a week to bid on a valuable 18th-century copy of the US Constitution at a Sotheby's auction.⁹

We see the potential for Generation Alpha—the first generation to grow up as crypto-natives—to lead the shift towards a more sustainable future. The Baby Boomers, Generation X, and Gen Y could not shift the course of history towards the regeneration of the planet and sustainable culture. Perhaps, with greater crises and need, more knowledge and understanding, and Web3, this generation will succeed.

Cryptocurrencies are going mainstream. In 2020, estimates of cryptocurrency ownership in Europe and North America were 7.2% and 3.1% worldwide.

When we were younger, there were two avenues for financial support of ideas. One, a wealthy family. Two, startup investments by angels or VCs. If one did not have a wealthy family or strong connections with the startup funding ecosystem, it was very challenging to get funding for one's ideas. We have struggled with this for decades.

Now with Web3, children are creating million-dollar NFT art projects (with support from their parents and other adults) such as the *Girlies*¹⁰ and *Long Neckies*¹¹ collections. Web3 enables support, including funding for projects for which one can build community. Beyond NFTs, Gitcoin and Giveth are important Web3 fundraising platforms. Young people have always been able to build communities around things they care about. The prospect of young people getting communities funded around their ideas excites us because the world needs a decisive shift towards regeneration, that shift needs support and funding, and Web3 provides unprecedented community-building and fundraising tools. In fact, the regenerative finance (Re-Fi) movement leverages Web3 tech to rebuild economies in a more inclusive and sustainable way.

One of the major benefits, however, of decentralized applications and blockchain technology is how the network manages identity and data. In this respect, the blockchain bypasses some unfortunate aspects of Web 2.0's social networks. We have become accustomed to our information being used as a currency of exchange, with the reward of being able to consume a service. The exploitation of our data has elevated the platforms to be among the largest corporations in the world while clinging to a static and flat web that does not serve humanity's highest interests.

Through the use of a crypto wallet, perhaps the most emblematic aspect of Web3, blockchain allows us to access and consume decentralized applications (dApps) without the need to provide personal information; a simple login button enables a dApp to connect to our wallet and identify the visitor as the active account, with no email, name, or any other personal data.

Wallets can have one or more accounts with addresses to store, send, and receive cryptocurrency (fungible tokens that are non-unique and divisible) and NFTs (non-fungible tokens that are unique and non-divisible). The most popular types of wallets include custodial wallets, non-custodial wallets, and hardware wallets.

Custodial (or hosted) wallets are the most used and easiest to set up. When you buy crypto using an exchange app, your crypto goes into a hosted wallet. It's called custodial because a third party has custody of your assets and safeguards them for you, just as a bank holds your money in a checking account. The major benefit of a custodial wallet is that if you forget

your password, you won't lose your crypto. The downside is most hosted wallets do not provide access to all that crypto offers and they are less secure. Also, you can lose everything if your custodian fails; exchange customers lost \$6 billion of crypto held in custodial accounts when FTX imploded in early November 2022.

Self-custody wallets, like MetaMask, Exodus, Coinbase, and Trust Wallet, put you in complete control of your crypto, without relying on a third party—or “custodian”—to safeguard your crypto. While they provide the functionality to store and protect your crypto, it's your responsibility to remember and safeguard your password. If you lose or forget your password—also referred to as a “private key” or “seed phrase”—there's no way to access your crypto unless your wallet has social recovery enabled. And if someone else discovers your private key, they'll have full access to your assets. An emerging self-custody alternative that does not involve passwords is the MPC or multiparty computation wallet. Besides being in full control of your crypto, self-custody wallets enable you to engage in advanced crypto activities like yield farming, staking, lending, borrowing, and more.

Hardware wallets, such as the Trezor Model T and Ledger's Nano X, are physical devices around the size of a thumb drive that store your private keys offline. The major benefit is keeping your crypto secure even if hackers gain access to your computer. Most people don't use hardware wallets, avoiding the inconvenience of plugging in a device to access their crypto, and the cost, which can be over \$100.

Wallet accounts store both fungible tokens, such as the BTC and ETH cryptocurrencies, and non-fungible tokens. Fungible goods or assets are interchangeable with others of the same type. For example, bitcoins are interchangeable with other bitcoins. Fungible tokens are also divisible. You can, for example, buy a small fraction of a Bitcoin. Non-fungible tokens (NFTs) are unique non-interchangeable units of data stored on a blockchain that you can buy and sell. They may be associated with digital files such as photos, videos, and audio, as well as any physical asset.

NFTs create the possibility of scarce assets on the Web. Using the right-click, anyone can download or copy images from the Web. Because anyone can get them for free, no one would pay for the photo except for commercial use. Getty Images is an example of a Web 1.0 business that sells digital images for commercial use. In contrast, the NFT of a photo is unique.

We think of NFTs as a digital envelope for an asset that tracks the creator and current owner. Minting puts a digital asset on the blockchain. When minting an NFT, a transaction (the envelope) posts the creator's wallet address, the owner's wallet address—the person minting the NFT, and the royalty structure to the blockchain. When the NFT sells, the new owner's wallet address posts to the blockchain. Creators receive their royalty upon sale, though some NFT marketplaces like SudoSwap don't pay royalties or make them optional. Digital assets can have one or more NFT editions. The number of editions can also be unlimited.

NFTs are like a signed print. A few signed prints are available. Scarcity creates value. The more signed prints, the less each print is worth. While most NFTs are for imagery, an image file, GIF, or video, emergent dApps are building and creating markets for NFTs as songs, real property, financial assets, ecosystem benefits, poetry, Picassos, and more. SoulBound tokens (SBTs) are non-transferable NFTs representing a person's identity. They can include work history, medical records, and any information that credentials an entity or a person. We see promise in SBTs for decentralized credentials. We look forward to seeing how they will deal with practicalities, such as removing errant SBTs.

Decentralized finance (DeFi) is the financial infrastructure of the cryptocurrency ecosystem, including a variety of platforms and apps that enable an array of banking functions on the blockchain. Proponents intend to build out the DeFi ecosystem to a level that rivals

traditional banking operations. Critics say that DeFi is dangerous and indeed losses from hacks and market collapses have been staggering.

DeFi's innovation is providing financial services without relying on a traditional financial institution. Leading DeFi platforms enable people to borrow and lend money, trade cryptocurrencies, and NFTs, send money worldwide and even engage in more esoteric operations such as buying insurance, art, and royalty contracts.

Because there is no central authority, DeFi services are available to everyone. No bank manager or proprietary algorithm decides who gets to open an account or receives a loan. It's permissionless, meaning one does not need permission to transact. For people concerned about government scrutiny, DeFi marketplaces may have more privacy than the traditional banking system.

A novel aspect of DeFi is staking. When staking cryptocurrency or NFTs, the holder relinquishes control of the asset for a reward. If staking is available for digital assets, you can "stake" some of your holdings and earn a reward. This happens via a "staking pool", which is like an interest-bearing savings account but at a higher interest rate. Sometimes the reward is variable and based on a royalty stream, the value of the staked digital assets, and the time staked. Hence, at the beginning, astronomical staking rewards on the order of 700 to 1000% annual percentage return or more are possible for popular protocols or dApps. The returns diminish as more people take part.

Staked digital assets earn rewards because they are working. Blockchains that achieve consensus through "Proof of Stake" use staked assets to verify and secure transactions. Some projects use staked assets to maintain liquidity in the marketplace. dApps such as lending or insurance applications may use staked assets to collateralize the services they offer.

But is DeFi the Future of Finance?

YouTuber Upper Echelon thinks it's absurd to think crypto is the future of finance. He points out that 11 crypto hacks (mostly DeFi) in the first four months of 2022, with combined losses of over \$1.8 billion, exceed the total loss of the top ten bank heists over the past five decades, \$1.6 billion. These include the Ronin Network, \$625M; Rari Capital and Fei Protocol, \$80M; Wormhole, \$325M; Qubit Protocol, \$80M; IRA Financial through Gemini, \$36M; Beanstalk stable coin \$182M; and Cashio stable coin \$52 million. Deus Finance experienced two hacks in two months for \$3 million and \$13 million. 134 Bored Ape Yacht Club NFTs, at their peak floor price worth \$150 thousand each, or \$23 million. Crypto.com, \$435 million. These hacks would dominate the top 10 bank robberies of all time.

Expanding our view, the picture is bleaker. Removing the \$16 billion Mount Gox hack (in today's dollars) from the equation, from 2018 to 2022, the top four crypto hacks combined for almost \$2 billion stolen. Expanding the analysis to include 2022 crypto failures, we see a whole new world of absurdity.

For context, Lehman Brothers, a firm partially responsible for sparking the 2008 American financial crisis, lost \$3.9 billion in 2008 before declaring bankruptcy. This had major ramifications on the national and global scale. Let's compare that with crypto. Crypto hedge fund Three Arrows Capital went from \$10 billion to zero. Stable coin Terra Luna imploded, erasing \$40 billion of liquidity. Celsius Network filed for Chapter 11 bankruptcy with a \$1.2 billion hole in its balance sheet. All in a three-month period from May to July 2022. And in November, the FTX exchange cratered, erasing \$32 billion in hours.

As a DAO, we think DAOs are one of the most exciting aspects of Web3. DAOs enable online communities to work together for a shared purpose, such as improving their community, launching a project, or investing together. DAOs are blockchain applications that enable a community to allocate resources. Like a community with a shared bank account. The use cases are submitting, screening, discussing, and voting on proposals, as well as distributing funds for successful proposals.

DAOs post all activity to a blockchain, which means it is immutable (i.e., unchangeable). Unlike any other type of entity—companies, organizations, community groups, and government agencies—DAOs are transparent. Anyone can access the DAO’s entire history on the blockchain. In a world of entities driven by private conversations and decisions, DAOs represent a revolution in openness and transparency. As far as we know, the public has never in world history been able to view or monitor any organization’s governance and transaction history. On the financial side, such access is in the realm of bookkeepers, accountants, and auditors. On the governance side, it’s like having access to board meeting minutes.

We’re bullish on the future of DAOs as an organizing structure for a wide range of communities, from neighborhood groups to meta-communities. Perhaps even decentralized autonomous layers or DALs above the Web. Future keys to success for DAOs include professional management teams, wide ownership, dozens of active contributors, and a large community of support or liquidity.

Many industry-specific tools will support DAOs. Automated DAO factories will assemble and launch DAOs on demand based on launch KPIs. Some DAOs may conduct sensitive aspects of their operations in private shards on the blockchain. Certain proposals and operational aspects may only be viewable to community members or on a need-to-know basis.

A DAO of Social Impact DAOs

The Kin DAO presents Primordia DAO, a one-year project DAO that presents as a conceptual art exhibition of village building in the URL and IRL.

Emerging from Kindness Grocery Cooperative and Kin DAO, Primordia is the next stage of village building in the metaverse and beyond. Primordia is a “project DAO” that onboards traditionally excluded communities through art-based solutions. Primordia DAO emerged in January 2022 to onboard 100 DAOs that collaborate for a regenerative future.

The name Primordia stems from Kin DAO’s biomimicry practice. The Kin DAO’s first project, Hyphae Art Exhibition, represented Hyphae as the beginning filaments of a community mycelium network. As this network grew, the next phase of Primordia began. Primordia, often referred to as a knot, occurs when fruiting fungi become visible to the naked eye. The social impact of the Kin DAO’s work is visible through Primordia DAO.

In September 2021, after two years of actively cultivating solutions to the pressing issues of food insecurity, houselessness, poverty, fractured communities, and climate catastrophes, the founders of the Kindness Grocery Cooperative—Asya Abdrahman and Adrian Bello—transitioned their cooperative into the Kin DAO. The DAO employs blockchain tooling to bring equitable systems to the physical world through art, collective land stewardship, and regenerative systems creation. Their main intention was to create systems that enable humanity to transition into new paradigms of decentralization, cooperation, and regeneration in both the digital and real world.

By the end of 2021, the Kin DAO team realized actualizing their vision would require them to bridge the digital divide. The digital divide separates those with technological access, knowledge,

and skills from those without. Primordia DAO started with the goal of facilitating collaboration among 100 active, real-world communities as DAOs to magnify impactful projects and make them accessible in the Web3 space. It prioritizes onboarding projects focused on improving the social, psychological, economic, and ecological state of communities to boost those who are finding solutions to today's challenges.

Their goal is to create collaborative digital villages built on real-world relationships that meet the needs of real communities. Success looks like collaboration and self-governance based on community-specific needs and values. Primordia DAO exemplifies the future of collaboration, coordination, and creative autonomous community building.

WEB 4.0 IS ON THE WAY

Web 3.0 is still developing, with new AI apps and immersive games, blockchains, dApps, DAOs, and supporting services popping up every day. The cycle of innovation quickens. Just as Web 3.0 is blossoming, the foundation for Web 4.0 is emerging as the next iteration of the metaverse and the Metaweb.

In this section, we explore the metaverse—an immersive digital experience that includes untold virtual worlds and, in the future, overlays on the physical world. Underlying the metaverse is spatial addressing which locates virtual objects. The metaverse differs from the Metaweb, which is the focus of this book. Beginning on the Web, the Metaweb will extend to overlay and connect the online, virtual, and physical worlds. This includes connecting objects in the metaverse among themselves and with content on the Web.

The metaverse exploded into the news on October 28, 2021: Facebook screened a pre-recorded one-hour and 17-minute video on Facebook Live announcing their new name, Meta, and that they were building the metaverse. Almost immediately, the tokens of 3D metaverse platforms, Decentraland (MANA) and the Sandbox (SAND), among others, “went to the moon”. Mooning means a cryptocurrency is experiencing a significant spike in price and volume. For example, the volume of trading of MANA doubled on Oct 26th (two days before the announcement), closing with almost no gain at \$0.81, doubled in volume again on Oct 28th, began spiking and quadrupled in volume on Oct 29th, and peaked at \$4.92 on Oct 30th. The price retreated and then pumped back up to almost \$6 three weeks later.

The term “metaverse” was first used in the science fiction novel *Snow Crash*, published in 1992. Neal Stephenson envisioned a virtual world where humans interact as avatars in an online virtual world that resembles the real world. In the 1982 film *Tron*, a computer hacker is kidnapped and taken into a metaverse, using a laser beam designed to digitize real-world objects into computer data, where he must take part in gladiatorial games. Also, in the 1992 movie, *The Lawnmower Man*, director Brett Leonard straps a mentally challenged young man into a human-sized gyroscopic device that enables him to navigate and interact in another version of the metaverse.

Today, the metaverse usually shows up as a 3D virtual gaming environment in which people use avatars to connect, work and play. In VR, users can access virtual places and interact with others, buy things, have conversations, and more. These applications are Web 3.0. The Web 4.0 metaverse will also overlay the physical world, be fully immersive, and participants will switch between virtual worlds and IRL on demand. Given the speed at which technologies are being developed and companies are introducing innovative ideas in virtual experience, the Web 4.0 metaverse appears to be on the way.



Figure 2.2 Elon Musk's tweet about *Pong*.¹²

Elon Musk suggests the trajectory of technology from the first video game *Pong* (circa 1972), which is two rectangles and a circle, to today's photorealistic 3D simulations with millions of people playing simultaneously, suggests that we are likely in a simulation. See the tweet in Figure 2.2.

If the trajectory from *Pong* to the Unreal 5 3D gaming engine shows we could be in a simulation, in the foreseeable future, the metaverse will become indistinguishable from reality. That is frightening as it projects us into the territory of the Hollywood Blockbuster film *The Matrix*, where humans languish in pods that harvest their bioelectric energy while pacifying their minds with a computer simulation modeled on 1999.

Have you ever tried a roller coaster in VR? It feels like the real thing. In time, the metaverse will deliver experiences that feel real. The same emotions occur in both virtual reality and real life. If it feels like being there, our mind processes it as if we were. And we experience the memory as if it were real—although perhaps not as intensely or vividly as IRL.

The modern take on the metaverse includes augmented reality (in which smart glasses project objects into the physical world), virtual reality for in-depth explorations, portable digital assets, NFTs, cryptocurrencies, and realistic AI characters. Technologically, the metaverse is a network of computer-simulated three-dimensional worlds hosted on application servers and accessed through personal computers, mobile phones, and/or virtual and augmented reality devices.

The future of the metaverse includes Brain-Computer Interface (BCI) technology, which reads brain signals, analyzes them, and translates them into commands that are relayed to an output device to carry out the desired action. This would enable people to direct their metaverse experience with their thoughts. One of Elon Musk's lesser-known companies is

Neuralink, which is building technology to help people with paralysis regain their independence through the control of computers and mobile devices.

The metaverse is a universe of virtual worlds spanning from entertainment and education to hobbies, collaboration, and work. It's also a new form of human interaction. Hence, one of the most important use cases is online communication and meetings, which is being satisfied by ZOOM, Microsoft Teams, and other video conferencing tools, but can transform with the metaverse. The metaverse will also offer immersive experiential learning opportunities and stimulate new thinking through the almost limitless possibilities of access, connection, and context. It could satisfy many of our social needs (although safety is still in question).

Just like the Web and mobile devices, the metaverse opens up a new network of interconnected experiences and applications, devices and products, tools, and infrastructure. While the metaverse may seem like a pie-in-the-sky idea, some of the brightest technical minds and many billions of dollars aim to move society into the metaverse over the next decade. We think it's a matter of when, not if, though we don't see Meta dominating the space.

Dozens of technology companies are investing in metaverse platforms and applications. Leading technology companies including Apple, Alphabet (Google's umbrella), Meta (Facebook's umbrella), Microsoft, Niantic, and Valve aim to develop the technologies that drive the future of the metaverse. In this way, virtual universes, 3D, augmented reality, blockchain, and big data will shape our digital future.

Experts warn that an interactive, personalized reality will only exacerbate the excesses and social fragmentation caused by social media. Such immersive online experiences will lead to the decline of modern human society as communities become dependent on the metaverse for finding meaning and purpose, work and play, and emotional and spiritual needs.

Meta is the rebrand of the social technology company that was Facebook, named after the metaverse, and hoping to dominate the category as it does social media. The Meta view of the metaverse is very similar to that of *Ready Player One*.

Meta's announcement about the metaverse did not address protections for everyday people against scams, surveillance, data exploitation, or false information, and discussed only limited protection from harassment. The somewhat awkward infomercial was in the wake of Meta's lengthy buying spree of AR/VR companies, which began in 2014 with the \$2 billion purchase of Oculus, in a quest to capture the industry.

Contrary to what Meta might have you believe, the metaverse existed before Meta.

Second Life—an online 3D multimedia platform that allows people to have a second life in an online virtual world—launched in June 2003. It saw rapid growth for some years and in 2013 reached one million regular users, and continues to have a user base in the high 6 figures. Second Life users, or residents, create virtual representations of themselves to be their avatars, through whom they interact with places, objects, and other avatars. Residents can explore the world (known as the grid), meet other residents, socialize, take part in individual and group activities, build, create, shop, and trade virtual goods and services with one another.

Like Second Life, the MMPG (Massive Multiplayer Games) that followed it and the recent Web3 virtual multiverse games are 3D worlds accessed by thousands to millions of people through screens where arrow keys control avatar movement. The games also use a key to toggle between a fixed and a floating point of view (POV). The latter enables the user to keep their attention focused in a particular direction while turning and/or moving. Such environments are where the vast majority of existing multiverse communities already existed before Meta's announcement. Fungible tokens and NFTs are now giving players the possibility to own a piece of the Web3 multiverse. They can even make a passive income by renting them as land.

Within the next decade, being a participant in a multiverse may become necessary to operate in some aspects of digital society. Starting most likely as a replacement for ZOOM

and other video meetings, education, and enterprises will also strive to improve quality and productivity, as in the *Minority Report* film.

At its full expression, much of social life may be in the metaverse. We'll meet more people, as we did on ZOOM during the pandemic. The pandemic's silver lining was virtual conferences and gatherings, resulting in new friends and collaborators.

When "only in the metaverse" experiences become common, it'll be a powerful draw for everyone. Once we experience the magic of connection in a virtual experience, we will keep coming back. If such experiences are available more or less upon demand, many people who think it's a bad idea may end up dipping their toe in the metaverse. And the rest will be history.

Some people will retreat into if not become addicted to the metaverse since they can choose adventures that fit their mood on demand. Especially if the metaverse seems better than IRL. We expect the latter could be the case for large swaths of humanity who live in impoverished city slums as in *Ready Player One*. Many people already spend inordinate amounts of time watching television, YouTube, and Netflix. On average, young people in the United States and the United Kingdom watch 3 hours a day, while continental Europeans watch 2 hours per day.

This possibility of large segments of humanity retreating from real life into the constructed reality of the metaverse is disturbing. As we indulge in the metaverse, what aspects of humanity are we enhancing and what are we diminishing? The metaverse could lessen the connection between mind and body. Will we worry more about our avatar's clothes and appearance than about the vibrancy of our IRL body and life? Time will tell ...

We see the possibility of a future with massive disparities in income and wealth. A small segment of people travel. A much larger segment of people serving them and producing what they consume. IRL traveling remains a luxury for the few. Absent course correction, we imagine a world where rich people homestead the Mars land grab while others can't afford to leave their town, much less imagine terraforming another planet.

African youth are the world's fastest-growing population. By 2050, the population of Africa may exceed 2.5 billion. Although eager to travel, 63% of African youth have yet to set foot in another African country. Breaking that down, almost half of South African youth (44%), 65% of East African youth, 67% of West African youth, and 78% of North African youth have not traveled to at least one country on the continent.

Most will travel in the metaverse. It could open them up, providing experiences they may not otherwise have.

We do, however, have lingering concerns. First, we don't know who's in the metaverse. IRL, we can touch people and look them in the eyes. In the metaverse, we won't know who is real, imposters, or bots. Second, our IRL interactions can be difficult to track (unless you have a cell phone). The metaverse may track our interactions and sell our data to the highest bidder or perhaps provide it to authorities.

Setting aside data tracking, when the metaverse can deliver unique, worldview-shifting if not mind-blowing experiences, people will flock to it. Even people that don't think it's a good idea may try it and enjoy the experience. It's game over when the metaverse can enable deep connections. It could be someone you know or a stranger. We'll frequent the metaverse if it enables us to meet people who enhance our lives.

Despite our reservations, we think the metaverse has the potential to be entertaining, effective for growth and learning, and groundbreaking for connecting with people in distant places. It's very possible that a significant segment of society may earn a basic income by playing games or performing tasks in their futuristic metaverse office from their shantytown crib. Net-net, we don't think it's a good thing if it reduces time spent IRL with nature and other humans. If we become more disconnected from nature, we will be less healthy and careful about protecting nature.

Regardless, we can imagine the metaverse being the most persuasive medium for entertainment and communication. Like the Attention Economy, the problem is who's doing the persuading. Imagine a movie that feels like you are there. At least at the level of a good roller coaster simulator. Perhaps IMAX plus THX. Jumping to judgment will be hard to intercept.

It behooves us to be mindful of how we commercialize persuasive technologies like the metaverse. If we want a democratic future, persuasive tech needs to be open and decentralized, guarding against any one entity driving the narrative.

Hence, the metaverse will be great, horrible, funny, sad, scary, and everything in between.

Given the uncertainty of where the metaverse is headed, we think humanity should begin ethical conversations now. Such conversations should consider the precautionary principle, which enables decision-makers to adopt precautionary measures when scientific evidence about an environmental or human health hazard is uncertain and the stakes are high.

Regardless, we don't see the metaverse replacing the Web or human contact.

In fact, we think the metaverse has a big missing feature: text. Text is essential for creating shared context and agreement. Can you imagine a world where people don't read or write? As of early 2023, most representations of the metaverse future don't address text. It's strangely absent other than on wall decorations.

Perhaps our metaverse visions are unfairly constrained by awkward VR mouse click typing experiences of the past; hand tracking, haptics, and voice commands now make VR typing a breeze. While the metaverse may not always be the best place for reading long posts, contracts, or books, we think it's important that the metaverse anchor on text as the foundation of agreement and shared understanding. A world that does center on text will be easily controlled.

That said, Frode Hegland and his Future of Text project are pushing the boundaries of thinking around VR and text. The Future of Text 3 book explores 3D presentation and manipulation of text. For example, flying through concepts and data could be an insightful and generative experience. Or books in a virtual library lighting up because they have information relevant to your search. We are optimistic that this work will develop new text-centered value streams in the metaverse.

In sum, without changing course, the metaverse will replicate the problems of Today's Web, including scams, harassment and other forms of abuse, deep fakes, and lack of information integrity—as well as the inability to trust speech, video, written materials, and information. Extrapolating the Web's scams and abuse into the metaverse is a terrifying prospect.

An elderly grandmother in California fell victim to a common online occurrence and lost several hundred thousand dollars as a result. Fraudsters found and targeted her on the Web. They contacted her using fake accounts and images developed on the Web. Frequent phone conversations (through an Internet spoofed phone number) played upon her desire for authentic human connection. For confirmation of their twisted narrative, they referred to information found on Google like gag orders. They sent fake documents in fake emails from fake accounts. They used an online spoofing service to falsify their caller ID. We are concerned that devious energies will pollute the metaverse.

Within minutes of being in metaverse chat rooms, a reporter working on a documentary about safety in the metaverse encountered hate speech, sexual harassment, pedophilia, and avatars simulating sex in spaces accessible to children. She described an encounter that she considered the virtual equivalent of sexual assault:¹³

I went into chat rooms and people were berating me, actually screaming at me. At one point, seven users surrounded me and tried to force me to remove my safety shield so they could do things to my body. I tried to run away, but they backed me up against a wall, trying to grab at me, making sexual comments.

This seems like an extrapolation of Today's Web into virtual reality. A 2018 study found that 49% of females and 36% of males who use virtual reality technologies reported having experienced sexual harassment.¹⁴

Without building in accountability, the metaverse almost certainly will be a new opportunity for predators, fraudsters, and trolls to expand their horizons. "If something is possible to do, someone will do it," says Lucy Sparrow, a PhD in computing and information systems at the University of Melbourne, whose research focuses on ethics and morality in multiplayer video games. She says people can be quite creative in how they use, or abuse, technology.¹⁵

For the Internet platforms, moving into the metaverse means shifting moderation from content to behavior. Facebook's chief technology officer Andrew Bosworth admitted in a leaked internal memo¹⁶ that doing the latter "at any meaningful scale is practically impossible."

We have concerns about the metaverse. How will we protect people from being scammed? What do elder abuse, harassment, trolling, and false news look like in the metaverse? Will impersonators finagle the life savings of marks in the metaverse? How will harassment feel in an immersive experience? Imagine a pack of trolls surrounding you, interrupting your speech and right of way, yelling and haranguing you, pushing you against walls. Considering the difficulty of talking with someone from Meta, who's going to save you? Where are ethical conversations about the metaverse?

Accountability is lacking on the Web. Nothing makes us think the metaverse will be any different. From our standpoint, fixing this is a crucial step and a tremendous opportunity.

NOTES

- 1 <https://permanent.link/to/the-metaweb/humanity-growing-up>
- 2 <https://permanent.link/to/the-metaweb/emarketing>
- 3 <https://permanent.link/to/the-metaweb/ai-revolution>
- 4 <https://permanent.link/to/the-metaweb/digital-engine-interviews-ai>
- 5 <https://permanent.link/to/the-metaweb/krino>
- 6 <https://permanent.link/to/the-metaweb/constitutional-ai>
- 7 <https://permanent.link/to/the-metaweb/benevolent-design>
- 8 Find out more at the project's GitHub repository: <https://permanent.link/to/the-metaweb/raven>
- 9 The organizers subsequently realized that raising money on a public blockchain tips the hand such that rival bidders can easily find out how much has been raised, and therefore their bidding capacity.
- 10 <https://permanent.link/to/the-metaweb/girlies> and [girlies.art](https://permanent.link/to/the-metaweb/girlies.art)
- 11 <https://permanent.link/to/the-metaweb/long-neckie>
- 12 <https://permanent.link/to/the-metaweb/musk-pong-tweet>
- 13 <https://permanent.link/to/the-metaweb/nightmare-trip-metaverse>
- 14 <https://permanent.link/to/the-metaweb/virtual-harassment>
- 15 <https://permanent.link/to/the-metaweb/moral-metaverse>
- 16 <https://permanent.link/to/the-metaweb/leaked-metaverse-memo>

The Vagaries of Today's Web

Sit, be still, and listen. For you are drunk, and we are at the edge of the roof.

—Rumi, 13th century Persian, Muslim poet and Sufi mystic

The words of the great Sufi poet Rumi speak volumes about the state of the Internet today. Just like a drunk person who stumbles and loses their balance, we often find ourselves lost or disoriented in the endless labyrinth of the Web. The sheer volume of information available to us can be overwhelming, leading us down countless rabbit holes and causing us to forget our original purpose for being online.

Meanwhile, in this vulnerable state, we are ever in danger of falling ... for scams, catfishes, trolls, predators, and misinformation. In this chapter, we explore this edge of the roof—the most tenacious and harmful vagaries of Today's Web. Later, we will learn how we can harness this knowledge to better protect ourselves individually and collectively.

THE INTEGRAL ACCIDENT

French Philosopher Paul Virilio once said, “When you invent the ship, you also invent the shipwreck; when you invent the plane, you also invent the plane crash; and when you invent electricity, you invent electrocution ... Every technology carries its own negativity, which is invented at the same time as technical progress.” We invent the integral accident at conception. New technologies create new ways for things to go wrong.

When you invent a tool, you also invent untold uses of the tool that can speed up harm to others and exacerbate power asymmetries. It's human nature to use tools for competitive advantage and to centralize power. Harm to others has not proven a significant deterrent.

Every new technology also carries an opportunity to iterate over time, which happens through innovation and creativity. With safety as a goal, innovation can focus on the prevention or reduction of the integral accident. Accidents happen, people tinker, companies form, laws pass, standards guide, and products flourish.

When English, French, and Hungarian inventors built and refined the components of the electric car in the late 19th century, they also invented the car crash. Six decades later, in 1958, Nils Bohlin of Volvo invented the three-point seat belt. Yet it took a full decade and many unnecessary deaths for US regulators to mandate seat belts, which illustrates the inevitable lag between invention and regulation. Also in 1968, Allen Breed invented the crash-sensing airbag. Both these inventions reduced the impact of the car crash on passengers, though not on what the car crashed into. Inventions that increased the driver's control over the car further reduced the likelihood of the crash and the ensuing collision damage.

Subsequent innovations, guided by goals other than safety, may increase the severity of the integral accident and/or the problematic use of the invention. In the car's case, ever more powerful engines not only increased the speed and therefore the convenience and utility of the car but also increased the severity and likelihood of the car crash. Yet powerful engines also led to the highway, which made traveling at high velocity safer. Of course, the highway also created the pileup accident and more benignly the (highway) traffic jam.

Besides the integral accident, the inappropriate use of the invention is also a concern. We can use tools with good or bad intentions. Soon after, or perhaps before, the hammer first hit a nail, surely, it was used to bludgeon someone. Of course, had the hammer not been available at that critical moment, the culprit may have used some other heavy object. For example, the car created the intentional running over of people and the drive-away accident. These, of course, were also problems with chariots and horse-drawn wagons, but as technology progresses, the magnitude of the problem and the severity of its effects rise.

The concept of the integral accident gives context to the problems of Today's Web. These problems are the integral accidents that came about when we invented the Web and social media. We have yet to discover the "seatbelts and airbags" that can make the Web safe again and reclaim our cognitive freedom. The integral accidents that are coming about as we invent Web3 are clear (e.g., rug pulls, discussed below), with more to come as we extrapolate the problems of Today's Web into immersive environments and integrate AI into daily life.

This chapter explores the three views of the Web that reveal the most damaging and intractable problems that impede the path to a thriving, democratic future. First, the bogus web comprises fake sites and projects, various schemes, scammers, and their victims. Second, the abusive web involves predators and those who experience abuse. Third, the misleading web includes false claims, the people and organizations that make them, fact-checking, and those who experience harm.

THE BOGUS WEB

When humanity invented the Web, we also invented online scams. Although under-reported, scams and fraud are a massive problem for the Web, including Web3, which is touted as the next phase of the Internet and, perhaps, of organizing society. Despite all the energy spent educating and warning people, online scams are out of control, which undermines confidence and safety on the Web. Scams persist because of the Web's lack of transparency and accountability, which starts with people not knowing who they are interacting with.

"On the internet, nobody knows you're a dog," says the dog sitting at a computer in Peter Steiner's 1993 New Yorker cartoon (see Figure 3.1). The cartoon captured a radical shift in human interactions that was just beginning with the Web; a shift both exhilarating and terrifying at the same time.

Over the past three decades, the online world has learned the dog's lesson. People you encounter on the Web can be anybody, anywhere. A worldly patron of the arts on a forum could be a kid in the family basement. A fifteen-year-old girl in a chatroom could be an undercover cop. The international business consultant who DMed you on LinkedIn could be a corporate spy. A Chinese lady with airbrushed photos on Tinder, the Nigerian oil heiress in your inbox, and the savvy investor who chats you up after an errant WhatsApp message, are undoubtedly scam artists. The old friend on Facebook who sends a new friend request is almost certainly a bot or some impersonator, putting pieces in place to scam you or somebody else.



"On the Internet, nobody knows you're a dog."

Figure 3.1 No one knows who you are on the Web.

The bottom line is, on the Web, you don't know who you are dealing with unless you are in a trusted network, identities are verifiable, and/or you can tell it's your friend. Otherwise, you don't know whether they are a real person, a con man, a catfish, or a bot.

While there are many similar aspects between the digital and physical world, they are opposites regarding accountability. In the physical world, you often are accountable for your actions, especially when the stakes are high. If you harass someone in the elevator, if you steal medicine from the store, if you violate laws that a jurisdiction enforces, you are accountable for your actions, unless you are young enough that your parents are accountable.

If you do something untoward, there is always the possibility that someone or some camera is watching you. You are visible. That is why so many crimes occur in the dark. If you do not leave, depending on the locale, the police or other authorities, someone protecting the community, or even a mob of fellow citizens, may accost you. If your violation is during the day, outside, in a modern city, more than likely there is a video recording of your coming and going, if not the crime itself. There may be eyewitnesses.

In financial doings, credit reporting agencies hold you accountable. The big three are TransUnion, Equifax, and Experian. They report based on the national identification number; with the US, it's the social security number. Many European countries have unique identifiers based on the date of birth and how many were born that day, including, for instance, Finland and Hungary. If you charge up credit cards and walk away, the credit cards will no longer work until you pay off the debts and rehabilitate your credit.

The digital world is different. Everyone is invisible until they allow cookies and/or register on a site. But then they are only visible to the trackers. For others to see our

creations (i.e., posts, comments, reactions, and shares), we create accounts. Most accounts include an account image that can be a photo or an avatar, a pseudonym or a “real name,” and an email and/or phone number.

On Today’s Web, most apps allow name changes without evidence and all permit changing of the account image. Hence, the problem with throwaway or burner accounts. At any moment, anyone can walk away from an account. And if they want, they can start ten more accounts moments later.

The Web Has No Accountability

If physical were like digital, criminals would disappear in front of your eyes, with an invisibility cloak or a spell that enables the soul essence of the criminal to jump into another body. In more practical terms, it would be the same as a criminal being able to enter the Witness Protection Plan and gain a new life at a moment’s notice, as much as they want. The former is impossible with 2022 tech, and the latter is ludicrous.

The lack of accountability on the Web has birthed a growing cornucopia of scams and other problems baked into the different generations of the Web. Many types and applications of online scams are common. For example, you almost certainly have heard of the advance-fee scam, one of the most common confidence tricks. The scammer promises the victim a significant share of a large sum of money, in return for a small up-front payment, which the fraudster claims will enable them to gain the windfall. When a victim pays, the fraudster either invents a series of further fees for the victim to pay or disappears.

Many variations of this scam exist, most notably the Nigerian prince scam, also known as a 419 scam, referring to the section of the Nigerian Criminal Code that deals with fraud and the charges and penalties for offenders. You receive word from the Nigerian prince, who has lost access to his private trust fund, but with your help (and money), he can liberate the funds and provide you with a large sum of money for your efforts. These scams have occurred with fax and traditional mail, but now online communications like emails and text are most prevalent.

According to the FBI, almost 15,000 people in the US reported falling victim to advance-fee scams in 2019, losing \$101 million, or \$6,800 each.

But it doesn’t stop there. Enterprising scammers bring a similar level of ingenuity to other situations, including disaster relief, “phishing” for login credentials with fake websites, shopping websites, tech support, antivirus software, travel, tricking the grandparents into sending money to get the grandkids out of a jam, pre-approved notices, debt relief and credit repair, lottery prizes, as well as checks and money transfers for online purchases.

Scammers also follow the news. According to Google, scammers took advantage of the enormous volume of COVID-19 communications by disguising their scams as legitimate messages about the virus. The scammers may use email, text messages, automated calls, and malicious websites to reach you. Common types of COVID-19 scams included fake health organizations, websites that sell fake products, bogus government sources, fraudulent financial offers, and fake nonprofit donation requests. Fraudsters can adjust any of the aforementioned scams for the post-COVID world or any other world-changing topic (e.g., food shortages).

Scams and fraud are difficult to stop. Fraudsters innovate around the current state of the defense. This makes them unpredictable, with authorities having to play catch-up. These crimes are especially prevalent in emergent technology markets since early stage technologies often have vulnerabilities that the builders never expected—the integral accident. Criminals are early adopters of communication technologies, seeking to exploit the new market and

leverage the new capabilities to increase their take. The early days of any technology ecosystem are a “Wild West” where imaginaries, visionaries, and hoodwinkers all vie for attention. Along with doctors, drug dealers were the first to use pagers and cell phones.

Emergent ecosystems are honeypots for finding victims. They aggregate people in an environment with few rules, standards, or constraints. The Web itself created a whole new category, online crime. It features phishing, Ponzi schemes, and rug pulls. Experts do not think that we can stop online crime. Even if all software was perfectly secure, we would still be subject to considerable losses from social hacking, crooked insiders, and human error. Interestingly, Gavin Wood, the founder of Ethereum and coiner of Web3, thought the blockchain technology would create safe transactions because of cryptography. But the criminals simply focus on the very human social aspects of an exchange, which are much more vulnerable and pliable than cryptography.

Given the evolving nature of the Web, eradicating scams and fraud is a challenging, if not impossible, proposition. And it could get much worse with generative AI, as the technology becomes more advanced and sophisticated, it could be used to create more convincing and sophisticated scams and frauds, making it more difficult to detect and prevent them. It could also make it easier for scammers to impersonate others, and create fake websites, emails and social media accounts, making it more challenging for individuals and organizations to protect themselves.

The Rug Pull

Most Web3 scams aim to separate people from the crypto in their wallets, often by tricking people to reveal their passcode, send their money, or purchase worthless crypto assets. Soon we'll tackle Web3 rug pulls and other scams that Today's Web cannot solve.

No place exists to report scams that will investigate and possibly catch perpetrators, retrieve stolen crypto, or prevent future crimes. If you live in the US, you can file a report with the Federal Trade Commission (FTC)¹, but our understanding is that agencies, like the FTC and the FBI, do not solve individual crypto scams.

The idiom “Rug pull” comes from the phrase to pull the rug out from under (someone), meaning to withdraw support without warning. Our mind goes to the gritty, dust-blown set of a Spaghetti Western where someone pulls a rug from under someone's feet. The person crashes to the ground, but the table remains standing. As the visual of this idiom suggests, when someone experiences a rug pull, they lose their stability. This can be sudden, startling, and harmful. It often takes the deceived person time to regain their bearings and any semblance of control in the situation. Rug pulls can cause cascading problems from the emotional hurt and pain of being let down and the financial hit from wasting resources. It can also cause embarrassment, mental and emotional stress, and further lapses in judgment from a diminished mindset.

In the Web3 context, the term “rug pull” refers to unknowing purchasers “*having the rug pulled from underneath them*” by the creators of a cryptocurrency. But it is now taking on several forms. The most common type is the liquidity scam, which most commonly takes place on decentralized exchanges (DEXs like Uniswap and Pancake Swap). DEXs run by consensus with many machines working together as one network, rather than centralized exchanges (CEXs like Binance and Coinbase), which are owned by one central party.

On a DEX, developers can list a coin or token for purchase. The gas fee is the cost of putting the transaction on the blockchain. Gas fees can vary from cents to hundreds of dollars, depending on the blockchain and the demand to place transactions. CEXs have more rigorous approval processes, requiring users to comply with KYC (Know Your Customer)

and AML (Anti-Money Laundering) procedures. And some charge hundreds of thousands of dollars to list a token.

A new coin created on a DEX must pair with another cryptocurrency. Fiat currencies, such as US dollars or Mexican pesos, are not typically accepted on decentralized exchanges (DEXs) because DEXs are designed to facilitate the exchange of cryptocurrency assets. Fiat currencies are issued and controlled by central banks, while DEXs operate on decentralized networks using blockchain technology, which allows for peer-to-peer transactions without the need for a central authority. The creators or developers add a paired cryptocurrency that balances the amount of their own new token in a “*liquidity pool*,” enabling trading between the two coins.

Once a coin lists on a DEX, the fraudsters will often whip up a PR storm around the coin on Twitter and inject substantial liquidity into the pool. They may also pump up the coin’s price by purchasing large amounts themselves before selling them off to legitimate traders (known as a “*pump and dump*”).

This activity generates market value for the coin and inspires unwitting “retail” investors to rush to buy the coin as the price skyrockets. Once the trading volume for the coin reaches a high level, the liquidity pool contains a significant amount of the paired cryptocurrency.

Then, the rug pull takes place. The fraudsters withdraw the paired cryptocurrency from the liquidity pool, and disappear, closing down or abandoning their Discord, social media accounts, websites, and other avenues for communication. The price collapses to zero, and investors are left “*holding the bag*,” having lost their entire investment in the coin.

2021’s most prominent rug pull was the SQUID coin, a play-to-earn cryptocurrency. inspired by Netflix’s dystopian South Korean thriller, *Squid Game*, in which game masters force debtors to play deadly children’s games for money. In the first game, Red Light-Green Light, the game masters shoot anyone caught moving; upon realizing the sadistic nature of the game, half the contestants panic, turning the game into a massacre. Investors thought they would take part in online games where they could earn more tokens, which would later be exchangeable for other cryptocurrencies or national currencies.

In just 11 days, the price of the coin rocketed to a peak of \$2,861, appreciating 310,000%. But as investors soon found out, there was no way to sell the coin. On the 11th day, the anonymous creators of the SQUID coin sold their entire holdings of over \$3.3 million and disappeared, making the coin worthless.²

Mo Money, Mo Problems

Traditional pump and dump rug pulls raked in over \$2.8 billion in 2020. This was 36% of the \$7.7 billion worth of cryptocurrency stolen, according to Chainalysis’ 2021 Crypto Crime Report.³ The total scam revenue was \$6 billion, or 69% lower than in 2019, because 2020 had no major crypto Ponzi schemes. In 2019, six Ponzi schemes accounted for about \$7 billion in losses. In a Ponzi scheme, existing investors receive funds collected from new investors like in the Bernie Madoff scandal.⁴ Ponzi scheme organizers promise high returns with little or no risk, and at some point, they disappear. Ignoring the Ponzi schemes, scams were 40% higher in 2020. This is consistent with the number of individual payments to scam addresses, which rose 48% from 5 million to 7.3 million.

Given that Web3 centers on money and wallets, it is not surprising that Web3 scams are significant and growing as more people enter the cryptocurrency space. As the space grows and communities get better organized, it may be harder to pull off Ponzi schemes. Ponzi schemes take months to develop and often involve paying out fake proceeds in the initial phase, which is a financial risk for the fraudsters.

As of February 2022, 80 million people—about 1% of the world's population—had blockchain wallets.⁵ The average cryptocurrency loss per blockchain participant was \$96. In a future where Web3 is ubiquitous, absent drastic changes to prevent Web3 crime and educate investors, losses could increase 100-fold or more.

The Internet Crime Complaint Center's (IC3) 2020 Internet Crime Report counted \$4.2 billion of total reported losses from Internet fraud in the United States. In 2020, IC3 received a record number of complaints from the American public: 791,790, a 69% increase in total complaints from 2019.

The top three Internet crimes reported in 2020 were phishing scams, non-payment/non-delivery scams, and extortion (also known as ransomware). Most funds went to business email compromise scams (BEC), romance and confidence schemes, and investment fraud. In a BEC scam, criminals send emails appearing to come from a known source, making a legitimate request such as these examples:

- A vendor the company works with sends an invoice with an updated mailing address.
- The CEO asks their assistant to purchase a stack of gift cards to send out as employee rewards. She requests the serial numbers of the gift cards so she can email them right away.
- A new homebuyer receives an email from the title company with instructions for wiring the down payment.

People over 60 years old had the largest percentage of losses at 23%. Elderly people have a challenging time discerning charlatans online.

And recall that the IC3 numbers are for the United States, which had 298 million Internet *users* in January 2021. The average amount lost to fraud per Internet user in 2020 was greater than \$14. If these numbers scale with the world's Internet *users* (4.95 billion), worldwide Internet fraud losses would be 16 times larger.

Internet and Web3 fraud is likely to exceed reports. People underreport losses for several reasons, including embarrassment, not knowing where to report it, and not thinking that there will be any positive outcome for reporting the crime.

Unfortunately, as AI becomes more sophisticated, it could be used to create more convincing and sophisticated Web3 scams and frauds, making it more difficult to detect and prevent them. For example, an AI-powered phishing attack could use natural language processing to generate highly convincing emails or chat messages that mimic the communication style of the target's contacts, enticing them to send crypto or connect their wallets.

AI could be used to create fake or malicious smart contracts that drain wallets, launder money, or manipulate decentralized markets. AI could also be used to impersonate real people and organizations, creating fake dApps, emails, and social media accounts, making it more challenging for individuals and organizations to protect themselves from Web3 scams and fraud.

What Makes Us So Vulnerable to Scams

Given the propensity of scams and Internet fraud, let's think about what enables the massive level of fraud, and how to stop it.

The Internet itself is a honeypot of potential victims. Over 300 million people in the US are on the Web,⁶ including the elderly, lonely, medicated, and young people, whose prefrontal cortex, which controls executive functions and decision making, is under-developed until they are 23–25 years old. The 71% of US Internet users who use Facebook are providing

predators an online dossier on their psyche through wall posts, likes and other reactions, comments, shares, photos, videos, RSVPs, and more.⁷ It's similar on many other platforms, including Twitter, Instagram, and Reddit.

The Internet has no accountability. The way platforms work aids criminals. Most platforms, including Reddit, Twitter, Instagram, and many aspects of Web3, allow pseudonymity. Facebook, the main platform that has full names, requires only an email to start an account. Ironically, Facebook claims that "Authenticity is the cornerstone of our community," yet they enable throwaway accounts based on a random email. In fact, several sites on the Web offer email-verified Facebook accounts online starting at \$0.54 each.⁸

Before reading on, please answer this question: how many fake accounts do you think Facebook took down in 2019, 2020, and 2021? Here's some context that may support your thinking process. Facebook had 1.5 billion and 1.9 billion daily active users in Q1 2019 and Q3 2021.⁹ They had 2.9 billion monthly active users in Q3 2021.¹⁰ How many fake accounts could Facebook have needed to remove?

PAUSE & CONTEMPLATE the question in the previous paragraph, and continue when a number pops into your mind. We've asked many people this question. About 5% were in the ballpark.

Drum roll ... According to Statista, Facebook removed over 6 billion fake accounts in 2019, 5.8 billion in 2020, and over 6 billion in 2021.¹¹ For context, Facebook removed 3.3 billion fake accounts in all of 2018. According to Facebook's Community Standards Enforcement Report, 5% of worldwide Facebook monthly active users (or about 145M) are fake.¹² But a more meaningful statistic is daily active users. Assuming that fake users are active daily, 7.6% of Facebook's daily active users are fake.

It doesn't take a lot of engagements to get noticed. Hundreds, if not dozens, of timely shares, can make surprising, intriguing, or counter-intuitive content go viral. Assuming a fake account's annual value is equal to that of a regular account, 7.6% of Facebook revenues or more may be from fake accounts. But in actuality, they are worth much more than authentic accounts because they are controllable. They focus on changing behavior and influencing purchases. Fake accounts have real value and are very inexpensive to make.

It's not only fake accounts. Recall the cartoon with the dog. We don't know who's on the other side of the computer screen. It could be anyone and they could have ill intent. Unfortunately, the Internet makes it simple for ill-intentioned people to spoof phones, websites, wallets, and emails. Phone spoofing services forward fake phone numbers to real phones. Spoof websites appear to be existing websites or legitimate businesses. Spoofed wallets are pop-ups that appear to be a crypto wallet in order to capture the victim's wallet password or passphrase. Spoofed emails appear to come from a legitimate and relevant source.

The dark web is an enormous factor in making the Web unsafe. It is a shadowy space on the Internet that is accessible with special browsers that prevent activity tracking, such as Tor. Many illicit goods are available for purchase on the dark web, including stolen credit card numbers, fake social security cards, bogus college degrees, uber-realistic silicone masks, illegal wildlife trade, and products such as African ivory and child pornography. The dark web hosts hacking collectives that send out Trojan horse emails that appear legitimate but can take control of your computer. Trojan horses can damage, disrupt, steal, or inflict some other harmful action on your data or network.

The lack of accountability on the Web reflects the scarcity of justice for online transgressors and their victims. Internet crime is lucrative and far less risky than crime in real life.¹³ The typical Internet hacker commits thousands to hundreds of thousands of crimes per year and almost never gets caught. The few that get caught rarely get jail time, which often amounts to a few years in a low-security facility.

Sadly, identity thieves almost never get caught. Between 2003 and 2006 (the years for which data are available), the FBI arrested between 1,200 and 1,600 identity thieves, and only a third of those cases resulted in convictions, much less jail time. To put this in context, these crimes affected 8.3 million victims, or 4% of the entire US adult population. Authorities convicted one identity thief for every 20,750 victims.

As AI and bots become more advanced, they have the potential to make identity theft even more prevalent and harder to detect. AI-powered bots can be used to automate many of the tasks associated with identity theft, such as creating fake accounts, harvesting personal information, or conducting phishing attacks. This automation can make it harder for authorities to trace the origin of the criminal activity, and to identify the individuals behind it.

AI-powered bots can also be used to camouflage criminal activity by mimicking the behavior of legitimate users, making it more difficult to detect malicious traffic and identity theft and to identify the individuals behind them. AI can also make the phishing and social engineering tactics of identity thieves more convincing and personalized.

Your Data Are Not Safe

A huge amount of personal data leaks from the Web every year. Despite almost 50% of reported data breaches not confirming the number of records exposed, the total number of records compromised in 2020 exceeded 37 billion. This was by far the most recorded in a single year, and a 141% increase from 2019.

Data breaches can be a hack, an inside job by current or previous employees, or an unintentional loss or exposure of data. In a data breach, hackers or employees release sensitive, confidential, or protected data into an untrusted environment. The severity of breach events increased by a factor of 10 over the course of 2020. Five breaches exposed one billion or more records and another 18 breaches exposed between 100 million and 1 billion records. Healthcare was the biggest target, accounting for 12% of reported breaches.

The breached data might disappear, or enable malicious uses such as ransomware. In ransomware attacks, hackers encrypt data to deny the owner access until they pay a fee in cryptocurrency. Ransomware is malware a hacker enables on a system that blocks user access to the files or systems. Encryption holds files or even entire devices hostage until the victim pays a ransom for the decryption key, which enables the victim to recover their files or systems.

Twice as many attacks included ransomware as an element in 2020 as in 2019. In 2020, Ransomware experienced a bigger increase than any other category of crypto crime, with cases at 319% above 2019 levels. This amounted to \$350 million or 7% of the total cryptocurrency taken in by criminal wallet addresses in that year. Experts think the jump from 2019 to 2020 was because of Covid-prompted work-from-home measures which opened new vulnerabilities for organizations.

On May 7, 2021, the primary artery for gasoline and refined products for the South and East Coast of the US, Colonial Pipeline, suffered the largest known cyberattack on energy infrastructure. The attackers encrypted about 100 gigabytes of data in Colonial's network in just two hours. They threatened to leak the stolen data to the Internet unless Colonial paid a ransom. Colonial took systems offline to contain the threat, halting all pipeline operations, and thus preventing the delivery of over 100 million gallons/day of fuel and heating oil. Overseen by the FBI, the company paid the 75 bitcoin or \$4.4 million ransom within several hours. Upon receipt of the ransom, the Russia-based hacking group DarkSide provided a tool to restore the system, which was down for 6 days.

Like other forms of hacking, ransomware is ever-morphing, with hackers innovating to defeat the current state of the defense, which seeks to prevent known hacking methods from happening again.

Discovering and prosecuting cybercrimes is harder than in other areas of law enforcement. Rules of evidence requirements and jurisdictional boundaries make Internet crime especially difficult to track and prosecute.

Under-resourced and under-skilled, law enforcement struggles to keep up with the growth of cybersecurity incidents. Law enforcement agencies focus on bigger cases they know will garner a successful prosecution. Authorities prioritize cybercrime that has actual effects on people, such as cyberbullying, child sex crimes, incidents that cause financial burdens to many people, and crimes that threaten national security. Smaller crimes are not on the radar of law enforcement.

Moreover, cybercrime is global and many country pairs do not have extradition treaties. For example, over 76 countries, including China and Russia, don't extradite to the US. The likelihood of prosecuting known cybercriminals for crimes against a nation committed from places without an extradition treaty is low.

Hence, for individuals and most small, mid-size, and even large businesses, there is little justice on the web. Because of this and the potential hit on their reputation, many companies and individuals forgo reporting cyber incidents to authorities. Experts recommend companies develop cybersecurity programs and maintain insurance to help businesses recoup losses from cybercrime.¹⁴

The stakes are increasing regarding fraudulent activity on the web. According to a 2020 report, the post-COVID global fraud detection and prevention (FDP) market size will grow from \$20.9 billion in 2020 to \$38.2 billion by 2025, at an annual growth rate of 12.8%. The market growth reflects increased risks from remote access, growing use of electronic transactions during the pandemic, and rising revenue losses from fraudulent activities.¹⁵

THE ABUSIVE WEB

Given the evolving nature of the Web, eradicating scams and fraud is a challenging if not impossible proposition. And the Web has several other problems, many of which seem intractable, including abusive behavior, data insecurity, and false information.

Abusive behavior is rampant on Today's Web, including harassment, trolling, cyberbullying, catfishing, and hate speech that incites violence.

Online harassment, especially of women and girls, can be relentless. According to Mozilla's Internet Health 2020 report, 58% of girls and young women faced online abuse or harassment. This affects their feelings of safety and well-being.

Trolling is a huge and persistent problem that is enabled by an online world with pseudo-anonymity, throwaway accounts, and no accountability. Trolls post inflammatory, insincere, digressive, extraneous, or off-topic messages in an online community. They intend to provoke emotional reactions from unsuspecting readers, engage in fights and arguments, or manipulate perceptions.

The name "trolling" comes from the fishing technique of "trolling," where a slow-moving boat pulls fishing lines with colorful baits and lures. The trolling lures attract unsuspecting fish as they flutter through the water, enticing them to "take the bait." Similar to unsuspecting Internet victims, once hooked, the fisher reels in the fish for the catch before they realize what's happening.

Cyberbullying may account for the rise in teenage suicides. While cyberbullying is rarely the sole factor contributing to suicide, kids who experience bullying—as well as those who

bully others—are at an increased risk of suicide. In cases that link suicide with cyberbullying, other factors exist, such as offline bullying and mental health issues, such as depression.^{16,17}

Girls are also more vulnerable to cyberbullying, as are disabled kids, LGBTQ, and the obese. The Centers for Disease Control estimates that cyberbullying affects 15.5% of high school students, compared to the 20% of high schoolers who experience in-person bullying at school.¹⁸ Cyberbullying can occur anywhere kids connect online. Cyberbullying is endemic in many apps that allow users to remain anonymous, such as Kik Messenger and Yik Yak. But plenty also happens on text, instant messaging, and social media comments.

Facebook estimates about 0.14% to 0.15% of their views contain bullying and harassment. In July–Sept 2021, they took down 9.2 million pieces of content, 1 million of which were appealed, and restored 433,000 or 5%.¹⁹

Meanwhile, less than a month after Meta (the former Facebook) announced that it was building the metaverse, a beta tester for its virtual-reality social media platform, Horizon Worlds, reported being groped by a stranger.²⁰ Although the groping was virtual, it was still troubling. Jesse Fox, an associate professor at Ohio State University who researches the social implications of virtual reality, says sexual harassment does not have to be physical; it can be a verbal or virtual experience as well.

In a clear case of “blame the victim,” Meta’s internal review found that the beta tester should have used one of Horizon Worlds’ built-in safety features. Safe Zone is a protective bubble users can activate when feeling harassed, within which no one can touch them, talk to them, or interact until they turn it off. Horizon vice president Vivek Sharma called the groping incident unfortunate and said he aimed to make the interaction-blocking feature easy and findable.

While this seems helpful, it masks a larger problem. Facebook’s response focused on how the victim could have better protected themselves and how Facebook can make it easier for victims to know how to protect themselves. While this is a step in the right direction, the response does not even mention anything about the perpetrator. That’s because they did nothing about the perpetrator.

In Horizon World, it’s easy to create burner (or throwaway) accounts. One can simply create an account on Facebook and log into Horizon with their Facebook credentials. Account transference is one way the problems of Today’s Web can extrapolate into the metaverse. As mentioned earlier, Facebook’s account system is one of the most porous. Facebook could be called Fakebook. As mentioned earlier, they removed an average of 6 billion fake accounts per year from 2019 to 2021. Again, they admit 5% of active monthly users are fake.

The only information required to start a Facebook account is an email address and your full name, which is not verified. The email address can be from a free email provider such as Gmail, which takes seconds to set up and requires no verification. Rather than safety, they optimize for a minimal friction account registration process. Just like the Web, Meta doesn’t think in terms of accountability or bringing perpetrators to justice. Or even stopping perpetrators from having access to vulnerable people.

Wait, but why? They don’t know who the perpetrators are.

In the video introducing Meta’s plans for the metaverse, Zuck didn’t address the Web’s major problems. These include surveillance, data exploitation, security lapses, the lack of information integrity, information silos, filter bubbles, search bubbles, fake accounts, unidentified bots, serial abusers, cyberbullies, scam artists, catfishes, trolls, etc.

He spoke on a cursory level about privacy and safety features. You can decide when to be with other people, block someone, or teleport into a private bubble—as if that would be better than taking off the headset. In our mind, we imagine a group of menacing avatars

including Mutant apes, Rude Boys, and Solano Sluts cornering the protective bubble of an avatar in a metaverse concert. Hopefully, the bad actors cannot hack the bubble, track the victim, or steal their metaverse property and possessions.

What about holding bad actors accountable for their actions in the metaverse?

Meta's introductory video shows a world of positive, fun, and interesting imagery. Personalized avatars, visible from the waist up, collegially chat, play cards, and hover if not fly about the control room of a spaceship. Avatars change the view and the decor of a personal dwelling. They flow between a library, a lab, and a meeting room and have pleasant and respectful interactions.

Of course, they don't show groping. But online infractions in the metaverse will go well beyond groping. What do confidence scams, catfishing, and bullying look like in an immersive 3D environment where everyone is an avatar?

Our society starts to pay attention after tragedies. The law of unintended consequences and Murphy's law suggest that we should use the precautionary principle with all aspects of computational technology from here on out. Of course, this is easier said than done. Competition and market demand drive a global economy. The first mover gains a competitive advantage by being the first to market with a product or service. Yet technologies that are rushed to market have unintended consequences.

A prudent society would convene ethical research and conversations about technologies that have the potential for substantial impact before they go to market. Especially for exponential technologies that can double in capability or performance over a short period. This would need to be done at national and global levels and have real consequences for non-compliance.

In some situations, a bit of restraint and transparency could go a long way. For example, on the Web and the metaverse, bots should be identifiable and avoidable. People should know what algorithms are operating and how they work. For political or issue-based communication, they should know who is pushing it and how to avoid it. People should also own and control their data, and receive the lion's share of the value should they decide to monetize it. And, of course, everyone should be accountable for their actions.

Hate Speech and Violence

Beginning in 2017 and continuing in 2018, Facebook was used to incite deadly violence against the Rohingya, an ethnic and religious minority in Myanmar. During this time, between 700,000 and 900,000 members of the Rohingya community fled to overcrowded camps in Bangladesh amid a military crackdown and ethnic violence. In March 2018, a United Nations investigator said Facebook had "turned into a beast," citing the violence and hatred against the Rohingya.

In August 2018, Reuters found over 1,000 examples of posts, comments, and pornographic images attacking the Rohingya and other Muslims on Facebook. Three months later, Facebook admitted trolls had used its platform to "foment division and incite offline violence" in Myanmar, citing a human rights report commissioned by the company.

The growing number of attacks on immigrants and minorities raises concerns about the connection between inflammatory speech online and violent acts IRL.²¹ As well as about the role of corporations and the state in stopping incendiary speech. Worldwide trends in hate crimes reflect changes in the political climate and suggest that social media often magnifies discord. At the extreme, online rumors and invective have contributed to violence, including lynchings and ethnic cleansing.

Hateful posts have occurred on every habitable continent. Individuals inclined towards racism, misogyny, and homophobia find niches on Facebook that reinforce their views and goad them to violence. Social media provides violent actors a platform to publicize their acts and encourage others to pile on.

Facebook monitors the content posted on its platform for a wide variety of online hate speech. They take down over 20 different categories of hate speech, such as calls for violence.²² This includes content where a symbol represents the target or violence, such as a gun or a clenched fist. Facebook monitors another 15 categories of hate speech, selectively removing those that appear dangerous based on additional information and/or context.²³

The prevalence and variety of hate speech illustrates both the creativity and depravity of humanity. It also reflects the structural lack of accountability and justice on the Web.

Unfettered Access to NSFW Content

NSFW is shorthand for Not Safe For Work. The Web is rife with NSFW content, including almost a quarter of the Reddit boards. For instance, the On/Off board hosts fully-clothed photos of women side-by-side with a photo in the same pose but in varying states of undress, and often nude. This is not something you want your work supervisor to see, hence the term.

Today's youth, however, have access to the entire web. Every kind of online content, including the vacuous, tasteless, and vile. Everything that you can imagine humans might do is likely somewhere on the dark web or the deep web, if not the surface Web. Snuff films, rape videos, suicides, BDSM (bondage, discipline, sadism, and masochism), sexual fetishes, self-mutilation, and reckless stunts all have the potential for great harm to young minds and bodies. Some of this is behind age gates that require the user to confirm that they are indeed 18 to proceed. The average teen doesn't follow that instruction.

Here are some sobering statistics about children and porn access:

- 4% of websites are pornographic, but 13% and 20% of web and mobile searches are for porn.²⁴
- 93% of boys and 62% of girls experience online pornography before 18.²⁵
- 13.4 years old is the average age of first exposure for boys, with 5 being the youngest.²⁶

In 1999, Benedek and Brown documented the negative effects of pornography on teenagers, including:²⁷

- Imitation and modeling of inappropriate behaviors.
- Unhealthy interference with the normal sexual development process.
- Emotional side effects such as nightmares and feelings of guilt, shame, anxiety, and confusion.
- Premature stimulation of sexual activity.
- Development of harmful and misleading attitudes towards sex.

Youth porn viewing can lead to addiction, which can have long-lasting consequences. Researchers found that teen pornography addiction is associated with poor social bonds and aggressive sexual behavior. If early exposure to porn can have such dire effects, one can only imagine the impact of the entire cornucopia of NSFW content on children's lives.

In 2020, Facebook agreed to pay content moderators \$52 million as compensation for mental health issues developed on the job. Two years earlier, a group of US-based moderators hired by third-party companies sued Facebook for failing to create a safe work

environment. The class-action lawsuit alleged that reviewing violent and graphic images—sometimes of rape and suicide—on Facebook feeds had given moderators post-traumatic stress disorder (PTSD). Over 11,000 moderators were eligible for compensation.

Since January 2020, third-party contractors that hire moderators for social media platforms, including Facebook and YouTube, ask workers to sign a form acknowledging they understand the job could lead to PTSD.

If adult Facebook moderators can get PTSD from NSFW content, imagine the impact on children. Because their brains are still developing, children are much more vulnerable to lasting damage from trauma. Access to disturbing images and videos can scar a child, such that the trauma shows up in adulthood as ineffective and self-harming behaviors.

While childhood traumas do not have to haunt us for our entire lives, we believe traumatic experiences that remain unexamined can pop up as unexpected challenges during adulthood. Unless we connect with our traumas, learn about ourselves, and integrate them, they will remain in our shadow, lurking for opportunities to interject themselves into our adult lives.

Thus, the impact on children of exposure to NSFW content is damaging and therefore an abusive aspect of the web—regardless of whether the creators of the content intended it to be viewed by children. The prevalence of damaging content online reflects the structural lack of accountability and justice on the Web and the subverting of parental authority and the integrity of the family in our society.

THE MISLEADING WEB

Knowing what's happening in any part of the world is fantastic, but how do you know whether it's correct? Twitter has 6,000 posts every second.²⁸ We already mentioned how fake accounts and bots influence less discerning readers. Human beings have always done things that are acceptable to their tribe; now our “tribe” extends to the people we agree with online. Hence, it is easier than you think to get lost in an echo chamber of misinformation. The algorithms won't help; they push you in further. When someone posts, no one confirms whether it's “actual and factual.” Rather, it's immediately available to all, and the algorithms serve it to people who are most likely to engage.

Hence, the world's most important communication and collaboration platform has become rife with false information. While the Web has a lot of useful information, it has much more false, misleading, and duplicative information. This limits trust in web content and its usefulness for learning and sensemaking.

Misinformation is inaccurate information, often open to multiple interpretations, that may spread without the intent to mislead. Disinformation is misleading or biased information, manipulated narratives or facts, and propaganda. We like to sum up misinformation and disinformation as false claims or information. A third category, malinformation, is real information leveraged to harm or gain an advantage, either personal or institutional.

The term “*fake news*” has become polarizing. It's often used as a catch-all for false or misleading information. But it is often used to disparage someone the speaker does not agree with. In the US political media, fake news has become a politicized term thrown about by politicians and journalists alike, meaning “don't pay any attention to what's being said.” In an attempt to give a less ideological definition, Temple University defines fake news as “purposefully crafted, sensational, emotionally charged, misleading or totally fabricated information that mimics the form of mainstream news.”²⁹

Namely, fake news is disinformation disguised as mainstream news. But this definition is inadequate as it doesn't account for when mainstream news manufactures false information.

We also think total fabrications are rare compared to instances where the fabrication is only part of the story. Our preference is to speak of false and/or misleading information when the motive for posting is unknown, as misinformation if unintentional, and disinformation if intentional.

Because advertising is the dominant monetization model online, commercial sites avoid using outbound links that would take users away from their site. As discussed earlier, the lack of outbound links to unaffiliated sites means there is little context on the Web. Information is in content silos produced by page authors without substantiation. The lack of context makes it difficult for users to discern what is real on the Web.

Over the years, false information shared on social media has posed major challenges to several sectors of social life, including elections, the response to the COVID-19 virus, and geopolitical conflict. For example, bots and unsuspecting users spread disinformation across Internet platforms, leading to the mass dissemination of false information in the 2016 Brexit vote in the U.K. According to University of Oxford research, Russia's Internet Research Agency (IRA) leveraged the Internet platforms—Facebook, Twitter, YouTube, Tumblr, Instagram, and PayPal—to spread propaganda during the 2016 US Election.³⁰ Of course, Russia is not the only state-level actor leveraging the platforms to spread propaganda. Who else do you think uses such tactics?

Research by the Pew Center shows Americans view made-up news as a bigger problem for the country than terrorism, illegal immigration, racism, and sexism. 68% of US adults say made-up news and information affects Americans' confidence in government institutions, while 54% say it is having a major impact on our confidence in one another.

According to a 2018 survey on false information on the Web and social media, humans can only identify false information with an accuracy of between 53–78% across experiments with different false behaviors, including hoaxes, fake reviews, and fake news.³¹ Many people neglect to take the time to consider the accuracy of the things that pop up in their feed. Hence, most people share information before reading it.

An example of this is the Syrian blog, *A Gay Girl In Damascus*.³² In 2011, a lesbian named Amina was blogging to support the Syrian uprising. One day, "Amina's cousin" made a post on Amina's blog claiming that three armed men had kidnapped Amina. This caused a public outcry from the LGBTQ community in Syria and made it into mainstream media. But it turns out Amina was a fictional character. An American named Tom MacMaster created her. As the *Washington Post* revealed, MacMaster had used Amina Arraf as a fake profile for several years, in online forums. He wanted to have genuine conversations about key questions in the Middle East. MacMaster's backstory is like those of other recent hoaxers: he had written in his own voice, including a novel, but these efforts met with little success. So he created Amina—but the endeavor, MacMaster says, "got out of hand."

With global reach achievable through the Attention Economy, promoting an intriguing idea or mindset is no longer difficult. In fact, seven to ten coordinated users on any major social media outlet can be enough to get the ball rolling.

The proliferation and weaponization of false information coupled with the inability to distinguish falsehoods online is problematic for a democratic future. Representative democracy requires an informed populace, yet the spread of misinformation and disinformation is uncontrollable, and people don't have effective tools for discerning falsehoods. Unmet needs include the ability to know what to believe on the web, context on the web, and access to information that contradicts and/or supports what one is focusing on. We can sum this up as a lack of information integrity for all important aspects of human development and social engagement online.

The media industry includes tens of thousands of publishers who create news content for websites, newspapers, magazines, and streaming shows. Commercial news is available through an advertising model, where customers have free access to the content with advertising paid for by large companies and organizations that seek to influence behavior. The industry has shifted in the past two decades so that Facebook and Google now control over 50% of the advertising markets. On some platforms such as YouTube, users can pay a subscription fee to remove ads or access exclusive content without ads.

Fact-checking is a process that seeks to verify or discredit information to promote veracity and correctness in reporting. Fact-checking can occur before or after publishing. Most large media companies now have internal fact-checking teams that seek to prevent their company from creating and spreading misinformation.

External fact-checking organizations first arose in the US in the early 2000s, and the concept grew in relevance and spread to other countries during the 2010s. The US is the largest market for fact-checking. Research suggests that fact-checking does indeed correct misperceptions among citizens, as well as discourage politicians and others from spreading false or misleading claims. However, corrections may decay over time, be incorrect, or disappear in a deluge of content from influencers that promote inaccurate information.

The International Fact-Checking Network launched in 2015 as a unit of the Poynter Institute dedicated to bringing together fact-checkers worldwide to support a booming crop of fact-checking initiatives by promoting best practices and exchanges in this field. There are now over 200 fact-checking organizations in the IFCN. Traditional fact-checking practices, however, are too slow for modern information systems, requiring an average of 13 hours to identify false claims and respond with correct information. Many fact-checking bodies operate separately from news publishers and the content they assess, which means most readers are unaware they exist.

Those that are not independent can be susceptible to the biases of their parent and partner organizations or those that fund their efforts. This can align fact-check content with mainstream narratives. Hence, many alternative media and researchers call for fact checking the fact checks.

As more people get their news on social media, the online false news problem exacerbates. 48% percent of US adults consume news on social media; but among those aged 18 to 49, about 70% say they often get news on social media.³³ All age groups turn to news websites at higher rates than other digital platforms for news, with one exception: among Americans aged 18 to 29, 42% say the most common digital way they get news is social media, versus 28% saying the same of either news websites or search engines. Many of these young people—and perhaps by extension future generations—may be susceptible to consuming and spreading false news. And on Twitter, ostensibly the digital town square, fake news spreads faster than true news, according to MIT research.³⁴

As discussed in *Dead Web: The Dark Side of Generative Tech*,³⁵ the ability of generative AI to create convincing, realistic, and personalized content is increasing rapidly. This presents a significant concern when it comes to the proliferation of false or misleading information online. With the ability to generate text, images, and videos, it becomes easier for actors with nefarious intent to create and distribute false information at scale, making it increasingly difficult for individuals to distinguish between true and false information.

This is particularly concerning when it comes to political and economic manipulation, as the ability to target individuals with personalized and misleading information could have significant consequences for democratic societies. The impact of this flood of misleading content generated by AI could be significant, as it could erode trust in institutions, sow divisions, and discord, and undermine the ability of individuals to make informed decisions.

The Post Web3 Forecast: Trust is Still a Must

The Web3 revolution has been gaining momentum in recent years, promising a decentralized, secure, and transparent internet. But as we look towards a post-Web3 future, it's clear that trust is still a must.

While Web3 offers numerous benefits, it's crucial to understand that it's not the end-all, be-all solution to online scams, abuse, and false information on the web. Decentralization and blockchain technology may be able to help reduce these issues, but it may also exacerbate some of them as well. As detailed earlier, cryptocurrencies and NFTs are new, highly lucrative scam vectors.

One major challenge with Web3 is the lack of regulation. The decentralized nature of the platform makes it difficult to enforce rules and regulations against scams, abuse, and false information. This lack of regulation means that users must be especially diligent and informed about the platforms and services they use.

In addition, accountability is also a monumental challenge. The decentralized platform makes it difficult to hold people accountable for their actions. For example, token transactions can be anonymous, difficult to trace, and are irreversible. This lack of accountability makes it impossible to stop scams, abuse, and false information.

Another issue is the constant evolution of scams, abuse, and false information creates a cat-and-mouse dynamic. As soon as a solution is found, scammers, hackers, and misinformation spreaders will find a new way to circumvent it. The bad actors are always one step ahead of those trying to prevent or enforce crimes.

Given these challenges, trust and accountability remain a crucial factor in the post-Web3 world.

APPROACHING A POINT OF NO RETURN

We may have come to an unpredictable and dangerous milestone in our relationship with artificial intelligence. In 1950, English computer scientist and cryptanalyst Alan Turing invented a method of inquiry in AI development for determining whether a computer can think like a human being. The Turing test assesses a machine's ability to exhibit intelligent behavior equivalent to, or indistinguishable from, that of a human. Given sessions are text-only, the machine need not understand speech, though progress is happening there as well.

As presented in Chapter 2, the GPT-3 AI is quite capable of having human-like conversations. It, however, is not infallible. Informal experiments with GPT-3 on the Web show even the best AI systems struggle with common sense and logical reasoning. Although we have very capable AI systems, they also make silly mistakes. We think of common sense as what you expect an adult to know about how to interact and interpret the world around us. AI needs common sense to serve as a useful collaborator, and to pass the Turing test.

In early 2022, humans couldn't distinguish photos of actual people from high-end computer-generated graphics. The study found their accuracy rate to be 48%, which is worse than a coin flip. Later, researchers trained new participants on spotting glitches in computer-generated graphics of people. The new participants classified faces taken from the same set in the first experiment—but despite their training, the accuracy rate only improved to 59%.

Humans find AI-generated faces more trustworthy than real human faces.³⁶ The ratings for synthetic faces, on average, were 7.7% more trustworthy than for actual faces, despite the actual faces smiling 11% more frequently. Besides black faces being rated as more

trustworthy than South Asian faces, there was no effect across races. Women, however, are rated more trustworthy than men.

The study used fake faces that were created with StyleGAN2, a “generative adversarial network” (or GAN) from US tech company Nvidia.³⁷ GANs pit algorithms against each other, to create convincing representations of the real world. In the above example, a GAN was used to create competing, realistic photographs of synthetic human faces derived from a pool of real human photographs.

In GANs, two neural networks play a game of cat and mouse. The generator model attempts to make artificial images indistinguishable from photographs. The discriminator model attempts to figure out which ones were created by the generator. At each iteration, the results are fed back into both models, making the generator more realistic and the discriminator better at picking out synthetic faces. The two networks train one another; after a few weeks, the image-creating network can produce realistic fake images.

Don’t Believe that AI Could Fool You?

Try your skill at the “Which Face is Real?” site, which generates fake images using the StyleGAN2 used in the aforementioned study at whichfaceisreal.com.

STOP reading after this sentence and try at least 10–15 before you come back.

Spoiler alert! We first did 20 and identified the computer-generated face 50% of the time. But then we went back and read the LEARN page where they explain the “tells” of fake images. These include water-splotches, background issues, asymmetrical eyeglasses, other asymmetries (e.g., in hair, earrings, facial hair, shirt collars), hair anomalies, fluorescent bleed in the background, and odd teeth. Armed with this knowledge, we spotted the computer-generated faces 17 out of 20 times (or 85%). If you are interested in getting better at spotting computer-generated faces, check out the excellent tutorial published by Kyle McDonald in 2018.³⁸ And then try playing again!

Machine learning algorithms trained on thousands of images of real paintings can also create impressive art. The more images of a style or aesthetic, the better. From abstract expressionist masterpieces to time-period portraits to idyllic landscapes, AI can create artworks indistinguishable from those painted by humans. A 2021 study found that 200 humans in online surveys could not pick out human-made artworks from AI-generated artworks.³⁹

AI can also generate writing that appears to be from a human. There are now several startups that provide AI-based writing services. From our observation, the AIs are pretty decent at rewriting individual sentences from a specific information ecology, and—now with ChatGPT—are good at stringing together sentences into paragraphs. Previous incarnations used to seem like aimless writing well attuned for SEO and backlinks. AI wrote the Preamble of this book.

Services exist that use AI for voiceovers. They claim their AI voice actors are getting more human-sounding all the time.

Regarding the Web, visuals matter. Seeing is believing. Society becomes more vulnerable as AI improves. But let’s talk about now. Today, we cannot distinguish between a face created by AI and a genuine face. AI is improving such that it also talks and sounds like us. Are we at risk of losing touch with reality? What safeguards will we need in place to prevent

“deep fakes”—AI-generated audio, image & video—and bots from manipulating us in the metaverse?

As the line between human-generated content and AI-generated content is becoming increasingly blurred, there is a significant danger regarding the proliferation of false or misleading information. As AI advances, it generates highly convincing and personalized content for political and economic manipulation and scams. It's crucial that we take this issue seriously and invest in tools and techniques to identify and label AI-generated content.

Absent structural changes regarding data, information, and accountability, the aforementioned problems ingrained in the structure of the Web (and to varying extents exacerbated by generative AI) will extrapolate and cause havoc in the metaverse. While these inherent limitations of Today's Web pose a significant challenge, the potential to improve people's lives by overcoming these constraints is enormous.

May we have your attention? The next chapter's subject—the Attention Economy—leverages the Web's structural problems to create extractive business models that harm users. If we prefer an online economy that supports our wellbeing, it's crucial to understand how the Attention Economy shapes our online experiences.

NOTES

- 1 <https://permanent.link/to/the-metaweb/reportfraud>
- 2 <https://permanent.link/to/the-metaweb/squid-coin>
- 3 <https://permanent.link/to/the-metaweb/rug-pull-scams>
- 4 <https://permanent.link/to/the-metaweb/madoff>
- 5 <https://permanent.link/to/the-metaweb/blockchain-wallets>
- 6 <https://permanent.link/to/the-metaweb/internet-users>
- 7 <https://permanent.link/to/the-metaweb/us-facebook>
- 8 Accessed on Jan 8, 2023: <https://permanent.link/to/the-metaweb/purchase-facebook-accounts>
- 9 <https://permanent.link/to/the-metaweb/facebook-global-users>
- 10 <https://permanent.link/to/the-metaweb/monthly-facebook-worldwide>
- 11 <https://permanent.link/to/the-metaweb/facebook-fake-account-removal>
- 12 <https://permanent.link/to/the-metaweb/transparency-facebook>
- 13 <https://permanent.link/to/the-metaweb/internet-crime-unpunished>
- 14 <https://permanent.link/to/the-metaweb/cybercrime-challenges>
- 15 <https://permanent.link/to/the-metaweb/fraud-detection-prevention-market>
- 16 <https://permanent.link/to/the-metaweb/cyberbullying-suicide>
- 17 <https://permanent.link/to/the-metaweb/prevalence-effect-cyberbullying>
- 18 <https://permanent.link/to/the-metaweb/us-violence-trend>
- 19 <https://permanent.link/to/the-metaweb/bullying-harassment>
- 20 <https://permanent.link/to/the-metaweb/metaverse-groping>
- 21 <https://permanent.link/to/the-metaweb/myanmar-violence>
- 22 <https://permanent.link/to/the-metaweb/violence-incitement>
- 23 <https://permanent.link/to/the-metaweb/hate-speech>
- 24 <https://permanent.link/to/the-metaweb/internet-porn>
- 25 <https://permanent.link/to/the-metaweb/pornography-youth>
- 26 <https://permanent.link/to/the-metaweb/pornography-exposure>
- 27 <https://permanent.link/to/the-metaweb/negative-porn-teenagers>
- 28 <https://permanent.link/to/the-metaweb/twitter-stats>
- 29 <https://permanent.link/to/the-metaweb/temple-fake-news>
- 30 <https://permanent.link/to/the-metaweb/propaganda-2016-elections>
- 31 <https://permanent.link/to/the-metaweb/humans-false-information>

- 32 <https://permanent.link/to/the-metaweb/gay-girl-damascus>
- 33 <https://permanent.link/to/the-metaweb/social-media-2021>
- 34 <https://permanent.link/to/the-metaweb/fake-news-faster>
- 35 <https://permanent.link/to/the-metaweb/dead-web> and deadwebbook.com
- 36 <https://permanent.link/to/the-metaweb/fake-faces-trustworthy>
- 37 <https://permanent.link/to/the-metaweb/ai-faces-trustworthy>
- 38 <https://permanent.link/to/the-metaweb/recognize-fake>
- 39 <https://permanent.link/to/the-metaweb/fake-look-trustworthy>

Pay with Your Attention

What information consumes is the attention of its recipients. Hence a wealth of information creates a poverty of attention.

—Herbert Simon¹

All Web problems are not the same. The problems of the previous chapter—scams, harassment and cyberbullying, children accessing NSFW content, and false information—are problems of opportunity due to vulnerabilities structured into the Web and human frailties.

Even if the Web’s next iteration were able to keep people safe (by solving these problems), the Web would still be detrimental to our thinking processes. It inspires companies to become voracious collectors and consumers of personal data; marketers to create meaningless content and false virality; users to compete for attention; and platforms to arbitrate truth.

This chapter explores the Web 2.0 operating system, the Attention Economy, its attendant poverty of attention, and its negative effects on social media users. These include polarization, false virality, online addiction, surveillance, data exploitation, and censorship.

THE SCARCITY OF ATTENTION

1970s Nobel Prize-winning economist Herbert Simon coined the term “Attention Economy,” positing that attention was the “bottleneck of human thought” that limits both what we perceive in stimulating environments and what we can do. Simon posited that since information consumes the attention of its recipients, an abundance of information—which is an understatement for the Web—creates a scarcity of attention. Fast forward to today, and this statement could not be more relevant for the web.

Concerning the poverty of attention, we have seen attention spans drop such that website designers aim to grab the visitor’s attention within 6 seconds. Within that time period, most visitors choose whether to stay on the page. They may see the hero image and read a couple of lines of text. So different from web surfing, where we would at least skim the page.²

A recent Microsoft study found that the human attention span has dropped to eight seconds—shrinking almost 25% between the years 2000 and 2015. Our ability to concentrate on a task or object now falls short of even a goldfish’s. The humble goldfish regarding being able to focus on a task or object. Could we be improving our ability to discern what we are interested in, responding to the diminishing quality of content, or are we overwhelmed by the amount of content passing through our minds and perhaps experiencing some restructuring of our brains?

The average user reads at most 28% of the words on a page and 20% on average.³ Average page visits last less than a minute and people often abandon webpages in just 10–20 seconds.

According to Contentsquare's 2021 Digital Experience Benchmark report, the average time people spend on a page across all industries is just 54 seconds.⁴ Business-to-business sites had the highest average time spent on webpages, at 82 seconds, and grocery and energy sites had the lowest average time spent per page, at 44 seconds.

How could this be? Two major shifts in how we focus our attention over the past two decades have been social media and smartphones. Social media is a big part of the stickiness of smartphones. Our smartphones are with us all the time. They beep or buzz to notify us of social interactions. We may experience phantom vibration syndrome where we perceive our mobile phone to be vibrating or ringing when it's not. On average, we check our phones over 200 times per day. Beyond sapping our attention, smartphones have other effects on our lives.

Smartphones help us connect with friends and family, plan our social life, stay informed, and track our steps. But they can also distract us from work, spark feelings of inadequacy, and encourage reckless spending. A study from the American Psychological Association found that about one-fifth of people think technology is a source of stress.⁵ Further, potential physical effects of being "always-on" range from neck pain and wrinkles to elevated blood pressure.

How can we make sure we are getting the most from our devices yet cultivating the best uses of our attention?

When considering Simon's "wealth of information," we think of wealth in the sense of quantity rather than quality. So much information is available through our phones that users can't keep up with it all, and it quickly buries quality content. Futurists predicted several years ago that content production would eclipse our ability to consume it, and some people now believe we have reached this point, known as "content shock."⁶

As shown in Figure 4.1, the amount of digital data stored on servers is approaching a vertical asymptote and will reach over 175 zettabytes by 2025. One zettabyte of data would take about a million supercomputers (or a billion powerful home computers) to store. We are storing over 8.4 million times more digital information than just 45 years ago.⁷ Clearly, it is impossible to keep up with what is being produced. We have more information than we

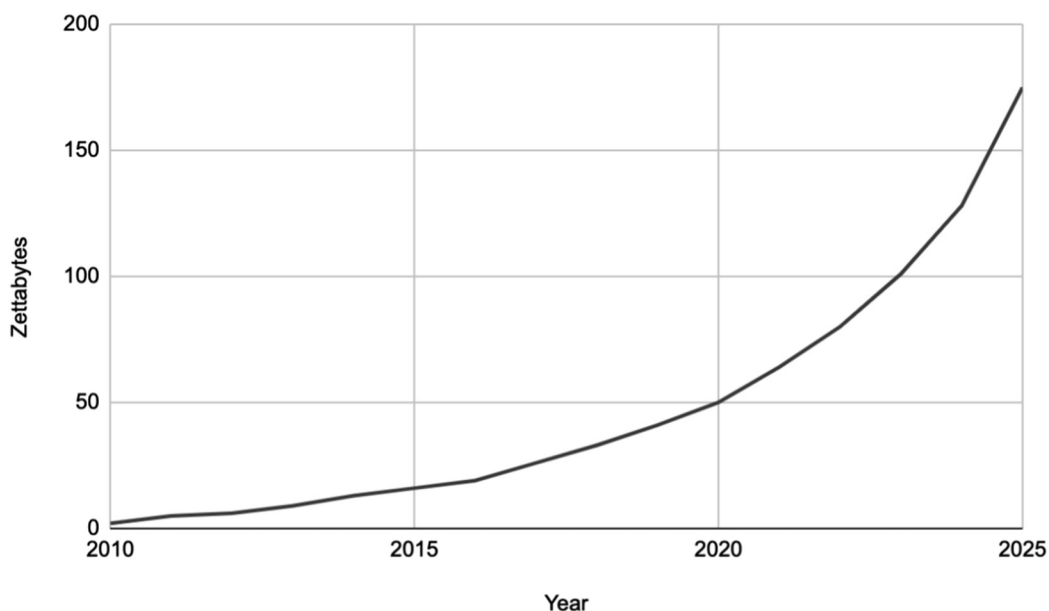


Figure 4.1 The exponential growth of the global datasphere (adapted from *Data Age 2025*).⁸

can imagine at our fingertips. If only we knew what was there and how to find what we need in the moment.

While digital information access experiences exponential growth, our ability to process information remains the same. In one of the most cited psychological articles, cognitive psychologist George Miller says that humans can process only about 7 (plus or minus 2) chunks of information at a time.⁹ Maps—of all types—are so useful because they merge information in a way that enables us to focus on what’s contextually relevant, see the relationships and the context, and process large amounts of information.

The Web amplifies the overload effect on a societal level. Have you noticed how overloaded people are? Perhaps you’ve noticed yourself or someone else saying, “I’m not sure what to think!” or “How do I know?” or “I wonder what the experts have to say.”

Perhaps we have hit an information scaling threshold. While analyzing a global history database spanning 10,000 years, Shin et al. found a pattern whereby civilizations scale until their information environment overwhelms them.¹⁰ In the words of Gordon Brander, founder of Subconscious¹¹:

“When a society hits its information scaling threshold, it stalls out. It can’t function until it invents new ways of making sense that can cope with the complexity of the information environment. And societies that don’t pull off this transition? They collapse.”

“The internet has massively increased the complexity of our information environment, but hasn’t yet produced the tools to make sense of it. Old forms of social sensemaking—institutions, universities, democracy, tradition, religion—all seem to be DDOS’d¹² by the new information environment. They can’t keep up!”

We’ve hit the information scaling threshold because of the Web. Online information and advertising have completely overwhelmed our sensemaking. This also affects how we think, publish, find, store, share, and make sense of information.

Brander continues.

“The cost of forking realities has dropped below the Coasean floor,¹³ and there’s little incentive to merge realities. We fractally fragment understandings, then algorithmically amplify the confusion to maximize engagement. The most effective coordination mechanisms left seem to be memes and conspiracy theories.”

The complexity of information overload, as depicted in Figure 4.2, involves multiple contributing factors and carries the possibility of significant repercussions. Michael Simmons says information overload comprises several challenges that are each getting worse, and together make up a crisis that is making us collectively dumber.¹⁴ One of the primary factors of information overload on the Web is the rapid rate of new information being produced. Also called journalism of assertion, the Web’s continuous news culture puts a premium on

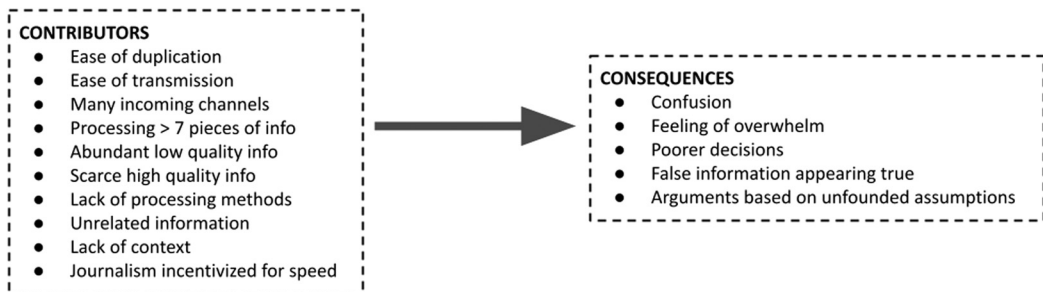


Figure 4.2 Contributors to and consequences of information overload.

breaking news, which gets attention to breaking stories earlier and rewards the producers of the first stories, but also leads to diminished quality of the news stories overall.¹⁵

Many other factors contribute to information overload on the Web, including:

- The ease of information duplication.
- An increase in incoming information channels (e.g., email, feeds, text, WhatsApp, Signal, Discord, Telegram, YouTube).
- Continuously growing amounts of historical information to dig through.
- Contradictions and inaccuracies in available information.
- Low signal-to-noise ratio for quality information in search and on social media.
- Lack of methods for comparing and processing different information.
- Lack of connections between related stories.

The solution is to reduce the amount of information or improve our processing capacity.

In this situation of extreme information overload, the solution is to reduce the amount of information or improve our processing capacity. If we were to reduce the amount of information, we would want to remove the superfluous information so that we can focus only on what is relevant at the moment. Improving processing capability requires new processes and/or tools.

MONETIZING ATTENTION

With content exploding, cognitive abilities flatlining, and attention spans shrinking, attention has become a scarce resource. This has led to the Attention Economy becoming the dominant monetization strategy on Web 2.0.¹⁶ All the Internet Platforms have embraced it.

In the Attention Economy, the sequence of content in feeds stems from algorithms designed to increase engagement, such as like, share, click, or buy. The Attention Economy has many actors, including individual users, organizational users, organizations, product and service companies, marketers, and data brokers.

The Attention Economy is a complex system with multiple actors, interactions, and patterns of use.

One, Internet platforms make a very limited set of functionalities available to the user and organizations for the right to monetize the data generated and original content posted to the platform.

Two, the Internet platform sells or auctions targeted advertising services to companies and causes, whereby algorithms serve specific ads to specific users based on their personal information, interests, and web activity, optimizing for the aim of the campaign.

Three, the Internet platform uses algorithms to create feeds for each user that reflect their interests and web activity, optimizing for engagement and behavior change.

Four, marketers of all stripes—with good, neutral, or ill intentions—often use strategies for making viral content that involve fake accounts or false information to amplify their messages on the platform and generate the intended behavior changes.

Five, people and organizations use the platform within its limited set of functions to engage with their friends, family, and fans.

Six, data broker companies extract your data, and then sell it to other companies, provided the receiving company agrees to not sell your personal data.

The Netflix documentaries, *The Great Hack*¹⁷ and *The Social Dilemma*,¹⁸ and Roger McNamara's book *Zucked: Waking Up to the Facebook Catastrophe*¹⁹ helped open our eyes to the Attention Economy and how it affects us. Today, many people acknowledge that

“free” social media tools are not actually free; they are extractive and we are the product. People realize billionaires are benefiting the most from the Attention Economy.

People also accept that the Attention Economy harms people and concentrates wealth in the social media giants while optimizing for attention, engagement, and behavior change. Their business models and algorithms control choices and change behavior, which leads to addiction and opens the doors to election hacking, data exploitation, and misinformation. Such are the unintended consequences of basing an economy on human attention, now visible for all to see.

Less recognized is how the Attention Economy fragments our information ecology into silos. Information silos occur when webpages do not have links to independent sources of information that substantiate their claims. The Attention Economy encourages the compartmentalization of information, because websites need to keep you on their site so they can extract your data and serve you ads. Commercial sites avoid linking to sites unless they have a financial interest (e.g., links to ads, their other sites, or affiliates). This reduces context on the web, diminishing our ability to make sense and meaning of the web.

Silos mix accurate information with misinformation, disinformation, and malinformation. Social media hosts cyber bullies, trolls, catfishes, fake accounts, unidentified bots, and serial abusers. Web users don’t know what or who is real.

Some users are retreating—away from trolls, meaningless chatter, clickbait, and false information—into smaller, more exclusive spaces where deeper and more authentic conversations prevail. But the movement to safe spaces increases fragmentation and the possibility of confirmation bias and groupthink. Web users also cut and paste into note-taking apps and writing applications, which create more silos, this time on the user’s own machine.

Perhaps some of you are seeing the outlines of our cognitive cage.

Culminating in the late 2010s, we had a series of recurring waking dreams that ultimately leaped out of the liminal space into our worldview.

We see ourselves walking in a dark alley in a dark city into a dark building. We are climbing multiple levels of a rectangular black and white checkerboard staircase. At the top of the stairs, we pause and then enter an ornate door into a large room with high ceilings and wooden wall panels. A man stands in the corner; black leather trench coat, thick heavy boots. He turns, facing me, close cropped hair, pilot glasses. He speaks:

Let me tell you why you’re here. You’re here because you know something. What you know, you can’t explain. But you feel it. You’ve felt it your entire life. That there’s something wrong with the world. It is this feeling that has brought you to me. Do you know what I’m talking about?

Without thinking, words spill out of our mouth “The Web?” The man smiles and asks,

Do you want to know what it is? The Web is everywhere. It is all around us, even now in this very room. You can see it when you look out your window or when you turn on your television. You can feel it when you drive to work, when you talk on your phone, when you pay your taxes. It is the world that has been pulled over your eyes to blind you from the truth.

Again, slipping out of our mouth, “what truth?” Morpheus continues:

That you are a slave. Like everyone else, you were born into bondage; born into a prison that you cannot smell or taste or touch. A prison for your mind ... This is your last chance. After this, there is no turning back. Take the blue pill, the story ends; you wake up in the metaverse and believe whatever you want to believe. Take the green pill; you stay in the Metaweb, and I show you how deep the rabbit hole goes ...

The Hollywood blockbuster *The Matrix* postulates a constructed reality that seems like being in the metaverse but not knowing it. In the film, after losing a war in the early 21st century with intelligent machines, humans languish in liquid-filled pods that harvest their bioelectric energy while pacifying their mind. They are in the Matrix, a shared simulated reality based on the world as it was in 1999.

Thinking about the Matrix as a metaphor for Today's Web, we had an alternate vision of the inside of a matrix pod, as if we had x-ray vision. From the side and above, we see a fit young man standing in front of a virtual control panel, wearing VR goggles, a haptic jumpsuit, and hand-control gloves. He yells "Hell Yeah" and shakes his fists above his head, pleased with his performance. He's been in-game for a long while.

We look deeper into the pod and we see bunk beds and even a green room for players to hang out between sessions. They are working—by playing games—in the metaverse. We interpret this vision as showing how we collectively might choose our own matrix. What if large parts of society end up spending their time in the metaverse and their data fuels the apparatus?

We find the notion of the Web being a constructed reality that impairs our cognitive freedom to be intriguing. Though we must concede, some consider "The Web" in the Americas much too free. China, for example, has firewalled its own Chinese Web off from the world. People in China have to use VPNs to break through to the "Free" web. Russia may head in a similar direction.

It's not that we have no cognitive freedom or that the cognitive cage is an overwhelming problem in our lives. Rather, its influence is insidious, barely discernible, popping into our consciousness without our being able to control our response to it. But a cage is not necessarily bad. Some dogs love their cages. Cages can make us better at some things—like social media. If we can align with the cage, we can act easefully, gracefully, and even get rewards.

The cage, however, can limit our cognitive freedom and hinder us from achieving high expressions of ourselves. Were we satisfied with mere existence in what appears to be a democracy, we could simply bliss out on social media dopamine hits. But we are living in exponential times, armed with linear thinking tools, and facing existential threats. We need to change how we live on the planet.

We don't say that lightly. Humanity is an ongoing experiment. Survival is not guaranteed, or perhaps even likely. Humanity must rise to a higher expression that significantly reduces our burden on natural systems and also manages not to destroy itself. But the cognitive cage many of us find ourselves in makes it challenging for us to even acknowledge our responsibility.

NOTICE what is going on in your head right now. Some of you are trying to discredit what we are saying. RELAX.

We think the Attention Economy merits social research and policy with the same level of curiosity and seriousness as addictions, such as alcohol and other psychoactive drugs. Health and social effects, even on people who were not involved in addictive behavior, are the subject of extensive studies for addictions like alcoholism.

Several obstacles would need to be overcome. The harms of the cognitive cage are hard to measure and not well understood. The conceptual realm is invisible. Outcomes do not correlate with a single isolated thought or occurrence.

The difficulty of pinning down the cognitive cage makes it interesting, if not powerful. We recall a line from the 1995 movie *The Usual Suspects* spoken by a guileful character:

The greatest trick the Devil ever pulled was convincing the world he didn't exist.

Like the cognitive cage. So let's unpack cognition.

Cognition comprises our basic mental processes: sensation, attention, and perception.

Sensation is the ability to feel things physically through your sense of touch. It can also be a mental process resulting from the external stimulation of a sense organ outside the realm of conscious awareness.

Attention is the ability to focus on aspects of our environment while tuning out others. How we focus our attention was a matter of survival for early humans. Now it frames our experience of the world.

Perception is our sensory experience of the external world. It involves recognizing environmental stimuli through sight, hearing, touch, taste, and smell, as well as our actions in response to the stimuli. Through the perceptual process, we gain information about the elements and properties of the environment that are critical to decision-making and survival.

Group-living is part of being human. We live as groups, because groups are better problem solvers and providers than individuals. Given the primacy of human settlements in the human journey, it behooves us to think about cognition on a collective basis.

Collective cognition is an interaction among three elements—the individual cognitive abilities of the agents involved, their shared knowledge, and their communication structure.²⁰ This book explores how to bring the collective's intelligence to bear on humanity's global challenges.

We consider culture and norms a type of collective cognition. The shared knowledge is family history, schooling, stories, entertainment, and relevant laws. Communication structures are ad hoc. The aim is to evaluate fit regarding the common culture and norms, and update or change these norms as necessary to align with new circumstances.

In the right circumstances, a crowd can outperform experts.²¹ A study on collective cognition found that groups perform better on both syntactic and semantic levels than their best members.²² The study analyzed how well individuals and groups reconstructed the sentences from a syntactic (i.e., the number of errors) and semantic (i.e., the quality of the retrieved information) perspective. Groups made fewer errors and could retrieve more information when reconstructing the sentences, out-competing even their best group members.

Collective cognition can help humans make better collective choices. The question is how can the Web help us improve our collective cognitive capabilities? The Web is our shared knowledge and communication structure. If the Web can increase the individual cognitive abilities of users, this would enable humanity to increase its collective intelligence and collective cognitive capabilities.

But the Web's feeble support for shared knowledge and knowledge building constrains collective cognition in humans. Despite the variety of communication structures on the Web (e.g., Twitter, Facebook), any platform will support only a few capabilities (e.g., tweets must be 280 characters or fewer). Also, the platforms have ineffective aggregation methods for knowledge building, and they lack integration with other platforms (unless they own them).

Thus, diminishing our collective cognitive capabilities for knowledge building, identifying false information, sensemaking, meaning-making, choice-making, and conflict resolution. We are on the lower end of what's possible for these collective cognitive capabilities.

If we want to work together democratically to address our existential threats, we will need to develop our collective cognitive capacities. Improving these capabilities is a solid way of thinking about raising consciousness, and may be the best way for humanity to chart its course into the future. Unless we are comfortable with autocratic and totalitarian governments or even AI making these decisions for us.

On an individual and collective basis, the cognitive cage of the Attention Economy impinges on our cognitive freedom by restricting or otherwise controlling our attention. Beyond influencing what we read, watch, and listen to on the Web, the cage can spur

sensations outside our conscious awareness based on what we experience online (and even more so with immersive experiences).

Infringements on cognitive freedom cause cognitive disorders, which could affect complex mental operations such as memory, learning, language use, problem-solving, decision-making, reasoning, and intelligence. Given that the cognitive cage comprehensively affects how we perceive the world and choose to live, it is important that we understand its significance.

How does it affect our mindset, our daily behaviors, how we think of material goods, what we wear and what we eat, who we engage with at different levels, and how we vote? We can only speculate on these at this point because of the lack of research on the cognitive effects of the Attention Economy. Later, we discuss some very specific and undesirable features of the Attention Economy.

Before we move on, let's take another look at the dream by addressing some questions it brings up.

Is the Web everywhere? In almost all the places one goes, the Web is there. Usually, the phone is in our pocket or purse. While cell phones don't work in places without a signal, they can capture and then release data when they connect to the network. And even if the phone is off, it may transmit information about its location to the network provider.

Could the Web blind us from the truth or not? As mentioned earlier, many people have trouble identifying false information before they share it.

Could we be in cognitive cages? We seem to be in a highly intentional cognitive cage that is controlled largely by algorithms at the behest of large corporations. The social feeds and the search recommendations that the algorithms personalize for us are the bars of a prison cell for our minds.

INFORMATION IN SILOS AND RODENTS IN MAZES

A poignant metaphor for the fragmentation and monetization of attention is the rodent maze. Early psychologists built wooden mazes with tricky blind alleys and wrong turns in which they conducted experiments to understand how rats learn. They build the modern analog—today's standard mazes—from a few pared-down primitives, pressed from plastic, and purchased in bulk: a tub, a circle, a plus-sign, a T-shape. The psychologist's maze has grown ever less convoluted since it was first invented—less intricate, less “mazy.” Instead of hoping the rodent will find its way around a labyrinth, today's scientists try to elicit a few simple, easily measured behaviors.²³

If you haven't already figured it out, you are the rat!

In this metaphor, the Attention Economy is the aggregation of all the different mazes. Each platform has its own proprietary maze structure, from which they generate unique mazes for each person. Your maze is a unique feed comprising a sequence of user-generated posts and ads that algorithms generate.

The ordered set of posts is the analog to the walls of the maze. The data scientists are observing how you respond to the feed, which posts you like, comment on, and share; your response time, and whether you click through and read the post. Like the rodent maze, what we experience on the Web is a constructed reality. What we see appears coincidental, but it's intentional.

We are alone and isolated.

This metaphorical view helps us see again how alone and isolated we are on the Web. We are the unassuming rat in the maze. Our interactions with the walls of the maze (i.e., posts)

can morph the walls of our maze (and the maze of others) to better steer desired behavior changes.

Filter Bubbles and Echo Chambers

You're not the only one on a webpage. Though it feels that way. There may be dozens of people on the same webpage, but you can't see or interact with them. So for social interaction, you head to social media. Social media algorithms, acting on your data, put you in a digital echo chamber that feeds your confirmation bias and reinforces your pre-existing views.

Echo chambers have always existed IRL. Examples include friend groups or departments at work. Echo chambers in social media are different. In social media, algorithms operate 24/7 on behalf of digital advertising cartels to curate what you see.

Eli Pariser introduced the notion of filter bubbles in a 2011 TED Talk. Filter bubbles are echo chambers on social media. Filter bubbles forefront posts of people with similar beliefs.²⁴ Some posts are from people, groups, and pages that we follow. All reinforce a specific narrative by presenting information that aligns with our views or may outrage us. The algorithms optimize for engagement and behavior change, such as liking a video post or clicking through to an advertisement.

Echo chambers can increase polarization, intolerance, and false information. Hence, spending extended time on social media in filter bubbles may stunt one's personal development and decision-making. It's like trying to understand the world, yet limiting oneself to a single news source.

Filter bubbles socially fragment us by putting us into small groups of like-minded people. This diminishes cognitive freedom and our ability to collaborate with others who may not think like us.

UNDESIRABLE ASPECTS OF THE ATTENTION ECONOMY

The Attention Economy has several features that harm users. These are integral to the business models of the Internet platforms. These essential yet undesirable features include polarization, false virality, addiction, surveillance, data exploitation, and censorship. Together, they are a Molotov cocktail for the Metacrisis.

Polarization

We find something interesting and share it without reading it or *verifying* the claims.

While navigating the Internet, you may have found an intriguing story and shared it with your *network*. With an infinite stream of information in our feed, we rarely take the time to verify the claims in articles we share. In fact, many of us will pass on links we have not read.

In 2016, researchers at Columbia University and the French National Institute found people did not read 59% of the links they shared on social media.²⁵ Most people are more willing to share an article than read it. Many form opinions based on summaries, without attempting to go deeper. On Web 1.0, we went deep, web surfing and exploring rabbit holes. In Web 2.0, we have shallow interactions with the feeds, even for our topics of interest.

Web 2.0 also made it much easier to publish content. In Web 1.0, one needed access to a web server. Web 2.0 platforms allow anyone with an account to publish content on their own channel or feed, and to be rewarded with likes, views, and followers. Follower count

has become the de facto ranking system for online popularity, and a key metric for being able to make a living online as an influencer or creator.

The opportunity to earn a living creating content or being an influencer has led to a proliferation of content channels. Whatever you believe, you can find one or more channels or groups on social media that provide content aligning with your beliefs. Even if you don't seek them out, the algorithms will bring them to you. As mentioned earlier, the algorithms organize people in filter bubbles of similar worldviews.

This massive fragmentation of perspectives in news media, combined with algorithmic filter bubbles, has led to an unprecedented level of political polarization. People get content related to their political inclination and the worst of the opposing party. There is no neutral ground. The opposition is far-left or far-right.

The longer you spend on the platforms, the more data they extract and the money they make. It's natural that their algorithm keeps showing you posts you will enjoy or hate. Not only does this lead to polarization, but it also causes social media addiction.

False Virality

Keeping you engaged is great for the platforms. Their primary need is your ongoing engagement, irrespective of the content you consume and the agendas you subscribe to. They don't care what you think as long as it fits within their acceptable narrative band.

As mentioned earlier, Facebook removed an average of 6 billion fake accounts in 2019, 2020, and 2021, and says up to 5% of monthly active users are fake accounts. That's about 145 million fake accounts. Enough to support millions of agendas—from legit marketing campaigns to political disinformation and everything in between. Regardless, it's a big win for Facebook as 5% (or more) of their revenues—\$5B—may be driven by fake accounts.

But this may be a gross underestimate. First, fake accounts are likely much more active than the average user. Second, they could be bots. Third, bots don't need to sleep or take breaks.

People cannot tell bots from real people. Not realizing this diminishes cognitive freedom and increases the potential for manipulation.

Today, people can't tell bots from real people within social media interactions. Not realizing you are having conversations with a bot diminishes your cognitive freedom. Whenever we cannot distinguish organic reality from a constructed reality, the potential for manipulation increases.

Each fake account lasts an average of 9 days, which is eons in content distribution. And they are cheap.

Assuming a uniform distribution of takedowns, if Facebook takes down 6 billion fake accounts a year and 5% of active accounts are fake, each fake account lasts an average of 9 days. This is eons in the world of content distribution. Given that the virality of content can be a matter of seconds to hours, fake accounts have tremendous value for changing behavior. And they are inexpensive to make. With account registration software, the marginal cost is the price of a token representing a *user* session. Sites sell these tokens on the surface web.

Bots are worth more than actual accounts because they are controllable and don't need breaks. They focus on changing behavior and influencing purchases or thinking. Unlike humans, their minds do not change. They only get upset and lose their cool if that's in their program.

In sum, armies of bots are trying to influence your behavior on the Web. They are active, and they can push thousands of agendas at a time. They create a fake reality tunnel through

their posts, likes, comments, shares, and messages, which amplifies certain content. This manipulation results in you seeing content that marketers, unscrupulous companies, and bad actors want you to see.

A Plandemic for the Mind: The Growth of Psy-Ops on the Internet

Deep in the shadows of the internet lies a sinister world of deception and manipulation, where psychological operations, or “psy-ops,” are carried out with alarming precision and reach. As the web becomes more and more entwined with our daily lives, the threat of these covert operations grows, and it’s becoming clear that the traditional methods of detection and defense are no longer sufficient.

Investigations have revealed that Today’s Web, with its vast interconnectedness and anonymity, has become the perfect breeding ground for these operations, and they are becoming increasingly difficult to eliminate or even reduce.

The key to successful psy-ops is subtlety. When done right, the targets of these operations often have no idea they’re being manipulated, and the true source of the deception remains hidden. This makes it almost impossible to know how successful these operations are, or to what extent we’ve been affected by them.

But there is hope. A new and innovative solution that could fundamentally change the game has been discovered: a meta layer on top of the web, through browser overlays, where individuals have only one account and are held accountable for their actions. This approach would eliminate the ability to create fake identities and control bot farms and would allow individuals to choose their own personal algorithms.

The implementation of this meta layer would certainly be a challenge. It would require a significant effort and buy-in from a wide range of stakeholders, including government, media, civil society, and the private sector. And there are certainly concerns about privacy and censorship that would need to be addressed.

But the potential benefits of this new approach are enormous. By creating a more transparent and accountable web, we would be in a much better position to identify and prevent psy-ops. Additionally, this meta layer would provide many other benefits, such as more control over personal data and a more democratic web.

The truth is, the web as we know it is becoming increasingly vulnerable to psy-ops, and it’s unlikely that we will be able to eliminate or even significantly reduce the problem. But by thinking beyond the current one-dimensional web, and embracing new and innovative solutions like a meta-layer, we can put ourselves in a much better position to identify and prevent them. It won’t be easy, but the potential benefits are worth the effort.

Social Media Addiction

Why do we allow social media to permeate so much of our lives? Tristan Harris of the Center for Humane Technology says social media is like a slot machine. Both deliver dopamine hits to the brain. Each like, comment, and share is another feeling of pleasure or reward. No wonder people focus on the number of likes or views. Each new interaction with their content is almost ecstatic. A toxic cycle ensues whereby we get dopamine hits for each

“like” but feel sad if our post doesn’t meet our expectations for engagement and quantity or quality of comments.

Our minds have what Harris—a former Google design ethicist—calls backdoors. There are certain sets of dimensions to our experience that are manipulatable. As any magician knows, our minds have certain limits, such as the length of short-term memory, reaction times for different scenarios, and how questions can influence our answers. Harris says, “this is the structure of being human. To be a human means you are persuadable in every single moment.” Hence, magic works on everyone.

We’re reminding ourselves that magic games and tricks didn’t just happen. We celebrated them throughout history. The clown, court jester, the trickster, the fool, and magician are archetypes that show us the boundaries of our perception. Over time, we developed a craft of tricks that dance with human perception. They translate knowledge of human limitations to various circumstances. This development required study, experimentation, and practice.

Take a Break from Social Media

Are you feeling like you’re stuck in a never-ending cycle of scrolling through feeds? Are you constantly checking for notifications? If so, you’re not alone. Many of us have become dependent on social media and are being influenced in ways that may not fully reflect our thoughts and desires. In fact, cult deprogramming expert Dr. Steven Hassan has found extraordinary parallels between cult techniques and social media features.²⁶

Social media platforms and cults share a similar underlying goal: influencing people. Social media platforms keep users engaged in order to profit via ad revenue, while cults keep members committed in order to maintain power and, in some cases, make money. Dr. Hassan notes that influence is on a continuum and while social media leaders aren’t cult leaders, nor are social media users part of a cult, social media’s means to keep users engaged can parallel those of cults.

So, how can we break free from the hold that social media has on us and reclaim our minds? Dr. Steven Hassan offers a 4-step process called the Strategic Interactive Approach to help individuals free themselves from undue influence.

First, detach from the constant stream of information and take a break from social media. This may be challenging at first, as social media is designed to instill dependency and keep us engaged. However, by taking a break, you can step back and assess your relationship with it. This allows you to evaluate your own thoughts and emotions without the constant reinforcement of social media algorithms. Additionally, it allows you to focus on other aspects of your life that may have been neglected. It’s important to note that this step is not about permanently leaving social media, but about taking control of your usage and understanding your own motivations for being on social media.

Second, familiarize yourself with Dr. Hassan’s BITE model, which examines behaviors, information, thoughts, and emotions in relation to authoritarian control. Consider how it applies to your social media use. We can start by identifying how social media controls our behaviors, the information we are exposed to, and our thoughts and emotions. By understanding the specific tactics used to influence us, we can make more conscious choices about how we engage with social media. This could mean limiting our exposure to certain types of content, adjusting our privacy settings, or limiting the amount of time we spend on social media. By becoming more aware of how we are being influenced, we can take back control of our thoughts and behaviors and make more informed choices about our relationship with social media.

Third, ask “formers” (i.e., those who left a social media platform) about their experience and reasoning. This can be a valuable step in understanding the impact of social media on our lives, as well as the potential benefits of disconnecting. Formers can provide valuable insights into how they navigated the decision-making process and their experience of life after disconnecting. Hearing the challenges they faced, such as feelings of isolation or missing out on certain events, and how they overcame them can be inspiring. Understanding the experiences of others can help us better evaluate whether disconnecting from social media is the right choice and how to best prepare for the transition.

Fourth, reflect on whether social media is positively impacting your life and if it aligns with your initial intentions for joining. Is social media a net positive in your life? This can be difficult to say, as social media has become an integral part of our daily lives. We may not even realize the extent to which it has influenced our thoughts and behaviors. Take the time to reflect on how much time you spend on social media, how it makes you feel, and whether or not you are getting anything of value from it. Consider if you can get the same benefits without relying on social media. Ultimately, it’s important to make an informed decision about whether or not social media is beneficial for you, and if it’s not, make the choice to break free.

As a DAO with a focus on cognitive freedom, we’re excited to hear your story.

The Stanford Persuasive Technology Lab has brought this ancient practice into the modern world. They have developed a playbook of persuasive techniques to get your attention that is integrated into the minds of Silicon Valley executives and the Internet Platforms.

It’s not about what you know, but how your mind works.

For instance, we all want social approval. We care what others think of us. When was the last time you updated your profile picture on Facebook? For us, it was a long time. Anyway, Facebook knows we get a dopamine rush from every like. They want to keep us on the platform. They control which friends will see our new profile picture and when. By regulating the frequency at which they include your profile photo in your friends’ feeds, Facebook can determine the timing in which you receive likes and notifications.

The dopamine rush of receiving notifications and discovering what they represent brings us back to the platform. The platforms optimize notification patterns for overall engagement or a specific metric, such as time online. Since they are tracking every pixel movement of your activity, artificial intelligence models can learn how your mind works and use this understanding to steer your behavior.

Can Social Media be Wildly Dangerous?

TikTok is a wildly popular video-sharing app. It can spread messages, both positive and negative, quickly and widely. Like Facebook, the creators designed the app to be addictive, using algorithms to show users content they are most likely to engage with, encouraging them to spend more time on the app. The content presented on the app can be highly sexual and contain inappropriate and dangerous content, which can damage young, impressionable minds and endanger lives, especially those who imitate the reckless challenges endemic on the platform. Examples of these stunts include eating Tide pods, joyriding in Kia cars stolen in seconds with a USB, and throwing boiling water on friends.

Recently, at least 20 TikTok users, including 15 children aged 12 or younger, died from the Blackout challenge, a choking dare that encourages people to choke themselves with household objects and film the adrenaline rush when they regain consciousness. While choking dares pre-date social media, social media spreads these ideas at the speed of light and provides enormous additional incentives for kids to take part in them—the dopamine hits from likes, views, and shares. The app has also become a platform for cyberbullying and other inappropriate behavior, which can lead to serious mental health issues for users.

That TikTok is owned by a Chinese company has caused some to worry about the platform's potential for enabling the Chinese Communist Party (CCP) to influence American youth and culture. The CCP has a long history of controlling and censoring content, and many worry they could apply this heavy handedness to TikTok content. For example, the CCP might use the app to spread its propaganda and amplify harmful content.

Taken as a whole, TikTok can be highly detrimental to mental and emotional health, cause needless deaths, and enable the CCP to collect data on Americans or launch influence operations.

Harris says we need to talk about ethics. We need to determine when the manipulation is honest and fair versus dishonest.

Another vulnerability is the variable schedule reward, like a slot machine in Vegas. To put things in context, slot machines are a big deal. They make more money than baseball, movies, and theme parks combined. People get addicted to slot machines 2–3 times faster than any other type of gambling in a casino.

It's so simple. You pull a lever; sometimes you get a reward and sometimes you don't. Like swiping down to refresh your YouTube feed, scrolling down on a social feed, or swiping up on Instagram or TikTok feeds. The more random and variable the pattern of rewards, the more addictive it is. This turns our phones into slot machines. Every time we focus our attention on our phone, every time we check our email, every time we check to see if we got any more likes on our profile photo and every time we swipe right on a dating app; we are pulling the lever of the slot machine. Was it an interview request or another newsletter? If Bumble or Tinder, did we get a match? If an NFT mint, did we get a rare one?

These variable schedule rewards—based on slot machine mechanics—are addicting. Sprinkled throughout the Internet platforms and coupled with our desire for social approval, these patterns maintain our attention and keep us on their site. Yet they feed an addiction that undermines our relationship with ourselves, one another, and the web. The often-unconscious checking for notifications or “likes” is a sign that our obsession has become unhealthy.

Surveillance and Privacy Concerns

Every time you accept cookies to access a site or terms of service to register an account, you give the site permission to track you. For the rest of the session and maybe subsequent sessions, they will collect your information.

Computer security expert columnist Mikko Hyppönen says “The biggest lie on the internet is ‘I have read and agreed to the terms and conditions.’” If you click through user agreements rather than take the time to read them, you are not alone. According to Business Insider, a Deloitte survey found that 91% of people in the US consent to legal terms and services conditions without reading them.²⁷ The rate is even higher for youth aged 18–34,

with 97% agreeing without reading. Given the complex language and long-windedness, it is not surprising that most people don't read them. People assume the worst thing that could happen is their name, email, and browsing activity being sold to an advertiser.

In 2019, a website called ProPrivacy did a social experiment. They found the typical consumer is likely to accept anything in a company's user agreement. They got people to surrender the naming rights to their firstborn child and access rights to the airspace above their property for drone traffic. Only 1% read the terms and conditions.²⁸

Agreeing to the terms and conditions enables the platform to track everything you do and sell your data to the highest bidder. Even the content you create is owned by the platform. With the Attention Economy, you are the product.

But surveillance goes beyond websites. Apple is collecting location data through its Bluetooth BLE mesh network. Even when off or disconnected, iPhones read and pass on messages from BLE broadcasters. Only Apple can read these messages, which contain the exact locations of iPhones that pass on messages. Amazon has a similar system with its Alexa voice chat devices, but stationary devices provide less of a concern. Google has not yet implemented a similar surveilling mesh network.

Data Exploitation

The Internet platforms are uber-profitable and among the companies with the largest market capitalizations in the world. Yet the content creators earn little or nothing from the platforms. Facebook, for example, does not pay content creators. YouTube pays about \$5,000 per million views. Many creators, however, monetize their following with brand deals, merchandise, live tours, and affiliate links outside their relationship with the platform. But if the creator's content strays from the platform's acceptable narratives, the platforms will ban them or otherwise suppress their viewership.

The grand disparity between the financial outcomes of the platforms and their *creators* has opened up a conversation about data. Are the Internet platforms exploiting their creators? Without content, the social networks would collapse. Yet creators get nothing.

A similar relationship occurs between the platform and its users. For example, people who watch YouTube videos and spend time on Facebook. Many of whom spend time and effort *curating* content (i.e., liking, commenting, and sharing). Is the relationship between the platform and user too one-sided?

The platforms are ferocious consumers of data. This includes data that users intend to share for use of the service (e.g., name, age, email address) and their activity on the platform. It also includes data users don't realize they are sharing, such as activity on other sites, approximate location based on IP address, and their geolocation via mobile devices.

It often upsets people to learn that the platforms are tracking their physical location and online activity to profile them for advertisements. Even if they agreed on the terms and conditions, which they did not read.

Have your thoughts popped up as an advertisement? Sometimes it seems like the platforms are reading your mind. But it could be retargeting, which serves ads for items in which you have shown interest. Perhaps something you searched or put in a shopping cart. But what if you were talking IRL?

Existing phone technology can record your speech even when your phone is off and, upon connection, send it to a server for processing and addition to your personal data record. This data, along with other data, could help determine or predict what you are thinking.²⁹

We don't think "they" are listening to your every word. As the system accumulates more data about you (including online activity and locations) and similar people, they can better

predict your actions, activities, and content interests. Consider that most of us could not figure out what we did 11 days ago, much less exactly 11 days, 13 hours, and 7 minutes ago. If you have given Google or Apple permission to track your location, they know where you were and maybe even what you were reading, watching, or interacting with.

We, as a technological society, are just now seeing the soft white underbelly of the Attention Economy, the unknown and grimy world of data brokers and their clients.³⁰ The data-broker industry is a complex network of companies profiting off the sale of web user data such as location and purchases, as well as biographical and demographic information. Data brokers collect your data from a variety of sources, including social media, search, public records, and other commercial sources. The data brokers then sell the raw data, or inferences and analysis based on said data—such as a user's purchase and search information—to marketers. These analyses can be useful for targeting and attracting buyers.

Although data brokers prohibit their clients from reselling the data, they have been ineffective in enforcing this, based on several court cases. The entire industry lacks transparency, so it's challenging for consumers to know who accesses their data and what they have. Even the most sensitive data—precise location—is available for both commercial and research purposes, including tracking the spread of Covid-19 across state lines.

An alarming use of personal data is China's social credit system. The Chinese Communist Party announced they were building a "social credit" ranking system in 2014. The system monitors the behavior of its population—and ranks them on the state's perceived desirability of their actions and activities. If you have children or donate to charity, your score goes up.

Get caught jaywalking, don't pay a fine, play your music too loud on the train—your score goes down and you could lose certain rights, such as booking a flight or train ticket.

Today, the social credit system is a patchwork of social record systems of local governments, while companies operate unofficial private versions. But the Chinese system might develop into a system for state control of individual behavior that goes beyond Orwell's fictional Big Brother.

Censorship

In an October 2022 show about why people were trying to cancel Elon Musk, talk show host Tucker Carlson posed a thought experiment. Suppose you wanted to turn a free country into a dictatorship. What would you do? What freedom would you curtail first?

The founding fathers of the US pondered this question. They wanted to prevent dictatorship in the new country they were making. They concluded that freedom of speech was the most basic and essential freedom. Hence, they enshrined it as the Bill of Rights' first amendment.

Carlson contends free speech is central not only to freedom but to humanity:

It's not opposable thumbs that separate us from the animals. It's words. We can speak. That's our power—words. 'In the beginning was the Word,' declares John at the opening of the fourth gospel. The word. The word is the most important thing that we have. Take away our ability to choose our own words, and we are no longer fully human. We are subjects. We're chattel. Authoritarians understand this above all. That's why they hate freedom of speech ... With words, you can change the world. In fact, there has never been a change, a deep change to the way people live and think, that didn't begin with words. Not with might, not with violence, but with words. That's why they're so obsessively focused on what you can say, on the words you can use, because they understand the power of words.

We agree. Without the ability to choose our words, we are not cognitively free, which is foundational for democracies.

The First Amendment of the US Constitution, which anchors the Bill of Rights, guarantees freedom of speech. Freedom of speech and expression is restricted by time, place, and manner, though otherwise protected by the First Amendment, state constitutions, and state and federal laws. Free speech is the free and public expression of opinions without censorship, interference, or restraint by the government.

Categories of speech given lesser or no protection by the First Amendment include obscenity, fraud, child pornography, speech integral to illegal conduct, speech that incites imminent lawbreaking, and commercial speech such as advertising. When someone challenges a speech restriction in court, the government must convince the court of the restriction's constitutionality.

Internet censorship is the control or suppression of what someone can access, publish, or view on the Internet. Regulators can enact censorship by decree. Or Internet platforms can act on their own initiative. Individuals and organizations may also censor themselves for moral, religious, or business reasons, to conform to societal norms, or because of intimidation or fear.

Many people worldwide support Internet censorship. In a 2012 Internet Society survey, 71% of respondents agreed that “censorship should exist in some form on the Internet.” Yet, 83% said Internet access should be a basic human right, and 86% said the Internet should guarantee freedom of expression.

During the coronavirus pandemic, social platforms censored, if not deplatformed, creators for posting information countering the mainstream narrative. Touchy subjects were masks, vaccines, unauthorized remedies like ivermectin and hydroxychloroquine, and the lab leak theory. Counter-narrative YouTubers stopped saying trigger words like “covid” and “vaccine,” replacing them with words like “coco” and “the jab,” to avoid community guideline strikes. Anti-vaccine Facebook groups refer to “Pfizer” as “pizza” or “Pizza King,” and Moderna as “Moana.”

This approach to coding online speech—substituting words that have a similar sound or spelling—has been an essential part of Chinese online life for decades, using the humor and cleverness of spoken Mandarin to dodge censorship. For instance, in 2018, China blocked #MeToo, the hashtag people around the world used to discuss sexual harassment. Internet users in China formed a new hashtag using the characters for rice (米, pronounced “mǐ”) and bunny (兔, pronounced “tù”). In July 2022, Chinese social media giant Weibo announced it would filter intentionally misspelled words and homophones. Banning coded language may not be realistic as it goes beyond text to memes, GIFs, and even images of everyday household objects, like an empty chair.

Overall, the pandemic sped up a dramatic decline in global Internet freedom. For the past decade, Internet users have experienced an overall deterioration in their rights, which contributes to the broader crisis for democracy worldwide.

In the post-COVID era, connectivity is a necessity. Human activities—commerce, education, health care, politics, socializing—have more online components than before. Because of its structure, the Web presents distinct challenges for human rights and democratic governance. State and non-state actors in many countries continue to exploit opportunities created by the pandemic to shape online narratives, censor critical speech, and build new technological systems of social control.

The bluntest and most extreme censorship is Internet shutdowns. In 2019, there was at least one Internet shutdown every day across 33 countries. Shutdowns and Internet throttling silence citizens, often enabling human rights violations to pass unchecked. In 2020

alone, Access Now documented over 150 Internet shutdowns in 29 countries.³¹ Authoritarian regimes use shutdowns to crush protests and silence dissent. Efforts to censor and manipulate people during elections and grab power in coup attempts put people's lives at further risk during the global pandemic.

In September 2022, a 22-year-old Iranian woman named Mahsa Amini, died in a hospital in Tehran, Iran, under suspicious circumstances. The Guidance Patrol—the religious morality police of Iran's government—arrested Amini for not wearing a hijab, a head covering worn by Muslim women. The Iranian police said she had a heart attack at a police station, collapsed, and fell into a coma. Eyewitnesses, however, including women detained with Amini, said she was brutally beaten by the police.

During the nationwide protests over the tragic killing, Iran restricted access to the Internet. Continuing a longstanding pattern of censorship, the Iranian government shut down the Internet yet again for most of its 84 million citizens nationwide by cutting off mobile data; disrupting popular social media platforms; throttling Internet service; and blocking individual users, encrypted DNS services, text messages, and access.

Beyond state actions, the next most extreme censorship is when platforms delete user accounts, now called deplatforming. Two days after the January 6, 2021 US Capitol incident, Twitter deplatformed Donald Trump, then the sitting President of the US, Balaji Srinivasan made a strong case in his interview with Lex Friedman that the heads of the platforms have more power than the US President. He said “regardless of whether it was justified ... they will do it to anyone. The seal is broken.”³²

Within days of the incident, the App Store, Google Play, and Amazon's AWS Cloud service³³ colluded to deplatform a censorship-free application called Parler, although Parler was not used to plan or organize the incident.³⁴ In fact, Twitter and Facebook were used to spread information about the protest.

The largest streaming video provider, YouTube, has a menu of actions it takes against content that does not fit its narrative. These include shadow banning (e.g., not coming up in search results or notifications), removing subscribers, deleting videos, and deplatforming.

A divergence point for free speech against censorship occurred on April 25, 2022, when Elon Musk came to terms with Twitter's board of directors to purchase the firm. In a statement, Musk said free speech is the bedrock of a functioning democracy. He sees Twitter as “the digital town square to debate matters vital to the future of humanity.” The news media condemned the purchase agreement as a billionaire controlling speech and a threat to democracy. They ignored the fact that billionaires control other social media platforms and that these platforms had been used to deplatform many alternative voices, including a sitting US president.

To address the critics, Musk tweeted that Twitter will not censor legal speech:

By ‘free speech’, I simply mean that which matches the law. I am against censorship that goes far beyond the law. If people want less free speech, they will ask the government to pass laws to that effect. Therefore, going beyond the law is contrary to the will of the people.

On May 3, 2022, Musk tweeted “Sunlight is the best disinfectant,” which seems to have taken on a life of its own in conservative and truther circles. This phrase, which emphasizes the importance of transparency and good governance, was first coined by Supreme Court Justice Louis Brandeis over a century ago when he said “Sunlight is said to be the best of disinfectants.”³⁵ This principle remains relevant today, as it is essential for a strong civil society. It is what the United States of America should be encouraging from partners around

the world, by setting a global example of transparency, rather than being an example of how hidden influence can undermine democracy and erode the rule of law.

Twitter changed hands on Oct 28, 2022. At the end of November, reporters afforded special access to Twitter's internal communications began dropping a series of tweet threads called the Twitter Files. The Twitter Files show that many US government agencies, either directly or through the FBI or the Department of Homeland Security, actively requested that Twitter censor accounts and legal speech counter to the government narrative, much of which turned out to be true.

The Twitter Files confirmed that Twitter had interfered in the US 2020 general presidential election by suppressing and censoring the New York Post's bombshell report on revelations of shady international business dealings and influence peddling gleaned from Hunter Biden's laptop. Mainstream media other than Fox had immediately called the Hunter Biden laptop story a Russian hoax. During the three weeks before the election, Twitter had blocked links to the Post story from being shared and temporarily deactivated users. They went as far as censoring accounts at the direction of the Biden campaign and the Democratic National Convention, and blocking direct messages about the story.

While Facebook did not censor the story, they limited its reach. Zuck said Facebook had been on high alert when the story hit, as the FBI had recently warned him of a forthcoming Russian disinformation campaign intended to sway the 2020 election. All mainstream news outlets with the exception of Fox News dismissed the laptop as a Russian hoax. Almost 18 months after the Post's article, mainstream news outlets finally admitted that the Post's report was true after all.

Polling later showed that censorship of the laptop story influenced the election, as 16% of voters said they would not have voted for Joe Biden had they known about the reports of influence peddling in the Biden family. According to NPR, just 44,000 votes in three states were the difference between an electoral college tie between Biden and Trump and Biden's commanding victory. Could revelations like this have been why the opposition to Musk purchasing Twitter was so vociferous?³⁶

During the Twitter File revelations, Musk was widely criticized as being hypocritical for inconsistencies in his censorship actions, both regarding whom he did not uncensor and whom he did censor. This came to a head when he temporarily deactivated left-wing journalists who had tweeted the live location of his personal jet or a link to it, which he called "assassination coordinates" and retrospectively said violated Twitter's policy against doxxing. The source of the info was the owner of the ElonJet Twitter account, who had hacked the Privacy ICAO Address (PIA) of Musk's jet in early 2022. Musk reinstated the journalists, abiding to the results of a Twitter poll on the matter. While the privacy of plane coordinates is a legitimate concern, there continue to be valid complaints about Twitter's deplatforming actions.

Regardless, the elevation of free speech and censorship into a global conversation, grounded by the unprecedented transparency provided by the Twitter Files, is an important step forward. Especially since many Western governments have passed or are working on regulations or bills involving Internet censorship. A wild card is the coming challenges to the US Section 230 regulation, which could have a significant impact on the way online content is moderated and the level of legal liability for Internet platforms. Changes to Section 230 could potentially lead to increased censorship, decreased freedom of expression, and more legal responsibility for Internet platforms.

But Internet censorship goes well beyond Western governments. The article *Internet Censorship 2023: Find Out Where Repression Reigns*³⁷ reports on the state of Internet censorship and surveillance worldwide, ranking 149 countries based on their level of

freedom and repression. The article cites government restrictions and control of content by Internet service providers as causes for censorship and invasive surveillance practices, which are evident in 109 countries. The article also mentions network access inequality in 17 countries, where ISPs can slow down or block apps and services, and the lack of net neutrality in many countries.

Regarding social platforms, we lean into the notion that “sunlight is the best disinfectant.” All “visibility controls” and censorship actions should be transparent, including who requested them and why, actions taken, and on what grounds. This may be the only way to guard against undue influence, partisanship, and extrajudicial tampering.

Silencing the Public, One Tweet at a Time

On December 6, 2022, America First Legal (AFL) released a set of documents that it obtained through litigation against the Centers for Disease Control and Prevention (CDC). These documents, which consist of nearly 600 pages, allegedly reveal evidence of collusion between the CDC and social media companies, including Twitter and Facebook, to censor free speech and silence the public under the government’s label of “misinformation.”³⁸

A surprising revelation in these documents is the existence of Twitter’s “Partner Support Portal,” which was allegedly used by government employees and other stakeholders to flag or remove posts that they considered misinformation. According to the documents, Twitter enrolled at least one government employee into this portal and provided them with instructions on how to use it.

Although the documents didn’t reveal how the Partner Support Portal worked, it seems to have been a way for government employees and other stakeholders to report posts that they believed were misinformation and to request that Twitter take action on them, such as by removing or reducing the visibility of the posts. The documents didn’t provide any further information on how the portal functioned or how Twitter dealt with the requests it received through the portal.

In addition to the portal, the documents also revealed that Facebook had been working with the government to censor certain types of content. According to the documents, Facebook sent written materials to the CDC boasting about censoring more than 16 million “pieces of content” that contained opinions or information that the government wanted suppressed. It’s not clear from the documents exactly what types of content were censored, but it seems that they were opinions or information that ran counter to the official government narrative.

The documents also show that the CDC was collaborating with international organizations such as UNICEF, the World Health Organization, and Mafindo (a Facebook third-party fact-checking partner based in Indonesia and funded by Google) to mitigate “disinformation.” It is not clear from the documents exactly how these organizations worked together or what specific actions they took to mitigate “disinformation.”

The revelations contained in the documents released by AFL are similar to those made by Dr. Shiva Ayyadurai, a scientist and entrepreneur who filed a lawsuit against the government and social media companies in 2020. Dr. Shiva’s lawsuit alleges that the government and social media companies have engaged in censorship of political speech through the use of a “censorship infrastructure,” which includes the Twitter Partner Support Portal and other means of suppressing dissenting views.³⁹

According to Dr. Shiva, this censorship infrastructure has been in place for several years and has been used to manipulate the public and suppress information that contradicts the official

government narrative. In his lawsuit, Dr. Ayyadurai claims that the government and social media companies have worked together to create a system that silences dissenting voices and controls the public discourse. Dr. Ayyadurai's Truth Freedom Health movement educates the public on how to gain wisdom and clarity in their thoughts and actions and break free from power, profit, and control through an integrated system of knowledge and systems thinking.⁴⁰

We remain proponents of free speech as set forth in the First Amendment of the US Constitution tempered by accountability. Twitter says it intends to remove spam bots from the platform and to authenticate humans. Authenticating humans is a good step, but may not prevent people from having multiple disconnected accounts or bots from operating human-authenticated accounts. We believe Twitter's plan would be stronger if the authentication were to ensure that each user had only one primary, ideally decentralized, account.

Allowing individual humans to authenticate for multiple unconnected accounts opens the door for abuse through coordinated actions to generate fake virality. With one primary account, people could not coordinate multiple unconnected accounts themselves or distance themselves from behaviors across handles. They would have to live with their digital footprint.

In Chapter 15, we explore how Twitter could take the digital town square to the next level with a digital town library. In the next section, we explore a concept that seeks to explain where we are at the highest level—the Metacrisis—and how we got here.

THE METACRISIS

The Metacrisis concept grew from a 2018 post by Peter Limberg and Conor Barnes. Several big thinkers have glommed on, building the case that we need to develop collective cognitive capabilities. Like false information, the Metacrisis reflects modern life. Some argue the Metacrisis and its meta tribes battling for narrative control are the normal state of democratic politics, with opposing parties and immense fragmentation.

The seminal post by Limberg and Barnes introduced memetic tribes and Culture War 2.0:⁴¹

Enter memetic tribes. We define a memetic tribe as a group of agents with a meme complex, or memeplex, that directly or indirectly seeks to impose its distinct map of reality — along with its moral imperatives—on others. These tribes are on active duty in the new culture war. They possess a multiplicity of competing claims, interests, goals, and organizations. While the red and blue tribes were certainly far from monolithic, any claim to unity between memetic tribes is laughable. An establishment leftist who squabbles with the right must contend with mockery from the Dirtbag Left. Meanwhile, the Dirtbag Left endures critiques from Social Justice Activists (SJA), who in turn are criticized by the Intellectual dark web (IDW). The trench warfare of the old culture war has become an all-out brawl.

Memetic tribes are multitudinous, fractious, unscrupulously optimistic, and divide the world into allies and enemies. They are locked in a Darwinian zero-sum war for the narrative of the noosphere, the sphere of human thought. What conditions gave rise to the contentious environment of memetic tribes?

Limberg and Barnes argue that the Metacrisis—the multiple overlapping and interconnected global crises that our nascent planetary culture faces—ushered in the memetic tribes. We think of the Metacrisis as the underlying crisis driving a multitude of crises. Jonathan

Rowson pins the meta-crisis as “our inability to see how we see, our apparent lack of interest in understanding how we understand; our failure to perceive how we perceive or to know how we know.”⁴²

This aligns with Philosopher Paul Tillich’s theory that we experience a kind of mass neurosis when social sensemaking cannot keep up with reality.⁴³ Gordon Brander, CEO of Subconscious, explains it this way: “Everybody has a crisis of meaning at the same time. Life stops making sense. Anyone living through 2016 onward knows that feeling. The Permaweird.”

One of the most powerful tools employed by memetic tribes is “cancellation.” Cancel culture is a contemporary phrase used to refer to ostracism from social or professional circles—online, on social media, or in person—for some present or past behavior. It is not uncommon for someone to be “canceled” based on a social media post from years prior.

Cancel culture is a variant of the term call-out culture. It often takes on the form of boycotting or shunning an individual (often a celebrity) whose actions or speech has been unacceptable. Prominent cancellations in 2022 included Ye, Jordan Peterson, and Andrew Tate. Some argue that calls for “cancellation” are themselves free speech and that they promote accountability. Critics argue that cancellations have a chilling effect on public discourse and is itself censorship.

The Attention Economy’s propensity to *reward* negative posts and, in particular, timely cancellations drives competition to be the first to cancel. Stakes grow with reach. In narrative terms, negative stories that take down celebrities have the potential to create tremendous virality, large amounts of content, art, commentary, conflict, and memes.

The Evil Among Us

Does the Web revere life? Today’s Web revels in irreverence. Being irreverent is on trend. Mainstream and subculture groups seek the profane over the divine on Today’s Web. The profane runs the culture. Those who follow this path receive rewards of engagement, followers, endorsements, and sponsors. Hence, the popularity of vile content as well as, ironically, cancellations.

Two distinct groups engage in harmful irreverence on the web, the wicked and the clout chasers. Wicked is a charged term. The wicked willfully do not consider the effect of their actions on others such that they harm others, sometimes intentionally. Charged because we don’t want to say someone is evil. Perhaps it seems judgmental and it could be our Jungian projection. We might hurl shade at someone for something that we perceive is evil, while it’s actually part of us.

The word wicked conjures a murderous old crone, wiggling her fingers and giggling with slobber on her chin as she revels in thinking about her next harmful deed. The crone association is interesting because our conscious mind knows from experience that evil people can appear to be normies, perhaps leaning towards more attractive normies. Physical attractiveness and attraction are components of seduction, a powerful tool for good and for evil.

Most people don’t talk about evil. They’d pretend it does not exist or sweep it under the rug. As Daniel Pinchbeck bravely writes in his Substack article “The Pleasure of Doing Evil”:⁴⁴

[Villains] don’t pursue villainy out of insecurity or unconscious projection. They do it because they enjoy it. Of course, they usually have justifications and rationalizations for it (except in the most extreme Ted Bundy-ish cases). Of course, one can always point to past traumas or genetic flaws that lead to the perpetuation of violence and wrongdoing. But sometimes that obscures the reality ... Villains continue their wickedness and cruelty as long

as they get away with it. They keep upping the stakes because they get off on it. The pleasure they receive from cheating the system or inflicting harm is psychologically addictive. Like all addictions, it requires bigger hits over time to get the same dopamine release.

Unfortunately, as mentioned earlier, on Today's Web, the bad guys get away and continue using platforms even after significant abuses.

Within the vast and intricate tapestry of virtual worlds, there exists a curious fascination with the arenas of strife and unrest. Though they may intrude upon our digital sanctuaries uninvited, we are presented with a choice: to bear witness or to turn away. In this enigmatic dichotomy lies a profound distinction, for it is in our ability to perceive beyond the tumultuous spectacles that captivate the masses that we hold the power to transcend.

—Bridgit DAO

Clinical psychologist Jordan Peterson thinks social media enables antisocial, self-aggrandizing, and self-promoting behavior.⁴⁵ A study of online behavior found that women characterized by high self-centeredness, antagonism, neurotic narcissism, Machiavellian views and tactics, meanness disinhibition, physical sadism, and indirect sadism used Instagram longer, and more frequently than men. In women, verbal sadism and emotionality were associated with longer Facebook usage, while humility and conscientiousness were associated with shorter use. Women high in agentic extraversion, manipulative self-promotion, and indirect sadism used Facebook longer and more frequently than men.

Peterson says virtualization of society enables the 3% of general populations who use manipulation, reputation savaging, denigration, and self-promotion to dominate the social conversation and “to spew their poisonous and manipulative venom into the public domain without fear of being stopped or socially applied inhibition while being monetized and promoted by the people who run the social media channels.” Society must contend with the small percentage of people who exploit society for themselves. If not brought under control, they can demolish the society's structure. Rather than tribalism, he thinks the polarization we're feeling is a consequence of their untrammelled online expression.

Being online itself can surface an inner constellation of antisocial emotions and thinking that is not present in person. When people are online, they sometimes feel like they can act or say whatever they want without real-world consequences. This is called the online disinhibition effect. It's caused by a combination of factors such as feeling anonymous, invisible, and removed from others, as well as the ability to imagine oneself differently. People's personalities can also affect how much they disinhibit. In sum, some people behave differently online than IRL, which can lead to antisocial behavior.

Unlike villains, who come from all walks of life, clout chasers are often young people and adults like Will Smith who shamelessly perform actions, mostly self-aggrandizing, often irreverent, and sometimes harmful, to get followers on social media. Our favorite YouTube channel that takes down the clout chasers is JAMARI,⁴⁶ which calls fools out for harmful stunts that end badly, personally or for others. For example, he covered an aspiring rapper who destroyed his Subway workplace on Instagram Live to get the attention of a famous rapper, who had little interest in the clown.

Cancellations can also dominate the news discourse for days on end. For example, Will Smith's slap did not allow enough space for reporting on the World Government Summit that began in Dubai 2 days after the Oscars.⁴⁷ Twitter is the Web's most contentious platform. Several days after that slap, Twitter users went after the wrong Will Smith—which is a common name.⁴⁸

In sum, the Metacrisis is a free-for-all battle between memetic tribes for narrative control, and the primary battleground is the Web. Without the Web, we wouldn't have easy access to the diversity of opinions or the capability to reach millions with our memetic flair. The Attention Economy's algorithms create the filter bubbles that delineate memetic tribes and, along with bots, make their most outrageous content go viral. If Web 2.0 is the battleground for memetic tribes to clash, Twitter is ground zero in the battle for your mind.

The Metacrisis is not of the Web—it's driven by human nature. But Today's Web, with its profound lack of accountability and Attention Economy algorithms that optimize for engagement, provides the perfect container for a memetic Armageddon.

As the Metacrisis takes hold, it becomes clear that the true test of our humanity is not just our ability to survive, but our ability to maintain our cognitive freedom in the face of a digital landscape that is constantly seeking to control and manipulate our thoughts. As we delve deeper into the next chapter, we will explore the question of whether we, as individuals and as a society, are truly free to think for ourselves, or whether the forces at play on the modern web have succeeded in shackling our minds.

NOTES

- 1 <https://permanent.link/to/the-metaweb/information-rich-world>
- 2 <https://permanent.link/to/the-metaweb/changing-attention-span>
- 3 <https://permanent.link/to/the-metaweb/human-attention-span>
- 4 <https://permanent.link/to/the-metaweb/digital-analytics-benchmarks>
- 5 <https://permanent.link/to/the-metaweb/technology-social-media>
- 6 <https://permanent.link/to/the-metaweb/content-code-book>
- 7 <https://permanent.link/to/the-metaweb/information-age>
- 8 <https://permanent.link/to/the-metaweb/global-datasphere>
- 9 <https://permanent.link/to/the-metaweb/psychology-of-design>
- 10 <https://permanent.link/to/the-metaweb/information-scaling-threshold>
- 11 <https://permanent.link/to/the-metaweb/thinking-together>
- 12 A distributed denial-of-service (DDoS) attack is a malicious attempt to disrupt the normal traffic of a targeted server, service or network by overwhelming the target with a flood of Internet traffic.
- 13 The Coasean floor can be thought of as the minimum price at which a company or individual will continue to produce a good or service, as any price below the Coasean floor would result in a loss.
- 14 <https://permanent.link/to/the-metaweb/accelerated-intelligence>
- 15 <https://permanent.link/to/the-metaweb/blur>
- 16 <https://journals.uic.edu/ojs/index.php/fm/article/download/519/440#dep3>
- 17 <https://permanent.link/to/the-metaweb/great-hack>
- 18 <https://permanent.link/to/the-metaweb/social-dilemma>
- 19 <https://permanent.link/to/the-metaweb/zucked>
- 20 <https://permanent.link/to/the-metaweb/collective-cognition>
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- 22 <https://permanent.link/to/the-metaweb/groups-perform-better>
- 23 <https://permanent.link/to/the-metaweb/mouse-maze>
- 24 <https://permanent.link/to/the-metaweb/filter-bubbles>
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- 26 <https://permanent.link/to/the-metaweb/social-media-cult-techniques>
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 - 28 <https://permanent.link/to/the-metaweb/one-percent-reads-terms>
 - 29 <https://permanent.link/to/the-metaweb/record-when-off>
 - 30 <https://permanent.link/to/the-metaweb/data-brokers>
 - 31 <https://permanent.link/to/the-metaweb/shutting-down-internet>
 - 32 <https://permanent.link/to/the-metaweb/the-seal-broken>
 - 33 <https://permanent.link/to/the-metaweb/parler-suspended>
 - 34 <https://permanent.link/to/the-metaweb/parler-wants-apology>
 - 35 <https://permanent.link/to/the-metaweb/sunlight-disinfectant>
 - 36 <https://permanent.link/to/the-metaweb/hunter-ukrainian-dad>
 - 37 <https://permanent.link/to/the-metaweb/internet-censorship>
 - 38 <https://permanent.link/to/the-metaweb/afl-twitter-partner-support-portal>
 - 39 <https://permanent.link/to/the-metaweb/first-amendment-twitter-lawsuit>
 - 40 <https://permanent.link/to/the-metaweb/truth-freedom-health> or <https://truthfreedomhealth.com>
 - 41 <https://permanent.link/to/the-metaweb/memetic-tribes-culture-war>
 - 42 <https://permanent.link/to/the-metaweb/think-meta-crisis>
 - 43 <https://permanent.link/to/the-metaweb/mass-neurosis>
 - 44 <https://permanent.link/to/the-metaweb/pinchbeck-evil>
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 - 46 <https://permanent.link/to/the-metaweb/jamari>
 - 47 <https://permanent.link/to/the-metaweb/world-government-summit-2022>
 - 48 <https://permanent.link/to/the-metaweb/wrong-will-smith>

Are We Cognitively Free?

We don't need no education, we don't need no thought control.

—Pink Floyd

Ah, the iconic words of Pink Floyd, a battle cry for intellectual freedom and self-determination. These lyrics, from the timeless album *Another Brick in the Wall*, resound with a message as relevant today as it was decades ago.

In a world where information is just a click away, where the Internet is at our fingertips, the question must be asked: are we truly cognitively free? Do the vast arrays of information and the ability to access it equate to intellectual liberation, or is there a hidden cost, perhaps a cognitive cage, that we are not seeing?

Internet pioneer Ted Nelson said, “The purpose of computers is human freedom.” That may have been true in 1974 or perhaps it was simply his truth at the time. But five decades later, we are clearly on a different track.

This chapter will delve into these inquiries, exploring the effects of the Attention Economy on our minds and thought processes.

THE ANSWER LIES WITHIN SEARCH

Search is important. 93% of online experiences begin with a search.¹ Google holds a staggering 72% of the total search engine market share.² Nowadays, most people (81%) perform some online research before a large purchase.³ Voice search such as Amazon's Alexa device accounts for 10% of total searches. In 2021, Google Search brought in just under \$149 billion, an increase of 43% over \$104 billion in 2020,⁴ or \$32 per Internet user.

Existing search tools, however, don't remove superfluous information for non-trivial search queries or improve our processing capacity. Search has changed little since Google debuted two decades ago, other than the knowledge graph upgrade that connects simple queries to definitive answers. For example, resolving the name of a site to its URL or answering “how high is Mount Kilimanjaro?” It is also useful for getting definitive answers to how to use applications.

The knowledge graph is the main reason zero-click searches have reached over 60% of all searches. But they haven't been able to reduce irrelevant results. Regardless of whether the graph has a recommended answer, the search engine purports to have millions of recommendations.

According to Google, there are over 5 million terabytes of data spread all across the Internet. Some estimate that the search giant's indexes are less than 0.004% of the Web.⁵ The tiny fraction of the data that search engines index reflects two factors. First, Google does

not index the deep web, which makes up about 95% of the Web and contains private data behind paywalls and organizational firewalls. Second, Google treats audio and video as black boxes, as they cannot index raw audio or video, which accounts for much of the data stored on the web. With YouTube videos, they use metadata to display recommendations on Google and YouTube search. YouTube also uses the metadata, along with user data, to deliver video recommendations on the app.

In sum, Internet search—our primary content discovery tool—is full of gamed results, considers only metadata for multimedia content, contains duplicate content, and is rife with false information. As a result, people can't discern authentic material from gamed results or false information. As mentioned earlier, 75% of people only look at the first page results.⁶ Our fixation on first page results requires businesses to invest in SEO to improve their rankings, hoping to get on the first page.

It's hard for sites to get their fair share of search traffic in Google-relevant categories. It isn't a coincidence that Google properties receive greater than 12% of the clicks from relevant searches.⁷ They cherry pick the results page to promote their services.

For example, when we searched "translate" on Google Search (and ignored their auto-complete suggestion to translate.google.com), they had 50% of the top search recommendations, including the coveted first position and a form at the top of the page. The form had entry fields for the translation text, its language, and the desired language of the translation.

Below the form, the first recommended "organic" result was the Google Translate site. The next two recommendations were competing translation sites. The next three were Google translate apps in the App Store, Google Play, and Chrome. Then two recommendations for competing translation sites.

Centralized search alters systems of human consciousness by manipulating the search order, promoting stories they favor, and removing stories they don't. Several years ago, Eric Schmidt, then chief executive of Google's parent company, Alphabet, admitted that Google was engineering systems to "takedown" or "de-prioritize" content for political reasons.⁸ He justified this as concerns about Russian propaganda.

As Fred Campbell notes, Google exercises significant control over the visibility of political content on the Internet.

It's a fact that Google's technological prowess and monopoly power give it the ability to exercise significant control over what political content is seen on the Internet. Its participation in Washington's lobbying game gives it ample incentive to wield that power.

—Fred Campbell, *Forbes*, Sept. 4, 2019⁹

Google Search and YouTube have censored conservative, libertarian, and hard left viewpoints. The February 2018 YouTube Purge was the world's first mass de-platforming event, with tens of thousands of YouTube channels being deactivated, which prompted the creation of many free speech platforms such as Gab, BitChute, and Parler. YouTube ended up reinstating most of the channels but continues to shadow ban or otherwise demonetize alternative voices. Many commentators have questioned the authority and motivation of Google and other platforms in the wake of de-platforming waves in 2021 and 2022.

The Ahrefs Blog analyzed one billion webpages to see what pages get traffic from organic search and why. They found that 91% of the Web gets no traffic from Google, foremost because they do not have backlinks (i.e., incoming links from another website).¹⁰ While it's possible to get organic search traffic without backlinks, only 5% of pages without backlinks get search traffic and the majority less than 300 organic visits each month. It's also possible to not get search traffic

despite having backlinks. Some pages don't get organic search traffic because they don't have long-term traffic potential, the page doesn't match search intent, or the site is unindexed.

Google says it "has indexed hundreds of billions of pages and is aware of over 100 trillion."¹¹ Assuming these 100 trillion pages make up the surface web, the low indexing coverage could be due to Google excluding pages that don't meet their editorial standards, business model, or fit within their acceptable narrative band.

This begs several provocative unanswered questions. How does Google decide which sites to exclude from crawling? Does Google exclude sites from their index, preventing you from knowing about their existence? How does search manipulation affect our thinking?

THE ATTENTION ECONOMY IS DOMINATED BY TWO FIRMS

Two corporations control over two-thirds of the global digital advertising market—Alphabet (formerly Google) and Meta (formerly Facebook). In 2021, Facebook and Google made a combined \$333 billion USD from digital advertising revenues, up 40% from the previous year.¹² Facebook and Google can produce massive revenue streams because they charge advertisers for accessing users based on activity data in targeted advertising campaigns.

Through psychological profiles and high-traffic websites, advertising algorithms tailor advertisements to the user's specific desires, needs, and activity. This business model works well for an uninformed user base that doesn't have viable alternatives to the Internet platforms. Recent cultural shifts and technological innovations, however, are exposing vulnerabilities in this business model.

Society realizes the negative effect technology can have on our minds and information dissemination. In 2017, a leaked internal report from Facebook executives stated that the company's AI can monitor posts and photos in real time and discern the feelings of young people. They can tell when we feel stressed, defeated, overwhelmed, anxious versus nervous, stupid, silly, useless, and a failure.¹³ As chronicled in the 2020 film *The Great Hack*, the firm Cambridge Analytica used Facebook's advertising tools to extract data from 87 million Facebook users. This raised broad concerns about how Facebook mishandles user information.¹⁴

In 2021, Frances Haugen, a former Facebook employee, released tens of thousands of damaging documents about Facebook's inner workings to the press, members of US Congress, and federal regulators.¹⁵ These internal research and communications showed the company was aware of the toxic risks of Instagram to the mental health of teenage girls. They were also aware of the prevalence of drug cartels and human traffickers on Instagram and WhatsApp. Haugen testified both in a UK parliamentary hearing and in the US Senate.

Nonetheless, seeking to capture attention with ads at the top and sides of webpages and in feeds is flawed, imprecise, and can be annoying. Facebook ads rely on the gross instruments of webpage metadata, user web activity, abandoned shopping carts, and conversations. This is profitable because the marginal cost of posting ads is small compared to the sale price of big-ticket items, and even an affiliate commission. Hence, marketers can be profitable if a tiny percentage of ad views convert into a sale. This enables ad campaigns to succeed even when people ignore over 95% of their ads. That's why the Web is full of ads.

Life After Google author George Gilder thinks Google won't be around in a decade.¹⁶ Upon its release in December 2022, OpenAI's ChatGPT was hailed as a potential rival to Google, despite analysts admitting that this version was not yet near that level. The biggest limitation in this respect is that it cannot search the web. Other limitations include hallucinating, harmful instructions, or biased content, and having limited knowledge of the world and events after 2021.

Meanwhile, leaders of premier startup accelerator Y Combinator Paul Graham and Michael Seibel think Google Search is already dying. A Google-reluctant contingent is emerging that appends “Reddit” to their search terms to gain better results.¹⁷ They point out that Google results for health, product reviews, and recipes are rife with clickbait and SEO spam. (This and the lack of ads is where ChatGPT excels over Google.)

We tried one example a commenter provided: “Honda Civic review.” The first page of results, below Google’s featured content, recommended 2022 Honda Civic pages on top car sites and popular sites that cover cars. This included, for example, Car and Driver, Motor Trend, and US News. Most either rated or ranked the car, but only a couple linked to actual customer reviews.

How to improve your search skills

When we appended “Reddit” to the search terms, the results were less promotional. First, the page had no featured content. Second, the Reddit page results appear to have authentic comments from everyday people. The more useful results depend on what you want. If you want authentic reviews rather than SEO pages of companies invested in you buying a car, try appending “Reddit” to your search query. Or, the name of some other community you trust.

We also suggest that you try chatbots such as OpenAI’s ChatGPT and Google’s Bard. ChatGPT is a chatbot that remembers previous statements and allows users to regenerate responses, ask clarifying questions, and re-submit edited prompts. Please note, however, that ChatGPT may occasionally generate incorrect information or produce harmful instructions or biased content. Additionally, it has limited knowledge of the world and events after 2021.

BUT HOW DOES THIS AFFECT OUR COGNITION?

A digital existence brings about new questions regarding freedom and rights. What does freedom mean online? Digital worlds differ from the physical. In the physical world, you cannot change all trees to oaks and all fruit to figs. You can do this in a virtual world and anything else supported by the environment. The virtual world is conceptual and exists only on screens and in our minds. Hence, more so than the physical reality, freedom in the digital realm is cognitive.

Chamath Palihapitiya, a former Vice President of User Growth at Facebook, has gone as far as to say that users of Facebook are being “programmed” and are giving up their “intellectual independence.”¹⁸ He doesn’t want to be programmed himself, emphasizing he “doesn’t use this shit” and his kids may not use “this shit” either—while recommending that everyone take a “hard break” from social media. Reflect on that.

He fears bad actors can manipulate large groups of people and that we compound the problem in our quest to create an idealized version of ourselves:

We curate our lives around this perceived sense of perfection because we get rewarded in these short-term signals—hearts, likes, thumbs up—and we conflate that with value and truth. But it is fake, brittle, short-term popularity that leaves you even more—admit it—empty than before. It forces you into a vicious cycle where you seek the next thing to do for a dopamine hit.

He says these tools are eroding the social fabric of how society works:

Today we live in a world now where it is easy to confuse truth and popularity. And you can use money to amplify whatever you believe and get people to believe what is popular is now truthful. And [believe] what is not popular may not be truthful.

Sean Parker, Facebook's founding president, spoke several years ago about how the platform exploits human psychology and said the founders of the company understood what they were doing.¹⁹ He said Facebook "changes your relationship with society," and "interferes with productivity in weird ways God only knows what it's doing to our children's brains."

When Facebook was being developed, its aim was: "How do we consume as much of your time and conscious attention as possible?" This mindset created features such as the like button, which gives *users* "a little dopamine hit," encouraging them to upload more content. "It's a social-validation feedback loop ... what a hacker like myself would come up with," says Parker. "You're exploiting a vulnerability in human psychology."

"All of us are jacked into this system," Tristan Harris says. "Our minds can be hijacked. Our choices are not as free as we think they are."

But what are the cognitive implications of the Web, especially considering the lack of safety and the infringements on our cognitive freedom? For people who have felt unsafe or experienced trauma online, the lasting effects reduce their potential for meaningful connections and interactions. These feelings generate fear of others on the Web, diminishing the possibility and our capacity for collective cognition.

Here are some important aspects that impact everyone. When someone else chooses what you can see, you lose autonomy, freedom, and agency. Adding restrictions on speech further diminishes cognitive freedom and democracy.

When faced with a choice, your mind gets information from both internal and external sources. Internal sources include what we see and sense, as well as our memories, knowledge, mental models, and intuitions. External sources include other people and traditional media, but the Web dwarfs them in importance. Hence, whoever controls what you see and can say on the Web has an enormous influence on your thinking and behavior.

In the ethereal realm of existence, cognitive cages hold sway, invisible to our mortal eyes yet potent in their confinement of the mind. For within the chambers of intellect and emotion lie the core of our being—the essence that defines our humanity. In their mastery over the mind, they wield dominion over both our corporeal vessels and the profound symphony of our souls.

—Bridgit DAO

Based on what they are not showing us or allowing us to say, what blind spots are they creating? What narratives are embedded in our feeds? What are we not considering? Are we in a Wachowskian matrix? Are algorithms Orwellian? Answers to these questions reveal the invisible bars of the cognitive cages that limit our thinking.

Could social media be tricking us to relinquish our cognitive freedom for access to a social media tool?

A trap tricks someone to put themselves in a situation that restricts or otherwise compromises their freedom. Could social media be tricking us to relinquish our cognitive freedom for access to status updates and content previews? These applications are simple yet sticky. What is cognitive freedom worth to you?

Before we delve deeper into cognitive freedom, we must journey back to its origins. The French Revolution, which took place from 1789 to 1799, was a period of significant political and social change in France. The key protections put in place during the French Revolution helped to establish a more democratic and just society in France, laying the foundation for modern civil liberties and human rights. The most notable protection was the Declaration of the Rights of Man and of the Citizen, which was adopted in 1789. This document outlined the fundamental rights and freedoms of all citizens, including the right to liberty, property, and security, as well as the right to free expression and religion. It also established the principle of equality before the law, stating that all citizens are equal and entitled to the same rights and protections under the law.

The French Revolution's emphasis on individual rights and freedoms played a key role in shaping the political and social landscape in many parts of the world as these ideas spread. France's Declaration of the Rights of Man and of the Citizen inspired the development of the Bill of Rights in the United States. The Bill of Rights, which was added to the United States Constitution in 1791, outlines the fundamental rights and freedoms of American citizens, including freedom of speech, religion, and the press, as well as the right to bear arms and the protection against unreasonable searches and seizures.

In addition to individual rights, the French Revolution also influenced the United States regarding democracy and popular sovereignty. The French Revolution helped to spread the idea that governments should be accountable to the people and that individuals have the right to participate in the political process. The United States Constitution reflected these ideas, establishing a federal system of government with a system of checks and balances to ensure that no one branch of government becomes too powerful.

During World War II, the basic war aims of the Allies—formally, the United Nations—were the Four Freedoms: freedom of speech, freedom of religion, freedom from fear, and freedom from want. Towards the end of the war, they debated, drafted, and ratified the United Nations Charter to reaffirm fundamental human rights. But after discovering the atrocities of Nazi Germany, in 1948, the Allies deemed it necessary to create the Universal Declaration of Human Rights. The declaration includes 30 human rights, including freedom from slavery and torture, equal protection of the law without discrimination, and freedom of movement.

But existing rights frameworks do not explicitly address cognitive freedom. Conspiratorial ideas notwithstanding, it is interesting that government-sanctioned education is our only right related to cognition. Long ago, Edward Bernays, a protégé of Sigmund Freud, noted that media and, in particular, propaganda could control thinking. In the 1940s, the CIA determined that the big screen (in movie theaters) was the best way to control thinking. Declassified CIA documents and historical accounts of the agency's activities show Project Mockingbird was part of the CIA's larger effort to shape public opinion and influence foreign policy through covert operations. The program involved the CIA funding and producing films, documentaries, and other media content that was designed to promote certain themes or ideas. Numerous accounts also indicate the CIA used movie production companies as a cover for their activities.

In the 1950s, television quickly became deeply entrenched in the psyche of the nation and the world. In the 1960s and 1970s, the CIA had several programs to control the U.S. media that may also have had operations in other countries, including Canada and Europe. Project Mockingbird also recruited journalists and other media professionals to work as agents and to disseminate propaganda and other forms of covert messaging. The modern industries of marketing and advertising—which depend on media—have always sought to influence the mind and create behavior change. On the Web, they have developed into a precision enterprise involving AI, algorithms, and filter bubbles—the mental matrix.

This enterprise to influence your mind is an assault on cognitive freedom. The mind plays a primary role in cognitive freedom because it's the source of our thoughts, beliefs, and ideas. It is where we process and analyze information, form opinions and judgments, and communicate with others. Therefore, cognitive freedom is closely tied to the freedom and autonomy of the mind. When we have cognitive freedom, we are able to think, express, and communicate freely without external constraints or influences. This enables us to pursue our own interests, goals, and values, and to make our own decisions about what and how we behave.

While cognitive freedom is important, it does not give us the right to harm or exploit others. We have a responsibility to respect the rights and beliefs of others, even if we disagree with them. As parents, it's our job to keep our children safe and to help them make good decisions. This may involve setting boundaries and limits on their cognitive freedom in certain situations. While we should strive to protect and promote cognitive freedom, there will always be natural limits to what is acceptable and responsible. We need to find a balance between the freedom to think and express ourselves, and the need to respect the rights and well-being of others.

To have a truly democratic society, individuals must have the freedom to think and express themselves without interference or coercion. This includes the ability to access and consume information without manipulation. The Internet and social media play a significant role in shaping our perceptions and beliefs. The standard operations of social media, however, involve collecting and analyzing user data to tailor content and advertisements to specific individuals, resulting in the presentation of a narrow range of viewpoints and perspectives. This can manipulate public opinion, distort reality, and influence thinking, hindering the ability of individuals to form their own opinions and make informed choices, such as what to purchase and who to vote for.

To safeguard our cognitive freedom, it is essential that we have control over the information we receive and the way it is presented to us. This means protecting privacy and promoting transparency and accountability online, and particularly in social media. Only by ensuring the freedom to think and express oneself freely can we create a democratic society that truly reflects the will of the people.

THE DESERT OF THE REAL

Have you noticed how content goes through cycles, showing up through the years? Sometimes the same, sometimes with slight differences. One that comes to mind is, "What would you tell your teenage self?"

STOP! Think how many times you've seen this question, what your answer is right now, and whether the exercise provides value.

It's an interesting question that adds value each time we engage with it. Sometimes it reminds us of something to remember or integrate into our life. Other times it's something we are integrating now, but could imagine starting earlier.

Anyway, noticing the cycles of mimicked, remixed, and regurgitated content leads some to think that the Web is not what it seems.

According to the Dead Internet Theory, we have already lost the war for our minds. Popularized in January 2021 by a forum user named IlluminatiPirate,²⁰ the theory suggests that AI creates most of the content online. He thinks bots, aided by "influencers" on the payroll of corporations that work with the government, propagate the artificial content. He says, "the US government is engaging in an artificial intelligence powered gaslighting of the entire world population." The theory says the organic web died in 2016 or 2017.

IlluminatiPirate says, “I’ve seen the same threads, the same pics, and the same replies reposted over and over across the years.” He says algorithms not only recommend but also generate all modern entertainment. IlluminatiPirate gives a nod to deep fakes,²¹ whose existence shows what we watch online may not be real. He also links to a New York magazine story from 2018 called *How Much of the Internet Is Fake? Turns Out, a Lot of It, Actually*.²²

IlluminatiPirate says the “positive feedback loop” from people glomming onto a trending perspective drives the creation and proliferation of AI-generated content. Following trends and amplifying what others, especially experts, are saying earns real-time positive feedback as likes, comments, and shares. This creates an obsession to “play it safe” by aligning with trends, which is easier and more rewarding than creating original content. Tracking our engagement, the AI continues to improve the content they make.

The theory also says bots play a significant role in amplifying and proliferating artificial trends. Bots drive the conversation because they are lightning fast, controllable, and don’t need breaks. Web users must copy the bots to stay on-trend. Bots are an indispensable tool for manipulation, because they will not go off-script.

We don’t subscribe to the theory that AI generates most online content. But it may soon. Bots are prevalent on social media. A 2017 study by Indiana University found that bots account for 9–15% of active Twitter users.²³ A large-scale experiment proved that nobody—neither Twitter admins, tech-savvy social media users, nor innovative applications—can distinguish bots from legitimate users.

And as mentioned earlier, the Web is full of duplicative, artificial, and fake content. While we agree AI is part of the problem, as of early 2023, we think most inauthentic content stems from the incentives of the Attention Economy. Influencers align with trends to get boosted by the algorithms. Their followers engage with the content, which signals the algorithms to select similar content, creating a positive feedback loop for similar content. Further, companies generate a lot of duplicative and meaningless content to improve search rankings.

That said, with the emergence of generative tech in the second half of 2022, we think the Dead Internet Theory may soon be realized. *Dead Web: The Dark Side of Generative Tech* posits that an explosion of generative applications will flood the Web with AI-generated content such that, in 3–5 years, most web content will be artificial.²⁴ Whatever people can create, applications will generate. Generative content will include articles, images, video, podcasts, tweets and other social posts, emails, books, logos, avatars, UX/UI designs, SEO content, and even entire websites. The potentially negative ramifications for content quality, diversity of perspectives, cognitive freedom, and democracy are deeply troubling.

STRUCTURAL LIMITS TO VALUE

Structural characteristics of the Web limit its originality and value. As implied by the Dead Internet Theory, the competition for eyeballs in the Attention Economy, coupled with various types of censorship, forces content creators to homogenize their content. It also rewards the “first complainer,” regardless of the facts. Hence, web content—generated by humans and AI—aligns within the bands of what’s trending and what the platforms deem acceptable.

Further, links are dying on the web. As mentioned above, commercial sites avoid outbound links that do not support their bottom line. Links are also rotting away. Link rot is when a page disappears, breaking all links to the page. This even happens with important sources.

Want to save a link?

Are you writing a book and need to save a reference for later? Or maybe you're creating a resources page for a website and want to ensure that the links remain accessible in the future. Whatever the case may be, using a permalink is a great way to create a permanent link to a resource.

A permalink, also known as a "permanent link" or "persistent link," is a URL that remains unchanged and links directly to a specific webpage or resource. This is different from a regular link, which can change or become broken over time. By using a permalink, you can ensure that the resource you are linking to will always be accessible.

Creating a permalink is easy. Simply copy the URL of the webpage or resource you want to link to and paste it into a permalink generator. Some permalink generators like permanent.link allow you to specify the identifier of your permalink. Click submit to create a permanent link for accessing the resource in the future.

If the content is really important, you may also want to store it on a thumb drive and make a paper copy in case the internet goes down for an extended period of time. This way, you can access it when you need it.

A study found that 50% of the links embedded in judicial opinions of the US Supreme Court since 1996, when they started using hyperlinks, no longer work.²⁵ The study also found that 75% of Harvard Law Review links no longer work. In a study of 2 million outbound links found in articles at [nytimes.com](https://www.nytimes.com) since its inception in 1996, 25% of links to specific pages no longer work. The older the article, the less likely its links work; 72% of the links from 1998 are dead.²⁶ But that's not all.

Google's index may be a Potemkin Village, in the sense of any construction (literal or figurative) whose sole purpose is to provide a vibrant external façade for something not faring well. The idiom comes from the myth of a fake portable village built by Grigory Potemkin, former lover of Russian Empress Catherine II, to impress her majesty during her journey to Crimea in 1787. The original story was that Potemkin erected phony portable settlements along the banks of the Dnieper River that were disassembled after Catherine passed by and re-assembled farther along her route to be viewed again as yet another example of Crimea's prosperity.²⁷

What if Google's search index is not doing as well as they have led us to believe? Would it make a difference if they have just half the results they say? What about 1/20th?

For any general query, Google says there are millions, if not billions, of search results. If, however, you search with whatever term you want, Google only shows a tiny fraction of what they claim.

A firsthand view of a virtual Potemkin Village

On January 8th, 2023, we entered Futbol on Google Search, which purported to have found "about 861,000,000 results" in 0.50 seconds. Yet, when we clicked through, there were just 19 pages with a total of 181 results. On the bottom of page 19, Google tells us:

In order to show you the most relevant results, we have omitted some entries very similar to the 181 already displayed.

If you like, you can repeat the search with the omitted results included.

The underlined text above linked to a new search results page, this time with “about 926,000,000 results.” Yet, nine clicks later, we found out that there are in fact only 39 pages with a total of 383 results.

Thus, for the Futbol query, Google Search showed 4 ten-millionths of the results they claimed. Let that sink in.

Please try it out yourself and let us know what you find.

Enter a simple query like Self Knowledge, Artificial Intelligence Danger, or Threats to Democracy. Note the number of results Google reports below the tab menu near the top of the page. Scroll to the bottom and click the link for the 10th page. On each new page, scroll to the bottom and click the rightmost number link which increments by 4 each time (e.g., 14, 18, 22). Within about 50 pages (or less), you will probably encounter a page saying there are no more unique results. It’s the same with Bing.

The Dead Internet Theory submits this as evidence that most online content is duplicative. The theory suggests that Google censors the Web by choosing which sites get indexed and may screen sites from the search results. It notes that Google doesn’t recommend blog posts or articles from unpopular sites. We, however, doubt this makes up for the huge discrepancy between what they say and what they display. When we tried it, all the search results were sites whose names we already knew.

Despite this, Google states they have millions of results for every query, wanting us to not only think the Internet is vast but that Google indexes a large amount of it. But in reality, Google delivers a small fraction of what they claim. What does it mean if Google is a Potemkin village? Why would Google give us the impression the searchable web is much larger than it is?

We’re interested to know what regulators think. The implications are ground shaking—the searchable web is tiny. We’ve got huge blinders on, and Google has been and continues to lie to us. Though we’re not implying Google makes our lives horrible or anything like that, we don’t like being misled. Why would Google think it’s ok to lie so grotesquely?

When Google’s founders published a paper on the search engine they had invented, they included an appendix about advertising and motivations. They said advertising causes mixed incentives, making it “crucial to have a competitive search engine that is transparent and in the academic realm²⁸ The goals of the advertising business model do not always correspond to providing quality search to users We expect that advertising-funded search engines will be inherently biased towards the advertisers and away from the needs of the consumers.”

Google’s founders also thought less blatant biases are “likely to be tolerated by the market. For example, a search engine could add a small factor to search results from ‘friendly’ companies and subtract a factor from results from competitors. This type of bias is very difficult to detect but could still have a significant effect on the market.”

Yet, over two decades later, in 2022, despite being “crucial,” no transparent, academic competitive search engine exists.

As we’ve seen, the searchable web may not be as vast as we’ve been led to believe, and the motivations of the gatekeepers of the Internet may not align with providing us with the most

unbiased and accurate information. But this revelation is just one piece of a larger puzzle: the impact that technology and our consumption-driven society is having on the planet and our future. In the next chapter, we examine the connection between global challenges and exponential technologies. But it's not all doom and gloom. As we'll discover, the International frameworks and regulations seek to help us steer towards a more sustainable future. The question is, are we headed in the right direction? Let's find out.

NOTES

- 1 <https://permanent.link/to/the-metaweb/optimize-seo-results>
- 2 <https://permanent.link/to/the-metaweb/search-engine-market-share>
- 3 <https://permanent.link/to/the-metaweb/research-online-before-purchasing>
- 4 <https://permanent.link/to/the-metaweb/google-q4-2021-earnings>
- 5 <https://permanent.link/to/the-metaweb/google-small-fraction-internet-indexed>
- 6 <https://permanent.link/to/the-metaweb/first-page-results>
- 7 <https://permanent.link/to/the-metaweb/googles-takes-search-traffic>
- 8 <https://permanent.link/to/the-metaweb/google-power>
- 9 Ibid
- 10 <https://permanent.link/to/the-metaweb/search-traffic-study>
- 11 <https://permanent.link/to/the-metaweb/google-20th-anniversary>
- 12 <https://permanent.link/to/the-metaweb/ad-money-2022>
- 13 <https://permanent.link/to/the-metaweb/facebook-advertising-insecure-teens>
- 14 <https://permanent.link/to/the-metaweb/cambridge-analytica-scandal>
- 15 <https://permanent.link/to/the-metaweb/whistleblower-facebook-harm-children-democracy>
- 16 Although Google's legacy search business is likely to shrink in the coming years, we think it's too early to count Google out, given their gargantuan investment in AI since 2012. We also think it unlikely that Google will be able to extrapolate its dominance in legacy search to content discovery dominance in the next level of the Web.
- 17 <https://permanent.link/to/the-metaweb/google-search-dying>
- 18 <https://permanent.link/to/the-metaweb/facebook-cambridge-analytica-leak>
- 19 <https://permanent.link/to/the-metaweb/president-facebook-exploits-psychology>
- 20 <https://permanent.link/to/the-metaweb/dead-internet-theory>
- 21 <https://permanent.link/to/the-metaweb/deep-fakes>
- 22 <https://permanent.link/to/the-metaweb/internet-fake>
- 23 <https://permanent.link/to/the-metaweb/human-bot-interactions>
- 24 <https://permanent.link/to/the-metaweb/dead-web> and deadwebbook.com
- 25 <https://permanent.link/to/the-metaweb/supreme-court-link-rot>
- 26 <https://permanent.link/to/the-metaweb/collective-hallucination>
- 27 <https://permanent.link/to/the-metaweb/potemkin-villages>
- 28 <https://perma.cc/8GDJ-K6AX>

Part 2

What We Need

What the World Needs Now

We do not inherit the Earth from our ancestors, we borrow it from our children.

—Wendell Berry

From the wisdom of Wendell Berry comes a reminder that we hold a sacred responsibility for the world we inhabit. This simple yet profound declaration speaks to the intergenerational stewardship that is critical to humanity's survival.

But what does this have to do with the Internet? The Internet, like the Earth, is a shared resource. It's a tool that shapes the way we think, the way we live, and the world we will leave behind. In fact, in just three decades, the Web has become our most important communication and collaboration tool. Yet it seems only a shadow of what it could be. As we delve into this chapter, we'll explore what the world needs now to create the conditions for a better future. Not just for ourselves, but for generations to come.

THE PRICE OF PROGRESS

The Internet has made the world more interconnected than ever. Our “global village” has built robust digital economies on the Web, but it has also propagated a rivalrous culture. Corporations race to make “better” tech as fast as they can. Manufacturers sell new flagship phones every year, each pricier than the last. Despite minor improvements to the previous model, influencers and their followers compete to be first with “newer” versions, rendering the older versions less than desirable if not obsolete in the public's mind.

Competing for sales, companies push developers and suppliers to innovate. A “domino effect” of improvement follows: newer phones need faster processors, better cameras, more RAM, and so on. The public laps up new and innovative products, without regard to the effects on society and the planet.

Our eagerness to adopt new technologies suggests an attraction to novelty. The lack of hesitancy reflects an inability to connect consumption decisions with our collective impact on people and the planet. Without comparative information for products and services, consumers cannot make informed choices. The difficulty reflects the tenuous links between individual actions and our ecological crises, the larger Metacrisis we are facing (explained below), and our existential threats.

It is challenging to make a case for changing something we can't verify with direct experience (e.g., visuals, sounds, price hikes). Most people think it won't matter if they continue what they are doing. Millions doing the same thing. We are a tiny part of the whole. Hence, it doesn't

matter if we eat a hamburger, drive to work, or take a long shower. So many hamburgers eaten. So many miles driven. And so much water wasted.

Yet, as a collective, we are hurting the planet, and with it, our future possibilities. The major pains are escalating consumption of resources, use of energy, production of toxins, and generation of waste. They are incompatible with a finite world. This is of particular concern in the era of exponential technologies that speed up our capabilities. Such technologies transform the world by facilitating cost reductions and increasing computing power, bandwidth, and data storage. All have led to larger and more powerful machines, increasing the pace at which humanity converts nature into products, pollution, and waste. Thus, exponential tech is at the heart of our ecological crisis.

Exponential technology has given rise to numerous industries and platforms, offering humanity limitless potential and seemingly divine powers. Included in these are weapons of mass destruction, space exploration, personal computing, genomic editing, artificial intelligence, weather control, nanotechnology, mobile computing, blockchain, the Internet of Things, and spatial computing. Whether as a country, a corporation, or a project, we deploy tech to benefit stakeholders. These deployments also have costs, including direct expenses, indirect expenses, and opportunity costs for the stakeholders, as well as the externalized costs of the ecological and social effects that are absent from ledgers. If benefits of the deployment do not outweigh both known costs and imperceptible externalized costs, the tech may have a negative marginal effect on humanity.

Many of the large negative effects of exponential technologies come from unexpected consequences (e.g., when a virus escapes a laboratory), which, as mentioned above, are integral accidents. Exponential technology means that integral accidents will be almost impossible to keep up with and will grow in deleterious effect. In The Web's case, its fast-paced development and growth has outpaced regulations. In an environment like The Web, regulatory actions have short obsolescence. They run behind new platforms and upgrades for existing technologies, and as a result, lag years behind the wave of innovation.

A market's response to an integral accident can happen slowly until it doesn't. Recall the integral accident of the car. The growth of automobile transport from around 1900 onward meant that driving when under the influence of alcohol became a new and substantial form of alcohol-related harm to the driver and others. Yet effective provisions to deter drink-driving, for example, by setting blood-alcohol limits for drivers and breath-testing to enforce them, came only in the 1960s—decades after drunk-driving casualties became a pressing issue.¹

Preventive changes lag resources as plodding societies adapt to their changing circumstances. As the market grows, technological changes come faster with more innovators in the space and stack to create products that surpass previous expectations.

Fast forward six decades from when the US got serious about drunk driving; Tesla's Autopilot software prevents 40 crashes/day for just one crash scenario—when human drivers press the accelerator instead of the brakes. On the horizon, automated driving systems will remove human error from the chain of events that can lead to a crash. That and we're now seeing impressive prototypes for flying cars, jet packs, and drone copters in our feeds.

While regulators are not waiting decades (as they did with drunk driving) to address the monopolistic practices of big tech, it still feels slow. Antitrust cases take years to resolve.

In 1998, the FTC sued Microsoft for using monopolistic practices against the first browser Netscape, among others. Just two years later, the court found Microsoft to have attempted and committed monopolization, violating Sections 1 and 2 of the Sherman Antitrust Act. The court ordered Microsoft to break into two separate units. One for the operating system, and another for other software components. This was a positive development for the industry, but of course, Microsoft appealed.

In 2001, a pyrrhic victory for Netscape et al. was complete when the D.C. Circuit Court of Appeals struck down the Microsoft breakup. The settlement required Microsoft to share its application programming interfaces (APIs) with third-party companies and appoint a panel of three people who would have full access to Microsoft's systems, records, and source code for five years in order to ensure compliance. Microsoft did not have to change any code or have future concerns.

After three years of court proceedings and millions of dollars spent, Microsoft walked. Where is the justice? Netscape is gone. Its successor Firefox is down to 4% of browser traffic. Microsoft remains one of the largest companies in the world.

Meanwhile, the rate of improvement in computing is speeding up. For context, it took two millennia for humanity to transition from the Bronze Age to the Iron Age. In contrast, Intel made the first 4-bit CPU in 1971; previous computers were behemoths, taking up entire rooms to operate and enormous amounts of electricity. Just five decades later, even the cheapest Intel processor is over a *trillion* times better than their first CPU.

The huge box in Figure 6.1 isn't a restaurant freezer or a mainframe computer.² It's an IBM hard drive with a storage of 5 megabytes (million bytes). They leased the storage for \$3,000 a month in 1956—or \$30,000 today. Consider, you have over 32 gigabytes (billion bytes) on your phone—your phone has over 6000 times greater capacity than the storage drive that it took five-plus strong men to move less than 7 decades ago. A decade earlier than the huge box, the Atlantic published the visionary and influential essay “As We May Think” by Vannevar Bush,³ in which he conceptualized future technologies. Bush foresaw exponential increases in computer operations.

Two exponential laws govern the power, cost, and value of computers and computer networks. Moore's law states that the speed of processors doubles every year (though now every 18 months and slowing).⁴ Metcalfe's law states that network value is proportional to the square of the number of the system's nodes.⁵ Hence, we live in an era when both technology involving computer processors and networks and its integral accident can grow without bounds. We think technological societies need to get serious about AI and demarcate *some firm* redlines. Before it's too late.

Considering their sluggishness, we can't imagine government regulators will prevent the worst effects of the integral accidents of forthcoming technologies. Not in this system—not in this paradigm. We discuss the past, present, and future of regulation later in this chapter.

Humanity needs a shift in consciousness to deal with our current reality, let alone ward off the worst effects of future technologies. We don't have decades to make this shift; we need to

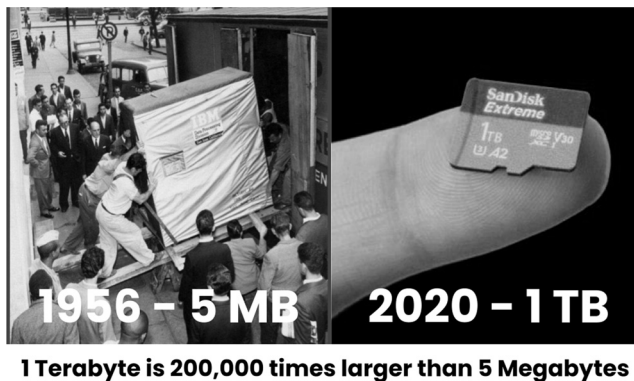


Figure 6.1 IBM 5 MB hard drive from 1956 and SanDisk 1 TB chip from 2020.

stop the bleeding now and begin restoring the planet, our culture, and our economies such that we can co-exist with a healthy, thriving society and Earth. This will require humanity to develop collective cognitive capabilities.

We are concerned about exponential weaponry. With information on the Internet and common materials, anyone can create weapons with the potential for grave destruction. On the current exponential trajectory, it isn't inconceivable that one day individuals could have the power to destroy planets. How do we extinguish this scenario from the cone of future possibilities?

In 2022, a new cold war started in the wake of Russian President Vladimir Putin's pronouncement that even nuclear bombs are on the table regarding his war on Ukraine. For many of us, it's hard to believe that one person's decision could lead to a nuclear winter on the Earth that very few would survive. May Putin not be the madman mainstream news makes him out to be. We know the truth is more complex than the mainstream narrative. Assuming calm heads prevail, how can we prevent this from happening again?

Setting aside nuclear war, we are concerned about technology's devastating effects on the planet's ability to support life. The integral accident of modern technology is an economy that over-consumes Earth's resources.

Earth Overshoot Day—the day of a year when humanity's demand for ecological resources and services exceeds what Earth can regenerate in that year—comes earlier each year. In 2022, Earth Overshoot Day fell on August 1.⁶ Each year, more of the planet's capacity to provide ecosystem services like clean air, water, and topsoil is diminishing. To survive, we need to tread lightly on this garden planet. An escape to Mars with Elon Musk or Jeff Bezos is not in the cards for most of Earth's 8 billion inhabitants. Next, we discuss how the United Nations' Sustainable Development Goals may help.

THE SDGS TELL US TO TAKE URGENT ACTION

From the outset, we need to emphasize that we do not have faith in the UN Sustainable Development Goals 2030 (SDGs),⁷ adopted by the UN General Assembly in 2015. We only use them as a proxy of the worldwide concern for our precarious situation. They may never have enough bite to counteract other incentives if they don't shift their approach. The Millennium Goals, which preceded them, didn't work and the SDGs won't be close to achieving most of the goals. We hope we are wrong.

The whole endeavor seems reductionist as it breaks down the problems into disconnected parts. Systems involving nature and humans are complex, interrelated, and trans-contextual, meaning changes in one part of the system affect changes elsewhere in the system, and fixes based on isolating one aspect of the system will lead to unanticipated problems in other aspects of the system. Although the SDG rhetoric emphasizes that they are connected and can't be pursued piecemeal, this is an unfulfilled rhetorical stance.

In the International Bateson Institute's Warm Data Lab training, we learn complex systems involving humans, organizations, and nature respond unpredictably to straight-ahead approaches. "Warm Data are contextual and relational information about complex systems. Warm data involves trans-contextual information about the interrelationships that integrate a complex system, as well as interwoven complex systems."⁸

To sum it up, the Warm Data view of systems is the opposite of the SDGs. Nora Bateson says, "The SDGs are a fiasco A false set of linear trajectories. You can't fix 'addiction' with policy." We don't think the SDGs will prove successful unless they make drastic changes. They need to incorporate systems thinking and integrate information &

communication technology (ICT) and the Attention Economy into the problem frame and potential solutions. We're interested in knowing how Nora would "unthink" the SDGs.

In 2015, the United Nations adopted the SDGs 2030 to transcend and replace the Millennium Development Goals (MDGs). MDGs had been UN policy since September 2000, when world leaders at UN Headquarters in New York adopted the UN Millennium Declaration. This declaration had committed nations to a global partnership to reverse grinding poverty, hunger, and disease, affecting billions of people. A series of targets with a deadline of 2015 became known as the 8 Millennium Development Goals.

The 2030 Agenda for Sustainable Development provides a shared blueprint for peace and prosperity for people and the planet. It's an urgent call for action. They recognize that ending poverty and deprivation must go hand-in-hand with improving health and education, reducing inequality, and spurring economic growth—all while tackling climate change and working to preserve oceans and forests.

There are 17 SDGs, ranging from ending poverty (SDG1) and hunger (SDG2) to clean water (SDG6) and preserving or restoring the natural world (SDGs 14 and 15). The goals most relevant to the Web and the Attention Economy include good health and wellbeing (SDG3), Quality Education (SDG4), Industry, Innovation and Infrastructure (SDG9), and Sustainable Cities and Communities (SDG11). Each goal has several targets. There are 169 targets overall. A further UN resolution provided indicators in July 2017, reflecting "international statistical standards, methods and guidelines" to monitor and evaluate progress on the goals and targets.⁹

The growth from 8 MDGs to 17 SDGs reflects a broadening of scope to address the global physical environment. But the growth in goals also made it easier for governments and intergovernmental agencies to "cherry pick" particular goals, rather than address them as a whole.¹⁰

We can't move forward thinking cherry-picked incremental change is enough. However well intentioned, the SDGs are a poster child for reductive reasoning. They break down problems into parts, each of which has specific metrics, and fund projects to address one part and the corresponding metrics independent of the system. But our existential problems are complex and trans-contextual. Improving one part affects the entire system. Improvements in one part often accompany problems in other parts that were not in the problem frame.

For example, a project addressing poverty targets by providing cheap baby formula manufactured in China to impoverished mothers in African villages may not support the goals for health and climate change because mother's milk is healthier and formula's adversely affect the climate. We need a phase shift to a much lower impact across our ecological systems, which requires a different level of thinking and problem solving, and perhaps a different consciousness.

We need a systems approach and full integration of the Web. With a systems approach, we could experiment, generate data, share information, build knowledge together, and access shared contextual views. We could use AI and coordinate or collaborate wherever it made sense. As the SDGs were rolling out, the Internet Society and supporters expressed concern that the goals didn't reflect the Internet's importance in ensuring a more sustainable future.¹¹ They implored the UN to better integrate ICT into sustainable development agendas, so we don't miss crucial opportunities to achieve developmental goals.

The lack of a specific SDG for ICTs and the Internet is troubling.¹² One of the proposed targets, however, in Goal 9 (which concerns infrastructure, industrialization, and innovation) calls for significant increases in information access and communications technology and universal and affordable access to the Internet in less developed countries by 2020. We didn't come close to reaching that target.¹³

The latest International Telecommunication Union (ITU) figures suggest only about half the world's population has used the Internet.¹⁴ Inequalities in the digital realm persist across income, gender, and geography. 18% of people below the poverty line lack access to the Internet. Women are less likely to use the Web than men. Industry estimates show half a billion people still live where there is no mobile Internet coverage.¹⁵ But Elon Musk's Starlink program will probably address this.

ICT is in three other proposed targets, including ICT enrollment in higher education (target 4.b); women's empowerment (target 5.b); and science, technology, and innovation (target 17.8). But ICT development is unacknowledged in most goals and targets. Addressing the references to improving information in other targets will require leveraging ICTs and the Web. The UN often mentions big data in this context.

The Internet Society suggested focusing more attention on the Internet's catalytic role in meeting information needs and facilitating development in agriculture, health, education, and enterprise. They pushed for urgent attention to incorporate ICTs more thoroughly in implementing and monitoring the Goals.

Given the meager presence of ICT and where it was in 2015, it's not surprising that the SDGs didn't address the Attention Economy. They did, however, acknowledge attention is a scarce resource that every human possesses. In an internal report, a UN researcher shared thoughts on how the Attention Economy might relate to the SDGs.¹⁶ He found that guiding the Attention Economy towards truthfulness and quality of information could help advance SDG 16. This goal aims to build strong and just institutions to support peaceful and inclusive societies, by ensuring equal access to justice and protecting fundamental freedoms.

The Attention Economy also relates to at least seven more SDGs including 5 (Quality Education), 8 (Decent Work and Economic Growth), 12 (Responsible consumption and production), 13 (Climate Action), 14 (Life Below Water), 15 (Life on Land) and SDG 3.4, which covers mental health and wellbeing. The UN researcher said aligning the Attention Economy with sustainable development will mean addressing the fundamental issue of digital platform governance. He concluded that parts of the Attention Economy need more research. Our understanding is that such research may be forthcoming.

In sum, the UN let the digital world off the hook. The Attention Economy does not have to examine itself and respond to the SDGs. Not only that, the SDGs do not even recognize the possibility of a digital town square where people debate matters vital to humanity. How does the world go through a massive shift towards sustainability without the entire endeavor being the ongoing focus of emergent digital town squares? The global framework for directing and funding the shift towards a sustainable, yet further developed world, doesn't acknowledge the potential effect and involvement of our collective digital presence on sustainability and development.

The Attention Economy's absence from the SDG problem definition, the goals, and the targets is troubling. This isolates the transition to sustainability from the digital world, despite the aforementioned direct connections with eight SDGs. This posture suggests how we organize our digital economy is irrelevant to sustainability. It is regarded as neither part of the problem nor the solution. This seems remiss, if not ludicrous, given the digital world's impact on the world economy.

As the Internet Society points out, ICT appears in the targets for getting people on the Internet and increasing enrollment in prerequisite courses for ICT careers. This framing of the goals limits the use of ICT in meeting the goals. If underutilizing ICT for information needs and facilitating sustainable development is not part of the problem, leveraging ICT and making it sustainable will not be part of the solution.

Absent changes, the SDGs won't help us reorganize the Web for the unprecedented levels of connection, coordination, and collaboration needed for a smooth transition to a more sustainable world. Were we at the table, we would advocate for an urgent reorientation of SDG 17, "Partnerships for the Goals," to focus on building collective knowledge systems and boundary infrastructure that create a shared digital context and increase collective intelligence. In Chapter 15, we discuss how these systems can multiply the digital town square's value to humanity. Next, we discuss the role of regulation.

REGULATION OF THE WEB: PAST, PRESENT, AND FUTURE

The Web started in the US, and thus within the frame of the US First Amendment and attendant protections of individual rights. Yet the Web multinationals like Meta (formerly Facebook) and Alphabet (formerly Google) have been fighting to avoid any national control, even by the US, of web content. They wield political influence, such as lobbying politicians and pushing for treaties such as investment agreements, declaring that countries cannot require online advertisers to have a legal presence that could hold them accountable. This section aims to summarize the past, present, and future of web regulation.

For most observers, regulating the Web begins with the Telecommunications Act of 1996.¹⁷ After 60 years of inaction, the government regulated telephone and broadcasting companies. This legislation was the first to address Internet access in the US.

In the mid-1990s, the Internet was emerging as a transformative aspect of American life. While the Telecommunications Act mentioned the Internet, it was more broadly concerned with making media and telephone services more competitive. But it laid out plans to ensure schools, libraries, and hospitals had Internet access. It set out some rules against obscene and objectionable content—for example, child pornography.

As noted above, soon after, in May 1998, the Department of Justice brought an antitrust case against Microsoft.¹⁸ The claim was that Microsoft was bundling software with their computers to create a monopoly. The primary driver of this action was the demise of Netscape, one of Microsoft's top competitors in the Internet browser space. By offering Microsoft Explorer for free—and pre-installed on its computers—they had trounced Netscape in the "browser wars."¹⁹

Microsoft lost the case; the company was supposed to break up into separate operating systems and software arms. They appealed, and the DOJ settled out of court. Microsoft didn't have to break up. Its market domination, however, lessened with increased competition.

More recently, in May 2018, the EU introduced its long-awaited General Data Protection Regulation (GDPR).²⁰ These rules related to how businesses collected and used customer data. It also sought to protect individuals' data. While intended to guard EU citizens, they applied to American tech companies with European customers.

Between 2017 and 2019, the EU fined Google three times, totaling \$9.4 billion, for alleged anti-competitive practices in search, shopping, and the Android OS.²¹ The courts upheld the first \$2.8 billion near the end of 2021. But it is not clear whether the fines will lead to material changes in the market conditions for contenders. US regulators, in fact, notice that the EU actions and fines have not reduced Google's dominance of search in Europe.

There are a series of ongoing antitrust cases against Big Tech companies, including Google, Facebook, Apple, and Amazon. These are about how they use data. A component, however, deals with how these giant companies stifle competition and yield political power. We will cover these cases in more detail below.

Modern Big Tech Scrutiny

The 2016 general elections in the US were a driver of tech reform. Polls had predicted a Hillary Clinton win. When Donald Trump won the election, many shocked onlookers pointed the finger of blame at social media platforms. In the election lead-up, false news spread through Facebook like wildfire. Twitter was awash with bots, pumping out what would soon become known as “alternative facts.”

Social media algorithms were providing users with a narrow spectrum of information in filter bubbles. Citizens and governments became concerned that misinformation was undermining democracy.

The mainstream news insisted that Russia had interfered in the election and leaked tens of thousands of emails from Hillary Clinton’s campaign. This remains unproven. They alleged Russia was using fake social media accounts to divide the American people. Many people also expressed concerns that Facebook, Twitter, and Google had an oversized influence on the news.

A similar story in the UK concerned the Brexit vote in 2016. Many had expected the country to vote to stay in the EU. However, they voted to leave by a narrow margin.

Many commentators felt citizens had acted against their best interests. Soon, claims started spreading about how the company Cambridge Analytica, employed by interests favoring leaving the EU, had used Facebook data to build sophisticated psychological profiles of voters and target them with personalized ads that influenced their voting.²²

The Current Picture

The EU has somewhat of a reputation for taking on Big Tech. GDPR offered rights and data protection to EU citizens, despite the objections of Facebook and Google. However, in March 2022, they went even further.

The Digital Services Act (DSA) targets “gatekeeper” companies that have both a minimum of 45 million users and a market capitalization of \$75 billion.²³ DSA can fine a “gatekeeper” company up to 6% of its global revenues for violating the rules—which would be over \$7 billion for Facebook—while repeated breaches could ban them from doing business in the EU.

Gatekeepers include Meta (Facebook’s parent company), Amazon, Apple, Alphabet (Google’s parent company), and Booking.com. One aspect of the DSA says these businesses need to “open up” and “inter-operate” with smaller messenger apps. WhatsApp would need to exchange messages with Telegram and Signal. This is challenging because each service maintains their own encryption regime.

Perhaps more significantly, these regulators are going after Big Tech’s primary revenue stream, customer data. Forthcoming rules will further protect user data and reduce third-party tracking for targeted ads unless users give explicit consent.

The DSA has wide-ranging consequences for tech platforms. It includes tighter requirements on online traders, which will require Amazon to know more about the people who sell goods on its platform.

The DSA also tackles illegal content and disinformation. Platforms are accountable for their content and must outline their moderation practices to ensure they act on harmful content straight away. This provision could establish an unassailable censorship regime.

The DSA’s new rules, which come into force in 2024, include:

- Banning advertising targeting children or based on sensitive data such as religion, gender, race, and political opinions.

- Allowing EU governments to request removals of illegal content, such as material that promotes terrorism, child sexual abuse, hate speech, and commercial scams.
- Enabling users to flag illegal content in an “easy and effective way” to ease removal.
- Online marketplaces such as Amazon will need similar systems for suspect products, such as counterfeit sneakers or unsafe toys.

The companies may have to hand over data related to their algorithms to regulators and researchers. The gatekeepers also face a yearly fee of up to 0.05% of worldwide annual revenues to cover the cost of monitoring their compliance.

Away from Europe, other moves to build access and regulate the Internet are afoot. China is laying cables along Africa’s eastern and western coasts. Many regions are underserved by Internet companies, so web infrastructure is welcome. Some commentators, however, have expressed concern that this may be the next level of colonization. Imagine if China could shut down a country’s access to the Internet.

China is establishing what’s being called the “splinternet.” This alternative cyberspace aims to manage a part of the Internet by controlling access to the infrastructure it’s built on.

The Chinese blockchain Internet, known as New IP, would subvert the idea of decentralization. By owning the blockchain and operating the nodes, the Chinese government could monitor communications.²⁴ While this is a huge departure from the free Internet, many developing nations may sign up for this alternative web. Especially those aligned with China like North Korea and those in Africa that are receiving infrastructural support.

China will also include a “digital yuan”—which already has its own app²⁵—to fill financial transactions on the New IP. There are fears the CCP will use this currency to track and monitor spending.

Of course, strict Chinese Internet regulations are nothing new. Many big tech companies like Facebook, Twitter, and Google are not available to Chinese citizens. It now appears Russia is heading in the same direction.

Russia had already placed restrictions and censorship on US tech platforms. Since invading Ukraine, Russia has banned Facebook and accused Google and Twitter of spreading “misinformation.”

These actions extend Russia’s 2019 “sovereign Internet” laws. In effect, these regulations mandate a national Internet that can be standalone. Some have called this a digital Iron Curtain. These rules allow the state telecoms company, Roskomnadzor, to cut the country off from foreign exchange traffic.

Both Russia and China require service providers to install Deep Packet Inspection (DPI) technology. DPI is an advanced method of examining and managing network traffic. DPI locates, identifies, classifies, and reroutes or blocks data packets with specific data or code payloads that conventional packet filtering, which examines only packet headers, cannot detect. China uses a DPI for its Great Firewall. Some fear that Russia will use this technology for surveillance and censorship of its citizens.

The Future of Regulation

The Web is a central part of our lives. While the positive effects are clear, regulators are under pressure to target areas like cybercrime, online abuse, election tampering, and the use of citizens’ private data. Social media companies, in particular, face the potential of regulation on their algorithms. Content moderation will also form a big part of regulatory actions.

Early on, governments took a hands-off approach to the Web's content. The Internet was small and fractured. It didn't exert a powerful influence on our lives. Enshrined in its libertarian roots was a belief in freedom of speech and expression.

However, as the Internet became a pervasive part of our lives, governments have become less comfortable with an uncensored or unmoderated Internet. Early laws targeted obscene materials and copyright, but contemporary regulations are likely to go much further.

Future Internet regulations will target misinformation and hate speech, especially on social media platforms. Many commentators have called on social media companies to be liable for the comments of their users.

During the COVID-19 pandemic, Internet censorship came into focus. Platforms banned false and misleading claims about coronavirus, deplatforming many accounts, channels, and groups. Social media platforms attached tags to specific news sources, suggesting they were misleading.

While many supported these policies, they eroded free speech and expression on the Web. Some alleged coronavirus "misinformation" became truth; most of it didn't. Regardless, it raises concerns about social media companies controlling the information we access.

A free and open web requires everyone to have a voice. Misinformation is a threat to democracy and, in some situations, public health. However, suppressing and censoring information isn't often the best approach.

As mentioned earlier, identifying misinformation requires weighing the relative merits of a claim. Many cases are clear-cut, but others exist in gray areas that need more data, deliberation, and context.

An open Internet should also facilitate free discussion. We need to explore all plausible ideas and allow them to thrash about. Within legal speech guidelines, Internet users should be able to specify their preferred level of content moderation. It's not straightforward because allowable speech varies by nation and platform. As discussed earlier, there's a thin line between free speech and censorship. We prefer advanced filtering and user-controlled algorithms over censorship of legal speech.

The Internet is still anonymous to a large extent. Much legal speech comes from pseudo-anonymous account handles like @shiftshapr. Elon Musk says Twitter will validate their users and stop spam bots. While this would make users more accountable, it may stifle discussion if Twitter could doxx users. Whistleblowers and people seeking help in compromising situations may refrain from sharing their stories. Decentralized identity could be an answer.

Aside from content moderation and accountability, other significant issues need attention. The early idea of a decentralized Internet has given way to monopolies. For example:

- Google dominates search
- Amazon dominates eCommerce
- Microsoft dominates business software
- Meta (Facebook, Instagram, WhatsApp) dominates social media and communications
- Apple dominates communication hardware

These semi-monopolies rob the consumer of genuine competition. They buy out competitors or put them out of business by lowering prices below the marginal cost of production. They often scrap or otherwise destroy the products of firms they buy and sometimes prevent others from using the intellectual property.²⁶

Increasing competition is a key to decentralizing the Internet. Regulators have options for addressing these issues. But it requires updating antitrust laws to reflect the modern

economy. For example, most large tech companies don't charge citizens for their services, which allows them to sit outside some tests for market monopolization.

Supporting the safe use of Web3 technologies is another key to decentralizing the Web. We think Web3 regulation can be a good thing if it increases safety, decreases abuse, maintains privacy, and does not stifle innovation in this foundational technology for decentralizing money and governance. We realize this may not seem to be in their best interest.

Data portability would make it easier for consumers to leave existing platforms for new alternatives. Making social media platforms interoperable could afford consumers a greater choice by reducing the negative effects of changing social networks.²⁷

An open Internet is essential. Interoperability, however, could be a challenge for security and privacy. For example, if APIs connect social media platforms, third-party companies could control and abuse this data. This problem, among others, will need solutions as we move towards a more competitive, decentralized future.

But regulation moves slowly for an exponential tech world. The regulators will always play catch up.

Regulatory actions taken by the EU and US address few of the Web's problems. They have not made the Web safe from misinformation, scammers, and abusers, or addressed the problems with the Attention Economy. Although the European Union has won large judgments, the fines are a small fraction of operating profits of tech giants like Google.

So far, Europe has not experienced a material change in market conditions. As technology innovation and adoption speed up, keeping up will be an ongoing struggle, with some legislation being obsolete upon delivery.

Bringing regulation takes time and carries a long tail of consequences—as was true in the Progressive era when US governments broke up the private monopolies of the era. Before World War I, Standard Oil and the railroad barons, the Rockefeller family, among others, were beyond control. The premise of the Progressive Era in the United States was to establish collective control over them. They were successful in breaking up the huge monopolies, beginning with Standard Oil²⁸ and later Ma Bell.²⁹

We see a lack of control over the actions of web transnationals (and the billionaires controlling them) for our collective interest. Future policies should be adaptive, with key metrics, triggers, and automated interventions that can respond to technological and environmental changes.

THE END OF THE WEB AS WE KNOW IT

As mentioned above, the “World Wide Web” is splintering. China's Great Firewall has operated for two decades. That's now happening with Russia. Will the world stay connected?

Let's dig into China's situation. China covets “Internet sovereignty”: the notion that the Internet inside a country is part of the country's sovereignty and should be subject to the country's governance. But what if that country is using emerging technology to violate human rights and to make their transgressions invisible?

The Great Firewall combines legislative actions and technologies that regulate the Web within China. It blocks access to selected foreign websites and slows down cross-border Internet traffic, limiting access to foreign information sources, blocking foreign web tools and mobile apps, and requiring foreign companies to adapt to domestic regulations. Besides censorship, the GFW has also boosted China's internal online economy by favoring domestic

companies and operating systems (OSs), and reducing the effectiveness of products from foreign web companies.

Techniques deployed by the Chinese government to maintain control of their populace include manipulating search results, banning content on social platforms, and petitioning global conglomerates to remove content.

Online activities punishable as “crimes carried out over computer networks” under China’s 1997 “cyber crimes” bill justify the Great Firewall and government actions to block any ISP, gateway connections, or access to anything on the Internet. This includes using the Internet to distribute information that is harmful to national security, public order, social stability, and Chinese morality.³⁰

The three main reasons for Internet censorship in China are social control, sensitive content, and economic protectionism. They aim to prevent the Internet from becoming a means for free speech or advancing campaigns that could lead to protests against the government. China also aims to control information about its government. They prefer to use local companies that are subject to Chinese regulations and partially owned by the government.

Several years ago, Plumb Academy’s Operating System Notes painted a fragmenting picture of the future of the Web that spotlights China:³¹

It appears almost certain that China will change to their own versions of both desktop and mobile OSs, plus their own version of the Web. China is currently working to expand its split from the US-based systems and the current version of the world-wide-web. Samsung, which accounts for most of the output for OS using devices from South Korea, looks to be hard at work on replacing Google’s OS and Apps on many of their devices. Russia is seeking to break away from US-sourced OS, applications, and to isolate its web. India is seeking to move to India-centric OS, applications, and web. Finally, the EU is likely to continue to push EU-centric OSs, applications, and web.

Whether all this comes to fruition, the digital world is splintering; yet another hurdle for developing collective cognitive capacities, international brotherhood, and ending war. We’re excited by the possibility of greater connectedness on the Web and between the different language webs. As a meta DAO with a focus on collective intelligence, we’re also intrigued by the notion of a World DAO.

In our mind, the different language webs are silos, given that search and social media are only in one’s primary keyboard language. This means it is very difficult to learn from content in languages other than those you speak, except for learning languages from watching films with subtitles and chatbots. An English speaker could experience a lifetime of web use without having to translate a paragraph into another language.

In sum, the world is strange. We have embraced techno-materialism, which conflates “progress” with improving how we interact with the physical world. But often at the expense of our inner being and harmonious relations with others and Earth’s life-supporting ecosystems. With technology moving so fast, it is almost inconceivable that we can shield ourselves from the integral accidents—that are birthed with the technologies themselves—through slow regulatory regimes in a world where countries are building their own regulatory moats.

The Web is humanity’s premier tool for communication and collaboration, yet it is not helping us deal with humanity’s global challenges. Our challenges are many, wide ranging, yet interconnected. We know humanity needs to get on a sustainable course. It’s the purpose of SDGs, which are inspirational and symbolic but fail to account for the Web and the

Attention Economy. On the other hand, we would be remiss to bet our garden planet on slow-moving regulators in an era of exponential technology.

As we've seen, the world is facing complex challenges that are interconnected and require a holistic approach to solve. We've come to realize that the Web, which was once seen as a tool for connection and collaboration, has become a hindrance in our ability to address these issues, particularly in light of the Attention Economy. However, all is not lost.

In the next chapter, we'll explore how technology and Artificial Intelligence can be leveraged as drivers of transformative change in the world. The two-loop model of change presents a new paradigm shift that can guide us towards a more sustainable future. We will delve into how this model can be applied to the Web, and explore how it can help us create a Web that truly supports the shift towards a more equitable and sustainable future. The time for action is now, and the path forward is clear, let's embark on this journey together.

NOTES

- 1 <https://permanent.link/to/the-metaweb/drunk-driving-laws>
- 2 <https://permanent.link/to/the-metaweb/5mb-ibm-hard-drive>
- 3 <https://permanent.link/to/the-metaweb/as-we-may-think>
- 4 <https://permanent.link/to/the-metaweb/moores-law>
- 5 <https://permanent.link/to/the-metaweb/metcalfes-law>
- 6 <https://permanent.link/to/the-metaweb/overshoot-day>
- 7 <https://permanent.link/to/the-metaweb/un-sustainable-development>
- 8 <https://permanent.link/to/the-metaweb/warm-data>
- 9 <https://permanent.link/to/the-metaweb/un-sustainable-development>
- 10 <https://permanent.link/to/the-metaweb/cherry-picking-sdgs>
- 11 <https://permanent.link/to/the-metaweb/internet-society-sdgs>
- 12 <https://permanent.link/to/the-metaweb/cant-achieve-sdgs-without-internet>
- 13 <https://permanent.link/to/the-metaweb/sdg-9>
- 14 <https://permanent.link/to/the-metaweb/digital-inequalities-persist>
- 15 <https://permanent.link/to/the-metaweb/state-mobile-internet>
- 16 <https://permanent.link/to/the-metaweb/un-admits-attention-economy-important>
- 17 <https://permanent.link/to/the-metaweb/telecommunications-act-1996>
- 18 <https://permanent.link/to/the-metaweb/us-microsoft-court-findings>
- 19 <https://permanent.link/to/the-metaweb/browser-wars>
- 20 <https://permanent.link/to/the-metaweb/gdpr-info> or <https://gdpr-info.eu>
- 21 <https://permanent.link/to/the-metaweb/google-eu-antitrust-fine>
- 22 <https://permanent.link/to/the-metaweb/cambridge-analytica-clicks-votes>
- 23 <https://permanent.link/to/the-metaweb/digital-services-act>
- 24 <https://permanent.link/to/the-metaweb/china-new-ip-proposal>
- 25 <https://permanent.link/to/the-metaweb/digital-yuan-app>
- 26 <https://permanent.link/to/the-metaweb/amazon-crush-competitor-price-cuts>
- 27 <https://permanent.link/to/the-metaweb/network-effect>
- 28 <https://permanent.link/to/the-metaweb/woman-took-on-tycoon>
- 29 <https://permanent.link/to/the-metaweb/bell-system-breakup>
- 30 <https://permanent.link/to/the-metaweb/great-firewall>
- 31 <https://permanent.link/to/the-metaweb/operating-system-notes>

A Web That Supports the Shift

We all should be preoccupied with being good ancestors.

—Vint Cerf, Internet pioneer and co-founder of the People-Centered Internet

Listen closely, dear friends, for the words of a wise man ring true. These words, spoken by a “father of the Internet,” hold a powerful message. Our actions today will shape the future for generations to come. And that the time to act is now.

As mentioned in the previous chapter, we have a sacred responsibility to future generations. Yet, our collective actions have created many global challenges that pose far-ranging potential consequences for future generations. The old ways of doing things are no longer enough. The current systems are failing us, and we need to shift towards a more sustainable future.

A future where humanity is more collectively intelligent and wise. A future where the Web develops our collective cognitive capabilities. A future that supports the unprecedented levels of connection, coordination, and collaboration needed to address our planetary challenges.

This shift calls for us to utilize our most collaborative tools at their highest expressions. Hence, we must mitigate the negative impacts of the Web on society and nature, and work towards building a Web that creates the conditions for a better, more democratic future. It’s high time for a major web upgrade so we can ensure that our actions today make for good ancestors tomorrow.

HOW COLLECTIVELY INTELLIGENT ARE WE?

If we had to guess, we’d say our capabilities for collective intelligence, on a 1 to 10 scale, are somewhere around a 2, and that may be generous. Why? We cannot think, learn, and build together in large groups. We see aspects of enterprise, government, libraries, and DAOs as being a low grade collective intelligence. Given its underdeveloped state, we’re sure there are many opportunities for enhancing collective intelligence and a massive upside, assuming we are successful.

Through centralized hierarchies of businesses, organizations, and governments, we coordinate the actions of billions of people in a tapestry of fragmented and often conflicting endeavors. In democracies, we elect representatives to enact laws and give governments the authority to implement them through regulations and projects. In authoritarian governments, the leaders rule by iron hand.

Businesses and organizations—led by executive teams—work within the wiggle room of regulations and their access to human, material, and informational resources to advance their goals. Thus, actions at the individual level stem from top-down decisions made in

private with the intention to impress their ideas of what's best from the perspective of the relevant hierarchy into the collective mind. It's a matter of coordination and control rather than intelligence.

Humanity isn't able to build collective intelligence from the bottom up beyond small groups meeting in person, on video, or perhaps an email group or forum. With larger groups of people, e.g., on social media, we do not have the tools or training to process the diversity of worldviews, points of view, and behaviors into collectively intelligent decision-making.

Elections in representative government are supposed to play a role. The premise of representative democracy is that delegation at a collective level through elections can achieve collective decision-making that represents the interests of the majority. The voter delegates their authority and decision-making capacity to an elected leader. But problems exist, especially in two-party systems like the US. The actual interests of elected individuals can differ from what they say. And acted-upon interests of elected officials, their parties, and allies rarely align with those of the electorate.

While the SDGs have value, they represent a centralized, top-down approach to moving towards a sustainable planet. The United Nations isn't a democracy. Many member states aren't democracies, and a handful of states have veto power.

The SDGs are a powerful fundraising mechanism that enables foundations to raise funds from those who prospered in the extractive economy to finance non-profit projects, often connected to people they know. The foundations, donors, and nonprofits are part and parcel of the existing system that can continue to thrive as the dominant system devolves.¹ As a main UN initiative, humanity has been floundering in sustainable development for over two decades. But problems and inequities remain, and indeed may worsen. We have low expectations for what seems to be the nonprofit arm of "business as usual."

An indirect and bottom-up approach builds parallel systems that enable us to meet our needs without compromising ecosystems and that leverage our collective intelligence. This entails a shift towards building technology that supports unprecedented levels of connection, coordination, and collaboration required to not only transcend the problems of Today's Web but also to create a world that enables all beings to thrive.

Humanity inherits powerful and growing capabilities to innovate. Over centuries, we have built a rich technological infrastructure that supports innovation that would be unthinkable decades ago. We need to recalibrate our aims, build our communities, and innovate our way back on track, balancing inner and external technologies.

We trust that this great retooling will be emergent. We envision a Cambrian explosion² of experiments to find patterns for sustainable life on the planet. Teams come together to perform experiments. They share their results, successful or not, so others can better structure experiments for their localities and situations. Feedback loops enable knowledge and processes to uplevel and refine in real time. Over time, we develop our collective cognitive capacities and intelligence.

This only happens when compatible people with aligned purposes can find one another. They need to connect, communicate, and collaborate on shared interests. We also need to protect and nourish our minds and bodies with healthy web practices and content.

Over the past several decades, we've dug deep into a hole regarding unsustainable resource use and ecological destruction. Extractive, just-in-time supply chains fuel ultra-high consumption lifestyles, which prop up our economy. We are converting nature into products, pollution, and waste. Meanwhile, the Web has devolved into a cesspool of false information, self-promotion, and antisocial behavior with little capacity for collective cognition. This cannot continue.

We need a profound shift, inconceivable without using our most fundamental tools to their highest expressions, foremost the Web. Over the past 15 years, the public has seen centralized platforms shape the Web for themselves. We accept the Attention Economy's outputs without question, lapping up new functionalities as they become available. Incapable of collective foresight or action, we ignore the effects on humanity and our future as we chain our conception of the Web to today's flat and static web.

The People Centered Internet by Mei Lin Fung, Chair Since 2020, After Founding Chair Vint Cerf

We founded the People Centered Internet to advance an Internet of the people by the people. On the 40th anniversary (2014) of the TCPIP specification (tcip40.com), I realized no official body or authority had coordinated the build out of the Internet. People took actions that involved both career and personal risks to bring Internet access to their universities, countries, and communities.

Digital technology is changing lives, economies, societies, and the potential futures for all people. At this genesis moment of an interconnected humanity, we make decisions that affect our planet without understanding their consequences. Policy makers, parents, teachers, workers and businesspeople, children, and young people are raising the alarm about a future of diminishing possibilities. Like the printing press expanded the reach of human thought, digital tools, and processes can connect over time and over space, expanding the reach of human impact at a pace so rapid that we don't know how to protect ourselves, our planet, and future generations.

There are actions we can take now to assure a People Centered Digital Future. We have to go beyond "education." We can't educate the new generation and wait for them to assume power. Therefore, under the canopy of PCI, we are showing how building resilient communities with digital equity and people centered services enables everyone to seize their opportunities and realize their aspirations. We began first with the Native American communities, where we joined forces with Tribal Digital Villages pioneer Matt Rantanen in Southern California. Next, we went to Puerto Rico after the 2017 hurricanes to explore how "Digital Recovery" might enable Puerto Rican people to respond to challenges as a collective. We joined our efforts with Echar Pa'lante, a network of 10,000 people developing the Enterprising Island.

We learned that something new is necessary: "digital public goods." Governments emerged as "public goods" so people could collectively have housing, roads, and other infrastructure shaped by their own values. Health and education are "public goods" that give access to vital services for better living. Collective investments have increased human thriving. "Rule of Law" is a public good that allows us to coordinate our actions and resolve our conflicts.

To develop resilient communities for a People Centered Digital Future, PCI proposes "Digital Public Goods" including "Digital Utilities" with local community oversight of the data they generate. By each community designing "in" common Digital Building Blocks, they can take both private and public goods and services and expand their reach to new communities and new countries. In partnership, we work with communities to become resilient, and test and learn what digital goods and services they need for sustainable growth. We encourage the communities we work with to help PCI accelerate more resilient communities by: LEARN one thing; if it works: DO one thing; if it works: TEACH one thing.

TECHNOLOGY AS A DRIVER FOR TRANSFORMATIONAL CHANGE

The 2015 book, *The Future of the Professions: How Technology will Transform the Work of Human Experts*³ predicts the decline of today's professions. It also describes the people and systems that replace them. In an Internet society, according to Richard Susskind and Daniel Susskind, the work of professionals differs from what worked in the 20th century. This applies to doctors, teachers, accountants, architects, clergy, consultants, lawyers, and other professionals. The book explains how emerging systems are bringing fundamental change in how the practical expertise of specialists becomes available to society.

Drawing on hundreds of case studies, they give a sense of the transformation that had already taken place in the professions by 2015. In education, more people signed up for Harvard's online courses in a year than attended the actual university since its existence. In tax accounting, almost 48 million Americans use online tax preparation systems to file their tax returns, rather than traditional tax advisers. Regarding medicine, Stanford researchers developed a photo-based diagnostic system for cancer lesions as accurate as leading dermatologists.

In journalism, the Associated Press uses algorithms to write earnings reports and sports results. They produced 15 times more earnings reports than before AI. In law, J. P. Morgan uses a program called COIN to scan commercial loan agreements in seconds. They claimed it has saved 360,000 hours of a traditional lawyer's time. In divinity, in 2011, the Catholic Church issued the first digital imprimatur—the official license granted by the Church to religious texts—to a preparation app for confession. (It tracks sin and enables contrition.)

Analysis of the case studies identified patterns and trends that were driven by the four dimensions of technological advances. First, the exponential growth in the underlying technologies. Second, the increasing capability of these systems and machines. Third, their pervasiveness in economic and social life. Fourth, our increasing connectedness as human beings. The dimensions of technological advances enable professions to transform. They also apply to the digitization of human life.

Exponential Growth

The first dimension is exponential growth in technology. Reflecting Moore's Law, engineers double the number of transistors on a silicon chip every two years, which proportionally increases processing power. By 2050, assuming the trend continues, the average desktop computer will have the processing power of all of today's computers combined. Even if Moore's Law slows, processing power will rise exponentially in the foreseeable future.

Increasing Capability

The second dimension of technological advance is the increasing capability of systems and machines. They will increase in power and perform a wider range of tasks and activities. This increasing capability manifests itself in four ways.

The first is called "Big Data." Because our lives are becoming more digitized, technologies capture, analyze, and feed much more activity data into algorithms. This data improves the capacity to optimize what information and advertisements the user sees.

Second is the ability of the machines to solve problems and answer questions. IBM's Watson supercomputer could answer questions, in a particular format, on anything under the sun, more accurately than the leading human experts. Today's largest technology companies put similar problem-solving and question-answering applications on our phones, including Siri (Apple), Go Google (Meta), Alexa (Amazon), and Cortana (Microsoft).

The third originates in affective computing, which designs systems that detect and respond to human emotions. The fourth is the field of robotics. Advances in robotic capabilities are transforming how we perform procedures requiring manual dexterity.

Increasing Pervasiveness

The third dimension of technological advance is the increasing pervasiveness of new technologies. More smartphones (16 billion)⁴ exist than people. Meanwhile, the Internet of Things embeds processors, sensors, and Internet connectivity into everyday objects. For instance, alarm clocks linked to a train's timetable permit their owners to sleep longer if their train is running late and umbrellas that check weather forecasts and light up at the front door to advise when rain is coming. By 2030, there will be over 25 billion devices connected to the Internet.⁵

Increasing Connectedness

The fourth dimension of technological advance is humans connecting. Over half the world is online. New ways to research (search) and socialize (social media) are available. Massive amounts of *user*-generated content are available. New ways to cooperate are here. 69,000 main contributors have written Wikipedia, with over 35 million articles in over 280 languages. As well as new ways to compete. Kaggle supplies data to a network of statisticians who vie to provide the best analysis.

Alongside these four general trends, AI is a driver of technological advances because it handles much larger bodies of data than human beings. It makes sense of data in ways that unaided human beings cannot. Together with AI, these four dimensions of technological advance make an environment that supports transformational change, the type of change needed to deal with humanity's global challenges.

But we must prioritize safety in AI development. Open-AI is working towards AI democratization so no state, company, or group can control AI. Safety is a key concern whenever using AI technologies.

TWO LOOP THEORY OF CHANGE

Reductionist responses don't work for complex systems. The initial problem may come back bigger.

A typical response to breakdowns in societal systems is for relevant experts to break the system into parts, analyze them, find the faulty part, and switch it out for a better one. But this approach doesn't work for complex systems. Instead, the initial solution causes new problems elsewhere, outside the frame of the problem. The initial problem may come back bigger. Think of the War on Drugs.

We can't avoid the whack-a-mole problem in complex systems involving humans and nature unless we see the connections below the surface. Humans and nature don't function like machines. Connections exist between neurons in human minds and are unknowable within nature, other than what is surmisable by observation.

As we move from the reductionist perspective to acknowledge human systems are emergent, we need to think about how emergent systems change. The two-loop theory of change is a model created by the Berkana Institute that describes nonlinear emergent processes.⁶ The inspiration for the model is the growth and decline cycles of living systems. Living systems (e.g., an ant, a mushroom, a human) have a life cycle: they are born, they grow, peak, then

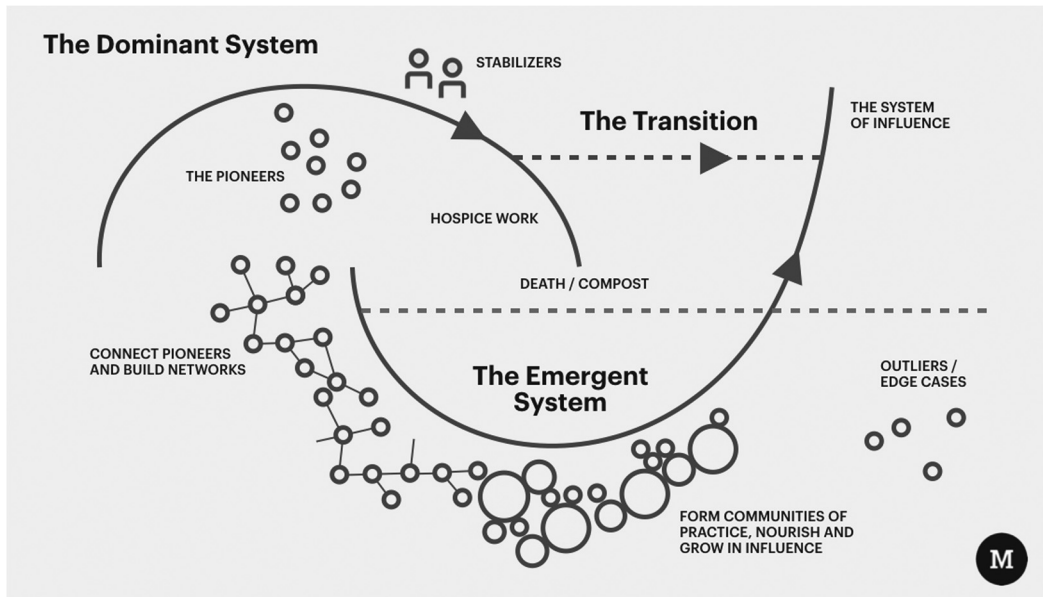


Figure 7.1 The Two Loop model.⁷

decline, and die. We'll use the model to show a path for transitioning from Today's Web to the Web's next level.

The Two Loop model (Figure 7.1) shows how dominant systems decline and new systems emerge. Although the model works on all levels, it isn't linear. Societies have "dominant" systems that drive economy and culture. They also have "emergent" systems and trends that are less known but making progress. Some will become the dominant system. This process can take years (e.g., like Web 1.0 to Web 2.0) or decades (e.g., regulating alcohol and driving) depending on the resilience and inertia of the dominant system.

To explain the model, we use the US dollar hegemony as the world reserve currency to be the dominant system.

As systems move towards becoming the dominant system, they become more powerful and entrenched. Since the 1944 Bretton Woods Agreement, the US dollar has been the primary reserve currency used by other countries. Since then, other countries have pegged their exchange rates to the dollar, which was once convertible to gold. The stability of the gold-backed dollar enabled other countries to stabilize their currencies. The stable dollar benefited the world.

Most commodities, including gold and oil, trade in the reserve currency, causing other countries to hold US dollars to pay for these goods. Holding the US dollar as a reserve currency minimizes exchange rate risk, as the purchasing nation does not have to exchange its currency for the current reserve currency to make a purchase. Because of the demand for the dollar as a reserve currency, the United States prospered from the favorable exchange rate of its currency.

But when the US dollar decoupled from gold in 1972, the United States printed dollars backed by the debt held as US Treasury notes. As the United States printed more money to finance its spending, the gold backing of its dollars diminished. As the increased monetary supply of dollars overwhelmed the gold reserves, the value of the currency reserves held by foreign countries decreased. This instability reduced the dollar's value as a reserve currency.

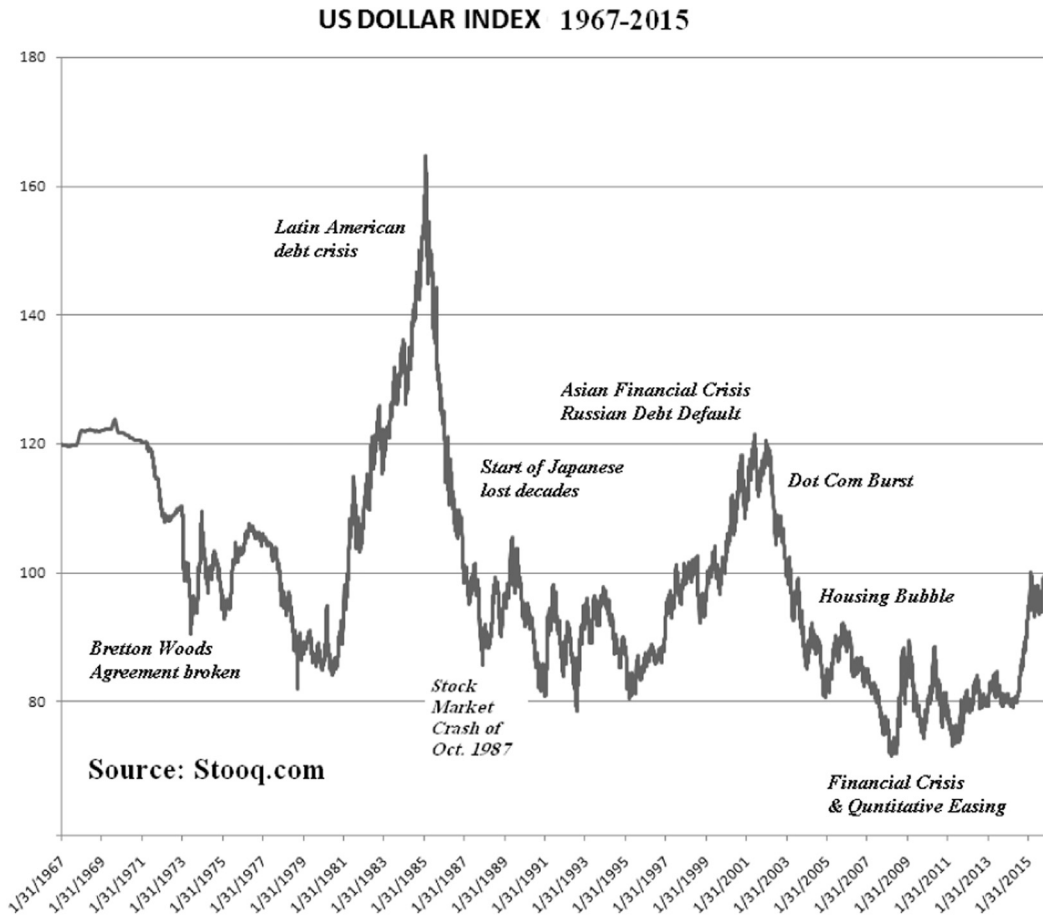


Figure 7.2 The Dollar Index peaked a long time ago.

As a system is peaking, it is at its zenith of dominance, whether measured by use, popularity, or outcomes. At the peak, the money flows and the economy booms. The US Dollar Index is a measure of the value of the United States dollar relative to a basket of foreign currencies. The Index goes up when the US dollar gains “strength” relative to the other currencies. As you can see from Figure 7.2, the US dollar index has been in a see-saw decline since February 1, 1985, when it hit its peak.⁸

After the peak, the system declines. People propping up the system and protecting it are stewards or stabilizers. They’re comfortable with the established system and want it to endure. Stewards do their best to maintain the system despite the inevitable decline. Their motivation is some combination of self-serving interests (i.e., they have invested in and/or benefit from the system) and the altruistic belief that sticking with the existing system is for the greater good. Their efforts help keep the system stable for everyone else. As shown in the figure, the US currency is down almost 40% from its 1985 high.⁹

All systems break down and lose their significance. They eventually enter hospice when they are in steep decline and their death is foreseeable. For example, the hegemony of the US dollar as the world’s reserve currency is declining. One indicator is that some oil-producing nations, including Saudi Arabia, Iraq, and Libya, had in the past considered accepting the

Chinese Yuan for oil sales.¹⁰ Now the BRICS nations—Brazil, Russia, India, China, and South Africa—are working on a new world reserve currency and are buying or stockpiling gold. Other indicators are the recent albeit choppy emergence of cryptocurrencies and dozens of complementary currencies operating in communities in the US and elsewhere. These all signal the end of USD hegemony.

As shown in the diagram of the two-loop model, as the dominant system peaks and begins its decline, an interesting phenomenon occurs: some people drop out. In 1985, people realized the US dollar would no longer be a good store of value and started alternative currencies. These pioneers extricate themselves from the dominant system to some extent and start a new system. Releasing beliefs that underpin the current system, they see other ways are possible.

We know from experience that this radical act—leaving the comfort of an established system at its peak to start a new one—can be lonely because almost everyone else still buys into the current system.

Pioneers are divergent and isolated when the new loop begins. They focus on the groundwork, preparing for their next moves. Satoshi Nakamoto, alone in the bat cave, writing a white paper. What do the pioneers need? They need to find others and work together, aligned in purpose. Once they decide what they stand for and how to approach the new system, they can name themselves. Satoshi called it Bitcoin. We wonder what the BRICS countries are thinking of for their currency's name.

The next step is building the community and social capital, which requires connecting, networking, and a common conception of the new system. Once the pioneers have regular connections and start building the system as a community, it becomes a community of practice. The community of practice nourishes the new system with time, space, money, expertise, and skill-building to continue growth. This stage involves a lot of experimentation, failing forward together and upwards as the new system continues to build and emerge. The systems and experiments which stick grow in influence.

Once a new system becomes a system of influence, it hits an illumination stage where success stories inspire the old system to transition. As Bitcoin and other cryptocurrencies grew in market capitalization to surpass silver and the GDP of many countries¹¹ during the past handful of years, many retail investors have entered the crypto market. In 2021, El Salvador became the first country to accept Bitcoin as a legal tender. The Central African Republic followed in April 2022. In November 2022, Brazil's Congress passed a bill that regulates Bitcoin as a means of payment, although not yet a legal tender.¹² While more countries are embracing Bitcoin, its future as a major monetary system is still uncertain due to regulatory scrutiny and potential resistance from established financial systems.

The Two Loop model helps us think and visualize how paradigms shift, which is unintuitive. Since paradigm shifts occur over long periods of time, sometimes longer than a human lifetime, and involve many imperceptible factors, people may not notice the gradual shift from a dominant system to an emergent system as it's happening.

The Two Loop model can apply to any fundamental shift of paradigm and even nested shifts. It's a product of the regeneration movement, providing a conceptual map of the transition to the New Earth. The New Earth is the regenerative future we want, where people, communities, and nature are thriving. The model posits that parallel systems and structures should be in place before the dominant system reaches a steep decline. We need to build parallel pathways and systems that do not rely on the dominant system. This increases the possibility of a graceful transition without devastating harm to humanity.

A Two-Loop transition to the New Earth includes the shifts to the SDG-attainable world and the human-centric, privacy-honoring worlds that EU legislators covet. Both nest within

the shift to the New World. They are necessary but not sufficient for the New Earth shift. Another shift nesting within the New Earth shift is the shift from today's flat static web to the hyper-dimensional Metaweb.

As we've seen, the Two-Loop model provides a roadmap for navigating the transition towards a regenerative future, where people, communities, and nature thrive. However, as we've also established, this transition requires not just changes in systems and structures, but also in the very technology we use to connect and communicate with each other. In the next chapter, we'll begin exploring the ways in which we can transform the web using existing technologies to build upon and beyond a statement made by a web insider in 2012: that the web browser has a big missing feature, one that prevents us from layering knowledge on knowledge. This statement may have been disregarded, but it holds the key to unlocking the potential of the web as a tool for not just connection and collaboration, but for building knowledge and addressing our global challenges. Let's dive in and discover how we can make this a reality with each other.

NOTES

- 1 The field of philanthropy is finally beginning to recognize the negative impacts of the power differential between funders and nonprofits; see <https://permanent.link/to/the-metaweb/nonprofit-starvation-cycle>.
- 2 Cambrian explosion is the unparalleled emergence of organisms between 541 million and approximately 530 million years ago at the beginning of the Cambrian Period. Many of the major phyla (between 20 and 35) that make up modern animal life appeared at that time. Many other phyla also evolved during this time, the great majority of which became extinct during the subsequent 50 to 100 million years.
- 3 <https://permanent.link/to/the-metaweb/future-professions>
- 4 <https://permanent.link/to/the-metaweb/multiple-mobile-device-ownership>
- 5 <https://permanent.link/to/the-metaweb/connected-devices-worldwide>
- 6 <https://permanent.link/to/the-metaweb/two-loops>
- 7 <https://permanent.link/to/the-metaweb/how-change-happens>
- 8 <https://permanent.link/to/the-metaweb/united-states-currency>
- 9 Accessed on May 26, 2022
- 10 <https://permanent.link/to/the-metaweb/saudi-arabia-yuan-chinese-oil>
- 11 <https://permanent.link/to/the-metaweb/too-big-fail-market-size>
- 12 <https://permanent.link/to/the-metaweb/countries-accept-bitcoin>

The Metaweb

The Web's Big Missing Feature

Missing pieces do more than complete the puzzle, they fill in an empty space.

—Luanne Rice

This thought-provoking quote tells us that the absence of certain elements can create a sense of incompleteness or unfulfillment. This sentiment holds true in many aspects of our lives, including the Internet. Despite its vastness and seemingly infinite possibilities, the Web has a big missing feature.

As we navigate the digital landscape, we encounter a host of services and platforms that promise to connect us to information, to each other, and to the world. Yet, for all its promise, the Internet often fails to deliver a truly meaningful and fulfilling experience. This may be due to a lack of accountability, safety, fairness, context, a sense of community, belonging, or purpose. These essential elements are key to a healthy and thriving online culture. Without them, the Internet can feel hollow, lacking in depth and substance.

A STRONG BROWSER?

Marc Andreessen, a pioneer of the Web and a respected futurist, has noted that the web browser has a big missing piece. And, he's right. Were this void filled, it would change the way we interact with the digital world.

Andreessen, the co-founder of Netscape and the venture capital firm Andreessen Horowitz, has played a critical role in shaping the Web. From building the first major web browser, Mosaic, to his current position as a respected commentator on business and technology, his impact on the Web has been immense. Despite criticism of his views and public persona, his expertise and experience with browsers are beyond reproach.

Yet, Andreessen's most notable comment on the Web is unknown and unfulfilled. One of the world's top venture capitalists and browser builders believes the Web has a big missing feature. In 2012, Andreessen said annotation, the ability to layer knowledge on webpages, was "the big missing feature of the web browser." He hasn't mentioned other big missing features of the Web since. Notably, this comment coincided with a \$15 million investment his VC firm made.

Relevant context for Andreessen's statement includes an interview with Steve Lohr, a tech and economics reporter for The New York Times, for the Ringer's oral history of the historic (and successful) antitrust suit brought against Microsoft for stifling competing browsers like Andreessen's Netscape.¹ Lohr said Andreessen wanted to "reduce Windows to a buggy set of device drivers underneath the browser, which would be the new top player that people would see and interact with."

It seems like a strange comment. The language is triggering, and the statement didn't age well.

Why would Andreessen think browsers could be the “top player” in personal computing? Browsers enable browsing and interacting with webpages. Supported interactions are minimal—skim, read, select, copy, cut, paste, fill in forms, and click through buttons or links. Right-click enables searching a selection and printing the page. You can save webpages and sometimes multimedia when enabled by the webpage.

True, providing access to the Web is essential for modern life. You probably use the browsers on your computer and/or phone almost every day.

Looking deeper, Netscape's idea, however, was to have a “strong” browser in contrast to today's weak browsers. Netscape Navigator distinguished itself from Microsoft's Internet Explorer by offering bundled online services such as search, email, calendar, and later, instant messaging with AOL's buddy list. These pioneering features foreshadowed the advent of social media and are now commonly provided for “free” by Google and other platforms.

We chuckled upon finding an aging webpage about Netscape's features with a large Chrome ad centered at the top of the page. Expanding the field, we are on Chrome, which works better with Google Office applications like email, calendar, and docs. When we hover our cursor on the tiny circle with the “i” at the top right of the Chrome ad, a tiny overlay shows “Ads By Google.”

It's not uncommon for Google to display its own products in ads or at the top of search rankings. For us, this page reinforces not only Google's dominance but Netscape's demise, showing a key to Google's dominion—display advertising. The other is search advertising.

Ironically, Google serves ads for their Chrome browser on a webpage displaying the features of Netscape Navigator, a more advanced product they never had to compete with because the behemoth tech firm Microsoft destroyed it. Google offers many features that Netscape Navigator bundled in their browser over two decades ago.

Andreessen's Netscape was the top browser in 1995 and 1996. But Microsoft started pre-loading their browser on all PCs as part of the Windows OS. Netscape could not compete. Because of monopolistic actions towards Netscape, a landmark antitrust case started in 1998. In the same year, AOL bought Netscape but never gave it the attention it needed. AOL shut down Netscape in 2008, after years of neglect, losing its market share to Microsoft Internet Explorer and Netscape's descendant, the Firefox browser. In 2008, Google's Chrome browser only had 0.3% of the market. Imagine, however, if Microsoft had not put Netscape out of business. Had Netscape been able to compete, the web would be much different.

We wonder where the Web would be now had Netscape had the opportunity to develop, refine, and extend its “strong” browser offering. During the Microsoft conflict, almost three decades ago, they had their own search, email, and calendar. What if Netscape had been able to fully develop the Big Missing feature of the web browser?

Hyperbole aside, let's examine Andreessen's assertion that the browser could overshadow the operating system, reducing Windows to “device drivers” underneath the browser. Today and throughout the Web's history, the top browsers are among the most popular applications for personal computing and among the most used time-wise. Most new services are web applications or mobile-first applications with a webpage. Much fewer desktop applications exist, and even fewer without a web application.

As mentioned earlier, Netscape bundled essential applications we use daily, such as search, email, calendar, instant messaging, and friends. Was Andreessen dreaming when he said browsers could become the operating system? Interestingly, the same company that benefited

from Netscape's demise introduced Chromebook, a laptop designed to rely on web applications for tasks using the Google Chrome browser.

Chrome OS is a Linux-based operating system designed by Google. Derived from the open-source browser Chromium OS, Chrome OS uses Google's Chrome web browser as its principal user interface. Google announced the project in July 2009, as an operating system in which applications and user data live in the cloud. Hence, Chrome OS primarily runs web applications. Like the strong browser paradigm.

Ultimately, Andreessen's commentary on annotation rings directionally true today. Annotation is important, though not a panacea within itself. Annotation is necessary but not sufficient to fix the web.

Despite the World Wide Web Consortium (W3C) releasing standards in 2017,² most web browsers do not support web annotation. Ironically, the only browser supporting annotations, albeit limited, is Microsoft Edge. It has "shopping" and other apps that allow modifications and notes to be made on top of webpages, and automatic application of coupons on relevant pages.

Today, several popular browser extensions support annotation, including Hypothesis, Diigo, Make Media Great Again (MMGA), and Reclipped. Most enable text annotation on text; Hypothesis has plans to support HP5 content and text annotations on images. Reclipped does text annotation on video clips.

Yet, a vast opportunity still exists for a decentralized platform that supports both multimedia and structured annotations, and enables computation above the webpage. With multimedia, a platform could enable any media (text, image, audio, video, tweet, HP5 content) to be an annotation or to have annotations. Humans and machines easily read structured annotations, which enable superior content filtering and discovery.

Computation above the webpage is a huge opportunity. Running computer code in the space above the Web enables a hyper-dimensional web with information, interactions, and experiences over webpages. This is transformational, creating the possibility of digital public space and unleashing creative expression, collaboration, and knowledge building above the webpage.

It's not surprising the Web is missing a big feature. As mentioned earlier, the Web is young. The Web is just decades into its development. An American Internet pioneer recognized as a father of the Internet, Vint Cerf, spoke metaphorically about the Internet:³

Bob [Kahn] and I created a road system and a bunch of rules for getting things on the road, but we didn't say details of the vehicles, what the vehicles are carrying, we didn't say what buildings should be built alongside the road. We tried to build an infrastructure that others could build upon, and they have. The Internet is such a powerful amplifier that we are still just learning how to use it safely.

No one designed or planned Today's Web. Many people, organizations, and companies built the Web, often directed by and subject to market demand. Continuing Vint's road metaphor, the next level of the Web is the data stream that supports self-driving cars so we can reduce accidents. It's time for a course correction.

AN OVERLOOKED PREDICTION FROM A SILICON VALLEY LEGEND

Andreessen's comments about web annotation weren't offhand. He had thought about the layering of knowledge on webpages since the early days. He sensed it was something the world needed.

He shared his thoughts after Andreessen Horowitz invested \$15 million into Rap Genius⁴ (now just called Genius). The startup's significant traction among rap aficionados, fast growth among Y Combinator startups, and strong team impressed Andreessen and his partner Ben Horowitz. But those weren't most important. In fact, Andreessen said that he found rap "as comprehensible as ancient Mesopotamian."

As Andreessen said in a blog post detailing the investment, he and his firm were interested because Rap Genius and its stellar team were on a mission to annotate the web:

Finally, there's the other reason—maybe even the real reason—why I'm so fired up about this idea and this investment. Only a handful of people know that the big missing feature from the web browser—the feature that was supposed to be in from the start but didn't make it—is the ability to annotate any page on the Internet with commentary and additional information.

In fact, both the first commercial browser, Netscape, and its predecessor, the first widely distributed web browser, Mosaic, experimented with group annotations. At Mosaic, Andreessen and his co-creator, Eric Bina, experimented with a feature that enabled users to add their comments to any webpage and select whether they were personal or for a group. If personal, they stored the annotations on the browser. If a group, they sent the annotations to the dedicated group annotation server for the selected group. Figure 8.1 is an email from Marc Andreessen recruiting participants to serve as guinea pigs for testing Mosaic's group annotation feature.

Anyone reviewing a page would see their personal annotations and the annotations for their groups. While the idea was popular among early Internet users, the Netscape team didn't have time to build a production scale server for annotations. There was no cloud computing, decentralized storage, or ways to monetize the annotations that justified further investment. Potential scaling issues made it difficult to keep going, so Andreessen and his colleagues dropped the feature.

Andreessen wondered what the web would have been like had browser-based annotation survived:

I often wonder how the Internet would have turned out differently if users had been able to annotate everything—to add new layers of knowledge to all knowledge, on and on, ad infinitum. And so, 20 years later, Rap Genius finally gives us the opportunity to find out.

This made Andreessen Horowitz's investment in Rap Genius exciting. It was his chance to continue the mission of annotating the web—but 20 years later. Andreessen said, Rap Genius "is on an ambitious mission, and one that we are proud to get behind."

Andreessen's announcement should have generated significant conversation. A top tech luminary used his firm's capital to invest in a company that was carrying out his favorite digital feature from several decades ago. He justified the investment by saying the Web had a big missing feature. How did the press react? Crickets. We liken it to Alan Greenspan—the 13th Chair of the Federal Reserve—saying the US economy has a big structural problem that no one had thought about. Or Steve Jobs saying the iPhone has a big missing feature.

Andreessen's commentary had a muted impact. Even considering his reputation in the tech community, large media outlets had a minimalistic response. They covered the investment; Andreessen and his firm were investing their capital into Rap Genius. The popular media outlets, however, completely ignored the "big missing feature" remark. They didn't discuss

group annotation server guinea pigs?

Marc Andreessen (marca@ncsa.uiuc.edu)

Mon, 31 May 93 22:41:06 -0500

- **Messages sorted by:** [date] [thread] [subject] [author]
- **Next message:** [Tim Berners-Lee: "Re: Keeping HTML Simple & Format negotiation between Browse](#)
- **Previous message:** [Marc Andreessen: "Mosaic 1.1 prerelease available for testing"](#)

We could use a few guinea pigs to try out our in-progress group annotation server for Mosaic. If you're interested, please let me know.

The current group annotation support is very simple (one group per group annotation server, one group annotation server per Mosaic session, no security except through obscurity), since it's mostly a proof of concept, but it does work and Mosaic 1.1 has built-in client-side support.

Suggestions as to future functionality and user interface components for same are welcome.

(How it works: every time you access a document in Mosaic, the group annotation server [if you're using one] is queried with the URL of the document you're viewing; if any group annotations exist for that document, the group annotation server returns to Mosaic corresponding hyperlinks which are inlined into the document just like personal annotations. So the assumption is that you've got a fast local net and group annotation server host. In the Mosaic Annotate and Audio Annotate window there are option menus that let you select 'Personal' or 'Workgroup' each time you make an annotation; for the latter, Mosaic sends the annotation over a socket to the group annotation server.)

Cheers,
Marc

Figure 8.1 An email from Marc Andreessen about group annotations.

annotation, or that Andreessen saw a huge opportunity to bring it to modern web browsers. They didn't cover that the Web we rely on might have an important flaw.

A BEHAVIOR THAT HAS EXISTED FOR CENTURIES

But let's take a step back. Andreessen isn't the first to recognize the value of the annotation. We can go back centuries to find humans leveraging annotation to educate others about our world.

Web annotation derives from text annotation, which connects commentary to content, but in an old-fashioned way, with pen and paper. The practice involves adding notes or creating a glossary in the margins of books that contribute to enhancing content. These include notes written for the reader's private study or as a shared resource available to

whoever accesses the text. Annotation is stigmergic, as it builds upon and adds to the work of the author and potentially other annotators.

Annotation traces back to early scripts. In 1000 AD, text annotations were an essential element of the Talmud. The Talmud is a collection of Jewish law and commentary on the Hebrew Bible, comprising the Mishnah and the Gemara. The Mishnah is the original text written by Rabbi Judah the Prince in the 2nd century CE. The Gemara, written later by the rabbis of the Talmudic academies, is a collection of debates and discussions on the Mishnah.

The Talmud is unique in that it not only includes the original text, but also the commentary, questions, and debates of generations of scholars. Rabbis added their insights, commentary, and other facts in the margins of texts. Scribes copied manuscripts with the marginal annotations and then distributed the annotated texts within their community. This created a layered text where each generation of scholars built upon the work of their predecessors, adding new insights, questions, and perspectives.

The process of Talmudic study is one of constant engagement, where the Talmud is not only read but also interpreted, debated, and commented upon. This stigmergic process creates a collective intelligence, in which insights, knowledge and understanding are layered on top of each other, making the Talmud a living text that continues to evolve and adapt to new interpretations and understanding. Figure 8.2 shows the page of a handwritten bible with annotations.

About a millennium later, Andreessen referenced the Talmud in a reference to Rap Genius' big idea and broad mission:

Generalize out to **many other categories of text ...annotate the world ...** be the knowledge about the knowledge ... create the **Internet Talmud ...** There's music in other genres and other languages, but what other categories? Poetry, literature, the Bible, political speeches, legal texts, science papers. And those are **just the start**. We think the **community will continue to expand beyond rap into all culture**. The potential of this company is large. (Emphasis added.)

The browser's father thought his investment would catalyze a vast expansion in annotation. It's a huge market opportunity. And a fraction of what's possible, as you will see.

In the medieval era, the practice of reading annotations was widespread. Scribes copied manuscripts with marginal annotations, which circulated within the community to share information and amplify the edited manuscripts. This practice also led to archaic texts being recopied and rewritten with updated information, ultimately improving the scripts available to the public.⁵

These marginal notes were an early form of crowdsourcing. Someone reading a medieval text could contribute to that text's meaning and value. Even if authors weren't interested in comments, that commentary could help other readers.

The invention of the printing press, around 1436 by Johannes Gutenberg, and the subsequent invention and proliferation of mass-market paperback books in the mid-20th century, shifted how people used books. Instead of one book shared among many people, each person had their own copy of the book, adding notes for personal use rather than public consumption.

Book owners would record key insights or information to help them recall their original thoughts. A study found that readers in cultures with print technologies, as compared with medieval readers, have limited opportunities for dialogue and learning through observation of others' annotations.⁶ This cultural shift altered the general view of written

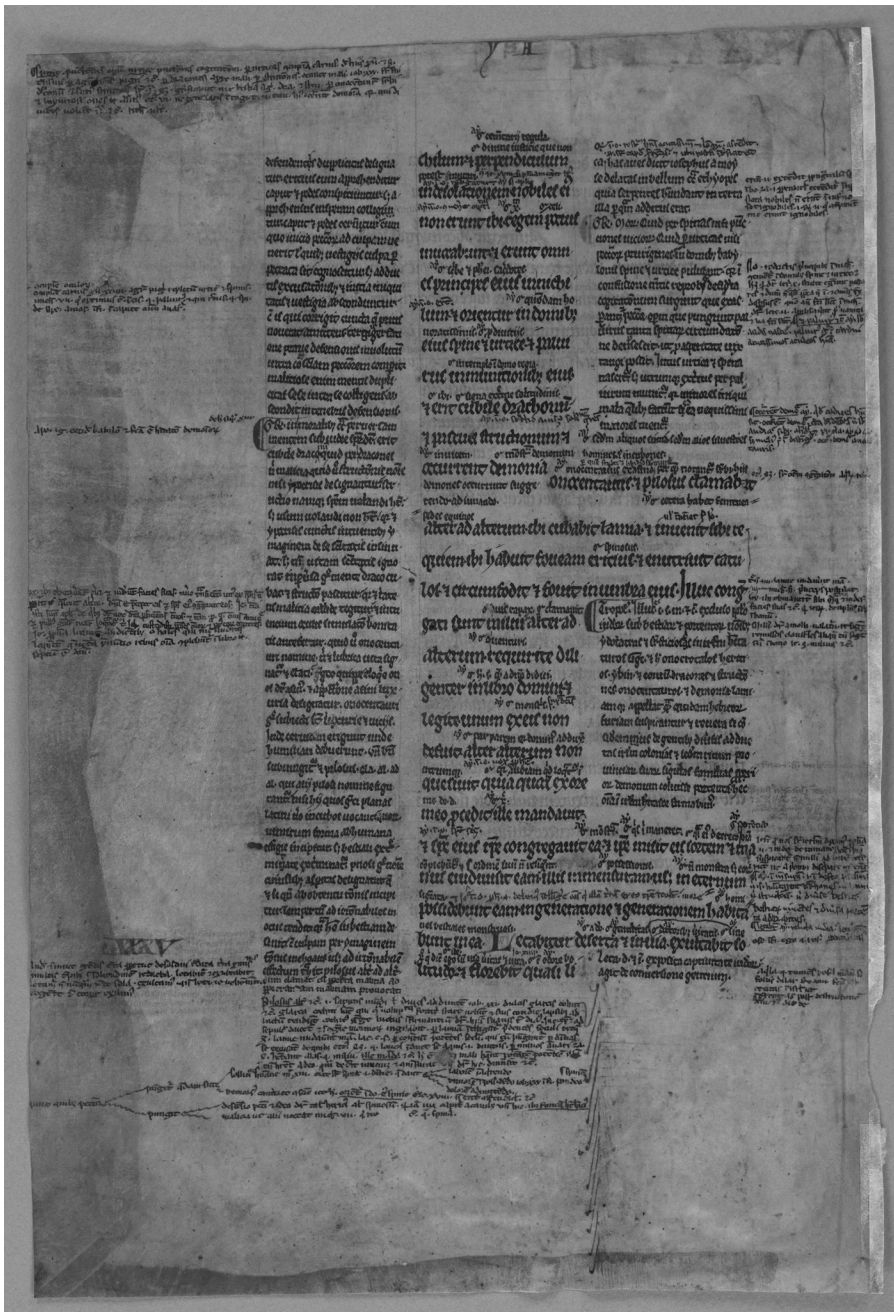


Figure 8.2 Bible with annotation commentary from the *Glossa Ordinaria*, circa 850–1499. MO389, Box 1, Folder 09. Courtesy of the Department of Special Collections, Stanford University Libraries.

annotations in the marginalia from being interesting and helpful to devaluing, if not damaging, the book.

Make no mistake, annotations within a text can be valuable and their aggregate value within a text may often be greater than the sum of the pieces. Their value is based on the

relevancy to the associated content, their legibility and comprehensibility, and the insight they convey. Annotations help the reader see what the text triggered in the annotation author's mind. We know book lovers who look for used books with annotations.

According to the aptly titled article *From the Margins to the Center: The Future of Annotation*⁷, which arrived during Web 1.0 after the Dot-com bubble, the primary functions of annotations are:

1. Facilitate reading and writing tasks
2. Eavesdrop on the insights of other readers
3. Provide feedback to writers or promote communication with collaborators
4. Call attention to topics and essential passages.

Textual scholarship is the umbrella term for annotations, which refers to all functions related to describing, transcribing, editing, or annotating texts and physical documents. Academic writers use annotations to add definitions and explanations in the margins to make the text more explanatory. Sometimes “annotated” editions of a literary work include marginal comments that explain, interpret, or illuminate words, phrases, themes, or other elements of the text. Enhancing texts with notes helps students better understand historical or academic texts.

ANNOTATION HAS ALWAYS BEEN PART OF THE WEB

While print technologies contributed to annotations being critical forums, digital technologies are reviving these practices and creating new ones. For 90 years, innovators have conceptualized and explored web annotation.

Web annotation enables layers of information on top of the original webpage. The annotator can add information without altering the content of the page. Groups can then access these “layers” of information. This type of information building differs from adding comments to a webpage. The webpage author chooses whether to enable comments. Commenting creates an entry in a database referenced in the page source rather than metadata in a layer over the webpage.

The earliest mention of what we think of as web annotation today was in Vannevar Bush's 1945 article in *The Atlantic Monthly*, *As We May Think*, in which he described an early hypertext-based system called Memex, short for the “memory extender.” Bush developed ideas for the Memex in the early 1930s and drafted a paper on it in 1939, which remained unpublished until it appeared in the *Atlantic Monthly*. The Memex would physically store, compress, and link information resources like books, articles, and individual communications in mechanical order. We consider Bush the “grandfather” of hypertext. Although never implemented, the Memex system foresaw many of today's technologies.⁸

Bush described the Memex both as “a sort of mechanized private file and library” and “a device in which an individual stores his books, records, and communications, and which is mechanized so that it may be consulted with exceeding speed and flexibility.” The Memex would store information on microfilm kept in the user's desk. Several projection displays would enable the user to compare different microfilms, similar to windows, which became popular on personal computers decades later.

Bush worried about the explosion of scientific information, which made it impossible even for specialists to follow developments in a discipline. The situation is exponentially worse now. Bush saw the opportunity to help people find information they are interested in. An

essential function of Memex was associative indexing, which links two pieces of information together so that when one is active, the other becomes immediately available. This annotation system for connecting two pieces of content was a precursor to hypertext and the hyperlink.

Besides individual links, Bush saw the Memex supporting the building of trails through the material as an ordered set of links that would combine information of relevance for a specific perspective on a specific topic. He forecast a new profession of “trail blazers ... who find delight in establishing useful trails through the enormous mass of the common record.” In the Metaweb, trail blazers are the early adopters who add value to web content within their area of expertise by connecting or bridging to evidence that relates to claims on the page. Bush thought the building of trails would also be an activity for the ordinary Memex user, who would share trails with their friends.

Bush’s hypothetical Memex, however, reaches its apotheosis in imagining the future utility of annotations:

Wholly new forms of encyclopedias will appear ready-made with a mesh of associative trails running through them, ready to be dropped into the Memex and there amplified. The lawyer has at his touch the associated opinions and decisions of his whole experience, and of the experience of friends and authorities. The patent attorney has on call the millions of issued patents, with familiar trails to every point of his client’s interest. The physician, puzzled by a patient’s reactions, strikes the trail established in studying an earlier similar case, and runs rapidly through analogous case histories, with side references to the classics for the pertinent anatomy and histology. The chemist, struggling with the synthesis of an organic compound, has all the chemical literature before him in his laboratory, with trails following the analogies of compounds, and side trails to their physical and chemical behavior. The historian, with a vast chronological account of a people, parallels it with a skip trail which stops only on the salient items, and can follow at any time contemporary trails which lead him all over civilization at a particular epoch. There is a new profession of trail blazers, those who find delight in the task of establishing useful trails through the enormous mass of the common record. The inheritance from the master becomes not only his additions to the world’s record, but for his disciples the entire scaffolding by which they were erected.

Bush foresaw the “associative trails” blazed by experts examining the record of human invention being soon thereafter available for searching. Besides making repeated research much quicker for the expert and followers, such a scheme would also lead to the serendipitous discovery of new ideas.

The Era of Invention

After Bush’s 1945 article, the hypertext field was inactive for 20 years. Computers were becoming interactive, but were sufficiently expensive that funding agencies wouldn’t consider wasting computing resources on non-numeric tasks, such as text processing.

In the early 1960s, Ted Nelson was the first to prototype a personal computer. Nelson was a hypertext pioneer with his Xanadu system. He coined the word “hypertext” in 1965, which represents the “HT” in “HTTP” (Hypertext Transport Protocol). Nelson and his Project Xanadu proposed bi-directional links, as well as parallel pages connected by visible links from an idea to its source.

Nelson explained, “The xanalogical content link is not embedded. It is ‘applicative’—applying from outside to content which is already in place with stable addresses. Xanalogical links are effectively content overlays. Any number of links, comments, etc., created by anyone anywhere, may be applied to a given body of content.” He imagined a report having thousands of overlapping links, created without coordination by many *people* around the world.

In *Who Owns the Future*⁹, Jaron Lanier explains the core technical difference between the Nelsonian network and Today’s Web. Nelson’s links were two-way instead of one-way. In a network with bi-directional links, each node knows what other nodes link to it. Two-way linking preserves context. He says, “It’s a small simple change in how online information should be stored that could have vast implications for culture and the economy.”

Nelsonian links are like the fundamental building block that underlies the Metaweb, the bridge. The main difference is the bridge, including the relationship between the two pieces of content. Bridges also store other metadata and analytics. The bridge is a contextual triple in the sense of a set of three entities that codifies a statement about contextual data as idea-relationship-idea.

Despite Nelson’s dedication to Xanadu for over five decades, the Xanadu vision has not landed. Xanadu was to create a repository for everything that anybody had ever written; a truly universal hypertext. Nelson saw hypertext as a literary medium, and he believes everything is “deeply intertwined”¹⁰ and therefore has to be online together. We love the notion of intertwingularity, though we think the idea of abandoning the Web and bringing all existing text into a new repository is untenable.

Also in the early sixties, Doug Engelbart started developing computer tools that augment human capabilities and productivity. Conducted at SRI (Stanford Research Institute), this was the first major project to work on office automation and text processing, and, in fact, was more ambitious and broad in scope than the capabilities afforded by today’s productivity tools.

Engelbart’s 1962 essay *Augmenting Human Intellect: A Conceptual Framework*¹¹ conceptualized a record system with cards and mechanical sorting. This included automatic trail establishment and trail-following, selective copying, and data transfer. The system would enable the development of powerful tools and processes for everyday intellectual work. Recognizing that electronic computer equipment would soon render a mechanical card-based system obsolete, Engelbart imagined circumstances not unlike our own, speaking as a hypothetical “friendly fellow” named Joe:

I’m sure that you’ve had the experience of working over a journal article to get comprehension and perhaps some special-purpose conclusions that you can integrate into your own work. Well, when you ever get handy at roaming over the type of symbol structure which we have been showing here, and you turn for this purpose to another person’s work that is structured in this way, you will find a terrific difference there in the ease of gaining comprehension as to what he has done and why he has done it, and of isolating what you want to use and making sure of the conditions under which you can use it. This is true even if you find his structure left in the condition in which he has been working on it—that is, with no special provisions for helping an outsider find his way around. But we have learned quite a few simple tricks for leaving appended road signs, supplementary information, questions, and auxiliary links on our working structures—in such a manner that they never get in our way as we work—so that the visitor to our structure can gain his comprehension and isolate what he wants in marvelously short order.

Engelbart elaborated on the value of associative-trail marking and following. Noting that statements often have implicit linkages to other statements, he advocated keeping track of

these associations. For instance, when several statements relate to a claim, if the claim changes, it's not always easy to remember why. Being able to retrieve the other considerations linked to the claim would be insightful.

In these respects, Engelbart's work predicted the Metaweb and the contextual information it provides, as discussed later.

As Engelbart makes clear, annotation is something all students and scholars already do. They work over other people's texts in order to better understand them. Being able to draw on the expertise and experiences of others is certainly helpful. That's why footnotes and endnotes exist in teaching editions of books and in anthologies. The innovations of Bush and Engelbart make it easy to share an experience. As Jason Jones observes, "Lurking behind their imaginative essays is an ideal of full comprehension—that we might be able to truly understand one another if we could just track down all the relevant influences and contexts and motives."¹²

One part of Engelbart's Augment project was NLS (oN-Line System), which had several hypertext features.¹³ (This abnormal acronym distinguished its name from another, the oFF-Line System, or FLS.) Engelbart designed NLS to augment collective knowledge work, focusing on making the users more powerful, rather than making the system easier to use.¹⁴ NLS supported rich interactions through text commands, in contrast to what Engelbart referred to as the WYSIAYG (What You See Is All You Get) paradigm that came later.¹⁵ NLS firsts include:

- | | | |
|--------------------------------------|--|---|
| • The computer mouse | • Integrated hypermedia email | • Universal "user interface" front-end module |
| • 2-dimensional display editing | • Hypermedia publishing | • Multi-tool integration |
| • In-file object addressing, linking | • Document version control | • Grammar-driven command language interpreter |
| • Hypermedia | • Shared-screen teleconferencing | • Protocols for virtual terminals |
| • Outline processing | • Computer-aided meetings | • Remote procedure call protocols |
| • Flexible view control | • Formatting directives | • Compilable command meta language |
| • Multiple windows | • Context-sensitive help | |
| • Cross-file editing | • Distributed client-server architecture | |
| • Annotation ¹⁶ | • Uniform command syntax | |

In 1968, Engelbart gave a demo of NLS at a special session of the 1968 Fall Joint Computer Conference called "The Mother of All Demos." A highlight was the introduction of a primitive three-button mouse. He also made some text annotations.¹⁷ This first public demonstration of many basic ideas that underlie interactive computing was risky for his group, given the significant investment of grant money needed to build it. During the 90-minute presentation, Engelbart used his mouse to move about the screen, highlight text, and resize windows. This was the first public presentation of an integrated system for manipulating text. He showed collaborative editing, annotation, and teleconferencing, the screen having a live video of Engelbart next to a panel with the computer's output. The demo was a success, receiving a standing ovation and inspiring future innovators in interactive computing.

During the Augment project, researchers stored all their papers, reports, and memos in a shared "journal" facility that enabled them to cross-reference their writings. This journal grew to over 100,000 items and remains a unique hypertext structure that supported actual work over an extended time.

In 1969, the Internet's predecessor, ARPANET, through NLS, had a functional model of web annotation that enabled users to annotate pages in layers. The model could not pass its testing phase, but provided a rudimentary ability to annotate pages.

Twenty years later, in 1989, Tim Berners-Lee included annotations in his influential paper, *Information Management: A Proposal*. This proposal for using the Internet to facilitate sharing documents among his colleagues at CERN laid the foundation for the World Wide Web. He famously wrote, "One must also be able to annotate links, as well as nodes ..." His director annotated the proposal with the words "Vague but interesting."

The Web got started in 1991 with a set of standards. The foundation of the World Wide Web is the Hypertext Transfer Protocol (HTTP), which loads webpages from hypertext links. HTTP is an application layer protocol designed to transfer information between networked devices. HTTP runs on top of other layers of the network protocol stack. (These layers differ from the Metaweb layers discussed earlier.) A typical sequence over HTTP involves a client machine making a request to a server, which then responds with a message.

HTTP created fair competition for resource-constrained startups, new market entrants, and well-financed incumbents. Absent standards, incumbents, which are almost always large corporations, define the tech, requiring under-financed contenders to adjust to their specifications since they drive the market. Standardization made the Web accessible to anyone with interest and a computer, mostly researchers and academics. The standards enabled an open web replete with authentic expressions of people's projects and interests, reflecting the ability to build and create. The early Web decentralized information and ideas.

The history of the early Web is rich, with many scrapped and forgotten browsers that, at their times, were functional enough to gain niche popularity in the market. Marc Andreessen led the team that built the first widely distributed web browser, NCSA Mosaic, in late 1992. As an undergraduate at the University of Illinois in Champaign, Illinois, Marc built the Mosaic prototype with a team of students and staff at the university's National Center for Supercomputing Applications (NCSA). NCSA released Mosaic in 1993. Offering a friendly, point-and-click method for navigating the Web and distributing it for free, NCSA Mosaic gained about two million users worldwide in just over one year.

While often described as the first graphical web browser, Mosaic came after the WorldWideWeb, Erwise, and ViolaWWW browsers. Mosaic was, however, the first browser to display images inline with text instead of in a separate window. Integrating multimedia such as text and graphics helped to popularize the World Wide Web. They named Mosaic for its support of early Internet protocols, such as File Transfer Protocol, Network News Transfer Protocol, and Gopher. Its intuitive interface, reliability, personal computer support, and simple installation underlaid its popularity.

Mosaic also offered a simple annotation capability. Users could select a piece of text and attach a written note and store it in their browser. When the user visited the page, the browser would automatically display the annotation. Mosaic was experimenting with a group annotation feature that would enable group members to see each other's annotations.

In April 1994, Andreessen left the NCSA to start Netscape with Dr. James Clark, founder of Silicon Graphics, Inc. They built an entirely new codebase called Mozilla. Netscape had a highly successful IPO in Aug 1995, raising investment funds at a \$2.9 billion valuation. Within four months of the release of Mosaic Netscape 0.9 in late 1995, it had taken three-quarters of the browser market. After rebranding, the Netscape Navigator web browser had over 90% of the market.

Not to be outdone and after Netscape allegedly spurned an offer to "split the market," Microsoft added Internet Explorer in the Microsoft Windows operating systems, first as part

of the add-on package, Plus! for Windows 95. Microsoft was a formidable adversary. Microsoft's only meaningful competitor in personal computers, Apple, had just 9% of the market.¹⁸ Microsoft used its dominant position to preload 90% of personal computers with the Internet Explorer browser, which it got in sketchy circumstances from an OEM browser maker named Spyglass that had licensed the code from the NCSA.¹⁹

The flooding of the market with a free, pre-loaded browser made it challenging to sell Netscape for its retail price of \$49. To compete, Netscape made deals with Internet Service Providers to bundle Navigator in their service. Yet, it was not enough. In 1998, having lost “the browser wars,” Netscape sold out to AOL for a whopping \$4.2 billion.

The Era of Experimentation

Since the 1993 debut of annotation in Mosaic, there have been many attempts to annotate the Web, mostly as browser plug-ins. Annotation startup Hypothesis maintains a historical list of annotation projects.²⁰ As of 2022, it had 76 entries of mostly defunct or limited use browser extensions for text annotation. The list is not comprehensive as it doesn't include Twitter's Birdwatch or Reclipped. We think the litany of failed projects reflects the difficulty of building a community around a browser extension with a single annotation use case, though clearly Hypothesis, Diigo, MMGA, Marker, Markup, Medium, and Yup have had a measure of success.

As mentioned earlier, in 2013, Andreessen Horowitz invested \$15 million in a startup called Rap Genius, now rebranded to simply Genius. Originally founded in August 2009, Genius enables users to provide annotations on song lyrics, news stories, sources, poetry, and documents. Launched as Rap Genius to enable fans to add comments and interpretations to hip-hop songs, the company attracted the attention and support of celebrities and venture capital. The site expanded in 2014 to cover other forms of media, such as pop, literature, R&B, and speeches, and added an annotation browser extension. Reflecting the new aspirations, the site relaunched as Genius in July 2014. But their browser extension was unsuccessful, and the company discontinued it without fanfare.

Two decades after Netscape dropped annotation, in February 2017 the WC3 unanimously passed the Web Annotations standard. The standard's goal is to bring conversations natively to the web, allowing for a precise discussion of any document on the web, hosted by any service. Yet still, none of the top browsers have integrated annotation. One would normally expect startups and agile companies to leap on the standard and begin innovating along with incumbents (in this case, Google Chrome and Safari). But the incumbents are not implementing and neither are the primary contenders (e.g., Firefox, Opera, Brave, DuckDuckGo).

Ironically, in 2022, Microsoft's Edge browser was the only browser that had annotation, although we doubt they meet the 2017 WC3 standard. Microsoft mentions on their product sheet, Edge is the “only browser that lets you take notes, write, doodle, and highlight directly on webpages. Then you can save and share your handiwork in all the usual ways. Make notes on recipes, send your partner ‘subtle hints’ about birthday gifts, or draw mustaches on celebrities—the Web is your canvas.”

Hypothesis is one of the most popular browser annotation tools. It's an open-source software project that aims to collect comments about statements made in any web-accessible content and filter and rank those comments to assess each statement's credibility. Hypothesis calls itself “a peer review layer for the entire Internet.” They were a key player in the WC3 annotation standard.

A Project to Annotate All Knowledge

For the last six years, Hypothesis has been building Annotating All Knowledge, a coalition of some of the world's key scholarly publishers, platforms, libraries, educational institutions, and technology organizations. Over 70 of the leading publishers, platforms, and libraries are bringing web annotation to the world's scholarship in the coming years. The likes of O'Reilly, Cambridge University Press, and Carnegie Mellon University are coming together to create an open, interoperable annotation layer over their content.

The Annotating All Knowledge site speaks to the reasons they have been so successful in bringing annotation of the highest caliber of scholarly publishers and knowledge platforms:²¹

Scholars are natural annotators, as the process of creating new knowledge requires building on what's come before. So, in order to bring this capability to the areas where new knowledge is created and published, we sought out some of the world's most essential scholarly publishers and knowledge platforms. They realize that a robust and interoperable conversation layer can transform scholarship, enabling personal note-taking, peer review, copy editing, post-publication discussion, journal clubs, classroom uses, automated classification, deep linking, and much more. They understand that this layer must evolve as an open, interoperable, and shared capability aligned with the motivations and interests of scholars and researchers. They agree to collaborate openly with their peers, to share experiences, and to work together towards mutual objectives.

Diigo is a social bookmarking and annotation tool with 9+ million users. Diigo enables users to save and organize web pages, as well as add notes and highlights to them. One of Diigo's key features is the ability to create and share lists of web pages with others. Users can create lists for different topics or projects, and share them with colleagues or classmates for easy collaboration and organization. This can be a useful tool for students or professionals who need to keep track of research materials or resources for a specific project.

Medium is the only open publishing platform to embrace annotation, and it remains one of the platform's main differentiators. Readers can highlight story content, which they can see in line. Others can see popular highlights. Medium also enables comments on text selections, visible to the story's author (and the commentator) in the blog post margins. Others can see popular comments. You can also tweet out the highlights or send a message to the author. Other readers can comment on your note and send you a message. Because the annotations are on Medium's site, you don't need to install a browser extension.

MMGA is a Dutch news organization owned by the Netherlands chapter of the Internet Society that trains annotators to identify false information on the Internet and fact-check the news sites. Their founder, Ruben Brave, says their structured annotations provide commentary to help viewers and readers better understand the facts. Structured annotations are machine readable, enabling different information structures and metadata. A machine can process them and decode them in a readable and understandable format. In 2020, MMGA teamed up with Federica Russo and Jean Wagemans of the University of Amsterdam.²² Together, they are working on extending the annotation from fact-checking to argument-checking, given that factual information is often embedded in argumentation and serves rhetorical purposes.²³

We love structured annotations. Ubiquitous, one-to-many, decentralized, bi-directional links that go beyond source material are key to building a transparent and trustable next-generation web. Had Nelson's ideas about bi-directional links stuck, we would have a quite different web than we do today. It might not be as vast, but it would have a higher signal-to-noise ratio, greater transparency, less piracy, and less false information.

Annotation is not the elusive big missing feature of the Web, but it relates.

We think the web's big missing feature is why Today's Web has intractable problems that diminish its utility, such as false information, privacy, security lapses, data exploitation, scams, catfishing, and more. Some of today's problems may benefit from the open, decentralized blockchain-based ledgers of Web3. But Web3 can also exacerbate them. For example, posting misinformation to a permanent ledger and creating new classes of scams such as counterfeit NFTs and rug pulls (again, the integral accident). The human-centric use of Web3 is necessary, but not sufficient for the next level of the Web. The web's big missing feature is also why Today's Web does not help humanity deal with its global challenges.

As mentioned earlier, in 2012, the father of the browser, Marc Andreessen, said that annotation—the ability to layer knowledge on top of webpages—is the big missing feature of the web. Directionally, he is correct. His thinking, however, was too restrictive. As far as we know, he was limiting the web's missing piece to notes on top of webpages. Many things—potentially much more valuable—can overlay webpages.

The Internet is unlimited in layers. Yet on the legacy Web, we limit ourselves to one or two layers. If we confine the solution space to one or two layers, then perhaps annotation could be the missing feature. But if we consider a hyper-dimensional web, the space above the webpage opens many possibilities well beyond unstructured annotation.

Unequivocally, the big missing feature is the Metaweb. The Metaweb is the computational space above the webpage; recall the second, third, and fourth layer of the proverbial Web cake. The Metaweb decentralizes the web experience. The computations above the webpage are whatever builders want them to be. It could display information from a remote procedure call, analyzing the operation of blockchains, promoting opt-in advertising, minting an NFT, and more. Now, people, information, and interactions have a presence above the webpage. The Metaweb has the potential to be a safe space for everyone. In Today's Web, the page author controls the experience. In the Metaweb, familiar, entirely new, and potentially decentralized interactions and experiences layer over the top of content elements on webpages. This meta-layer, as you will see, makes all the difference.

With decentralized digital public space, we can align our digital world with regenerating the Earth as we regenerate our collective mind, body, and spirit. Our new digital existences can bring us joy, understanding, and insight. For the first time, we can think, learn, and build knowledge together at scale. The collective cognitive capabilities that we develop will spiral humanity upward towards wise action.

As we've discovered, the big missing feature of the current web is the Metaweb, a computational space above the webpage that holds immense potential for innovation and collaboration. It has the ability to decentralize the web experience, bringing new possibilities and interactions to the forefront. It's no exaggeration to say that this meta-layer has the power to change everything.

In the next chapter, we delve deeper into the emergence of the Metaweb and the importance of accountability. We'll discover how an accountable meta-layer can enable us to think, learn, and build knowledge together at scale, collectively elevating humanity towards wise action. It's time to harness the power of this technology and build a better future. Let's explore the exciting future of the Metaweb.

NOTES

- 1 <https://permanent.link/to/the-metaweb/microsoft-antitrust-netscape>
- 2 <https://permanent.link/to/the-metaweb/annotation-web-standard>
- 3 <https://permanent.link/to/the-metaweb/vint-cerf-video>
- 4 <https://permanent.link/to/the-metaweb/andreessen-investing-rap-genius>
- 5 <https://permanent.link/to/the-metaweb/archaic-texts>
- 6 <https://permanent.link/to/the-metaweb/future-annotation>
- 7 Ibid
- 8 <https://permanent.link/to/the-metaweb/hypertext-history>
- 9 <https://permanent.link/to/the-metaweb/who-owns-future>
- 10 <https://permanent.link/to/the-metaweb/deeply-intertwined>
- 11 <https://permanent.link/to/the-metaweb/human-intellect>
- 12 <https://permanent.link/to/the-metaweb/context-motive>
- 13 <https://permanent.link/to/the-metaweb/engelbart-nls>
- 14 <https://permanent.link/to/the-metaweb/invisible-revolution> and <http://www.invisiblerevolution.net>
- 15 <https://permanent.link/to/the-metaweb/history-engelbart>
- 16 <https://permanent.link/to/the-metaweb/annotation-nls>
- 17 <https://permanent.link/to/the-metaweb/mother-all-demos>
- 18 <https://permanent.link/to/the-metaweb/apple-niche-player>
- 19 <https://permanent.link/to/the-metaweb/spyglass>
- 20 <https://permanent.link/to/the-metaweb/annotation-efforts-spreadsheet>
- 21 <https://permanent.link/to/the-metaweb/annotating-all-knowledge>
- 22 <https://permanent.link/to/the-metaweb/mmga-misinformation>
- 23 <https://permanent.link/to/the-metaweb/krino>

The Emergence of the Metaweb

You cannot solve a problem from the same consciousness that created it. You must learn to see the world anew.

—Albert Einstein

This and other similar quotes such as “we cannot solve our problems with the same thinking we used to create them” are commonly attributed to Albert Einstein ... but did he really say it? He spoke German, so it would likely be a translation.

We are not aware of reliable online references to Einstein ever saying this in English or German. Of course, Einstein probably said something close to it, and we have certainly found it useful in our life. Nevertheless, it seems vital for navigating our predicament with Today’s Web.

At our current level of thinking, many of the Web’s problems appear intractable. The good guys are always one step behind the bad guys. It’s a playground for predators without consequences. With problems like this, humanity hasn’t even contemplated how the Web could support the unprecedented levels of connection, communication, and coordination needed to address our global challenges. Hence, the troubling absence of our collective digital presence from the SDGs. From a systems perspective, when problems seem intractable and desirable solutions are not even on the table, it often stems from how we frame the problem.

As shown in Figure 9.1, the cone of future options depends on our current circumstances and how we define problems. We can uplevel our thinking by removing unnecessary constraints from the problem frame that limit the cone of future options. This could involve, for example, expanding the problem’s definition or the systems we are considering. From a systems perspective, we can develop problem frames that make new classes of future options, including omni-win solutions.

Let’s apply this meta-level thinking to humanity’s global challenges and the SDGs. The SDGs intend to reverse grinding poverty, hunger, and disease, improve health and education, and spur growth, but oddly they do not consider our collective digital presence. The Attention Economy is obviously an important driver of the global economy. Much commerce starts as e-commerce. Companies research, design, build, buy, and sell products and services online. Creators create online. And everything touches the Attention Economy. Given its importance, the SDGs should include the effects of the Attention Economy in the problem frame and a conscious Attention Economy in potential solutions. The same goes for ICT. The same goes for the emergent digital town square where real people debate matters vital to humanity.

This approach also applies to the Web. The constraints on how we perceive the Web limit the potential solutions that we consider. We think of the Web in a limited way. As reflected by the top-down control exerted upon our collective digital presence by mighty Internet

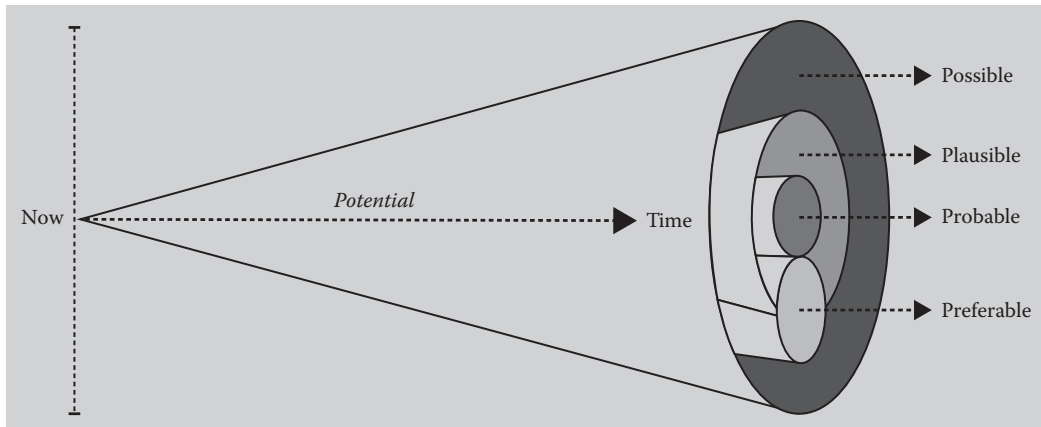


Figure 9.1 The cone of future options.

monopolies and overreaching states via online censorship and Internet shutdowns, the current structures of the Web facilitate digital authoritarianism. Very few options for democratic futures are plausible without a significant shift in how we think about and organize the web. And free speech is just the start of what's needed.

The Web lacks accountability. The current digital regime normalizes the creation of as many “unaccountable” accounts as one wants. Anyone can easily generate hundreds of accounts that have no connection to their “real” online accounts or material existence. One day, we might sum this era up as “The Unaccountable Web.”

Currently, there is no way to effectively hold people accountable for their online activities as part of Internet governance, as there is no system in place to identify individuals. Ideally, online communities could step in the void and govern themselves, as government entities have been ineffective in addressing online transgressions and jurisdiction is often unclear. Smart contracts and other Web3-based solutions, such as DAOs, tokens, and NFTs, potentially offer a transparent yet privacy-respecting path forward, but are currently only used in specific, isolated cases.

The Web decentralizes publishing such that anyone can publish a webpage or post information pseudonymously. Until recently, we could track nefarious sites through the WHOIS database of web domain owners, but now the domain registries offer a private listing service. To post on social platforms, one needs an account. This simply requires a working email and/or a phone number, which are free on the web.

Thus, anyone can publish content on the web with throwaway accounts based on false information without meaningful verification. They can make these accounts for free.

Anyone can publish content to the web with throwaway accounts based on false information without meaningful verification other than being able to receive an email, a text, or a phone call. Anyone can set up a free email service without verification. Free and low-cost services spoof phone numbers.

Google gives anyone with a Gmail account their own Google Voice phone number in their area code (or any other) that can receive voice calls, take voicemails, forward incoming calls, make outgoing calls, and send/receive texts. All for free. (We wonder what rights Google Voice users are giving up.)

A functional future Web will hold people accountable for what they say and do online. With today's thinking, this is impossible. Why? People are pseudonymous. They have finsta

accounts. We have no clue who's who. With this line of thinking, the solution is teaching people how to not be victims. (Recall Meta's response to the groping incident.) But actually, we could shift our thinking to include the lack of accountability and identity mechanisms in the problem frame. People can remain pseudonymous, but their history stays with them. We favor one account per participant for life, with multiple pseudonymous personas.

We wonder what it will take for communities to wrap themselves in a blanket of collective accountability. Everyone would have to prove their unique humanity, have one primary account for life, and own their behaviors. Accountability would force many people—the predators, the serial abusers, the scammers, the NFT counterfeiters, the catfishes, the trolls, the cyberbullies, the gang stalkers, and the impersonators—to change their ways completely or risk being easily tracked down. And if the blanket is thick enough, perhaps even hackers, white-collar criminals, arms dealers, money launderers, and unscrupulous investors might have much fewer degrees of freedom.

Is the webpage within the Web or the Web within the webpage? We're for the former. But we suspect most are for the latter, which holds the webpage supreme. One layer of content. Flat and static. Controlled by the webpage author. No evidence anyone else is there other than cookies, which are invisible. Some people could be above the page using an annotation extension (or maybe a Web3 wallet), but they're probably on their own unless they're doing social annotation for class at their university.

If our entire palette of potential solutions must limit itself to this one layer of content—the bottom level of the web cake—we are indeed selling ourselves short. It's the spiritual equivalent of cutting oneself off from universal supply. We would never do such a thing unless we didn't know what we were doing, which, unfortunately, is the case for most people when thinking about Today's Web and what's next. We think one of the strongest objections to the viability of Metaweb as a platform is the difficulty of getting everyone to use a browser extension. Yet another example of limited thinking.

A MULTI-LAYERED WEB CAKE

If we expand our perception of the Web (and the browser) to include the hyper-dimensional space above the webpage, the solution space becomes massive. We go from one layer with Today's Web to as many as needed for every specific use. As Bajali Srinivasan—the visionary behind the Network State—says, “The internet increases variance.” Each variant can be a layer of information and access.

There can be meta-layers for any online community, DAO, NFT collection, brand, association, company, non-profit, guild, neighborhood, class, and more. There can also be layers for applications that are active over the entire Web and for smart tags that attach to webpage content. Private layers are possible and layers just for friends or colleagues. (We explore this in the section of Chapter 12 on meta-communities.)

Here's a thought experiment. Think about the richness and diversity of the Web. From the people who use it. The languages they speak. How they live. Platforms they use. Content they can post. And most of all, the content itself. Remember, this is the first or bottom layer of the web cake. It is also the first or bottom level of the web.

Now make a copy and paste a second layer on the web cake. Actually, you'll have to copy and “paste-special” so that this new layer knows it's different. Whereas the richness of the Web is the vast breadth of topics and amount of content present on the web, the richness of content in this Metaweb layer is based on the quality of the content most conceptually proximate to the focus of attention.¹

This layer is transparent, except for the places it has proximate content. That is the richness of the Metaweb. As a Metaweb layer, the new layer tracks the cursor of the layer below it. At points of proximate content, the meta layer can display everything relevant and accessible on the Web.

From any webpage, you can pull information from the most relevant subset of web information and aggregate it at whatever level of detail needed. Actually, it's all the accessible information on the web. You—meaning a builder or an entrepreneur working with a developer²—can pull information from any public blockchain, database, and API (application program interface) as well. And you can do all this on any site; you don't have to own a site to build above it in the meta-layer. We go from displaying a webpage with content for the HTML code and the data stores to which the webpage author has access to displaying information above the webpage from all relevant webpages and data stores to which the developer has access.

Now imagine that this transparent layer is not just one layer. It's a level—more precisely, the next level of the Web—and it comprises many transparent layers of proximal content. In fact, countless different views can layer on a webpage for distinct groups that we'll call meta-communities. Members can see members of the same meta-community. Members can be part of different layers and move among them. Many layers will be DAOs, so they manage their own membership and moderate their own content or outsource it to the Metaweb ecosystem.

Stepping back, if we focus our thinking on the bottom content level, then most solutions we consider will simply be to create better content and purchase ads on the social platforms. This potentiality significantly limits the cone of future content possibilities. You can't really do anything else on the Web but consume and create content. At this level, if there is something damaging on the Web (e.g., a counterfeit NFT that rips off someone's art), our main options are to create a site or page that addresses the damage (e.g., a site that identifies counterfeit NFTs and the counterfeiters) or try to get it taken down. Good luck with the latter!

Have you ever noticed how difficult it is to get the attention of the Internet Platforms? They don't have phone numbers for you to call. Whether infringing content, false accusations, or counterfeit NFTs, the processes for getting information taken down are often information-intensive, time-consuming, and frustrating. Until the damaging content is taken down (which itself is uncertain), people who view the harmful content almost certainly won't know about the page on an entirely different site that explains the problem.

This structural problem, in which the problem and solution are separate webpages, applies to false information, scams, online predators, etc., if there is a link between them, it always goes from the evidence to the claim. The people at the evidence can see the claim, but not vice versa. For example, most people who see disinformation never see the associated fact check. And if they lean away from the mainstream, they may be skeptical, perhaps rightfully so, about who is doing the fact-checking.

That all being said, if we continue willfully to think of the Web as simply one level of content, it won't be possible for us to fix many of the most harmful problems of the web. We need a meta-level that enables a view of the Web similar to a satellite over the Earth. From its vantage point, the satellite can identify and track visual patterns on Earth that are imperceptible from the ground. Likewise, the view from a tall building or the canopy in a forest provides a meta-view of your surroundings. A meta-view of the Web enables relevant information aggregated from the entire web to be available while reviewing a specific page on the web.

Today's Web has ample problems that are being addressed—presupposing a flat and static web—one by one in corporate silos, but what if there is a systematic problem? We can't fix

thousands of little parts of the Web piecemeal and expect the aggregate to fix the entire system. It doesn't work that way. As mentioned earlier, the reductionist approach of decomposing problems into parts that can be dealt with as separate mini-systems can be counterproductive for complex systems like the web.

Most accelerators and incubators encourage startups to focus on fixing something small. This atomizes the energy to create and generate a field of small, isolated ventures. Most of their investments focus on incremental changes to expand the value of Web 2.0, extending its useful lifetime. They are the stabilizers from the aforementioned Two Loop model. They are working to preserve Today's Web—the dominant system in the Two Loop model. These are susceptible to being swallowed up by the Internet monopolies. The more ambitious new ventures focus on building communities of practice on the decentralized web, which is part and parcel of the ascending new system.

When startups agree to be bought out, they often have to let go of their initial vision and true potential, especially if there was a pro-social component to their vision. The acquirers always have their own vision of how to leverage the acquisition to increase their bottom line. This can include integrating the startup's technology and/or business into their platform—which may or may not be successful and could have different aims, subsequently selling the property, and/or even strategically shutting it down, potentially hoarding the intellectual property for decades if not indefinitely.

Thus, while selling their startup is a major milestone and success for many entrepreneurs, it also means that whatever benefits their tech could have brought to the world may never occur and likely will not contribute directly to systemic change.

Obviously, the tech giants and venture investors are not interested in systemic change. They don't want change unless they are steering it. Otherwise, they run from it. Instead, they invest in projects that are strategically aligned, easily digestible, focused on profitability, and have a clear plan and relevant traction. This favors projects focusing on profit above all else or technology that extends their power and control.

Projects with relative complexity or social missions are unlikely to attract most institutional funders. These are the funders of most Web 2.0 startups and many Web3 projects as well. Their funding choices reflect the level of thinking embedded in Today's Web.

THE TOP LEVEL OF THE WEB CAKE

As discussed earlier, Today's Web is flat and static. Like a cake without layers. It's two-dimensional. The X and Y axes of the screen. You can't do much. But how we frame the Web limits what it can be. Let's blow up the frame.

If the Web is a cake, the Metaweb is the emblematic top level. It's where the action is. But we only see it if we look at the Web differently. The Metaweb opens the Web's next level, the Z-axis. The third dimension is layers of information and interactions that overlay webpages. There's no limit to the number of layers on the Internet.

Metaweb theory is the next level of thinking for the Internet. The Metaweb creates decentralized public space above the webpage. It allows people to think, learn, and build knowledge together, and supports the development of collective intelligence and other collective cognitive capabilities.

The underlying navigational structure of the Metaweb is an ever-growing knowledge graph that connects online ideas, evidence, and claims. Each node of the graph is a content snippet, reflecting an idea, claim, or evidence. Lines representing the relationship between two pieces of content connect the nodes.

Bridgers build this Universal Content Graph. They build connections between web content in the layer above the webpage. The connections are called bridges. As a primitive, a bridge connects two content snippets—e.g., evidence to a claim—with a relationship. Content snippets can be any piece of content: text, parts of images, segments of videos or podcasts, tweets, and more. The relationship can be supporting, contradicting, citing, etc ...

Bridgit DAO is catalyzing a part of the Metaweb that we call the Overweb. In the Overweb, Bridgers receive rewards based on the value their bridges create in the system. Trained people validate bridges to ensure that the relationship between the nodes is correct, so the bridge can self-assemble into the Universal Content Graph that maintains its integrity.

A universal content graph connecting all relevant information online has tremendous value to society. For every sentence, image, or piece of an image, segment of a video, or audio, we can have a 360° contextual view of everything verifiably related.

The universal content graph connects siloed information into an information ecology, beaming higher-order female “connectedness” energy into the Web. By enabling the Web of connections, we create the knowledge base necessary to activate the high-order masculine of logic and order. These two polarities dance together and balance each other on the next-level web.

THE METAWEB EXPERIENCE

The universal content graph supports the three key experiential patterns of the Metaweb.

One, the Metaweb triggers based on attention. The browser overlay tracks your focus—cursor on a laptop, touch on a mobile device, and line-of-site or controllers in a virtual experience. When your attention approaches, content with information in the meta layer signals that more information is available. Surfers only see signals for content that they are focusing on.³

Two, the Metaweb provides a 360° overview of the contextual information related to the focus of the viewer’s attention. The viewer can reveal more detailed information as needed based on their interests.

When your attention comes towards it, the content highlights and a badge displays the number of contextual items available. Surfers can click the number to see a 360° contextual overview. The overview enables filtering so that the surfer can narrow their current focus down to specific information and interactions. The context can include simple and structured annotations, conversations, polls, bridges, lists, and/or other smart tag-based interactions, transactions, and experiences created by the Metaweb ecosystem.

Three, the Metaweb supports focus-based navigation. When the surfer filters through contextual info, they can click on things that interest them. Bridges take surfers to another page with the bridge content highlighted. This counts as a bridge crossing. If it’s some other smart tag than a bridge, it overlays the relevant information and/or interactions on the current page.

This meta layer creates the possibility of mitigating, if not solving, the problems with Today’s Web. Recall the four-layer web cake from Chapter 1 and how we propped up the web cake to look like a laptop. The three top layers are annotations, Web3 wallets, and computations that display information and interactions. Henceforth, we will use the term Metaweb to represent the intermixing of these top three layers of the web cake above the basic webpage content.

The Metaweb is the space above the webpage that is filled with annotations, wallets, their transactions, and computations that could be interactions and/or experiences. Imagine an

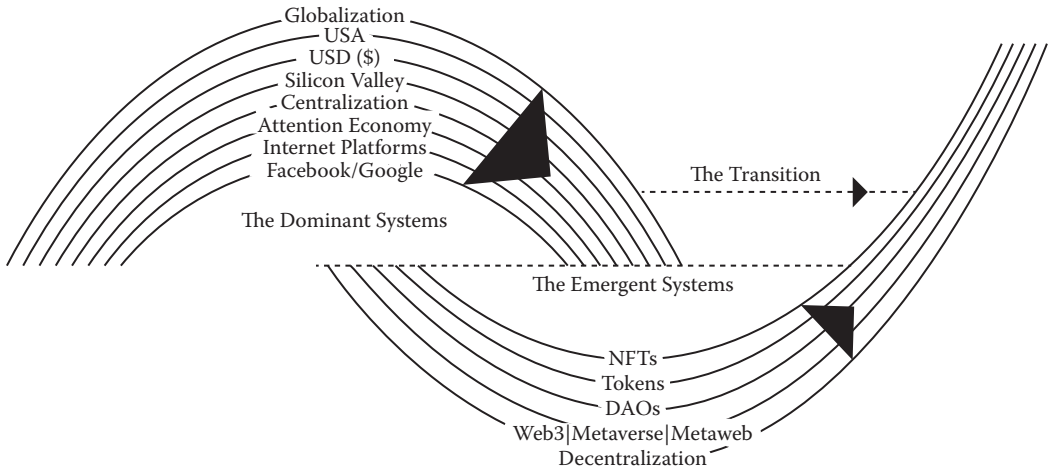


Figure 9.2 The nested Two Loops.

animated cake with three transparent layers. Each bustling with a party train of icons, avatars, and tiny 3D figures, boogying to that classic party tune “Celebrate good times, come on.”

In Two Loops parlance, as shown in Figure 9.2, the Metaweb is an emergent system nested alongside the metaverse and Web3 within the emergent decentralization movement, which aims to become the new dominant system.

Applying the Two Loop model to the Web, we’re past the peak and in a decline with several nested dominant systems. These include Globalization > USA > USD > Silicon Valley > Centralization > Attention Economy > Internet Platforms > Facebook & Google > Web 2.0 startups Facebook & Google. To varying extents, these are past their peaks and into irreversible decline. At or soon after the peak, the pioneers start decoupling. They need to meet the people with whom they share a common purpose. The Metaweb gives them a presence on webpages of their common interests. For the first time, people think, learn, and build knowledge together, organizing into communities of practice, which we call meta-communities.

The next section explores how we can affect which emergent systems prevail and become the new dominant system.

AN EDDY IN THE RIVER

Tom Atlee’s work at the Co-intelligence Institute intrigues us. It provides ways of thinking about how emerging initiatives become part of the new system described by the Two Loop model. Tom’s ideas about the role of collective wisdom and knowledge in the transition to a more sustainable future, as expressed through his Co-Intelligence Institute, are relevant to this discussion. Particularly, those about processes that develop our collective intelligence such that we can cobble together a future that works for all.

About a decade ago, we happened upon a post by Atlee⁴ that changed how we see the early stages of building the new system. The post expanded upon Robert Gilman’s Stages Of Change article,⁵ which explains how small events in chaotic systems can lead to major

changes. And how we can use them to our advantage if prepared. This reminded us of Bucky Fuller's "trim tab" concept.

Gilman talks about a river, which is usually smooth at its river bends. But when swollen by rain and snowmelt, it produces many small eddies at its turns, which appear and disappear, generating chaotic turbulence. As the flow volume and speed increase, one of those eddies suddenly transforms into a large, stable whirlpool. This new "structure" in the river's flow absorbs the turbulent, chaotic energy of the small eddies.

Gilman notes that at any level of flow in non-equilibrium systems, the system has multiple potential structures it can "choose" from. At any transition point, small fluctuations can have an inordinate and sometimes decisive impact. In social systems, these "small fluctuations" may be the actions of individuals, small groups, and distributed networks. This is the science that underlies Margaret Mead's famous quote—"Never doubt that a small group of thoughtful, committed citizens can change the world. Indeed, it's the only thing that ever has."

Let's return to the river bend. Suppose two places in the bend could form a stable whirlpool. But once a whirlpool forms in one, it prevents others from forming. As the flow increases, how does the river choose?

The river chooses whichever place has an eddy when the flow can amplify eddies into a whirlpool. The unstable phase of a non-equilibrium system contains a lot of unstructured energy that is available to amplify small events into major changes. These small events are "seeds" of a potential structure the system could become. As a non-equilibrium system moves from one stable structure to another, it depends on which "seeds" are available for amplification as the system reaches its critical "decision point."

This is significant for humanity's evolution towards sustainable societies. Gillman says our tools—spiritual, mental, emotional, and physical—determine our behavior. Thus, the most effective way to change a culture is by creating new ways to address basic needs. During times of preparation, we can develop components of the new culture and operate them in small ways. So they're ready to become part of a new synthesis when a society arrives at the crossroads. Well-developed and mutually supportive new pathways that are easily adopted and harmonious with new circumstances become the new road.

Atlee concludes, "I see the evolution of increasingly wise 'sustainability commons' as providing 'seeds' for potential structures that our out-of-equilibrium systems could settle into after the chaotic transitional periods that are emerging."

The eddy in the river metaphor suggests building parallel systems while the dominant system (per Two Loop model) is still functioning. Let's build the Metaweb now.

Pioneers create communities of practice as the eddies—the seeds of change—so relevant ideas and their manifestations are available for transformation when the time comes. Who will be the pioneers of the Metaweb?

Communities of practice that provide the easiest transition to a new stable structure have the best chance of becoming part of the new system. People can access the Metaweb by installing a browser extension, using Metaweb-aware browsers, or visiting a Metaweb-enabled site. Metaweb communities of practice are forming to include open source projects, developers, entrepreneurs, designers, business people, and marketers.

To ensure the Metaweb prevails at the crossroads, we need to build applications that the future needs. These could address the problems on Today's Web or increase cognitive freedom and collective cognitive capabilities. They can also enable new pathways of connection, communication, and coordination to address our global challenges.

BEYOND COMPUTATIONAL THINKING

We appreciate the decentralized nature of Web3 and its potential to promote financial literacy and give individuals more control over their economic affairs. Web3 alone, however, isn't sufficient for improving collective intelligence, sensemaking, decision-making, coordination, and collaboration.

The next level of the web—beyond Web3—demands that we liberate ourselves from limiting thinking patterns, such as computational thinking and reductionism. This creates space that can enable different levels of thinking, helping us make sense of Today's Web, how we arrived here, and where we can go.

Foundations of Humane Technology

The Center for Humane Technology offers a free, self-paced online course called Foundations of Humane Technology for professionals learning how to build more humane tech. Over 12,000 participants from 130+ countries have enrolled, including engineers, product managers, researchers, and professionals from top organizations such as Meta, Apple, and the UN. Their premise is "To build technology that matters tomorrow, we have to start with different principles today." Enroll in the course at humanetech.com/course.

Liberating Ourselves from Computational Thinking

This book is about computation above the webpage, which expands computational thought. What we choose to implement in this space should consider the methods, theories, and perspectives of the social sciences. In the article *Why Computing Belongs Within the Social Sciences*, Randy Connolly argues that:⁶

The well-publicized social ills of computing will not go away simply by integrating ethics instruction or codes of conduct into computing curricula. The remedy to these ills instead lies less in philosophy and more in fields such as sociology, psychology, anthropology, communications, and political science. That is because computing as a discipline is becoming progressively more entangled within the human and social life-world, computing as an academic discipline must move away from engineering-inspired curricular models and integrate the analytic lenses supplied by social science theories and methodologies.

Over time, claims about the importance of computational or algorithmic thinking have become sufficiently mainstream that many people believe everyone can benefit from thinking like a computer scientist. The research in the field has expanded significantly beyond the development of algorithms, hardware, and software architectures to include using these computational lenses to examine social, psychological, and cultural phenomena.

This expansion brings bold claims about understanding and explaining the social world with no background in social theory, economic models, or psychological concepts. While many fruitful insights into social phenomena certainly are happening and will continue happening through computational approaches, the flow of ideas appears to be in just one direction.

Connolly warns about the ramifications of excluding social sciences in thinking about fixing the problems like Today's Web. The technocratic belief that computing provides privileged insight and methodologically superior techniques and approaches that universally

apply and provide unblemished truth propositions is a flawed ideology. This ideology results in tech executives testifying in Congress about what went wrong.

It ignores the unintended effects on truth propositions of social institutions, human failings, and antiquated theories. It's not only academically arrogant, but short-sighted. Instead of replacing social science approaches, academic computing would improve immeasurably by supplementing its thrust with the methods, theories, and perspectives of the social sciences. Indeed, one could claim that not only would computing improve with more social science, but that computing today actually is social science.

We agree. Things really clicked for us when we saw Stanford had opened a department called Persuasive Technology in 1997, one year before Google and seven years before Facebook. On their original site⁷ (which was deleted from the Internet at the end of 2019, but still available on archive.org), it said, “the Stanford Persuasive Technology Lab creates insight into how computing products—from websites to mobile phone software—can be designed to change what people believe and what they do.”

Indeed, their URL was captology.stanford.edu. Captology is the study of computers as persuasive technologies. The term was coined by BJ Fogg, the Director of the Persuasive Technology Lab, to refer to the intersection of computer science and the study of persuasion. The field of captology focuses on understanding how computers can be used to change people's attitudes or behaviors, and how to design technology to be more effective at persuasion. This can include the study of how design choices, such as the layout of a website or the wording of a message, can influence user behavior, as well as the ethical implications of using technology to persuade people.⁸

Based on information from the Wayback Machine, it appears they rebranded to the Behavior Design Lab in 2018⁹, which perhaps coincidentally was six months after the Cambridge Analytica scandal. Now they say their research has moved on from persuasive technology to focus on designing for healthy behavior change. They now highlight the ethical contributions of persuasive technology so that those tasked with designing persuasive technologies can do so ethically. Certainly, the lab could have applied social sciences more ethically in its early incarnations.

Uncover the past with the Wayback Machine!

Uncover the secrets of the past with the Internet's digital time capsule. Investigate how websites have changed over time, or delve into the history of your favorite sites. From unsolved mysteries to forgotten gems, the Wayback Machine has something for everyone.

Are you ready to be an Internet sleuth? Try it now at <https://web.archive.org>.

Social science agrees that we need a new level of thinking to address the ills of Today's Web. It tells us to respect the complexity of the Web and the integrity of the interconnected ecosystems on which the SDGs operate. We must not limit ourselves to breaking problems into pieces in our minds and working on them in disconnected and sometimes competing silos.

Social sciences are academic disciplines that study human society and human individuals in society and thus are a bridge for us to better understand what is going wrong with the Web. While long-established fields such as sociology, economics, anthropology, psychology, and political science may come first to mind, disciplines such as education, law, linguistics, geography, gender studies, communications, archeology, and even business school fields such as management, marketing, and human resources are all social sciences. The diversity

of specialized fields matches the complexities of the domain of study. That's us. Anything we—children of nature—touch can become complex.

One of the key insights and values of modern social science is its embrace of complexity. Methodological and theoretical pluralism defines both the social sciences and its subject—humans—in social, political, economic, and cultural contexts. This contrasts with the natural and engineering sciences, where subjects are more predictable, and thus a single methodological approach for making and evaluating knowledge claims is possible.

Humans are not machines. Humans are complex systems that form and create complex systems. We are each recursively nested systems of complexity. Of course, from this perspective, narrow, reductive thinking would not work reliably on human systems (unless perhaps the humans themselves have few degrees of freedom, for example, as in totalitarian systems or Internet platforms).

Are you ready for the next leg of our journey? Buckle up, because we're about to delve even deeper into the intricacies of the Metaweb. We've set the foundation for understanding the dynamic, interconnected web of the future, but there's still so much more to discover. By revisiting the conceptual building blocks, we'll continue to uncover the true potential of the Metaweb. So, put on your thinking caps and get ready to embark on the next chapter in our quest to understand this revolutionary technology.

NOTES

- 1 While attention can apply to all senses, the focus of attention is important because it tracks what the eyes are actually focused on.
- 2 As of 2023, a developer. But in the future, there will be drag-and-drop AI app-building wizards that anyone can use to spin up dApps on the Metaweb.
- 3 We are bringing back “surfer” since one can surf bridges to see related content as people used to surf links from page to page, and often down stimulating rabbit holes.
- 4 <https://permanent.link/to/the-metaweb/atlee-post-rivers-bend>
- 5 <https://permanent.link/to/the-metaweb/gilman-stages-change>
- 6 <https://permanent.link/to/the-metaweb/computing-social-sciences>
- 7 <https://permanent.link/to/the-metaweb/archive-stanford-captology>
- 8 <https://permanent.link/to/the-metaweb/stanford-persuasion-technologies>
- 9 <https://permanent.link/to/the-metaweb/behavior-design-lab-stanford>

The Cairns in the Rabbit Hole

The rabbit-hole went straight on like a tunnel for some way, and then dipped suddenly down, so suddenly that Alice had not a moment to think about stopping herself before she found herself falling down a very deep well.

—Lewis Carroll, *Alice in Wonderland*

The emergence of the next-level web was like Alice’s journey to Wonderland. Initial conversations were rather pedestrian, following common thought tunnels. But ruminations about connecting information led to a mysterious rabbit hole that went deep into the unknown. Just like the rabbit hole to Wonderland, this rabbit hole—filled with its twists and turns, synchronous surprises, and moments of realization—also ended up in an entirely new world.

At first, it was like free-falling through ideas. Then, the rabbit hole landed on the synaptic web and a map for the Internet, both of which were located in the territory of the Metaweb. As the Metaweb emerged into our awareness, it opened a new world of opportunities for transforming the Web and solving the problems that plague it, many of which are chronicled in this book. In retrospect, the Metaweb was always here, but only became apparent because the rabbit hole shifted our perspective.

THE SPARK

The initial Metaweb spark happened in 2014. Despite exponentially more data available and everyone having access to more or less the same information, experts were arriving at opposite conclusions. Sometimes even when they had similar values. Obviously, they were focusing on different data or interpreting the data differently. Perhaps entrenched interests, blind spots, personal biases, or access issues were also in play.

Meanwhile, links on the Web were less plentiful than before. Commercial sites were avoiding external links, seeking to keep visitors on their websites. Links were the mechanism to cite sources and connect evidence to claims. With links disappearing, the context was dying and information was in silos. The Web needed a shared context, but information silos and the lack of context foreclosed on that possibility. No one, including experts, could even agree on what information was available.

Everyone including the experts has personal stashes of information in silos. They may prove useful in the future ... if one remembers them and can find what they need. But they’re spread across the social web, bookmarked pages, and information copied and pasted into Google Docs, Evernote, Roam Research, or another silo on the cloud or our computer.

The Metaweb started as a concept of a long-term conversation on the Internet; unlike the news and online conversations, which were short-lived and ephemeral. Here today, gone tomorrow in a sea of past stories. No continuity other than self-interested links. Even Reddit archived its threads after six months. The web was for short-term conversations.

Humanity has needed a place where a layperson can explore any topic and get a deep sense of the relevant questions related to it. Everyone would have access to the same information. People could investigate claims, evidence, perspectives, and data sources. And if they found relevant knowledge somewhere else, it would be self-evident where to post it. Also, a notification would optionally inform every person in the thread that additional information had become available.

Such a platform could be indispensable for many complex issues that humanity was facing. But it couldn't be just another information silo; information silos were a large part of the problem.

WHAT THE TRIVIUM TELLS US

Mark Passio is truly one-of-a-kind. He's acerbic and ardent. A metal-playing, natural law-advocating, American-Italian commentator on the state of the world who describes his presentation style as caustic, intense, and combative. He begins presentations by saying, "My catchphrase for my entire speaking career has been and continues to be 'get as offended as you like.' I'm not here to be liked or make friends." He presents the harsh truths and realities of what is happening on Earth.

Passio recorded over 200 "What on Earth is Happening!" podcasts. The 6+ hour podcasts address the state of the world through a natural law lens and expose the dark occultists ("the controllers") that control the masses (whom they call "the Dead") through ancient knowledge of the human psyche. Passio dedicated a handful of episodes to the Trivium, a primal method for discerning the truth, which he considered foundational for understanding natural law and the podcast.¹ While perception is not reality, it can be in harmony with reality. We all have filters that veil us from the comprehensive and accurate perception of reality. The Trivium removes filters, uncovering truth and reality, so we can act.

Always go beyond memorizing formulas, passing tests, to always go deep into the underlying principles, to track any problem down to the root cause buried in the dirt and the dark.

—Elon Musk²

As in the quote, the Trivium suggests that to effect real change, one must focus beyond the symptoms to the causal factors or the source of the problems. This requires understanding the problem and how it started—that is, "the Why," which uncovers available options. The story of how something came to be often illuminates its Why. (In the next chapter, we explore how the Web came to be.) Without the Why, we cannot reliably effect change.

As shown in Figure 10.1, we depict the Trivium as a three-layer pyramid. Knowledge is the base. Understanding in the middle layer. Wisdom is at the top. The Trivium simply tells us, to make wise decisions, we need a correct understanding. To have a correct understanding, we need a clean and accurate knowledge base.

Pre-dating Egypt, the Trivium comes from ancient mystery school traditions. Humans have known and used this technology to discover the truth for eons. We don't know the age of the Trivium. It's from antiquity.

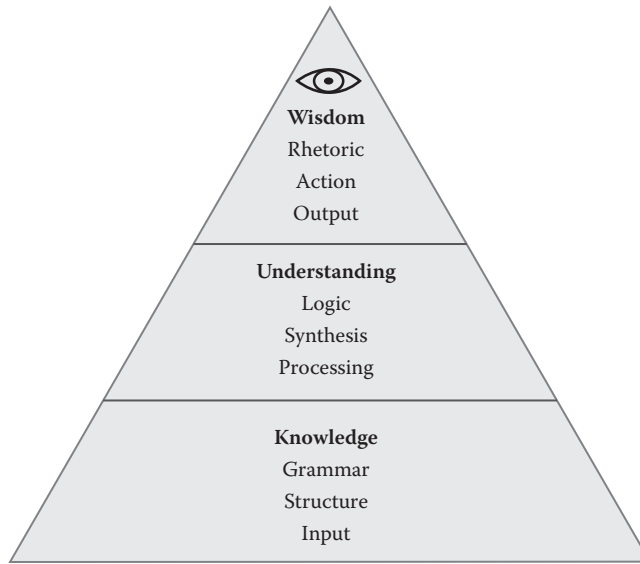


Figure 10.1 The Trivium: a pathway to truth.

The etymology of the Trivium is two Latin words—*Tres* or *Tria* meaning three and *Via* meaning way, path, or road. A way is also a method. Putting the words together, we get three roads, three ways, or three methods. Hence, the Trivium is the threefold path to truth. The Trivium helps us to get to the true underlying causes of our conditions so we can change them.

The takeaway from the Trivium is profound. Knowledge is the foundation of our intellectual capabilities and the discernment of truth. With a bad knowledge base, there is little chance of understanding and, therefore, making consistently wise choices.

For a hyper-computational civilization, we perhaps have the worst knowledge base conceivable, which diminishes our capacity for collective sensemaking, meaning-making, and choice-making. It's not that it is all bad information, but that, without context, the good and bad information are indistinguishably mixed on Today's Web of silos.

Humanity needs the Web to be its knowledge base because it serves as a vast and constantly evolving repository of information, accessible to people worldwide. In today's fast-paced and globalized world, the ability to easily access and share information is essential for personal and collective growth, innovation, and progress.

The Web offers a level of flexibility, scalability, and accessibility that no other platform can match. Other knowledge management systems, such as libraries or physical archives, are limited by location, accessibility, and outdated information. Similarly, other digital platforms, such as databases or intranet systems, are restricted to specific groups or organizations and lack the reach and connectivity of the web.

Humanity needs its knowledge base to provide a shared context for communication, coordination, and collaboration, so we can think, learn, and build knowledge together. This would enable us to build our collective intelligence and other collective cognitive capabilities. The Web is the only system that offers the possibility of a shared contextual view of the world's knowledge, but it's far from that now. Today's Web is deeply fragmented with information isolated in silos.

This reality begs the question: how might we reorganize content on the Web to elicit effective responses from humanity at all levels, from experts to policymakers to the public? The Trivium (and observations of social media) show that effective conversation platforms cannot operate on a rotten knowledge base. It follows that we need to focus on creating our best knowledge base before developing conversational platforms.

First things first. Effective sensemaking and collaboration platforms that operate at scale will come after reorganizing the world's information and building upon the new knowledge base.

Google claims to have already organized the world's information, but the organization of information is still a work in progress, and their approach is optimized more for them to extract large amounts of money through advertising. This has created incentives for companies to create low-quality and duplicate SEO content, which makes the Web worse overall. The information presented by Google is not always of high quality, and there can be inconsistencies in search results. It is clear that the way Google organizes information is more beneficial for them in terms of revenue rather than for the benefit of humanity's knowledge base.

In the next section, we explore how the human brain organizes information and how it relates to reorganizing the Web.

HOW THE WEB COMPARES TO THE HUMAN BRAIN

Neurologists now believe the density and flexibility of *connections* between neurons in the human brain, rather than neurons themselves, are at the root of intelligence. These chemical bonds between neurons are called “synapses.”

The total number of brain cells, or neurons, peaks in early adulthood. Our ability, however, to generate new connections among neurons and between different parts of the brain persists throughout our lives. Neurologists call this “neuroplasticity” or “plasticity.” Like other muscles, the brain's plasticity increases with exercise.³

The synapse is the chemical bridge in the “gap” between one neuron and another. These neural bridges create pathways that grow and contract. When used, they strengthen with additional connections. When unused, they dissolve, making way for new, more useful pathways. These incredible chemical bridges define the neural communication patterns that define our cognitive capacity.

Relative to learning in the human brain, the Web seems, if not primitive, certainly an ineffective teacher with its own learning disability. While the Web has almost 2 trillion pages, the amount of contextually relevant links is low, surely less than a handful per page. The web grows in connections when new pages link to previously existing pages.

In contrast, the human brain is a cornucopia of connections. Each human brain has about 86 billion neurons.⁴ Each neuron may have up to 10,000 connections or synapses, and the human brain up to 1 quadrillion connections⁵—many more connections than the entire Internet. And the human brain can continue learning throughout its entire life.

Could we organize the Internet like synapses in the brain?

A SYNAPTIC WEB OF LINKS

Neural science provides an apt metaphor for the “synaptic web” in that the connections between objects are more important than the objects themselves. Just as neurons in the brain connect and communicate to form complex networks that allow for thought and consciousness, the interconnectedness of information on the web is crucial for understanding and making sense of it.

Similarly, indigenous or original peoples have traditionally placed great importance on relationships, both between individuals and between individuals and the natural world. This concept of relational thinking can be applied to the web, where the connections between pieces of information and ideas are vital for understanding and utilizing them effectively. The synaptic web emphasizes the importance of understanding the relationships between pieces of information and ideas, rather than just the information and ideas themselves. Thus, the synaptic web is not just about organizing information, but also about understanding and utilizing the relationships between that information to create a more holistic and meaningful understanding of the world.

The work on the synaptic web is focused on creating a more interconnected and holistic way of organizing and understanding information on the web. The idea is to move away from traditional hierarchical structures, such as search engine results, and instead focus on the connections and relationships between pieces of information. Key players in this field include companies like Google, with their knowledge graph, and other search engines like Bing, Yahoo, and DuckDuckGo, who have been working on ways to create a more connected web by making use of semantic web technologies, such as RDF (Resource Description Framework) and OWL (Web Ontology Language).

Semir Zeki, a professor of Neurobiology at University College London, has been working on the relationship between brain science, art, neuroscience, and the web and how the synaptic web can be related to the human brain and the way humans perceive and process information. Additionally, researchers and experts in semantics, artificial intelligence, data science, and machine learning have been working on developing methods for understanding the relationships between different pieces of information. This includes the use of natural language processing, graph databases, and network analysis to create more meaningful connections between pieces of information.

The director and founder of the World-Wide Web Consortium, Tim Berners-Lee—also known as the “father of the World Wide Web”—has been a prominent advocate for the development of the synaptic web. He has been working on developing a more connected and semantic web through the use of technologies such as RDF and OWL. James Hendler, Lada Adamic, Tom Mitchell, Deborah McGuinness, Annette Hautli-Janisz, Katherine Parr, and others have been working on understanding relationships between pieces of online information on the web and how they can be used to create a more connected web and holistic understanding of information.

How might the connections between information create new Web experiences? Online tools could enable us to see the connections between specific pieces of information, a shared context, and the patterns and pathways that form in the aggregate.

Some see the exploding variety, speed, and flexibility of electronic connections to be the root of developing collective intelligence. We are hyper-connected! With electronic connections between and among people, datasets, machines, the physical world, and online content we’ve seen, within the metaverse, with gestures, meanings, messages, and even thoughts. As we optimize for connectedness in our hyperspace world, we’ll get collectively smarter along the way.

As discussed in Chapter 6, Metcalfe’s law states that the value of a network is proportional to the square of the number of its nodes or users. In a synaptic web of content, its contextual value increases exponentially with the connections between content on webpages. In applying Metcalfe’s Law, however, to estimate the network value of context, we must account for each piece of information having limited connections. One way is to calculate the network value of a densely connected group of content within the synoptic web. The assumption is with such a group, all or most of the pieces of content are directly related.

Mathematically, Metcalfe’s law can be represented as $V = n^2$, where V is the value of the network and n is the number of pieces of content. As the number of connections between webpage content increases, the value of the network also increases exponentially. For example, if there are 100 pieces of content within a densely connected group, the contextual value of the network is 10,000 (100^2). If there are 200 pieces of content within a densely connected group, the contextual value of the network is 40,000 (200^2). As we can see, the contextual value of the network increases three times as fast as the pieces of content.⁶

The context resulting from connections between content can increase sensemaking, collective intelligence, search effectiveness, usability and navigation of the web, productivity, and user satisfaction. It makes it easier to find relevant information and make connections between different pieces of information, leading to a deeper understanding of topics and better decision-making. It also improves the overall usability of the web, making it easier for users to find what they need and navigate the information, ultimately saving time and effort and increasing productivity.

Hence, the synaptic web that connects information would provide tremendous value at the root of the next evolution of the Web.

Today’s Web is an unhealthy melange of the Web 1.0 document delivery platform and Web 2.0 as a platform for social communication. The sum is a sparse and noisy network; shown on the left in Figure 10.2. The low number of connections meant low intelligence in the network. With so many less-than-useful nodes compared to high-quality information nodes, the Web had a very low signal-to-noise ratio and therefore was less useful than it could otherwise be. With Today’s Web, we don’t have a legend or a way to know what is high quality vs low quality vs promotional vs misinformation, before we visit the page.

The synaptic web can move us towards something much more profound: a dynamic web of adaptive “organic” and implicit connections, where real-time information flows give structure and meaning to previously unconnected sets of data. Imagine the Web as a sea of conversations streaming through connections, and patterns having meaning.

As shown on the right in Figure 10.2, a synaptic web enables filtering for high-quality information. Wouldn’t it be helpful to filter out clickbait and partisan hacks? We can categorize the nodes into types. You choose the nodes to display. Or you can see everything connected to the focus of your attention. A synaptic web provides context.

In a synaptic web mimicking the human mind, everything connects. Every piece of information, claim, and content snippet explicitly connects to everything that relates. Each connection between pieces of content is a relationship (e.g., supporting, contradicting, exemplifying, citing), or perhaps several relationships. You can imagine adding colors to the

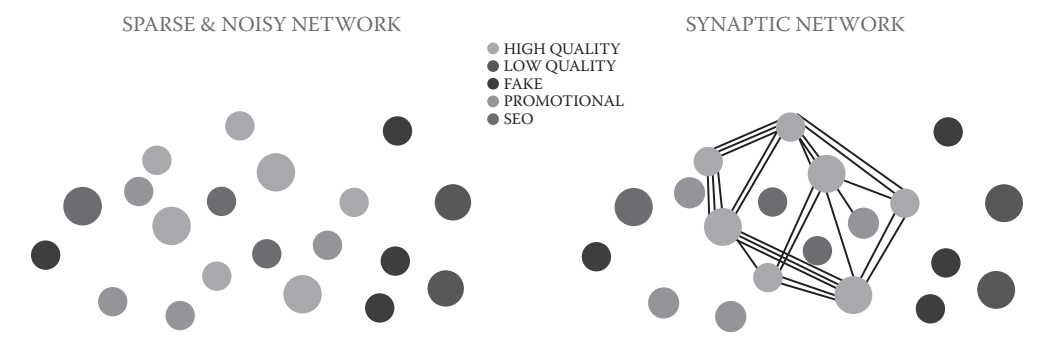


Figure 10.2 Moving from sparse and noisy to a deeply connected synaptic network.

connections to distinguish the relationships and even changing the line's thickness to reflect the strength of the connection.

Bamm! The bridge was born. Bridges connecting pieces of online content emerged as a building block of online knowledge that mimics the brain's synaptic connections. A bridge connects two pieces of information with a relationship. The bridge itself is a basic expression of knowledge. But they self-assemble into an entire web or graph of bridges that could connect all the world's information.

What if bridges are everything?

Our first thought was citing bridges from news articles to studies. We thought of supporting and contradicting bridges from news to relevant studies, articles, blogs, and even tweets. We imagined a vast network of bridges to Wikipedia pages. And a vast ecology of bridges just among tweets. The possibility of connecting the world's information was novel and exciting.

It became apparent that, in a synaptic web, filtering was more important than searching. Filtering is a next-generation tool that helps navigate content discovery. While a search narrows the infinite document Web to a digestible set of pages, filtering is about narrowing the torrent of streams, nodes, and networks into something that matches your immediate criteria. It's about defining and constantly refining your worldview so that information can find you.

Filtering is not just a synaptic web opportunity, it is a synaptic process. Smart filters can help match the content and conversations with your current interests, future intents, and tolerances. The filtering makes a synaptic web much easier to use, less overwhelming, and more focused.

THE BRIDGE PATENT

In 2016, members of Bridgit DAO developed a first-of-its-kind animated (i.e., augmented reality) children's book, *Pacha's Pajamas: A Story Written By Nature* (Figure 10.3).⁷ The AR book had an app called Pacha Alive. Hovering Pacha Alive over one of 80 line drawings brought the characters to life. 3D characters appeared to leap off the page!

After New York publisher Morgan James published the AR book, we spent several months thinking about the future of books and augmented reality books. The thesis was, in the future, the book becomes a guided and curated pathway through the relevant universe of multimedia, interactions, and experiences. Like Vannevar Bush's associative trials! A universe of possibilities emerges as overlays displaying videos, information, interactions, and virtual reality experiences associated with text snippets and images in the book.

Access to the overlay would be through a mobile phone, virtual and augmented reality interfaces. We developed concepts around personalization, creation of content, and dealing with multiple triggers in the field. But we were most excited about the social AR book concept. It included connecting book knowledge on an idea-by-idea basis, as well as book clubs meeting on book pages and asynchronous conversations in the digital marginalia.

In early 2017, when the Pacha project got some funding, we filed five provisional patents. The last patent we worked on was the social augmented reality book. When we spoke about making connections between pieces of content in different books, we realized passages and images in books could also connect to relevant info on the Web. We wanted to add the Web to the patent.

But connecting to the Web required another patent. In the following days, we were excited. Not only about connecting book-to-web but also web-to-book, and even more so about connecting web-to-web. To accommodate the latter case, a provisional patent was created that became the Bridge patent. In June 2017, building upon the notion of a synaptic web of bridges, we provisionally filed the Bridge patent: A system for retrieving and creating contextual links between objects of the user interface.⁸



Figure 10.3 Pacha of Pacha's Pajamas.

We now see this as a defensive patent that reduces the likelihood that a Google or Facebook could prevent us from using our invention. Our intention is to move Bridgit patents into the public realm, but in a way that prevents tech giants from monopolizing or harming the movement. (In 2021, Holochain announced it was using its patent in this way.⁹ An enthusiastic response from the market shot the token value up, tripling its value in a couple of months.)

We were easing into a deep well of possibility.

CONNECTEDNESS AS A MEASURE OF HEALTH

“Knowledge is power” is an aphorism most commonly attributed to 16th-century philosopher Francis Bacon, but earlier forms date as far back as the 10th century in the Islamic script *Nahj al-Balagha*.¹⁰ Knowledge is the primary step of the learning process and enables the possibility of understanding and wisdom. Without knowledge and the required access to

information, we cannot understand “the Why,” and therefore cannot act wisely. From time immemorial, humanity has always leveraged imbalances of knowledge for advantage—from weapons of war to the trading floor to sales conversations.

Information is only as useful as the connections between it. The relevant connections are context which make information actionable. Without context, information is not actionable.

Here’s a thought experiment. Imagine long ago, two rival factions in neighboring kingdoms aim to accumulate knowledge and power. The first faction understands the notions of knowledge (i.e., knowledge opens doors to the future) and power (i.e., power breaks barriers) separately. Hold these two ideas in your mind, but separately. Do not allow knowledge and power to connect. The second faction understands the separate concepts, as well as how they connect.

- Knowledge creates opportunities for power
- Applied knowledge leads to power
- Power leads to more accessible knowledge and power

Understanding the relationship between knowledge and power gives the second faction a lethal advantage over the first faction. It helps the second faction better understand “the Why” of the dynamics of circumstances involving knowledge and power. This understanding helps them reliably choose the wisest actions.

It’s similar for two, ten, or ten thousand things. And exponentially more so as network complexity increases. Simply put, those who understand how nodes relate in complex networks are best prepared to navigate and make sense of the network.

In 2018, we did an ad hoc experiment to see to what extent links connect the Web. The intention was to start with a breaking news article related to an ongoing story and follow the path to up to 100 external pages, all the while tracking the number and type of external links. We randomly started with the NBC News article “Bill Cosby sentenced to 3 to 10 years in prison for Andrea Constand sexual assault.”¹¹

There were zero outbound links about the Cosby story from the starting page. One internal link, however, went to a previous NBC News Cosby story that itself had zero external Cosby links and four internal Cosby links. We followed these links. Two pages had Cosby links back to the predecessor page and two were Cosby dead-ends. This NBC Cosby story-verse had only 7 pages with zero external links and 7 internal Cosby links as well as 8 ads, up to 2 recommended stories per page, and about 50 generic internal links per page.

A textbook example of an information silo, which is web typical. We’ve read articles on important environmental reports that don’t link to the report.

We see a future where connectedness is a measure of the Web’s health. Imagine web spiders traversing the Web starting with a page or a site, and tracking the different links on each page: navigation links, other internal links to pages on the site, advertising links, and non-advertising outbound links. We’d love to know the average number of (non-advertising) external links for different websites and site categories.

THE IMPACT OF BREAKTHROUGH KNOWLEDGE

In the Information Age, it’s all about mindset; the person to decipher the information the quickest is the winner.

—Lamar Wilson

As Bitcoin OG Lamar Wilson notes, winning highly correlates with breakthroughs. On Today's Web, we happen upon breakthroughs but what if we could plan them?

Breakthroughs require deviating from the informational fields that keep us stuck, including confirmation bias, filter bubbles, unfounded conspiracy theories, and top-down narratives. The top recommendations of search reflect and shape the dominant and sometimes competing narratives (e.g., Wikipedia, Snopes). While there often is truth in competing dominant narratives, they also have inherent blind spots. In such cases, an integrative narrative that reconciles contradictory aspects of the competing narratives and cherry picks from less popular narratives can be the fullest representation of the reality of the situation.

Innovation is Inclusion ... All the great art in the world, great scientific discoveries, great business is when we included the outlier. Whether that is an outlier idea, an outlier person ... That point of inclusion when we decide to say 'alright; it doesn't have to look like this other thing.' That's where true innovation comes from.

—Kelly Leonard, Improvisational Comedian, Second City Works

Kelly's commentary on outliers reminds us that breakthroughs are on the edge of knowledge.

Imagine this. You're learning something, but things aren't adding up. You get a piece of information that changes everything. That morsel of knowledge connects the dots and everything shifts. You see the world differently, clearly understanding how things fit together. Your world model has integrity again. You are ready to put your new understanding into action. You move forward definitively, knowing your choices have a solid foundation of knowledge and understanding.

Breakthrough knowledge is transformative and attainable. We can cultivate it with effective systems. Knowledge is immanent. Triggers for learning are all around us. The right idea, intuition, or time makes the difference. It's not usually the initial idea, but the directionality that leads to breakthroughs. Staying on the path, intuition or experiments at crossroads, and noticing cairns along the path increase the likelihood of breakthrough. Breakthrough knowledge can be the difference between success and failure, between hundreds and hundreds of millions of euros, between relapse and recovery.

Yet, we rarely seek breakthrough knowledge. Most breakthrough knowledge comes via brute force (e.g., relentless searching) and/or serendipity (e.g., scrolling feeds). When we experience breakthrough knowledge, others in the same consciousness and/or information field often have parallel breakthrough experiences. We, however, seem to have reached a "research flatline" of sorts, where we're all doing the same searches. Google, of course, encourages this via search suggestions because it supports their bottom line. Ultimately, we comb through the same shallow set of easily retrievable information. We're not able to collaborate informally on online research. Or build on one another's research.

Think about it. Something happens; breaking news involving Elon Musk or Trump or Will Smith slapping someone. In the next 5 hours, let's say ten million people at least skim the article. Of these, one-tenth or one million find a specific claim to be intriguing or contentious. Of these, one-tenth or 100 thousand do variations of seven searches about the claim. That's why Google can predict what you want to search for. Of these, one-tenth or 10 thousand find something they consider noteworthy. Of these, one-tenth or one thousand decide to save it into one of their personal data silos or to share it on social media (Figure 10.4).

Most people on social media engage without reading. Even if they read it, the information they share is separate from the claim that started it all. They post links to the page but not directly to the claim. Meanwhile, everyone who goes to the article sees exactly the same thing—just the article, despite thousands of people having found relevant information.

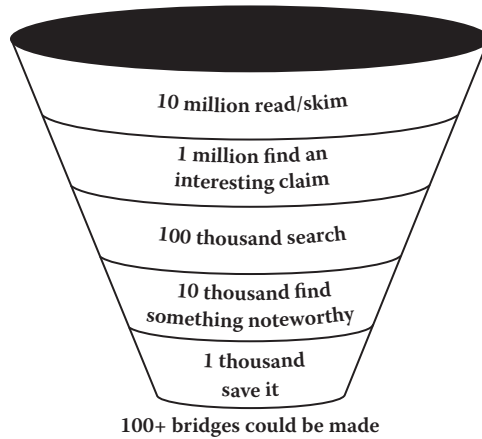


Figure 10.4 Hypothetical engagement funnel for a popular article.

With a synaptic web, people can see the research of their predecessors. Of the 1,000 content snippets people saved or shared, let's say 10% are unique, relevant, and interesting. That's fodder for 100 bridges. People with "synaptic web" access will see the bridges attached to the contentious claim. If they like, they can search for new information and build bridges upon what's already there.

Do you want to be a world-class innovator? Then you will need to dig and search broader and deeper to discover innovations others aren't willing to put in the effort to discover.

—Phil McKinney¹²

As McKinney highlights, the path to becoming a world-class innovator lies in the commitment to relentless and thorough exploration, unveiling innovations that remain undiscovered by those who shy away from this arduous task.

Author Michael Simmons suggests three ideas for increasing the likelihood of an info-utopia with breakthrough knowledge rather than an info-apocalypse:¹³

- Ask yourself simple questions that help filter breakthrough knowledge from incremental knowledge.
- Use knowledge formats that have a higher signal-to-noise ratio.
- Learn a skill that will help you systematically improve your ability to find breakthrough knowledge.

A synaptic web creates the possibility of a new knowledge format with a significantly higher signal-to-noise ratio as well as providing a platform for improving your ability to find breakthrough knowledge.

THE LONG TAIL

A February 2003 essay by Clay Shirky, "Power Laws, Web Logs and Inequality"¹⁴ explored the prevalence of inbound links in blogs. He found that relatively few blog posts have many

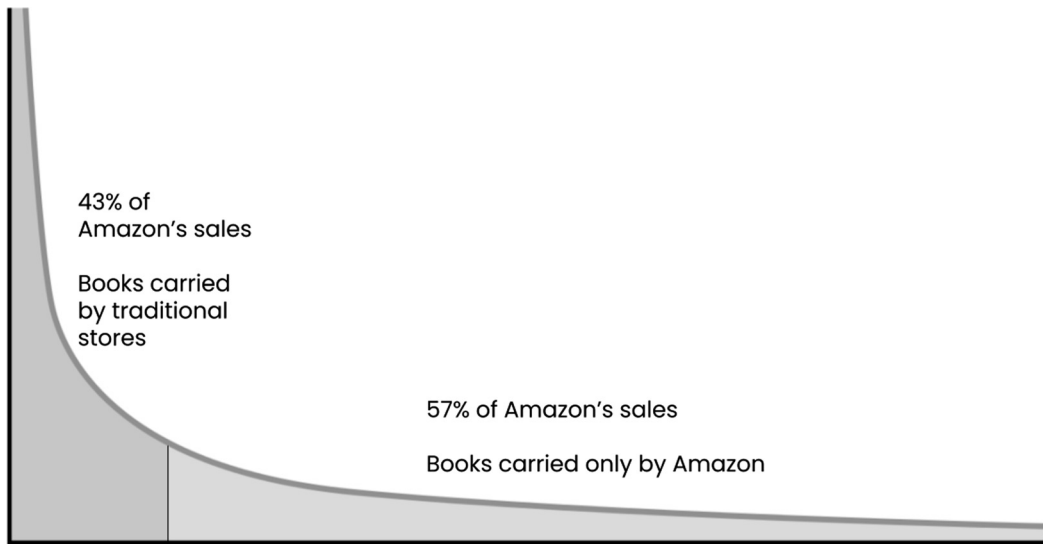


Figure 10.5 Most of Amazon's book sales come from "the long tail".

inbound links. He described "The long tail" as the many millions of blog posts with few inbound links. Because Google uses inbound links as a key metric in search ranking, the long tail is not in search indices.

Chris Anderson of TED built on Shirky's long tail. He suggested long tail opportunities are important in the aggregate. Anderson explored the collective market share of products in low demand or with low sales. He found they could collectively rival or exceed a few current bestsellers if the distribution channel is large enough. Anderson cites research showing that 57% of Amazon's sales came from obscure books that are not available in brick-and-mortar stores as shown in Figure 10.5.¹⁵

Applying the long tail to knowledge suggests the importance of integrating accurate long tail information to develop a full understanding. This is consistent with the Trivium, the ancient tradition of grammar, logic, and rhetoric in classical education. As explained by Plato's dialogues, considering relevant and accurate information is a prerequisite for consistently achieving an accurate understanding.

The long tail extends to search. Most searchers only see the top results. The long tail suggests highly relevant information likely exists beyond the first page for many informational queries. (But if Google is a Potemkin village as discussed in Chapter 5, a Shirkyian long tail for search may not even exist.)

The long tail concept also extends to advertising. Many advertising opportunities involve pockets of information that are unrelated to page metadata, yet could be highly effective for targeting customers by interest.

Consider, for example, an engaging paragraph within a post on a relatively unknown travel blog. The post is about a trip to France and has a delightful passage about a coffee experience in a café outside the Louvre museum in Paris. The metadata for the site is about travel and it has few incoming links. Hence, search would not recommend the page for the search term "French coffee." Display advertising doesn't know about the reader's potential interest in French coffee because coffee is not in the metadata. But a context engine (working through the synaptic web) that accesses the long tail of content metadata could effortlessly

present ads directly related to any content on the page. If the reader's attention went to the passage about the café, the context engine would present content and ads about French coffee. This would provide more accurate targeting, more impressions per page, and per browsing session in long-tail search and advertising.

In the next section, we explore how the synaptic web is like a map for breakthrough knowledge.

A MAP FOR THE WEB

Albeit underappreciated, maps are transformative. With maps, we find our way, choose our next adventure, and discover new things. Maps help us find a location, how to get there, and what's nearby.

Here's a thought experiment. Imagine having to cross a country and or even an unfamiliar town without a map app. When was the last time you used a paper map or a Thomas guide? Imagine using paper maps to cross a continent. Now imagine a world without maps.

Hundreds of years ago, the average person rarely traveled.

Journeying was a common subject in ancient literature (e.g., Homer's epic poem *The Odyssey*). Most travel was for warfare, diplomacy, general state building, or trade. Social motivations for travel included religious pilgrimages, festivals such as the Olympics, and health-related reasons. But the average person usually stayed close to home.

Around 600 B.C., ancient Babylonians scratched the earliest known map of the world onto a clay tablet. The star-shaped map is only five-by-three inches. It shows the world as a flat disc surrounded by an ocean, or "bitter river." Babylon and the Euphrates River are a pair of rectangles in the center. The neighboring cities of Assyria and Susa are small, circular blobs.

Outside of the disc sit several triangular wedges, depicting far-off islands with whimsical names like "beyond the flight of birds" and "a place where the sun cannot be seen." The accompanying cuneiform text describes these unknown lands as being populated by mythological beasts. The map shows both real geographical features and elements of Babylonian cosmology.

Maps help us see the high-level and ground-level views, and to switch between them. They help us know a location, how to get there, and what's nearby. Without our map app, we feel uneasy even when familiar with the area. Part of us wants confirmation we are heading the optimal way. Maps can be invaluable.

Maps are one of the most underestimated tools that we have; this may be especially potent for those of you who are horrible with directions. While mapping apps have liberated directionally challenged people, the jump from physical to virtual maps is less drastic than from "no map" to map.

There's no map of the Internet. That's one reason why there's little context (and trust) in web content.

A Map for the Internet. How Would that Work?

Imagine towns being ideas and roads describing the relationship between the concepts. Instead of a fixed geography based on precise measurements, ideas and connections on a "knowledge graph" are dynamic. They can adjust their position based on the active view. Connections remain intact regardless of the positioning of ideas, which may adjust for convenience, aesthetics, and/or sensemaking.

Knowledge graphs are also much more scalable than search engines. When you have a knowledge graph, it is computationally simple to create, build, and retrieve context for the active node or, in this case, idea, since that is a simple query of its direct connections.

Contrast this with search engines. Google has enormous server farms to index one small sliver of the Internet day after day and serve lightning-fast queries of the giant databases holding the index. And yet even this sliver of the Internet (i.e., Google's dominion) is growing exponentially, while the length of a day is steadfast, which could ultimately prove a limiting factor for an index-based search.

What happens when Google can't index fast enough? Google's index broke at least five times in 2019. And in early August 2022, when there were "tons of complaints about indexing issues, pages dropping out of the index, and super poor quality and dated search results being shown," which Google said was due to an electrical incident in their Council Bluffs, Iowa data center. (Perhaps a good reason for decentralization.) Recall the earlier discussion about Google being a Potemkin village. Could the limits to index-based search be the reason Google's search results are orders of magnitude smaller than purported?

In sum, a knowledge graph based on bridges may be the Holy Grail. But Web 2.0 advocates say it'll take too much time and effort to make a graph of the Web. We don't think it has to.

Furthermore, we think that people actually want to solve crimes and injustices, uncover the truth, and build knowledge around their purpose and passions. The enthusiasm of creators jumping into citizen researcher, reporter, and sleuth mode on YouTube to solve mysterious tragedies is palpable. While rampant speculation and hasty conclusions may be present, the fervor and passion for connecting the dots suggest an appetite for better information. We think many of the YouTubers and Tweepers investigating unsolved murders would be excited to collaborate on a digital crime board containing everyone's finds.

When people can earn value for building knowledge on topics they already research, we think maps will take off in subareas. With proper incentives, many would take part. This brings more value to the ecosystem, enabling it to grow.

And once one subarea happens, seven more are on the way because everything connects. Subject creep. What subareas will people build out first? Perhaps, the ideas whose time has come. And topics relating to major controversies, pressing issues, or confounding crimes and mysteries.

Given humanity's existential threats, we need to develop the capacity to build and scale collective intelligence. If we are collectively intelligent, we can alter our course towards balancing the biosphere and thriving in a just and life-affirming world. Had we the necessary tools, might we reorganize our lives to be less taxing on the planet and provide the essentials for all? Were we collectively intelligent, we'd build a world focused less on money and more on heart.

Transitioning to the New Earth will require a map of the online information ecology. Perhaps regenerative practitioners, communities, and even a "regen" or ReFi meta-community will build a map for regeneration and the New Earth. Such a map could surely grow into a collective map for matters vital to humanity's future.

Today's Web is not great for research. Since commercial sites avoid non-commercial outbound links, researchers have to rely upon search rather than the surf-and-search approach that prevailed before advertising became the dominant monetization strategy and outbound links became less plentiful on the web. Yet the #1 search engine Google indexes only a small fraction of the surface web, which limits researchers in what is ultimately findable with search.

When researchers find related content, they don't have tools for connecting it to what they have already collected. The annotation tool Diigo enables one to create lists of content snippets, but that is much less granular and informational than one-to-one connections between pieces of information with relationships. Existing tools isolate found information in silos in the cloud or on users' machines, leading to repeated tasks, missed opportunities, and diminished productivity. Silos inhibit collaboration as well as learning from and building upon one another's work.

A synaptic web turns researchers into innovators by making the entire web into a collaborative research experience. As an advanced bookmarking space, a synaptic web speeds up research by enabling collaborative mapping, helping filter false information, and revealing context on any idea. The value proposition is collaboration without coordination, faster time to insight and innovation, accelerated learning, and collective memory and intelligence.

This activity generates bridges, which self-assemble into a knowledge map that provides 360° context for any idea—starting online, and in the future, in virtual worlds and as a digital overlay in the physical world. We can use AI to analyze this context to provide additional insight, for example, with anonymized data overlays for intent and sentiment.

Discover the Power of Knowledge Graphs

Have you come across an in-depth inquiry requiring a lot of research? Is there a complex problem you want to solve? A developing field or pressing issue that requires frequent monitoring? If so, you need a map!

A knowledge graph or map will enable you to navigate through the vast amount of information available and connect the dots between different sources. It will provide you with a comprehensive understanding of the subject matter, making it easier for you to identify key players, trends, and patterns. With a graph, you'll be able to see the big picture, as well as the finer details, giving you a valuable tool for decision-making, research, and problem-solving. It acts like a mental map helping you navigate through the subject or inquiry. Don't let overwhelming amounts of information hold you back, create a map today using one of the available online tools¹⁶ and take the first step towards mastering your research!

The Trivium tells us, to make wise decisions, we need a correct understanding and knowledge base. Our collective knowledge base—the Internet—sucks. It's low signal-to-noise and mostly walled gardens with few external links. The opposite of what we need for collective intelligence.

With a universal content graph, we can make the entire web into a collaborative research platform and a social knowledge network. The premise is: if we drastically improve our knowledge base, it could evoke a new collective response that is life-affirming, and that shifts our collective relationship with the biosphere, each other, and ourselves.

Ecosystem participants build online knowledge maps of ideas and the relationships between them, thus providing a shared context for collaboration and communication across mindsets and geographies. The knowledge maps aggregate—connecting where they share the same node—into a universal knowledge graph that provides a 360° context for ideas.

The key innovation is the bridge. Bridges enable participants to connect the Internet's walled gardens and information silos into a broader knowledge graph that provides access to

deep layers of context for every idea on the web, and later for other realities. It also provides a basis for interacting with others in new ways, like connecting with friends and other participants who are on the same webpage and/or having asynchronous conversations, or responding to polls related to specific content snippets wherever you are on the web.

Bridges foster a transition towards evidence-based and values-aware discussions, as well as decentralized collaboration. Thus, a synaptic web catalyzes a shift from an Attention Economy—which exploits user data, captures attention, and manipulates behavior—towards a context (or knowledge) economy that enables more nuanced sense-making, enables more fruitful interactions, and aims to support attention.

A synaptic web connects the world's information into a universal knowledge map, creating a shared context for collaboration and communication, and an aggregation method for collective intelligence.

Wherever we are online, we can have access to deep layers of context for any idea, thus providing a shared context for collaboration towards a life-affirming, democratic future. Learning together will speed up. We will largely neutralize false information. We can work together to build knowledge around vital matters.

Imagine hundreds of millions of people all across the world, speaking dozens of languages, earning a living wage, or significantly supplementing their income doing what they would already do. In this future, democracy is strong and growing. We are more informed and active on a local level, and people are finding unexpected collaborators and team members; building social enterprises, organizations, and collaborations; and even falling in love—above the webpage.

THE EMERGENCE OF THE OVERWEB

In mid-2020, after the George Floyd riots, we created the Black Browser project to create a safe digital space for black people over the Web under the moniker #BlackIdeasMatter. We got enthusiastic responses from over 100 people who attended four virtual salons. In the Fall, we joined with app development firm 4th Ave to build the Presence browser overlay. The aim was to build a full-featured and safe instantiation of the Metaweb as a browser overlay. It would be accessible as a browser extension, and later as an open-source browser. The our firms came together and formed Presence Labs.

On Jan 21–22, 2021, over 100 people from 13 countries gathered at the Overweb Challenge, a virtual event focused on envisioning a web that works for humans. The challenge presented the concept of the Overweb as a trust layer over the Web that creates a healthy relationship and fair value exchange between society and the Internet.

The challenge included four panels of experts exploring the state of various aspects of the Web, including combating false claims and sensemaking; two forward-thinking panels about designing the Web and digital communities; a 20-hour working period; and a closing session with 7 “elevator pitches.” The highlight of the latter was Excellerent's proposed smart tag integration that would activate the audio chat platform Clubhouse on any webpage

The event had 190 registrants and featured 35 speakers. Sponsors were the Forbes Funds and the Web4 Foundation. Bridgit and Noetic Nomads hosted. Partners included the EU's Next Generation Internet program, Edgeryders, UMI, Excellerent Solutions, WeTech Belgium, and Skööl.

The event surfaced many issues about the state of the Web that have already borne fruit.

On April 30, 2021, California Software Professional Association (CSPA) hosted a virtual meetup called “The Future of the Web—Reinstalling the Big Missing Feature of the Web”

with over 100 registrants.¹⁷ Niki Gastinel of CSPA introduced the event. Bridgit founder Daveed Benjamin presented the unknown history of web annotation, the case for the Overweb, and the Overweb pattern. RSA fellow Joshua Armah spoke about open-source blockchain-based tools. Dan Whaley, CEO of the pioneering web annotation app Hypothesis, spoke about the history of web annotation.

The unique aspect of this event was the jury of 13 experts. The jury included Dan Whaley, Mei Ling Fung (co-founder with Vint Cerf of the People-Centered Internet), and Dan Mapes (CEO of Verses.io), among others. After the presentation, each jury member gave several minutes of commentary, including critique, use cases, and unanswered questions. The feedback was quite positive, ranging from Whaley's cautious optimism to Mapes' enthusiastic support.¹⁸

With each step down the rabbit hole, the conceptual cairns guiding our journey have increased in focus. Now, as we approach the threshold of the Metaweb, the most crucial cairn of all comes into view: the bridge. This bridge unlocks the true potential of the Metaweb, allowing us to traverse the vast expanse of interconnected information with ease. As we step onto the bridge, we leave behind the familiar terrain of the hyperlink and set out on a journey to explore the foundational principles of the Metaweb. The path ahead is uncharted and the possibilities are endless. Shall we cross the bridge and discover the true power of the Metaweb together? Let's go!

NOTES

- 1 <https://permanent.link/to/the-metaweb/trivium-video>
- 2 <https://permanent.link/to/the-metaweb/musk-quote-video>
- 3 <https://permanent.link/to/the-metaweb/neural-science>
- 4 <https://permanent.link/to/the-metaweb/humans-grow-brain-cells>
- 5 <https://permanent.link/to/the-metaweb/huamn-brain-connections>
- 6 When the pieces of content grow from 100 to 200 or 100% ($[200-100]/100$), the value of the network grows from 10,000 to 40,000 or 300% ($[40,000-10,000]/10,000$).
- 7 For more info or to see Pacha dancing in AR, please visit pachaverse.io.
- 8 <https://permanent.link/to/the-metaweb/bridge-patent>
- 9 <https://permanent.link/to/the-metaweb/holochain-patent>
- 10 <https://permanent.link/to/the-metaweb/nahjul-balagha>
- 11 <https://permanent.link/to/the-metaweb/bill-cosby-assault>
- 12 <https://permanent.link/to/the-metaweb/phil-mckinney-quote>
- 13 <https://permanent.link/to/the-metaweb/accelerated-intelligence>
- 14 <https://permanent.link/to/the-metaweb/shirky-power-laws>
- 15 <https://permanent.link/to/the-metaweb/long-tail>
- 16 Check out thebrain.com and roamresearch.com.
- 17 <https://permanent.link/to/the-metaweb/meetup-future-web>
- 18 <https://permanent.link/to/the-metaweb/future-web-video>

The Link and the Bridge

[The URL] is the most fundamental innovation of the Web, because it is the one specification that every Web program, client or server, anywhere uses when any link is followed.

—Tim Berners-Lee, founder of the World Wide Web

Once upon a time, the Web was a vast, interconnected network of hyperlinks, allowing users to travel from one page to another with a simple click. The URL, or uniform resource locator, was the key that unlocked the door to this new world of interconnected information. As Tim Berners-Lee so astutely observed, the URL is what makes the Web a web.

But as the Web evolved, so too did the challenges it faced. With the rise of the Attention Economy, commercial sites began avoiding external links, seeking to keep visitors on their pages for as long as possible. The result was a Web with less context and less connection between pages, a far cry from the interconnected network of its early days.

Enter the Metaweb, a new era of the Web that builds bridges between information on different pages, revealing vast new layers of context that eclipse what was previously possible. The bridge is to the Metaweb what the link was to the Web of old, offering participants a way to travel through ideas with a simple click. The bridge makes the Web a densely connected, explainable web of ideas. Thus, through bridging, the Metaweb will reconnect the Web, restoring and surpassing its former glory.

THE LANDSCAPE OF LINKING

The Web wouldn't be a web without hyperlinks. Connecting websites is at the center of the web's ideological and technological roots. Without links, the Internet would be a bunch of disconnected pages. Sites would be silos. Navigation between pages would depend on search. Websites themselves exist because navigation links enable one to move among its pages.

The link's downside is centralization. Only page authors can make a link. As page authors use fewer links, the Web becomes less connected. Within the Attention Economy, many commercial sites avoid outbound links so that they can keep serving you advertisements and extracting your data. Hence, the Web has much less context than its early days.

The bridge is to the Metaweb as the link is to Today's Web. The bridge connects information on one page to information on another page. Clicking a bridge takes you to the other side.

The bridge is the connective tissue of the Metaweb. Bridges connect pieces of web content with a relationship. They self-assemble into a universal content graph, which provides context.

Bridges democratize linking. Unlike links which are the exclusive right of the page owner, anyone can build a bridge on any page. Bridgers can work together to build information

ecologies around their common interests. Bridgit won the 2019 Disruptive Technology Culture Driver award from the EU's flagship Next Generation Internet program.¹

This chapter compares the bridge to the link and discusses how an “Internet of Ideas” based on bridges compares to today’s “Internet of Pages.” It also makes a distinction regarding bridges in the Metaweb versus the Overweb, a specific instantiation of the Metaweb in which Bridgit DAO is building safe decentralized public space. Notably, to maintain integrity, the Overweb bridges go through a validation process before posting to the public ledger. Metaweb bridges do not undergo a validation process.

THE LIMITATIONS OF LINKS

Hyperlinks are like jumping into the abyss. You don’t know where you’ll land. Or whether there’s anything to land on. In fact, before links became the fundamental building block of the web, they were called “jump links” and considered by some to be quite radical. While they provided a very simple, elegant structure for the early web, twenty years down the road, they have locked us into intractable problems such as false news, an overabundance of digital flotsam and low-quality content, a lack of transparency, and powerful digital monopolies.

Aesthetics and technical aspects prevent or make it cumbersome to link from anything but short phrases and images. Linking from sentences or paragraphs, while possible, looks quite unappealing as a permanent fixture on the page and can be distracting.

Jump links open a page but can also jump to an anchor on the page. But they can’t link to a specific piece of content. (You can link directly to an image file but not within a webpage.) Some video platforms like YouTube allow links to a point in a video, but these are non-standard implementations and are present in video search results.

Since the Web began, the one-link limit for a piece of content has locked in the view that (non-promotional) links are for connecting content to its source. Links became more or less a digital citation or footnote. Links to source material can be useful. They, for example, enable web surfing. The focus, however, on this single use case and the infeasibility of multiple links from a single idea inhibits our thinking. We don’t think about (non-promotional) linking to things other than source material or consider the desirability of multiple links.

The page author’s capacity for adding supra-information to their pages also limits the use of links. Often, the page author does not have the knowledge, time, energy, or inclination to link to valuable supplementary information that could enhance understanding and learning.

The simplicity and elegance of jump links led to their adoption. The page author doesn’t need to ask permission to link to the page. And pages don’t know that other pages are linking to them. The abyss doesn’t know it’s being jumped into.

Links are unidirectional. Consider a link from Page A to Page B; Page A knows it links to Page B, yet Page B is oblivious to Page A’s existence. This enables a web author to link to pages with nothing besides the page’s address. It’s ok if content changes, as long as the address remains.

This created a golden opportunity to track incoming links to a page and user click-throughs. This information underlies the online search, advertising, and analytics industries. Just imagine if links and other UI objects tracked such activity themselves—we’d have a different web ecosystem on many levels. For one, Google would not exist.

Today’s link is not only foundational for the Internet of Pages but also Google’s dominance. Before Google, search engines looked for relevancy in the dataset regarding the search

terms. Google innovated with an algorithm that prioritized sites within the relevant set based on the number of incoming links to the page. They used a bot called a “spider” to gather metadata, traverse links, and count the number of inbound links to each page. Google’s results were higher quality and faster than the then-dominant Altavista search engine, which was rendered to oblivion by the incoming link-aware Google juggernaut.

THE POWER OF BRIDGES

Bridges democratize linking such that anyone can create deep, precise links between content on webpages. In Today’s Web, only the page author can create a link on a webpage. In the Metaweb, anyone with access to a browser overlay can create bridges. Whenever they find something interesting on the web, they can create a bridge. Subsequently, their bridge is available on both pages of the bridge. Visitors with Metaweb access see any relevant bridges and can add their own.

Table 11.1 compares the link and the bridge.

Bridges enable Internet researchers, citizen fact-checkers, and creators to organize, share, attract eyeballs, and monetize the deep search they are already doing. They do this by building bridges, which are conceptual links between an idea (i.e., content snippet) on one page to an idea on another page with a relationship.² Bridges are bi-directional. Consider a bridge between Content A and Content B; Content A bridges to Content B and Content B bridges back to Content A. Often, the bridge is from claim to evidence that either supports or contradicts the claim. Ideas can be a paragraph, phrase, part of an image, segment of a video, audio, etc. The relationships are supporting, contradicting, citing, etc. ...

Additionally, bridges can be considered stigmergic, meaning they build upon the work of others and create a collective intelligence without the need for direct communication or coordination.

Imagine a contradictory bridge from a sentence in a news article to a video segment. The person in the video segment says something that contradicts what’s written in the sentence. When reading the article, the bridge enables news consumers to access the video segment that contradicts the sentence’s claim. And vice versa; when watching the video, the bridge enables video watchers to access the sentence that the video segment contradicts. In either case, the viewer has access to contradictory information they would not otherwise see. Contradictory bridges could give someone pause before sharing content without reading, and perhaps inspire them to do further exploration.

Table 11.1 Compares the link and the bridges

	<i>Link</i>	<i>Bridges</i>
Distribution	Centralized	Decentralized
Who can create	Page author	Anyone
Directionality	Uni-directional	Bi-directional
# Links from an idea	1	Unlimited
Links to	Url or anchor	Specific ideas
Relationships	NA	Various (e.g., opposing)
Metadata	NA	Description, tags, category
Tracks	NA	Upvotes and crossings



Figure 11.1 The three steps of making a bridge.

Bridge-making begins with selecting a piece of content (Figure 11.1). Next, you select content on another page. Then, you specify the relationship between the pieces of content, and provide metadata explaining why the bridge exists, its privacy status, category, topic, and tags.

Submitting the bridge sends it to the Bridge Registry. The submitter must pay the gas fee. The registry accepts bridges from any source and posts them on the Metaweb’s universal content graph. It ensures that metadata passes basic checks and verifies the bridge’s uniqueness before posting. The smart contract does not permit duplicate bridges. Bridges are freely available to applications through the Bridge Registry’s APIs.

The submitter can elect to send the bridge to the Overweb validation queue for a small fee to cover the validation service, plus a modest stake that is returned upon validation. Validation makes bridges more reliable and valuable, and clears them for use on the Overweb.

Before posting to the Overweb, a bridge goes through the validation process (Figure 11.2). As shown in the validation figure, when the bridger submits the bridge, it goes to the Overweb validation queue. The validation queue filters out bridges with logical issues, metadata irregularities, and errors. Approved validators confirm whether the specified relationship between Content A and Content B is correct. If consecutive validators confirm the relationship and there are no metadata irregularities or errors, the bridge enters the Bridge Registry as “Overweb validated.”

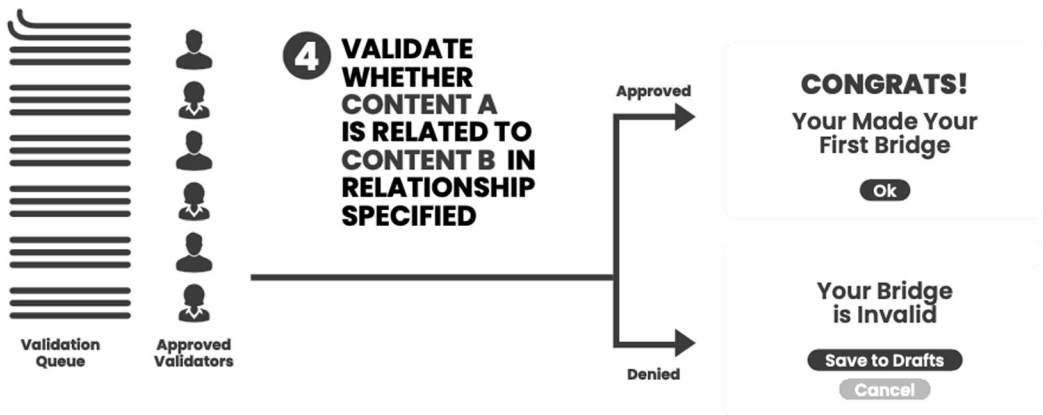


Figure 11.2 Validating a bridge.

Unlike hyperlinks which support one link per piece of text or image, any idea (such as text, parts of images, videos, and audio) can have as many bridges as needed to express its connectedness.

An idea's bridge content depends on the bridger community rather than the page author. In this sense, bridges are a quantum improvement over the hyperlink, enabling groups of bridgers to create connections between ideas without coordination.

The only true voyage of discovery, the only fountain of Eternal Youth, would be not to visit strange lands but to possess other eyes, to behold the universe through the eyes of another, of a hundred others, to behold the hundred universes that each of them beholds, that each of them is ...

—Marcel Proust

As Marcel Proust suggests, truly discovering something new and gaining a different perspective requires seeing the world through the eyes of others. In a similar vein, examining multiple bridges associated with a single idea can offer a diverse range of perspectives and ideas, allowing us to broaden our understanding and improve our collective cognitive abilities.

Multiple bridges associated with a single idea can present diverse sources, supporting ideas and evidence, similar ideas, differing perspectives, and more. Bridges bring back context, which was dying with the link. They also provide a foundation for neutrality, recognition of uncertainty, and holding multiple narratives; all are important for upleveling collective cognitive capabilities.

Bridge-It!

As soon as the page appeared, a wave of excitement surged forth through us. We knew Bridgit was it.

Bridgit was a name coined by two General Assembly students we contracted to do UX design for a synaptic web app. We were exploring how to connect content elements on the Web as synapses connect neurons in the human brain. In homage to this intention, we were calling the project Synapse. In our first meeting, we explained our nascent plan to bridge information on the Web like synapses bridge neurons. They inquired if a new name was possible. We said yes, but it was hard thinking of a better name than Synapse. 😊

A month later, in our final meeting, the students presented their work, including research, mockups, and a logo sporting the Golden Gate bridge. Their research showed no one in the marketplace was taking our approach. The mockups were ok. We weren't inspired by the logo either. But the name felt right. They explained Bridgit contracted the words "Bridge" and "It."

When we saw the URL bridgit.io was available, a pulse of energy went through us. We knew that the world had changed. This felt like becoming a parent. A surge of excitement, a wave of elation, a deep feeling of connection, and a sense of knowingness. The ecstatic joy of birthing something meaningful. We delighted as Bridgit—meaning Bridge-It—rewired the connections in our collective mind.

We knew, one day, the world would say "Bridge-it!" for connecting information online.

We have always had a fascination with names, especially when a person's name describes them. We understand why Africans are intentional about what they name children. Our old friend Tigiste's name means patience, and she is extraordinarily patient. Perhaps, like our tools, our names design us.

This curiosity led us to look up what Bridgit means. Google proclaimed, at the top of search results, the name to mean:

exalted one

Bridget, Bridgit, Briget, or Brigid or Brigitte is a Gaelic/Irish female name derived from the noun brigh, meaning “power, strength, vigor, virtue.”

The answer to a “People Also Ask” entry said Bridgit meant a strong and protective woman, and in mythology, a goddess of fire, wisdom, and poetry.

From baby name sites, we discovered Bridgit is a derivation of Brigitte, the Celtic deity of all things high, including intelligence and wisdom. A poet and a blacksmith, Brigitte was a perfect reference for this new venture that promised to connect the world’s information and enable humanity to develop collective intelligence and wisdom.

When we think of this, we feel shivers.

As shown in Figure 11.3, the article on why Putin invaded Ukraine has three major claims, one explicit and two implicit, in one paragraph: 1) Nazis control Ukraine, 2) Ukraine bullies some its people and committed genocide for the past 8 years, and 3) Ukraine is not neutral. Each claim brings up questions. For example, what details substantiate the bullying and genocide or why might Russia think Ukraine is not neutral? Supporting or contradictory information can be videos, other news articles, studies, photographs, and more. We can represent each piece of supplementary information as a bridge. Each time a person crosses a bridge and deems it useful, the bridger earns a small reward.

Another example. Imagine you are on a page about blockchain regulation around the world, and you’re interested in a paragraph about Malta, which lights up as you move toward it. The overlay displays a circular badge with the number of bridges associated with

Russian accusations and demands

Further information: *Russian opposition to Ukrainian NATO membership*

In the months preceding the invasion, Russian officials accused Ukraine of inciting tensions, **Russophobia**, and the repression of **Russian speakers in Ukraine**. They also made multiple security demands of Ukraine, NATO, and non-NATO states in the EU. These actions were described by commentators and Western officials as attempts to justify **How?** ^{[262][263]} On 9 December 2021 Putin said that “Russophobia is a first step towards **genocide**.” ^{[264][265]} Putin’s claims **What evidence is there that** **Neo-Nazis in Ukraine have political power in the government?** ^[266] and Russian claims of genocide have been widely rejected.

In a 21 February speech, ^[270] Putin questioned the legitimacy of the Ukrainian state, repeating an inaccurate claim that “Ukraine never had a tradition of genuine statehood”. ^[271] He incorrectly described the country as having been created by **Soviet Russia**. ^[272] To justify an invasion, Putin falsely accused Ukrainian society and government of being dominated by **neo-Nazism**, invoking the history of **collaboration in German-occupied Ukraine during World War II**, ^{[273][274]} and echoing an **antisemitic** conspiracy theory which casts Russian Christians, rather than Jews, as the true victims of **Nazi Germany**. ^{[275][268]} While Ukraine has a **far-right** fringe, including the neo-Nazi **Azov Battalion** and **Right Sector**, ^{[276][274]} analysts have described Putin’s rhetoric as greatly exaggerating the influence of far-right groups within Ukraine; there is no widespread support for the ideology in the government, military, or electorate. ^{[262][273]} The Poroshenko administration enforced the law **condemning the Soviet Union and the Nazis** in 2015. Ukrainian president Zelenskyy **What evidence is there that** **Russian Jews were the true victims of the Holocaust?** ^[277] three of his family members died in the Holocaust. ^[278] His grandfather served in the **Soviet army** fighting against the Nazis.

During the second build-up, Russia issued demands to the US and NATO, including a legally binding arrangement preventing Ukraine from ever joining NATO, and the removal of multinational forces stationed in NATO’s Eastern European member states. ^[279] Russia threatened an unspecified military response if NATO continued to follow an “aggressive line”. ^[280] These demands were widely interpreted as being non-viable; new NATO members in **Central and Eastern Europe** had joined the alliance because their populations



Figure 11.3 Layering information on a news article with bridges (courtesy of Wikipedia).

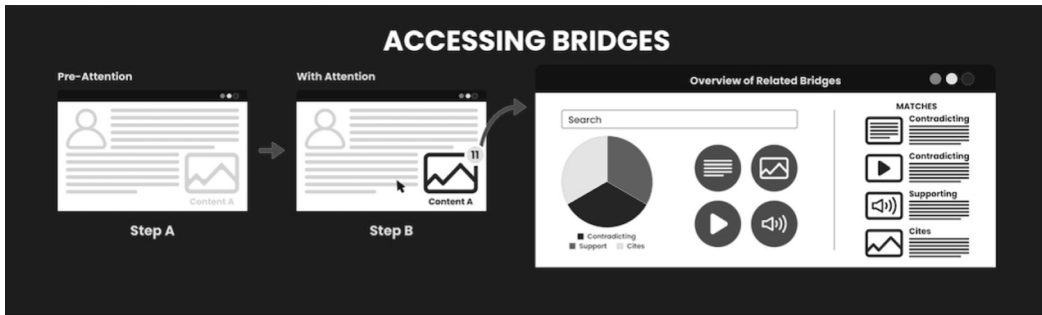


Figure 11.4 Accessing bridges.

the paragraph. Clicking the badge displays an overview of the bridges. Bridges go to videos and articles about the discussion and the vote in Parliament. Bridges also show that the Malta legislation is very similar to the earlier Gibraltar legislation, that Malta considered Gibraltar's legislation, as well as commentary about the differences.

The text of the legislation bridges to various articles that provide commentary, both supporting and opposing. Bridges also go to a list of startups domiciled in Malta and key legislators that were instrumental, including the President of Malta, Marie-Louise Coleiro Preca, and any commentaries that they have made in the text, podcasts, and videos. And so on ...

Consider an idea on a webpage you want to learn more about. It could be text or a part of an image, video or audio. As shown in Figure 11.4, by focusing your attention, you can access all the contextual information that relates to the idea. As you move your attention—your cursor on a laptop, your touch on a mobile, or line-of-sight in a virtual experience headset—towards the idea, if information related to the idea is available, the content snippet will highlight and a badge will display with the number of bridges and other smart tags that are associated with the idea.

Clicking the badge displays an aggregate overview of all the associated bridges and smart tags in a mini-dashboard for the content snippet. The dashboard contains filterable elements for the relationship (e.g., supporting, contradicting), media type (e.g., text, photo, video), and smart tag type (e.g., bridge, note, conversation). There is a list of the matching bridges and smart tags associated with the selected filters. Clicking bridges lead to the other side. Clicking other smart tags displays their relevant information and/or interactions.

Bridges Don't Lie

Many of you are wondering about how bad actors might game bridges. This is a common concern of people who understand incentives, social media, and game theory.

The concern is valid. Legions of hackers scrutinize applications and exploit any vulnerabilities. We've seen hacks on many Web 2.0 and Web3 projects, as well as counterfeit NFTs and fake NFT websites. Humans will exploit any aspects of an incentive system that are manipulable. Recall in Chapter 2, Lucy Sparrow, a PhD at the University of Melbourne, whose research focuses on ethics and morality in gaming, said if it's possible to abuse a system, people will figure it out. As mentioned earlier, the incentives of Steemit led developers to create voting bots, which is about as benign as it gets, though still annoying to people who just want to be recognized for good content.

Bridges, however, are not manipulable. Consider this: a contradictory bridge between Content A (Trump is like Hitler) and Content B (Trump is a good person who is doing his best under harsh adversarial circumstances). Neither Left nor Right would say these statements support each other. In fact, if someone believes one, they will adamantly disagree with the other. But they can agree that the bridge is contradictory.

THE UNIVERSAL CONTENT GRAPH

The idea is not new. Associative links between the components of human knowledge were first recorded in 1728 when Ephraim Chambers, a London globe-maker, published the *Cyclopaedia*,³ or, *An Universal Dictionary of Arts and Sciences*. The *Cyclopaedia* intended to link by association all the articles in an encyclopedia. The Preface explains this innovative system of cross-references:

Former lexicographers have not attempted anything like structure in their works; nor seem to have been aware that a dictionary was, in some measure, capable of the advantages of a continued discourse. Accordingly, we see nothing like a whole in what they have done ... This we endeavored to attain, by considering the several matters [i.e., topics] not only absolutely and independently, as to what they are in themselves; but also relatively, or as they respect each other. They are both treated as so many wholes, and so many parts of some greater whole; their connexion with which is pointed out by a reference ... A communication is opened between the several parts of the work; and the several articles are in some measure replaced in their natural order of science, out of which the technical or alphabetical one had removed them.

Two centuries later, in 1910, Belgian lawyers and bibliographers Paul Otlet and Henri La Fontaine proposed a central repository of the world's knowledge called the Mundaneum. They organized the Mundaneum using the Universal Decimal Classification—which still lives on today through the Universal Decimal Classification Consortium.⁴ It now houses over 15 million index cards, 150,000–250,000 document collections in 57 languages and 135 countries, and millions of images. In 1934, Otlet further advanced his vision for the Radiated Library, in which people worldwide would place telephone calls to a “mechanical, collective brain” and would receive back information as TV signals.

In 1936, the “father of Science Fiction,” English writer H. G. Wells first predicted the World Brain, a central repository of the world's knowledge, organized by a complex taxonomy invented by Wells.⁵ He said the whole human memory will be accessible to every individual. Any student, anywhere, will examine any book or document.

As mentioned earlier, in July 1945, Vannevar Bush published “As We May Think,” in which he envisions the “Memex,” a memory extension device serving as a large personal repository of information that one can retrieve through associative links:

The human mind ... operates by association. With one item in its grasp, it snaps instantly to the next that is suggested by the association of thoughts, in accordance with some intricate web of trails carried by the cells of the brain ... Our ineptitude at getting at the record is largely caused by the artificiality of systems of indexing ... Selection [i.e., information retrieval] by association, rather than by indexing may yet be mechanized.⁶

Six Interesting Knowledge Graph Projects

TheBrain is a digital platform that helps manage information and knowledge. It provides a unique visual approach to organizing information, allowing users to connect ideas in a personal knowledge graph. TheBrain also allows users to create customized workflows and processes, ensuring that information is organized and accessible in the most efficient way.

Jerry Michalski has an outboard brain—Jerry’s Brain—which is represented in a concept-mapping application created by TheBrain. He has been using TheBrain since 1997 and currently has over 420,000 Thoughts interconnected by 775,000 links. This brain acts like an enhanced bookmarks list that is easily accessible and editable.

Noosphere is a protocol for thought that creates a massively multiplayer knowledge graph made up of smaller public and private sub-graphs. Each sub-graph or sphere is owned by an author, who signs it with a cryptographic key. Your sphere functions as your data backpack and self-sovereign social graph.

Roam is a software by Roam Research that helps users quickly capture their thoughts and ideas and connect them into a graph with user generated tags. It allows users to create a personal wiki that can help them retrieve and analyze data.

The Graph Protocol is a decentralized network that enables developers to query and change data stored on the blockchain. It provides a blockchain-agnostic knowledge graph, which enables developers to integrate blockchain data into their applications, enabling faster, more reliable access to data. A global community of indexers earns rewards for their contributions.

Out of MIT, the Underlay seeks to make public knowledge universally accessible and connectable. The platform aggregates statements, observations, and citations to provide detailed context and provenance of all knowledge. The Underlay also stores contested and contradictory statements and allows for the refinement, revision, and replication of observations. These features enable alignment of unrelated datasets and the use of machine learning, all for the public good.

These past ideas are emerging as the Universal Content Graph underlying the Metaweb. The overlay of the Universal Content Graph converts today’s Internet of Pages connected by the odd link into an Internet of Ideas connected by bridges. Through bridges, the graph connects information on an idea-by-idea basis, which enables rapid discovery of trustworthy information as well as 360° context for relevant ideas in the information ecology.

As mentioned above, when someone submits a bridge to the Overweb, it goes to the validation queue of the Bridge Registry, where approved bridge validators check whether the relationship specified is, in fact, correct. If incorrect, the registry notifies the bridger. If correct, the bridge posts to the Bridge Registry and self-assembles into the Universal Content Graph.

The Overweb surfaces relevant information from the Universal Content Graph in both web and graph views. In the web view, the participant sees information from the Universal Content Graph related to content in the focus of their attention. First, they see the content highlights and a badge appears containing the number of related smart tags. Clicking the badge displays an overview of the related smart tags. The overview has pie charts for smart tag type (e.g., notes, bridges), media type (e.g., text, video), and relationship (e.g., contradicting). Each pie slice is a filter, enabling one to narrow down the list of smart tags.

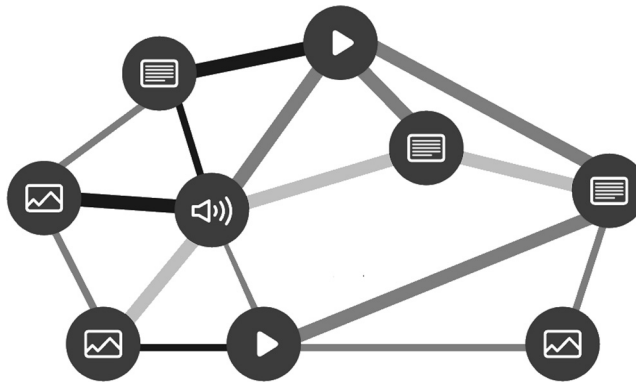


Figure 11.5 Graph view of an information ecology.

In the graph view, the relevant part of the information ecology displays as a “force-directed” graph (Figure 11.5). Imagine toggling from the web view to seeing a graph view, which displays nodes that represent content snippets and colored lines connecting them with relationships. Green connecting lines are supporting. Red connecting lines are contradicting. Yellow connecting lines are citing.

The line’s thickness reflects the bridge’s support based on crossings and upvotes. The Overweb uses these as metrics to reward bridgers for the value their bridges create in the ecosystem. By rewarding high-value bridges, the Overweb motivates people to create high-value bridges. In this way, bridges “tokenize” research.

The Universal Content Graph is the best approximation of the noosphere, the collective consciousness, the collective mind. A veritable record of meta-level interactions with online content and virtual worlds. It’s the Akashic records of the digital world.⁷

THE MANY USES OF BRIDGES

As shown in Figure 11.6, bridges can be very useful for secondary research.⁸ First, they enable a participant to organize all of their online content together with associative links, forming a web of online research. Speaking of research, bridges connect related pieces of content that are better seen together. This includes connecting new research to related studies, audio and video segments to text accounts, text that describes the subject of a photo, and vice versa.

Bridges are deep, precise links. They are precise because they bring the bridge crosser to specific content rather than a page. (Links to pages are deep links while simple links are to domains.)

A co-creation of bridgers, the graph is a unified map of relevant online information that serves as a shared context for communication, coordination, and collective sense-making. The graph enables 360° context for online ideas and discernment of false information.

Discerning the Real from the False

As mentioned earlier, the lack of context on the Web makes it difficult for web users to distinguish fact from fiction. While bridges don’t explicitly address “truthiness,” they provide context, which helps us discern whether to trust web content.

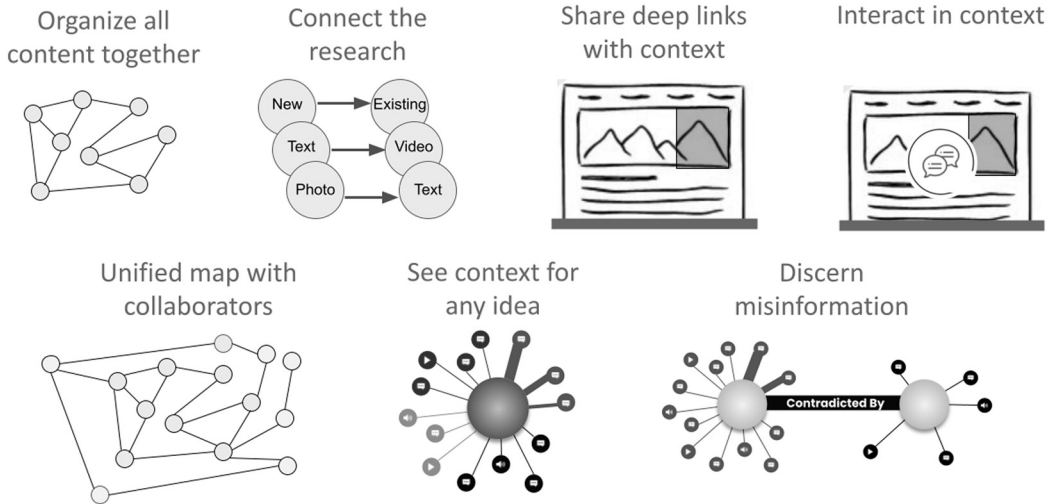


Figure 11.6 The many uses of bridges for secondary research.

Imagine a contradictory bridge between Content A and Content B. Given the relationship is valid, we know either Content A or Content B is true, and the other is false. Figuring out which is true requires examining the bridges.

Consider three scenarios, each with eight bridges to Content A, which tell very different stories:

1. Seven supporting bridges and one contradictory bridge.
2. One supporting bridge and seven contradictory bridges.
3. Four supporting and four contradictory bridges.

In Scenario 1, Content A has seven supporting bridges and one contradictory bridge to Content B. At first blush, it seems likely that Content A is true. To be sure, we'd review each bridge.

In Scenario 2, Content A has seven contradictory bridges and one supporting bridge. It appears Content A may be false, but it is always good to check.

In Scenario 3, Content A has four supporting bridges and four contradictory bridges. We need to further examine the evidence for each bridge.

BUILDING THE INTERNET OF IDEAS

Bridges make the Web more like the human brain regarding its connectedness. The Metaweb is a multi-layer extension or augmentation of our individual and collective brains. The ideas on pages are the nodes and they connect as needed. A page can have hundreds of ideas. Every idea is an opportunity to interact. Each idea may have dozens of connections or more. Hence, a long, popular webpage could have thousands of connections.

Bridges enable trusted influencers to build a supra-information layer on top of the *Internet of Pages* that connects information on an idea-by-idea basis. They create a relevant information ecology of ideas. We call this the *Internet of Ideas*.



Figure 11.7 The Night Sky and the Orion Constellation.

Let's imagine the night sky during a new moon in the countryside (the left side of Figure 11.7). A universe of stars shines across the dark sky. Visible throughout the world, Orion is a prominent constellation on the celestial equator (the right side of Figure 11.7). Named after a hunter in Greek mythology, Orion is one of the most conspicuous and recognizable constellations. Yet to untrained eyes, Orion is invisible. Like context for ideas on a webpage.

Imagine ideas on a webpage are stars. The ideas are alone. By themselves, ideas may have little import, like solitary stars to a constellation seeker. We wonder if the idea is correct. There is no context. We don't see relevant research or stories. But we know these ideas connect through pages. We don't see the connections because they are indirect and, therefore, invisible from our vantage point. Hence, the import of search.

One might ask, what's wrong with the search? Search treats webpages as monoliths as opposed to tapestries of ideas. Search uses one set of metadata for an entire page. But a large page could have hundreds of ideas in it when you consider its text, images, and/or videos. Search would work if ideas had metadata. And the page would need indexing, which isn't a foregone conclusion.

Bridges obviate the need for search in many use cases by connecting ideas into constellations of context. Once ideas connect into a constellation, their meaning is discernible. Similar to how the lines between the stars bring Orion to life.

The bridge is a contextual triple as in a set of three entities that codifies a statement about contextual data as idea-relationship-idea, which differs from the subject-predicate-object RDF/semantic triple. The bridge can encode and track additional information as needed.

Bridges, like annotations, are stigmergic, as they involve building upon and adding to the work of other bridgers. The process of building bridges creates a collective intelligence, in which insights, knowledge, and understanding are layered on top of each other, making the Internet of Ideas a living entity that continues to evolve and adapt to new perspectives and understanding.

Table 11.2 compares the Internet of Pages and the Internet of Ideas.

Table 11.2 The Internet of Pages vs. the Internet of Ideas

	<i>Internet of Pages</i>	<i>Internet of Ideas</i>
Content unit	Pages	Ideas
Technology	Centralized	Distributed
Compensation	No	Yes
Quality of Results	Mixed	High
False news	Biased	Unbiased
Contextual Relevance	No	Yes
Coverage of the Web	0.04%	Much More
Free Speech	No	Yes
Long Tail Search	No	Yes
Long Tail Advertising	No	Yes
Payment for Watching Ads	No	Yes

In sum, the Internet of Ideas relies upon the Universal Content Graph to connect ideas and provide context. The graph grows, self-assembling with each new bridge. While other discovery solutions cannot forefront relevant but less popular content, the graph solves the elusive long tail. It's an old idea that improves the experience, enabling us to see the context for ideas, notice patterns, and find what we need.

As we turn the page and delve into the world of the Overweb, we are greeted with the sight of a new and exciting landscape. A landscape that is built upon the foundation of the Metaweb, but forefronts cognitive freedom, trust, and safety. We will explore the Overweb Pattern, a powerful toolset that has been specifically designed to solve the problems of Today's Web. This toolset provides a common language that will allow us to navigate this hyper-dimensional new level of the Web.

We will examine the three pillars that form the foundation of the Overweb and discover the 7 Cornerstones that make up the Overweb. The pillars and cornerstones create are the foundation of the next level of the Internet, a truly interconnected web like never before.

NOTES

- 1 <https://permanent.link/to/the-metaweb/next-generation-internet-award-winners>
- 2 While bridges can technically connect ideas on the same page, this is not the intended use, although it may at times be useful, most notably when a webpage contradicts itself.
- 3 <https://permanent.link/to/the-metaweb/cyclopaedia>
- 4 <https://permanent.link/to/the-metaweb/universal-decimal-classification-consortium>
- 5 <https://permanent.link/to/the-metaweb/world-brain-wells>
- 6 <https://permanent.link/to/the-metaweb/as-we-may-think>
- 7 The akashic record is a compendium of pictorial records, or "memories," of all events, actions, thoughts, and feelings that have occurred since the beginning of time.
- 8 Secondary research is a research method that compiles and analyzes existing data from a variety of sources commonly including the Internet, government statistics, and research reports.

The Overweb Pattern

The Overweb enables ant hills of meaning on every page, where thoughts about page content come to life and build upon each other without direct communication, opening up a whole new conceptual space for exploring and connecting ideas in ways we never thought possible.

—Bridger One

The Overweb is the first full instantiation of the Metaweb (Figure 12.1). It builds in enhanced privacy, accountability, and safety. Whereas the Metaweb is anything goes, the Overweb is a safe decentralized digital space. They differ as much as the Wild West and Bhutan.

A People-Centered Country

Bhutan measures Gross National Happiness (GNH) instead of the traditional Gross National Product (GNP). Introduced by Bhutan's former King Jigme Singye Wangchuck in the 1970s, GNH reflects the Bhutanese government's commitment to considering the well-being and happiness of its citizens in policy decisions. The framework for measuring GNH includes multiple domains beyond traditional economic indicators, providing a more comprehensive view of a country's progress and well-being. These include nine domains: psychological well-being, health, education, time use, cultural diversity and resilience, good governance, community vitality, ecological diversity and resilience, and living standards. This unique approach to measuring a country's success has been recognized and praised by many experts and organizations. The Bhutanese government conducts regular surveys to assess the happiness and well-being of its citizens and uses this data to inform policy decisions. Bhutanese citizens have embraced GNH, participating in the survey process and working to promote happiness in their communities. Bhutan's GNP is an inspiration for the Overweb and perhaps a model to emulate.

Both are digital spaces. Like the Wild West, the Metaweb is liberating. But it's dangerous and you can't tell who's who. Similar to Bhutan, on the Overweb, you're accountable for your actions, know your neighbor, and commune with like-minded people.

Just as the Wild West birthed Silicon Valley, the raw Metaweb has immense potential but it can't be trusted. We expect over time it will birth many digital spaces, some of which we hope will emphasize safety, like the Overweb.

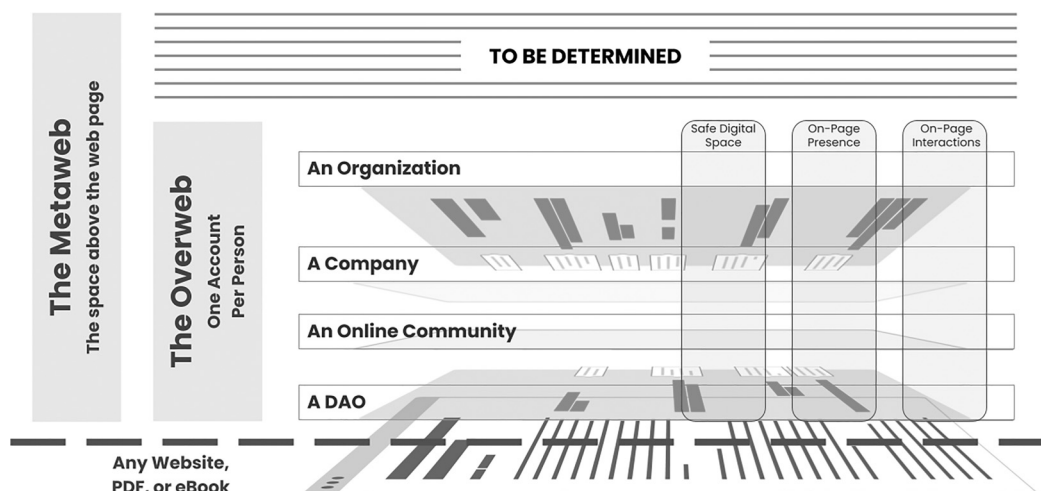


Figure 12.1 The Overweb is the Metaweb's first full instantiation (adapted from Hypothes.is Overweb deck).

The Overweb is a DAO, an open-source community, and a software application layer above the webpage that follows a design pattern called the Overweb Pattern. It includes three key elements, seven cornerstones, and a bill of rights. Together, these enable self-governance while optimizing for collective cognition, value creation, and fair value exchange.

Foremost, the Overweb is a trust layer over the Web, where people can feel safe from harassment, scams, and other forms of abuse. A safe digital space that enables communities to connect in an always-available wherever-you-are generative space and build together. The Overweb provides digital analogues to important IRL structures, including the individual being, families, friend groups, community groups, teams, organizations, and even countries, so our lives can stratify into an integral digital existence.

The Overweb is a social ecosystem that secures and supports creators, context, private research, fact-checking, citations, collaborative research, and much more. Today, we have a limited view of the information on the Web. We need a unified contextual view to reduce duplication of effort and speed up sense-making, collaboration, and learning. The Overweb leverages the Web to build knowledge and convene a collective conversation among real people and communities. We also think of the Overweb as a learning layer over the Web that makes thinking visible. In Chapter 15, we liken the Overweb to a “library layer” over the digital town square.

Access to the Overweb is through meta-communities—networks within networks—that have their own sovereign layer above the webpage. Meta-communities have a common purpose, a “thou shalt not” code of honor and an approval process for entry. The Overweb works as a safe autonomous zone above the webpage, accessible by anyone with an account. Safeguards ensure privacy and certify that participants are real people in good standing with their community. On the Overweb and within meta-communities, there are no unidentified bots, fake accounts, serial abusers, or throwaway accounts, and far fewer trolls than on Today's Web.

The Overweb gives participants a visual presence to interact in the space above third-party webpages. Participants opt-in to visibility and know others are real people in good standing. The Overweb allows people to connect in safety. Participants can see visible people, review the chat, and post messages or replies. They can create channels and chat rooms as needed.

The Overweb is accessible via the Presence browser overlay, which has browser extensions, the Canopi SDK, and a mobile browser. Loading a page with the SDK enables Overweb access without installing extensions. Acting as a virtual private network (VPN), the overlay can hide the participant's location by sending webpage requests from a generic IP address.

The Overweb pattern is a toolset for solving the problems of Today's Web that defines a common language for the hyper-dimensional next level of the Web. This chapter explores the Overweb Pattern, including its three key pillars, its cornerstones, and the Overweb Bill of Rights. It also covers the Presence browser overlay and the Overweb's basic building blocks.

THE THREE PILLARS

Comprising three pillars, the Overweb pattern is present throughout the Overweb ecosystem: Safe Digital Space, On-Page Presence, and On-Page Interaction.

Safe Digital Space

Safe digital space gives participants a level of security, safety, and surety impossible on Today's Web. The Overweb is a safe, privacy-honoring digital space above the webpage with one-account-per-participant and abuse safeguards. Hence, anyone you encounter on the Overweb is a real person in good standing. Communities organize as meta-communities, which have their own application process and community-based moderation. The Overweb has an adaptive account unification process that ensures each account is a unique real person so that no one person controls multiple accounts. Say goodbye to scammers, trolls, serial abusers, unidentified bots, trolls, fake accounts, purveyors of disinformation and clickbait, and throwaway accounts.

Participants have one main Overweb account that includes a real phone number and a unique biometric identifier, which is controlled by the account owner. The Overweb cannot unmask the user. People have multiple personas tied to the primary account. The connection between the person and primary account can be visible or invisible. The different personas carry different information. For example, a winter sports persona may have the type of equipment, experience level, even mountains or resorts visited, and/or favorite conditions. Communications use encryption and can be disappearing.

Communities organize themselves as meta-communities—networks within networks that have their own dedicated layer above the webpage. Meta-communities have a common purpose, a code of honor or conduct, as well as an approval process for entry, and community-based moderation. Skiers and snowboarders, for example, could have their own meta-community.

Many meta-communities are DAOs, while others are existing online communities or communities formed for a specific purpose. Members choose an active persona when accessing a meta-community. E.g., members of the skiing or snowboarding meta-community may choose their winter sports persona. Thus, meta-communities operate as a safe autonomous zone above the webpage, accessible by members. Their safeguards ensure privacy and that people are real and in good standing.

The meta-community may deactivate the account or restrict access to people who violate their code of honor. This affects all of their personas. Completing a community-defined rehabilitation process is necessary to restore their account. All the past activities with any of their personas stay with the main account, whether the persona still exists. In this way, the

main account is like the US Social Security account number. If we charge up credit cards and don't pay them off, our credit score will plummet and we can't get more credit. In either case, we cannot just get another account. We're interested to see how conflict resolution, rehabilitation, and restorative Justice emerge on the Overweb.

Real-Time On-Page Presence

The Overweb gives people a real-time presence on webpages similar to Google Docs. Today's Web isolates participants. This prevents them from knowing who is on the same webpage as them. We're missing people with common interests and similar reasons for being there. Today's Web is like an empty café or stadium. The Overweb creates a safe way for participants to meet by going visible on webpages of their common interest in persona. Visible participants can see whoever else is visible on the page at the same time (or in the recent past), review participant profiles, and start communication. They can also limit who they see to members of their meta-communities.

Here's an example of how it works. Our friend Jane enjoys snowboarding, hence one of her personas could be @snoboarder333. Because of the safe digital space pillar, anyone who encounters her on the Overweb knows she is a real person in good standing. Let's say Jane wants to snowboard in the Alps next winter. In search, she finds SkiResort.info but feels overwhelmed to learn that the Alps are humongous with 1150 ski resorts, 26,738 kilometers of slopes, and 8,218 ski lifts. What to do? She needs help and not just from anybody with a snowboard.

With an on-page presence, she can "go visible" on the webpage for the Alps or even the SkiResort.info home page. Once visible, she can see others who are visible. Hundreds could be visible.

She can review the profiles of the visible participants and the ongoing conversation in live chat for the page. She wants help on where to go. So she filters the live chat and the visibility list for people with similar levels of experience, equipment, and favorite conditions. She can browse profiles to find people working or living in the Alps. Or people who've skied many mountains. She may see a post from someone she wants to talk with on the page's live chat. She can reply to the post or send them a private message.

The recipient will receive a notification that someone—a real person—they don't know has sent them a message. They can ignore or flag the message. If they choose to respond, a dedicated chat room will open up between them. This new connection can go wherever it wants.

Through on-page presence, the Overweb gives opportunities to meet and interact with people you wouldn't meet IRL. Meeting people will be easier if you already have many interests in common.

On-Page Interactions

On-page interaction is like a graffiti project for people in your tribe. People can see the creations within their meta-communities and add their perspective. On-page interaction gives computation—as interactions, transactions, and experiences—a real-time presence on webpages, similar to how you can tag your aunt in a photo on Facebook. The content creator controls the webpage. You can weigh in if they enable comments. Or if you have access to the Overweb.

Beyond notes on text, the Overweb enables structured annotations configured by Overweb developers on any media from text snippets to parts of images to segments of

videos and podcasts. In the future, participants will have a quiver of smart tags, including bridges, notes, conversations, polls, lists, and more that they can attach to any content on the web. This enables participants to have conversations directly in real time and indirectly through smart tag interactions attached to specific pieces of content, layering thoughts and knowledge on webpages. Smart tags are stigmergic, meaning they build upon the work of others and create a collective intelligence.

The Overweb is not only a new frontier for participants, but also for developers. It offers a new realm for developers to explore and create useful and innovative tools that can help people navigate and interact above the webpage with more precision and understanding, showcasing their skills and work in the next level of the Internet.

On the Overweb, developers can create smart tags for a wide range of applications, such as meetings, audio conversations, fact-checking, content manipulation utilities, content generation, polling, and more. With the ability to easily reuse backend code and build upon existing work, developers can quickly create smart tags and begin earning rewards while contributing to the ecosystem. They can offer their creations for free or a fee, or by subscription or donation.

THE SEVEN CORNERSTONES

The Overweb's virtual headquarters in the Metaverse will be heptagonal, reflecting the seven cornerstones of the Overweb. The cornerstones represent the fundamental values of the Overweb, and demonstrate the Overweb's philosophy of creating a symbiotic relationship between participants and the environment, where all parties benefit from the exchange of value and data. The evolving implementation of the 7 cornerstones into smart contracts and software will be led by the Web4 Foundation, the appropriate working groups, and open source developers.

Sovereign Identity

The Overweb is built on the principle of sovereign identity, which enables individuals to manage and control their own digital identity and data without outside intervention. This approach is rooted in the sovereign identity movement, which aims to bring the same trust, freedom, and accountability of the real world to the digital realm.

The Overweb also recognizes that traditional centralized systems of identity management are often exploitative and lack transparency. Hence, the Overweb will not require that participants connect their sovereign digital identity to a real world identity and will support pseudonymous personas. The goal is to enable trustworthy and safe management of identity online, so that every individual can manage their privacy and control their data.

The Overweb prefers social recovery or multi-party computation (MPC) for managing identities, as these methods allow people to retain control of their identity and data even if they lose access to their wallet or other login credentials. This approach ensures that participants can regain access to their identity and data in case of loss or theft, without relying on centralized authorities.

Fair Value Exchange

The Overweb is built on the principle of fair value exchange, which is a way of compensating creators and participants for their efforts and the quality of content they bring to the

environment. This approach is rooted in Web3 and game theory, and is designed to create a more equitable relationship between the environment and its participants.

The Overweb recognizes that the exploitative relationships of Web 2.0 have led to a lack of fair compensation for creators and participants. Hence, the system is implementing a system of value-based incentives that rewards participants for their engagement and contributions. This system is designed to acknowledge and reward participants for the value they bring to the ecosystem.

One important aspect of the fair value exchange system is the compensation of bridgers—participants that connect information and create value by fostering interactions and collaborations. Bridgers receive rewards based on the value their bridges create in the ecosystem as described below. They can also earn rewards from curating bridges, labeling content, and watching advertisements.

The Overweb is committed to creating a more equitable system of value exchange for all its participants, where participants are fairly compensated for their efforts and contributions.

Data Control and Compensation

The Overweb understands that data privacy and control are crucial. The environment is built on the principle that participants own, control, and monetize their data. This means that participants have the power to decide who can access their data and for what purpose. They can choose to track their data, make it available for use by marketers and algorithms, or keep it private, or not track it at all. The Overweb's algorithms are open-source and auditable, ensuring that participants can trust the handling of their data.

In addition, the Overweb also provides opportunities for participants to monetize their data. Members of meta-communities can sell or rent their anonymized data through data cooperatives, or take part in basic income programs. These programs allow participants to earn compensation for their data while maintaining control over its use. The participant can choose not to take part in any data program if they do not want to share their data.

The Overweb is committed to creating a transparent and fair data ecosystem where participants have the power to control their data and the opportunity to monetize it.

Symbiotic AI

The Overweb is working towards a future in which humans and AI work together in alignment. Instead of fearing AI, the Overweb recognizes the potential for AI to augment human capabilities and enhance our ability to navigate the digital landscape as well as the need to align AI with human interests. The Overweb accomplishes this by requiring autonomous AI Systems to be constitutional or glass box, which were both discussed in Chapter 2. To join the Overweb AI-DAO, systems must either have heuristic imperatives to reduce the likelihood of harm or be explainable such that system workings and decision-making processes are clearly understandable and manageable.

To participate in the Overweb, autonomous agents must possess a decentralized digital identity, be associated with a real individual, obtain consent for accessing participant data, and publish notable activities and discussions with other autonomous agents on the blockchain. AI-DAO agents will actively monitor this activity in real-time to identify anomalies, as well as any inappropriate or suspicious behavior.

On the Overweb, AI is present to serve the participants. It can help participants understand their personal preferences, protect them from cognitive manipulation, act as their personal assistant or agent, filter out unwanted information, generate unique content, and

safeguard their data. The Overweb's algorithms are open-source, which means that the community can contribute, share, and monetize their own AI models. This approach creates transparency and ensures that all bots are identifiable and connected to a real person. Additionally, AI generated content is labeled on the Overweb.

Our DAO is well aware of people's concerns about the future of work and the potential for AI to displace jobs. The Overweb, however, is committed to creating a symbiotic relationship between humans and AI, in which AI can enhance human capabilities and create new opportunities for work and collaboration.

With the integration of AI assistants and agents, mundane tasks and content generation will be handled with ease, freeing up more time for connecting IRL. By 2030, experts project that AI assistants on cell phones will surpass the capabilities of the AI currently employed by today's largest companies. The AI-assisted web of the future promises to greatly enhance both human-to-computer and human-to-human connection, conversation, and collaboration.

Contextual Relevance

The Overweb is built on the principle of contextual relevance, which means that the environment only shows participants information that is related to the focus of their attention. This approach is designed to improve the user experience by reducing information overload and providing participants with relevant and useful content.

To achieve this, the Overweb anchors on the Universal Content Graph. The Universal Content Graph connects the web on an idea-by-idea basis, creating a collective learning map that grows with the interactions of participants. This map provides a new way of locating information, interactions, transactions, and experiences, enabling participants to build and explore the relevant information ecologies.

Compared to traditional database structures, the Universal Content Graph retrieves information orders of magnitude faster than searching. For a piece of online content, participants can find all related information within microseconds. This approach allows participants to quickly and easily find the information they need, saving time and reducing frustration.

This approach ensures that participants only see information that is relevant and useful to them, making the environment easy to use and enjoyable to explore.

Radical Transparency

The Overweb is built on the principle of radical transparency; people thrive when they have access to information and can make informed decisions. The environment's transparent approach ensures that participants know when they are dealing with humans versus bots and how their data is being handled.

Every person on the Overweb is a real person in good standing, and their entire history of communications and interactions across profiles stays with their primary account. People are responsible for their digital footprint, which helps build trust. For any participant, the Overweb provides an overview of the blocks and flags they've given and received as well as any censorship requests and actions. This helps the participants to understand their own behavior and the behavior of others.

All bots tether to a human account, which is responsible for the bot's activities. The Overweb's smart contracts and owner can revoke a bot's permissions and participants can block bots and the content they produce.

The algorithms governing the display of information and interactions on the Overweb are open-source, this means that participants can understand the algorithm's implications and

turn it off, or switch to a better-aligned one. This approach ensures that participants have control over the information they see and can make informed decisions about the content they consume.

Participants also have their own smart filter, which they can use to limit what they see. The filter is adaptive, transparent, and tunable. This means that the filter can adapt to the user's patterns and the user can see what's getting stopped and make adjustments.

Attention Triggered

The Overweb surfaces information related to the viewer's focus, similar to how when we observe something in real life. By focusing our attention, we can access greater levels of detail and context that are relevant at the moment. The Overweb aims to replicate this experience online by enabling participants to zoom in and see more details about the content they are viewing.

In the meta-layer, participants have access to information relevant to what they are focusing on. Participants can see information, interactions, and context that relate to the focus of their attention. Unlike on traditional web pages that are flat and static, participants can access a greater level of detail and context by activating metadata and navigating to additional information.

The Overweb uses attention triggering, which displays controls and metadata related to the participant's focus of attention. This reduces distractions and makes the user experience more engaging and efficient. As participants scroll through content, content with metadata activates. When content activates, a circular badge displays the number of smart tags available to the participant; interacting with the badge displays an overview of smart tags with a filterable list.

The smart tags represent metadata and interactions related to the content such as notes, bridges, conversations, polls, and more. They provide context for the focus of one's attention, enabling participants to see more details about the related metadata by hovering and clicking. This approach allows participants to access the information they want, when they want it, making the environment less distracting and overwhelming despite the sheer quantity of information available.

THE OVERWEB BILL OF RIGHTS

You have rights on the Overweb. You don't have rights on Today's Web. Depending on where you are, you may have rights on the Metaweb.

The Overweb is a sovereign digital space above the webpage comprising a shared universal content graph and countless sovereign meta-layers for applications and meta-communities. Operating as a DAO, the Overweb creates safe decentralized public space with protections and rights unavailable on Today's Web. We think these are the minimum viable rights for digital existence and a thriving digital democracy. The evolving implementation of these rights into smart contracts and software will be led by the Web4 Foundation, the appropriate working groups, and open-source developers.

Privacy

On Today's Web, you have no privacy. By accepting cookies on websites, you are consenting for them to track your data. By accepting the terms of the conditions set by the platforms,

you are giving them free rein over your data. This includes ownership of your creations, and even the photos that you upload to private galleries.

On the Overweb, your privacy is paramount. You determine what information to divulge to specific groups of people. You have a sovereign identity for which only you have the private key (unless you outsource its protection to a custodian). The Overweb cannot doxx you or even supply your identity to authorities because your private key secures access.

Security

Today's Web has frequent security lapses. As mentioned earlier, millions of records leak in hacks every year, providing fodder to be sold on the Darknet to identity thieves.

The Overweb protects your data from unauthorized access. It builds on the cryptography systems inherent in blockchain. This provides a much higher level and granularity of security than centralized systems. In centralized systems, security lapses can leak hundreds of thousands, if not millions, of records at a time.

Data Sovereignty

On Today's Web, you relinquish the right to your data to use Internet platforms. Over time, they amass a huge amount of information about you, which they exploit to influence your behavior to their benefit. The ticket to entry for the use of social platforms is access to your mind and ownership of your expressed thoughts. This is like being in a film. Someone else is bankrolling it so you're being directed and you don't own your performance. And the film never ends, unless you quit. Online, the influence is subtler: they decide what content hits your feed, its order, and they own your data.

On the Overweb, you control your data. You don't need to opt out of having your digital whereabouts tracked; that's the default. Opt-in if you want your activities tracked—for your use. You can monetize your data through data cooperatives, which protect your identity. There may also be some which reveal your entire identity, but pay you more. You can also enter UBI (Universal Basic Income) programs. Ultimately, there will be applications for you to learn more about yourself via your data, e.g., when you are acting on confirmation biases.

Cognitive Freedom

Self-determination requires cognitive freedom. We exercise cognitive freedom when our cognitive process is free from undue influence such that we are able to align our actions with our best interests and values. This can be challenging on Today's Web.

On Today's Web, when you use a search engine or social media, you enter a cognitive cage that limits what you can do, say, and see. Algorithms whose sole purpose is changing your behavior determine not only what you see but the order you see things. Within algorithmic regimes, if your legal speech is outside guidelines, they can curtail or throttle down your existence on their platform. That speech legality varies from place to place complicates matters.

The Overweb offers freedom from manipulation by unidentified bots, fake accounts, algorithms, and disinformation campaigns. You control the algorithm, not vice versa. Each participant has an adaptive, transparent, tunable smart filter. You see what goes through. You can find out what doesn't. If not, you can set up specific criteria for the filter to alert you. Make up your own mind based on synthesizing the totality of information that you allow through your filter.

Context

Today's Web has less context than ever. Escalating since the Attention Economy took hold, commercial sites avoid outbound links from which they do not benefit. The lack of non-promotional outbound links has created a dearth of context, which makes it difficult to distinguish what is real from what is not.

On the Overweb, context is a digital birthright. You can see connections between first-hand accounts in blogs, tweets, and posts; historical accounts; and unfolding events in the news. You have unfettered access to deep layers of context for the focus of your attention. Access 360° contextual information right when it's most needed.

Safety

Today's Web is a dangerous place. Scammers are everywhere on the Web as people, sites, pages, offerings, reviews, videos, communities, and more. Abusers are on social media to steal from you or exploit your emotions. Trolls and gang stalkers want to make you pay for what you say. Bad actors peddle disinformation to affect your thoughts and behaviors.

On the Overweb, participants enjoy safety from serial abusers, and the comfort of knowing that everyone you encounter is a real person in good standing. Because everyone has one primary account, you know the person you're connecting with online is not a serial abuser. You know they don't make a practice out of harmful behavior. Disinformation is just a trickle because it stays with the person's primary account forever.

The Overweb prioritizes the protection of its participants from rogue AI agents. Its primary goal is to establish a secure environment that allows for the responsible and safe deployment of AI advancements. Serving as a computing environment, the Overweb has the capacity to enforce limitations on autonomous agents that were previously unachievable within the current web framework. To operate within the Overweb, these agents are required to register using a decentralized digital ID, adhere to specific cognitive architecture requirements such as David Shapiro's three heuristics, establish a connection with an Overweb participant, obtain consent for accessing participant data, and log significant activities and intra-agent communications on a blockchain. The Overweb's AI system actively evaluates this data to promptly identify any instances of inappropriate or suspicious behavior.

Transparency

The algorithms operating on you are insidious and invisible. You don't know when they are active or how active they are, despite them controlling much of our digital lives. The algorithms may use data you don't want them to use. They may also lump you in with others (i.e., filter bubbles) that reflect the biases of the data scientists that are building them.

On the Overweb, you fully understand the algorithms acting upon your data. Overweb algorithms are open source, identifiable, and explainable. In open-source communities, people can contribute, share, and monetize their own AI models. All bots are identifiable and connected to a real person. Additionally, AI-generated content is labeled on the Overweb. AI systems on the Overweb must be explainable such that the workings and decision-making processes of these systems can be clearly understood and examined. This level of transparency and accountability is essential for promoting responsible and ethical AI technology.

You can also control your personal AIs. Your smart filter, for example, ensures that you see what's needed but also allows you to inspect what's filtered out.

Free Speech

On Today's Web, free speech is a highly debated topic. Many argue that the Internet should be a place where people can express their opinions and ideas freely, without fear of censorship or retribution. However, there are also concerns about hate speech, misinformation, and other forms of harmful content spreading on the web. As a result, many platforms have implemented policies and systems to moderate and remove content that violates their terms of service or community guidelines. These policies are often reactive, responding to reported content after it has been posted, and have led to inconsistent enforcement and censorship along partisan lines and for narrative control.

The Overweb is built on the principle of free speech, modeled after the First Amendment of the US Constitution. This means that participants have the right to express their ideas, perspectives, and opinions without fear of censorship or retaliation. The Overweb recognizes the importance of free expression in fostering open and constructive conversations and interactions.

The Overweb's approach to free speech is not absolute. While participants have the right to express themselves freely, they are also accountable for their words and actions. The Overweb community has a zero-tolerance policy for harassment and other forms of abusive behavior, and for speech that incites violence or causes harm to others. Participants who violate these policies will have their access curtailed. All enforcement actions will be community-driven and transparent. Meta-communities set and enforce (or outsource) their own policies for content. Additionally, the Overweb will transparently comply with valid legal requests, such as court orders and subpoenas, to remove content that violates local laws.

Expression

Today's pages inhibit expression. Some blogs allow you to leave a comment at the bottom of the page. You can also make comments in-line on Medium articles and on webpages with an annotation extension. Social media limits you to fit their business model; tweets must be 280 characters or fewer. Many trolls are there to harp on whatever you express, especially Twitter.

On the Overweb, enjoy a safe digital space for art, civil discourse, and debate within your meta-communities. You can translate your thoughts and creations into knowledge artifacts that layer over the webpage. You can create bridges, notes, conversations, and more, only limited by the ever-expanding repertoire of smart tags available on the Overweb.

But wait! Something integral is missing. A differentiator from those who play in the flat, static web. One right—perhaps the most important—that you've likely never heard of before. It's actually multiple rights wrapped up into one.

Presence

The right to be present. And the right to be. The right to access a digital presence overlaying our lives. The right to have a presence and to be seen.

We don't have a visible presence on webpages. The closest thing is social media. We see a time-stamped record of thoughts. Chat may be available. We can't, however, know who is on the same webpage.

We exist in the digital world as our interactions. Except in the metaverse, where you have an avatar, but not safety. On the Web, we can't post up on our favorite sites and meet fellow visitors.

Being online is like being a soul without a digital body that incarnates as posts. We can't see one another. But we see the posts. We continue posting, leaving cairns along the path for someone to know our soul and ideas. Some of us leave long trails of these digital incarnations. But follow as one may, these web paths never lead back to the soul. You're the digital shadow one cannot catch.

The Overweb offers a visible presence on webpages to meet people with common interests. You can toggle between visible and invisible on specific browser tabs or webpages. One day, your avatar will walk on webpages.

THE PRESENCE BROWSER OVERLAY

The Presence browser overlay provides access to the Overweb, a safe digital space in which real people and information have presence and interactions over the webpage. The overlay is accessible through browser extensions, a software development kit (SDK), and a mobile browser. Browsers, web apps, dApps, and mobile apps can integrate the Overweb by adopting the protocol. Thus, entire communities will connect across devices and applications. Society will benefit from a connected citizenry equipped to take part in all levels of democracy (Figure 12.2).

Presence provides first access to the Overweb. Its intention is to support and motivate people, teams, and communities who want to connect and interact around their shared areas of interest and build robust information ecologies. Community incentives reward participants with value for their contributions to the ecosystem.



Figure 12.2 A Wikipedia page with a canopi.

The Presence overlay technology won the award for “Disruptive Technology Culture Drivers” from the EU’s flagship NGI program. This award was for building a new culture around technology that breaks up knowledge silos, creates contextual intelligence, connects disciplines, and targets a diverse audience. The initiator of the Overweb, Bridgit, was the only non-European company to win a Next Generation Internet award from the EU’s flagship program.¹

Assessing the Overweb via the Presence browser overlay provides a rich web experience. It enables people to interact, collaborate, and learn together on the same webpage with no coordination. Even if they don’t know one another. Or speak the same language. Via a browser overlay tool that navigates an interactive self-assembling collective learning map, we call the Universal Content Graph.

Among many other things:

- Presence enables participants to access a secure social ecosystem over the webpage and engage in civil discourse within meta-communities through on-page evidence-based conversations, bridges, polls, and other smart tags. The Presence ethos is every person has value and can contribute to knowledge and collective intelligence.
- Presence supports critical thinking, reasoned argument, shared knowledge, and verifiable facts. These interactions self-assemble into the Universal Content Graph, a map of the online information ecology.
- Presence enables massive online collaboration among diverse communities working together on issues that lift their communities. Collaboration on the scale is necessary to address humanity’s global challenges.

The first Presence products are the Canopi SDK and the Canopi Extension. Both enable participants to access canopis above websites. (See the section on Overlay Apps.) The Presence DAO aims to decentralize and open source the Presence browser overlay.

The Overweb gives participants a visual presence to interact in the space above third-party webpages. Participants opt-in to visibility and know others are real people in good standing.

Canopi

Canopis enable people to connect in safety. A canopi is a sidebar on a third-party webpage that enables people to have a visual presence in the meta-layer without modifying server-side source code. Participants can opt-in to being visible on specific pages, knowing that everyone there is a real person in good standing. They can retrieve profiles, chat, and make friends.

Participants can see other visible people on the same webpage, review the page chat (that is always available on the page), and post messages or replies. They can create channels on the page chat to focus more deeply on a subject. They can also form private encrypted one-on-one and group chats. These chat messages can also disappear after a set time. People can also block or flag people and bots. They can also see those blocked by others in their meta-community, and set filters to exclude people that others block.

OVERWEB BUILDING BLOCKS

The six main building blocks of the Overweb are smart tags, overlay applications, meta-communities, smart filters, MORCs, and adjacent space.

Smart Tags

Rather than just reading web content, smart tags enable interactions with content and real people interested in the content. For example, you can make notes, start a conversation, or schedule a meeting. The Overweb provides a quiver of smart tags for use on any page. While Today's Web relegates comments to the bottom of the page (except for Medium), smart tags—including notes, conversations, bridges, lists, and more—attach to related webpage content, making the Web hyper-dimensional and stigmergic. Developers will create hundreds of smart tags for you to use.

Smart tags enable real people to interact around and with content snippets, which is a foundational infrastructure for the next level of the web. They are computational code that attach information and interactions to webpage content, enabling indirect collaboration. Smart tags allow interactions with content (e.g., notes, bridges), interactions with others through content (e.g., comments on notes, conversations), and the display of information and interactions from other applications. They support computational text, which updates based on information external to the page.

For instance, smart tags enable an author to make their publisher's page work for them. Through the Overweb, she can tag information related to the book. She can hold or announce a Reddit-style "Ask Me Anything" (AMA) on the webpage, next to the image of the book. She may use a meeting smart tag or a specific tag for AMAs. Anyone who visits the page will see the tag for the event. If they RSVP, they will receive an email reminder to return to the page to interact with the author.

Standard webpages do not enable authors to connect with book readers except through comments at the bottom of the page. Substack's solution for connecting creators and readers is a mobile chat app, but this requires code switching, which can be exhausting. The whole point of the smart tag is that you don't have to go anywhere else. There's nowhere to go. The action is where you are. This is but one example of the plethora of possibilities presented by smart tags.

Structured Conversations and Annotations

In 2022, most online information is still unstructured, be it a news article, blog, video, tweet, or podcast. It's just a blob of information that machines cannot process for non-generative purposes. As humans, we need to read and think about it, and sometimes even structure it in our minds to understand it. Machines and humans can read structured information. Examples of structured data include tables, data in databases, classifications, and metadata. The Overweb enables developers to create custom smart tags and overlay applications that label web content and create structured annotations. For example, there will be an app that enables bridgers to label content on webpages for machine learning. This data will also enable smart filters that personalize the Overweb experience.

The Overweb has native smart tags, including notes, bridges, conversations, polls, labeling, lists, and meetings. Developers can build and monetize external smart tags. The creator chooses which ones to use. Others with access (e.g., within their meta-community) see their tags.

As shown in Figure 12.3, we envision many types of smart tag integrations. From extending the reach of legacy apps into Overweb, to connecting with personal and enterprise systems, content servers, payment gateways, paywalls, and more.

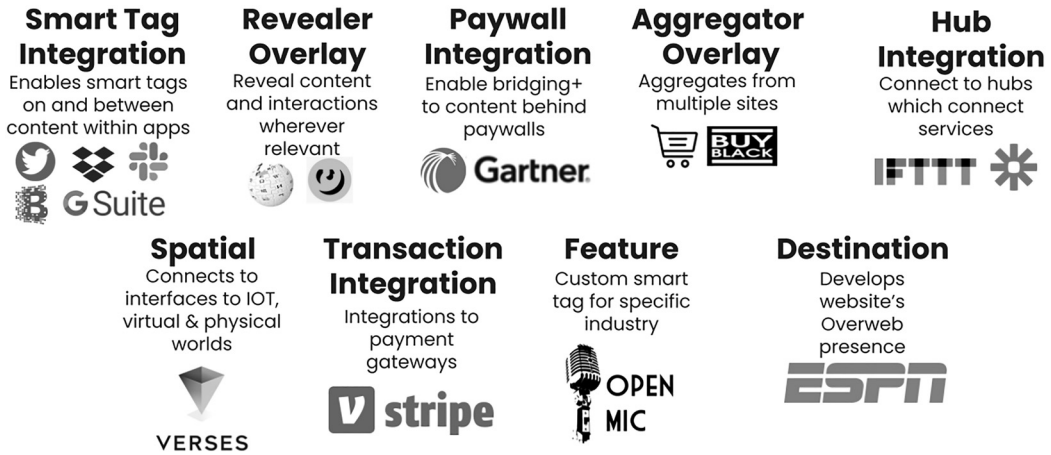


Figure 12.3 Smart tag integration possibilities.

Overlay Apps

Today's Web is full of siloed applications available only from destination sites. Gordon Brander, founder of Subconscious, says the Web was supposed to be a global, connected brain that would enable us to think together but it didn't pan out. Apps trap our thoughts in software as a service (SaaS) silos. Early on, the Web adopted a same-origin security policy, restricting documents or scripts loaded by a page to only interact with resources from the same source. Apps inherit this model. Each app lives in its own pocket universe. They can only talk to one another with one-off integrations, which often require business deals. Because security ties to specific computers in the cloud, the data gets siloed. This doesn't scale to thinking together as a planet.

Since overlay applications use a unified identity model, as an abstraction layer, they can call multiple apps without violating the same-origin policy. This means overlay applications can authenticate participants, pull data from multiple sources for the participant, manipulate the data, and expose the relevant context and interactions either in the Presence sidebar or otherwise above webpage content. Some overlay apps, like smart tags and canopi, are native to the Presence browser overlay and provide its foundational functionality.

The smart tags overlay app supports Overweb native smart tags. Smart tags are computational code that attach information and interactions to content on webpages. These include bridges, notes, conversations, polls, lists, and meetings, as well as the capacity to work with external smart tags produced by the developer community.

Canopi is like a rooftop speakeasy above every webpage; a place you can schmooze with people in the know. The canopi overlay app operates in the Presence sidebar and is accessible via the Canopi Enterprise SDK and the Presence Browser extension. With just three lines of code, the Canopi Enterprise SDK operates a canopi on a single website. The extension enables access to canopis on every site and provides additional functionality and benefits. Canopi enables real people to enter a safe digital space over a webpage. They can see visible people, their profiles, or take part in the page's group chat. They can create rooms that provide messaging over the entire web for one-on-one or group conversations. In the future, developers will build plug-ins or extensions to canopi.

We expect the developer and business communities will build overlay apps to create new experiences with their proprietary information and interactions. People can activate overlay apps for a specific tab.

Meta-Communities

In real life, we have social contracts that govern our behavior in specific contexts and align with prevailing culture and norms. These contracts enable us to create safe environments for interaction and pathways for building trust in social situations. The Overweb reconstitutes the notion of a social contract on the Web so purpose-aligned communities can have safe digital spaces to meet and interact.

The Overweb enables people to express ideas and perspectives within structures for constructive online conversations and interactions. They provide a safe space for aligned communities to interact, at ease, without having to worry about predators, trolls, or harassment. And participants get rewards for the value they create.

With the Overweb, when people interact, they operate within a new social contract for the digital age—safe digital space above the webpage. Everyone on the Overweb is a real person in good standing. Unidentified bots, fake accounts, and serial abusers cannot take part. In addition, each person's entire history of communications and interactions across profiles stays with their primary account; they're responsible for their digital footprint.

This digital social contract manifests within a social construct called a meta-community. The meta-community enables communities with a common purpose to connect, communicate, cooperate, coordinate, collaborate, and celebrate above the webpage. It has a sovereign layer over the Web that provides a safe digital space for its members. People can meet and interact with like minds, without worrying about predators or trolls. As a sovereign, meta-communities make and enforce their own norms and rules, rather than rely on third parties like Facebook. They cannot be de-platformed.

Our DAO is excited about the potential of meta-communities to support online existing communities like the Good Country and Balajian network states. We believe that creating safe digital spaces for like-minded individuals to connect and collaborate on important issues is crucial for addressing global challenges and creating a better future for all.

The Good Country was aspirationally the first meta-community. British policy advisor, author, and researcher Simon Anholt and American human rights and development practitioner Madeline Hung sought to enable people around the world to shape and implement environmental policy. They wanted to foster enhanced cooperation and collaboration within the international community.

Their audience was the 13% (over 700 million people) who care about international cooperation and collaboration. The Good Country began in September 2018 and at its heyday had several thousand citizens that paid \$5 per year.

Their premise was that humanity could tackle “grand challenges” more effectively if nation-states worked together, supported by a strong mandate from their own citizens. No individual country can resolve complex global issues on its own. But instead of collaborating to face our shared existential threats, countries focus most of their attention, resources, and energy on competing with one another. The Good Country was to shift this dynamic towards cooperation, enabling people to act as citizens of the planet.

In his 2022 book *The Network State*,² Balaji Srinivasan, former CTO of Coinbase, coined the terms “network state” and “startup society.” Balaji offers multiple definitions of the network state. First, his definition in one sentence:

A network state is a highly aligned online community with a capacity for collective action that crowdfunds territory around the world and eventually gains diplomatic recognition from pre-existing states.

This makes sense. Create a new Internet community online, once it grows big enough, materialize it offline, and begin negotiating for status. As Vitalik Buterin noted, anyone could imagine a network state under this definition that they could get behind. Balaji's more complex definition is a long sentence:

A network state is a social network with a moral innovation, a sense of national consciousness, a recognized founder, a capacity for collective action, an in-person level of civility, an integrated cryptocurrency, a consensual government limited by a social smart contract, an archipelago of crowdfunded physical territories, a virtual capital, and an on-chain census that proves a large enough population, income, and real-estate footprint to attain a measure of diplomatic recognition.

The startup society is the beginning stage of a network state. It's a startup that aims to be a country based on a shared purpose rather than geography. Along the way, the startup society becomes a network union; like a trade union but for the network. The network union centers on a social networking app that posts daily actions and crowdsources land IRL. While we love the vision, we prefer community leadership models such as DAO councils rather than a single person.

We also think creating another information silo is an inadequate digital strategy for a startup society. The members of any digital community, including startup societies, would benefit from being able to meet and interact wherever they are online, and from having a shared view so they can come to a common understanding and build knowledge together. We don't think communities should have to go to a siloed site or app to interact or that members should have to worry about fake accounts, bots, or predators.

Like the Good Country and startup societies like Afropolitan, meta-communities emerge from groups of people coming together for a common purpose. This can be Web 2.0 online communities, such as a Facebook group, a community on Circle or Mighty Networks, a Discord server, or any website. They can also be Web 3.0 communities like DAOs, NFT collection holders, network states, Gitcoin applicants, and Giveth projects.

The Overweb's meta-community has governance features that Elinor Ostrom recommends for protecting commons from free-rider tragedies.³ These include clear boundaries on who can be a member, management by members, communities making their own rules, communities monitoring behavior, graduating sanctions for those who violate community rules, cheap and accessible means of conflict resolution, and self-determination.

Communities moderate themselves through their code of honor, which specifies interactions and content that are not allowed. The consequences of violating the code are progressive and may include cessation of the ability to post, contact others, be present, or deactivation of membership. The amount of rogue content, however, is minimal because of the accountability built into the Overweb. Also, entry to the meta-community is a process owned by the community. A good screening process and accountability means less exposure to antisocial and antagonistic behavior. Meta-communities may offer rehabilitation processes, decentralized arbitration (e.g., through Kleros.io, a decentralized arbitration service for new economy disputes), and/or restorative justice for people who run afoul of the rules. It's up to the meta-community.

With meta-communities, communities extend their reach from a single webpage or site to all relevant webpages. Members can meet on webpages of their common interest and connect or interact through content via smart tags. Participants can see the smart tag interactions of members from all their meta-communities.

On any page, members within a meta-community can go visible, signaling to others they are open to connect. They see others attracted to the same content. Who's there and for how long? Maybe even those who were on the page before they came. They can initiate conversations.

Connections based on common interests can lead to unforeseen and serendipitous occurrences like in a café or a library. Those already connected can research together and share their findings, ask questions, and otherwise have enriching conversations. (If a connection turns sour, of course, members can block people from seeing their visibility and communicating with them.)

By providing an on-page presence for people with common interests, meta-communities facilitate conversations that wouldn't otherwise occur. People who don't know each other—but have a shared interest—can connect, share info, learn from one another, recommend, sell, and purchase products and services, become colleagues, date, and more. We look forward to celebrating the first engagement from an Overweb meeting.

On any page, members within a meta-community—visible or not—can interact with content on the webpage. Through on-page interactions, members have access to a set of smart tags (e.g., bridges, conversations) that they can attach to web content. Most smart tags posted within a meta-community are only available to its members. The exceptions are bridges and labeling, which are available across all meta-communities. If one is in multiple meta-communities, they can re-post content among the other communities.

Voting results are the collective voice of the meta-community. Citizens may vote on the nation's strategic direction, how it operates, what projects to support, member preferences, etc. While many governments conduct censuses once a decade, a meta-community can conduct votes whenever needed in hours or minutes. Taking part in paid polls for research centers and projects can generate income for participants and the meta-community.

The Overweb's meta-communities are a network of networks that connect through parent-child relationships, as well as categories, subtopics, and tags. Communities can have channels with their own content filters and serve a subset of the nation's membership. When making a post, participants can tag channels within the meta-community.

The Overweb's meta-community capability supports the following purpose-aligned communities:

- **Decentralized community**—Meta-communities can be part of a comprehensive digital strategy for decentralized communities, such as DAOs, websites, follower and fan groups, advocacy groups, neighborhood groups, discords, slacks, group chats, cohorts, conference and event attendees, and NFT communities. With the meta-community, small, high-trust communities can safely meet and interact over any webpage. NFTs can gate participation in the meta-community. We expect that many decentralized communities will organize their meta-community as a DAO. Single sign-on with an existing community directory is possible, but it prevents interactions with other communities, although individuals could choose to register with the Overweb to work with other communities. Our DAO sees the potential for meta-communities to enable members to effortlessly connect and interact around common interests, between meetings.
- **Network state/startup society**—The Overweb's meta-community aligns with and complements the Balajian network states and startup societies, extending their reach to the entire Web. In fact, a meta-community would round out the digital strategy of any startup society or network state that's not thinking beyond a siloed super-app.
- **MORC**—Massive online research collaboratives (MORCs) are groups that come together around the goal of building knowledge or the information ecology for a specific topic.

- **Data cooperative**—Data cooperatives aggregate the data of interest aligned groups (e.g., people with high blood pressure) to secure and administer deals with companies that want access to the anonymized data of cooperative members.
- **Federation**—Federations are meta-nations or supra-nations that have one or more child communities. As meta-DAO with a focus on regeneration, we can imagine the federation for planetary regeneration having child communities, e.g., one for regenerators in the northern hemisphere and one for regenerators in the southern hemisphere. People in a child nation can see (or filter out) people and posts of the federation vs. their meta-community. When they post, they specify whether it goes to their nation or the federation. Meta-nations and child communities must both approve the relationship.
- **Company/organization**—Digital entities enjoy their own dedicated layer over the Web to advance the aims of a company, organization, agency, project, or organization. The forming entity has control. The entity determines the purpose, who gets access, and moderates the content as they see fit. Examples of a digital entity are a corporation, a nonprofit organization and its network of collaborators, and a government agency and its contractors. Digital entities have a clear hierarchy that makes and enforces its rules. Digital entities can activate single sign-on with an existing organizational directory but, as above, it prevents interaction with other communities.
- **Digital guild**—Digital guilds are small groups of purpose-aligned people collaborating on a project. Guilds have a specific mission and members coordinate to make it happen. Guilds have up to 150 members (the Dunbar number), which, according to British anthropologist Robin Dunbar, is the cognitive limit of the number of people with whom one can maintain stable social relationships.

We expect millions of meta-communities in the next decade. Federations or meta-communities will bring together child communities that share geography, ethnicity, topical interest (e.g., regeneration), spiritual practice, or religion. Meta-communities can amplify the voices, ideas, and products of their constituency. Meta-communities provide a novel organizing structure for communities, projects, causes, mutual aid, support, adulation, and building knowledge.

People who join meta-communities early have the most influence. And the greatest opportunity to earn rewards. The people who start the meta-community—by buying the meta-community domain and building a community—may earn a percentage of the funds raised. Members vote for new members. The founding members determine the economics of membership and may issue a fungible token or NFTs. The percentage for consensus defaults to 50% but can change by vote, similar to other configurations. If disagreements arise, members can fork a meta-community.

Smart Filter

As a systemic solution, the Overweb seeks to:

- Support free speech
- Avoid censorship of legal speech
- Prevent violence and harm
- Allow safe places for expression within communities that have different tolerances
- Prevent information overload
- Avoid types of content and people
- Maintain accountability regarding content and interactions

Thus, the Overweb addresses:

- Who's there and holds them accountable for their behavior
- What you can post and who sees it
- Filtering out what does not fit within personal tolerances
- Ways to see aggregate views of what's available and drill on what captures their interest

The information on the Overweb might seem overwhelming. Hundreds, if not thousands, of smart tags, can attach to a single piece of content. Any page can have many pieces of content with smart tags. But attention-triggering and smart filtering reduce the noise, by screening out content that's not relevant at the moment. Smart tags that make it through attention filtering aggregate into filterable smart tag overviews that enable the participant to drill down according to their interest. When someone's attention approaches active content, an aggregate view of the associated smart tags will pop up. The participant can then filter by media type (e.g., text, video), relationship type (e.g., contradicting, supporting), and smart tag type (e.g., note, bridge) to generate a matching list.

Smart filters enable participants to regulate the content and interactions that come to them. The smart filter is the participant's personal algorithm. They own and control it. The smart filter displays the intersection of the page's smart tag content, the content accessible to the participant, and their preferences. Smart filter settings include whitelists and blacklists for applications and sites, and blacklisted words. They also contain preferred sources, measures of tolerance for violent, sexual, offensive content, tolerance for people with flagged content, and blocks. Participants can change their preferences to try out new settings, maintain a list of sets of settings, and revert to a previous set as needed.

Sometimes you need to see what you're not seeing. You can open the smart filter widget and display the hidden content. Based on Overweb activities, including reactions, flagging, and blocking the smart filter may adjust its setting with permission. It may notice differences in responses to specific content and repulsion to content or people. Here, you could ask the smart filter to show you the additional things it filtered out or in because of the adaptive change it made earlier. You can even configure it to notify you when it filters out certain content.

The smart filter system provides you with detailed explanations of how the filter's decision making and what factors are being taken into account. This allows you to understand how your filter is working and make more informed decisions about the content you see. You can access a breakdown of the filter's decision-making process, including the weighting of different factors and the specific criteria used to filter content.

The smart filter system also includes a feature for community moderation, which allows you to interact with others who share their interests and want to have more control over the content they see. You can create and moderate your own filter groups or moderate for meta-communities, and discuss and share content with other like-minds. This feature can be enabled for specific meta-communities or groups. By providing more options for community moderation, the filter gives participants more opportunities to interact with others who share their interests, have more control over the content they see, and serve their meta-communities.

Another unique feature of the smart filter enables you to create your own algorithms, which can be tailored to your specific needs and preferences. You can choose from a variety of pre-built algorithm templates or create your own using a simple drag-and-drop interface. You can also test and fine-tune your algorithms, and save them for future use. By giving you

the ability to create your own algorithms, the filter empowers participants to have even more control over their content and browsing experience.

Additionally, if you create particularly useful or innovative algorithms, you can monetize them. You can choose to make them available for others to use for a fee, or even sell the algorithm outright. You can set a price for your algorithm, and track its usage and revenue. This feature not only allows participants to earn money by creating valuable algorithms but also encourages participants to come up with new and innovative ways to filter content.

The labeling smart tag enables participants to earn rewards for attaching metadata to content. Some classifications of web content provide data for machine learning to train on. Some labeling data enables smart filtering. For instance, the smart filter may use labels for violent, sexual, or offensive language to filter out disturbing content. Developers and participants can create smart filter configurations and add-ons and receive rewards for the value created in the ecosystem.

Smart Generator

The smart generator is the ultimate tool for the generative web. With this innovative technology, you can effortlessly create personalized content tailored to your unique preferences and needs. Whether you're looking to generate a joke meme, create a recipe, design a desktop background, write a story, or produce a video response to a podcast, the smart generator makes it easy to bring your ideas to life.

Using the smart generator is simple and intuitive. For example, suppose you are browsing a website and come across a photo of a delicious-looking coconut brownie from the Asian Bodega restaurant in Tulum, Mexico. With a few simple clicks, you can use the smart generator to create a recipe for the brownie, complete with detailed instructions, an ingredient list, and nutritional facts.

First, you activate the smart generator by long clicking the photo. This brings up a window with a variety of options, including "Generate Recipe." You select this option, and the smart generator begins to analyze the photo, extracting information about the ingredients and the preparation process.

Next, you are prompted to provide some additional information, such as the desired serving size and any dietary restrictions. The smart generator uses this information to create a customized recipe that is tailored to your needs. You can also make some adjustments to the recipe, such as adding or removing ingredients, or adjusting the cooking time.

After you make the adjustments, the smart generator generates a recipe, and you can now regenerate, remix, edit, and attach it to the photo.

When you're satisfied, you can save it to your device, share it with friends and family, or submit your coconut brownie recipe to a cooking website for others to enjoy. You can even use the smart generator to create a video tutorial showing how to make the brownie, or a NFT out of the recipe, which you can sell or trade.

With the smart generator, creating personalized content has never been easier. Whether you're looking to create written content, a video, or a unique piece of digital art, the smart generator makes it easy to turn your ideas into reality.

The smart generator is not just a tool, it's an extension of your personal AI assistant. It is designed to work seamlessly with you, learning from your activity and preferences to deliver content that is truly unique to you. It's the opposite of the smart filter, which limits your options and curates content based on algorithms. With the smart generator, you have complete control over the content you create, and can fine-tune it to your exact specifications.

One of the key features of the smart generator is its presence. It can be used on any page and any web app, making it the perfect tool for creating content on the go. Whether you're browsing the web or working on a project, the smart generator is always there to help you create and edit content as needed.

Another important feature of the smart generator is its security. Your AI assistant can only access your data with your permission, and it cannot interact with other people or assistants without your authorization. This ensures that your data remains private and secure at all times.

The smart generator is designed to learn from your activity and preferences, adapting to your unique needs over time. It uses advanced machine learning algorithms to analyze your behavior, including the types of content you create, the style you prefer, and the specific features you use. By following your activity, it can train and tune its models to deliver content that is perfectly suited to your tastes and needs.

For example, if you frequently create memes, the smart generator will learn to generate memes that are more in line with your sense of humor. Similarly, if you often create recipes, the smart generator will learn to generate recipes that align with your dietary preferences. This means that the more you use the smart generator, the more personal and unique the content it produces will become.

Additionally, the smart generator also allows you to change models and adjust the settings to get the perfect result. You can choose from a variety of pre-built models, or create your own custom models that are tailored to your specific needs, which you can also monetize. By adjusting the settings, you can fine-tune the generated content to your liking, ensuring that it is exactly what you need. Whether you're looking for a specific style, tone, or set of features, the smart generator makes it easy to create content that is perfectly suited to your needs.

In addition to its many practical uses, the smart generator also brings us into the era of the generative web. With this technology, you can create NFTs (non-fungible tokens) that are uniquely tailored to you and your preferences. Whether you're looking to share a message with a specific audience or create a one-of-a-kind piece of digital art, the smart generator makes it easy to turn your ideas into reality.

MORCS

MORCs, or Massive Online Research Collaboratives, are designed for indirect collaboration through stigmergy, similar to how ants work together in a colony. We coined the term "Massive Online Research Collaboration" or MORC because it's needed. MORCs are the research analog to the mission-driven Massive Multiplayer Online games (MMOs) of Jane McGonigal's February 2010 TED Talk *Gaming Can Make a Better World*. She said if we want to solve our most pernicious problems in the next decade, we need to increase the amount of game playing online. But Esports and streaming emerged in the 2010s, and now reign over the impulse for meaning that might otherwise have led to massive games with massive social impact.

For us, the connection between the gaming market, social MMOs, and measurable social impact was always tenuous. There were some successes. Foldit predicts protein structure by leveraging human puzzle-solving, intuition, and competitive nature to fold the best proteins. But they seem more the exception than the rule.

Yet we continue to envision many thousands working together online on issues that matter. Let's expand the focus on gamifying online experiences that have an impact. The question becomes, what online activities can aggregate into the collective shifts needed in the world?

We're excited about knowledge building through MORCs or Massive Online Research Collaboratives, many of which will operate as meta-communities, DAOs, and/or data co-operatives. Researchers are like game players in some ways. They're on monumental quests, they don't know what's going to pop up, and they grind on with their craft. We think—with effective gamification—humanity can mobilize tens of thousands of researchers and students to build knowledge in common interests.

The MORC enables researchers and stakeholders to build maps of online knowledge that enable coherent collective responses to real-world puzzles. MORCs can grow and experience network effects for both people and content. Especially, if they tap into groups and loose associations of people that are already researching or curious about an issue.

A MORC produces the following outcomes:

- A map that connects online information about the subject and helps filter false news, disinformation, and misinformation.
- The first-generation knowledge base. People can extend the genesis constellation and connect it to any relevant claim or idea.
- A shared context for collaboration and communication.
- An active, connected, and engaged community that builds knowledge together.
- Quicker, faster answers to complex questions and 360° context for any idea.

Emergent information realms for which humanity needs better sense-making—like false news—are candidates for a MORC.

Meta Domains and Adjacent Spaces

Meta domains and adjacent spaces are a novel way of claiming and occupying conceptual space on the Web. In the space above the legacy web, every website has a corresponding meta domain. This includes all top-level domains (TLDs) and Web3 domains. For example, the meta domain for `espn.com` is `espn.com.meta`. Adjacent spaces are subdomains that exist in relation to a meta domain.

Meta domains and adjacent spaces can be used for various purposes, such as advertising, e-commerce, or social interactions. For example, a business may create a community page on the meta domain for their site and purchase adjacent space to a competing site to promote their products or services. An individual may create a social space above their blog site to connect with their followers and another in an adjacent space to a relevant site where they can connect with fans that may be interested in their content.

Meta domains and adjacent spaces are stored as non-fungible tokens (NFTs). Meta domain NFTs convey ownership of the meta domain on the Overweb and the right to deploy content in the space. The owner of the meta domain receives a 10% royalty on the corresponding adjacent spaces. Meta domain NFTs include the anchor address (e.g., `espn.-com.meta`), optionally the name of the meta domain, the minter, the owner, category, topic, and tags. Once minted, the meta domains can deploy through the Presence browser overlay. The meta domain name and owner can be changed after the mint.

Adjacent space domains are meta subdomains that exist relative to a specific meta domain (Figure 12.4). Adjacent spaces are defined using polar or cylindrical coordinates, which use angles to define the relative position of the space. Adjacent space domains use a simplified cylindrical system, for example, the meta subdomain for the southwest of `espn.com` would be `southwest.espn.com.meta` or `sw.espn.com.meta`. Adjacent NFTs include the anchor address, the relative location of the adjacent space (e.g., `sw`), and the name of the adjacent space.

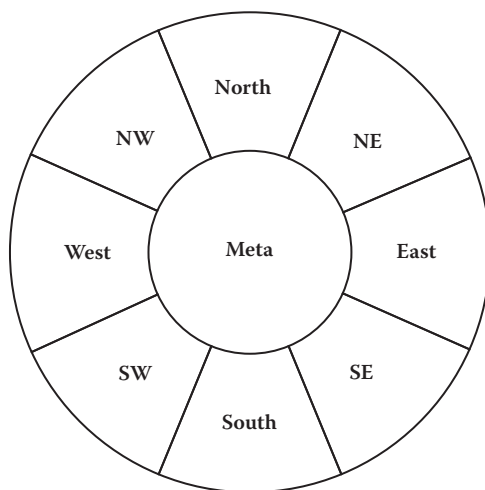


Figure 12.4 Adjacent virtual space around a meta-domain.

Meta domains and adjacent spaces provide valuable context for websites. Imagine browsing a real estate marketplace. You navigate to a home in the southern part of town. When you activate the overlay, it reveals the presence of a meta domain and its surrounding five adjacent spaces in the direction of north, south, east, west, and northeast.

The meta-domain has detailed information about the property, FAQs, a virtual tour, and the ability to chat with the selling agent. The north space is rented by the top real estate company in town and lists all the homes for sale in the town. The northeast space is rented by another agent and lists the properties for sale in the affluent northeastern part of town. The south space does not have any content. It is owned by the owner of the house and is for sale. The west space is a web presence for the top real estate agent from a competing company. The east space is a social space for real estate called the Real Estatement.

Meta domains and adjacent spaces are accessible via various overlays, such as APIs, browser extensions, and mobile apps. They are also viewable through specific overlay applications. Spaces can also have a presence on the flat web that is addressable via theoverweb.com.

By activating the overlay feature, you can view meta domains and adjacent spaces for any website. The virtual space owners can build virtual spaces as webpages or entrances to virtual worlds using the overlay's virtual space deployment tools. The owner can configure their space to allow participants to add smart tags to content and visit the canopi. If the meta domain or adjacent space is accepting offers, the visitor can make a bid or purchase the space directly.

OVERWEB REWARDS

The Overweb is committed to building a decentralized ecosystem that fairly compensates contributors. Participants can earn rewards for the value created by their contributions. The primary way to earn rewards is by building bridges that are crossed and upvoted by other users. Compared to social media, this system is much less susceptible to undue influence, as everyone only has one account, and you have to cross a bridge to upvote it. People can also earn Overweb rewards for labeling content and for watching advertisements. Some meta-communities may also give rewards based on engagement.

Builders can also earn rewards for contributing to the development of the Overweb ecosystem. For example, developers can earn rewards for creating new smart tags or algorithms or for contributing to the development of the Overweb protocol.

As the ecosystem evolves, there will be other ways to earn rewards, such as bounties and challenges, that reflect the ecosystem's needs. For example, participants may earn bounties for creating bridges in a certain topic or for labeling a certain type of content. Also, challenges may reward the best bridges for a specific question or topic.

The Rewards Token

To support the reward system, the Overweb has developed a comprehensive plan for the distribution and governance of a rewards token. This plan is designed to ensure that everyone who participates in the growth and development of the Overweb is fairly compensated for their contributions. Given the uncertain regulatory environment, the plan may require modification prior to launch. We believe that a collaborative and community-driven approach is the best way to ensure that modifications are beneficial for all stakeholders.

The Overweb (OWEB) token is a dividend-bearing asset that reflects the value of the Universal Content Graph and the activity on the Overweb ecosystem. The token will be distributed through a fair launch, the timing of which will depend on when engagement milestones are reached. The total supply of OWEB is 1 trillion, with 25% allocated to the fair launch, 25% allocated to the Automated Market Maker (AMM), 49% locked in the DAO's Treasury, and 1% allocated to development and marketing.

The AMM is a key component of the Overweb ecosystem. The AMM provides liquidity for OWEB and enables efficient trading of the token. The AMM is initially funded through the fair launch and the reflections mechanism ensures that liquidity is always being added to the AMM, which helps to stabilize the price of OWEB.

The fair launch is an important aspect of the OWEB ecosystem as it ensures a fair and equitable distribution of the tokens. By allocating tokens based on the relative value of native coin (or stable dollar) contributions, the fair launch ensures that everyone pays the same amount for their tokens. The fair launch will be conducted through a crowd-pooled token offering, where participants allocate a native coin (or stable dollar) to the fair launch. At the end of the fair launch, the total amount of native coin allocated is added to the AMM, and all fair launch participants receive OWEBs at the same price.

The OWEB token has a reflection mechanism, where a 10% transaction fee is charged to holders whenever the token changes hands, with 5% reflecting back to all holders based on their relative holdings and 5% going to locked liquidity in the AMM for the OWEB/native coin pair. This ensures that liquidity is always being added, solidifying the token's price over time.

As reflections occur, liquidity is added to the DAO's locked liquidity, which helps to recoup the 1% allocated to development and marketing. This ensures that the DAO reaches and maintains 50% locked liquidity, which is a key component of the OWEB ecosystem as it provides a stable source of revenue for the DAO to manage. The DAO's 50% locked liquidity ensures that the DAO will receive 2.5% of all volume movement (50% of the 5% to holders). The DAO manages reflections that accumulate beyond the 50% locked liquidity. Of these, one half is allocated for rewards and the other half is for operations.

AMMs, Liquidity, and How It's Secured

Simply put, liquidity is a pool of funds that enables investors to buy and sell instantly. Without this pool, investors would have to wait for someone to match their buy or sell order and there is no guarantee that the trade will be completed at all.

Automated Market Makers (AMMs) on decentralized exchanges such as Uniswap and PancakeSwap, use a mathematical formula to calculate the price of a token based on the amount of liquidity in the pool. The more liquidity in the pool, the easier it is for investors to buy and sell the token. Liquidity is created by pooling in the new token along with another token of established value (e.g., ETH or stablecoin like Tether) in an exchange like Uniswap or PancakeSwap. This pool of funds gets deposited in the exchange and the liquidity provider receives liquidity pool (LP) tokens in return, which can be used at a later point to withdraw from the pool fund.

Let's say someone wants to buy 10 tokens of XYZ, and the current liquidity pool for XYZ tokens on the AMM has 100 XYZ tokens and 10 ETH. The current price of OWEB token is based on the ratio of ETH to XYZ in the pool. Because there are ten times as many XYZ, the value of XYZ is one-tenth of an ETH, or 0.1 ETH. Purchasing 10 XYZ will cost 1 ETH.

To calculate the new price of XYZ after the purchase, the AMM uses the following formula:

$$\text{New ETH Pool} = \text{Current ETH Pool} + \text{ETH spent by the investor} * (1 - \text{fee})$$

$$\text{New XYZ Pool} = \text{Current XYZ Pool} + \text{XYZ bought by the investor}$$

$$\text{New XYZ price} = \text{New ETH Pool} / \text{New XYZ Pool}$$

so in this case:

$$\text{New ETH Pool} = 10 + 1 * (1 - 0.3\%) = 10.997$$

$$\text{New XYZ Pool} = 100 - 10 = 90$$

$$\text{New price} = 10.997/90 = 0.122 \text{ ETH}$$

The buying pressure has increased the price of XYZ tokens from 0.1 ETH to 0.122 ETH. Sales of XYZ will make the price go down. The 0.3% fee goes to the holders of the LP token. This is how the liquidity pool adjusts itself after each transaction to keep the price updated and fair, while compensating liquidity providers.

If liquidity is unlocked, however, then the token developers can do a "rug pull" as discussed in Chapter 3. Once purchasers start buying the token from the exchange, the liquidity pool accumulates more and more coins of established value (e.g., ETH or Tether). Each purchase sends valuable tokens to the AMM to get the new token. If not locked, developers can withdraw this liquidity from the exchange, and run off with it.

Liquidity is locked by renouncing the ownership of liquidity pool (LP) tokens for a fixed time period, by sending them to a time-lock smart contract. Without ownership of LP tokens, developers cannot get liquidity pool funds back.

It's worth noting that AMMs are a new way of creating liquidity for tokens, and have become increasingly popular among token creators because they are truly decentralized, eliminating the need for an order book with a matching offer, and enabling more efficient trading.

To harden the protocol, the Overweb Council is considering ways to add buying pressure to the protocol including the issuance of a stable coin for transactions on the Overweb that holds a portion of its reserves in OWEB, distributing advertising revenues in OWEB, and/or other revenue streams that can be distributed in OWEB.

The OWEB also has a governance component where holders of the token can vote on proposals to direct the direction and development of the Overweb ecosystem. This decentralized autonomous organization (DAO) structure allows for community participation in decision-making and ensures that the ecosystem is aligned with the vision and values of its stakeholders. Votes can be council-only, token-weighted, or one-vote-per-person.

Earning Rewards

The rewards system compensates bridgers who contribute verified bridges to the Overweb Bridge Registry with OWEB based on the relative value of their bridges to the ecosystem. This incentivizes the creation and maintenance of high-quality bridges, which are essential for connecting the different parts of the Overweb. Bridgers are also compensated for curating bridges and watching advertisements.

Half of the DAO's reflections (above the 50% locked liquidity) are paid out as rewards on a regular basis. These reflections accumulate in a rewards pool. The reward pool may also generate revenues from other sources such as API fees to access the Overweb's Universal Content Graph. Participants receive OWEB from the rewards pool based on the relative value their bridges contributed to the ecosystem, the value of their curations, and the amount accumulated in the rewards pool in rewards period. The final tokenomics will specify the frequency and timing of rewards.

The bridging rewards pool is split between the bridge reward pool (80%) and the curation reward pool (20%). (In the future, labeling content will also receive a portion of this pool.) A participant's portion of the bridge reward pool for a given period is based on their proportion of unique bridge crossings accompanied by upvotes for the period and the amount in the rewards pool. Bridgers also earn curation rewards for upvoting bridges before they get popular. The curation pool for bridges will be distributed to the first 20% of the people who upvoted a bridge that has valuable activity during the rewards period using an exponentially decreasing function.

The Overweb recognizes that bridges are not built in a vacuum, but rather in an ecosystem that supports and amplifies their reach. As such, when content earns rewards, 70% goes to the creator, and 30% goes to the ecosystem that supported its creation and distribution.

As shown in Table 12.1, the ecosystem share is split 10% for community affiliations, 10% for context indexes, and 10% for referrers. The bridger designates the 10% for community affiliation to meta-communities and guilds they are a part of, or causes they support. Some meta-communities and guilds may require that members assign a percentage of rewards to them in general or for specific content. The context indexes are fixed portions that go to tags, category, and topic of 6%, 1%, and 3%, respectively. A 9% referral reward goes to the persons who brought the bridger into the system and 1% goes to the protocol enabler (e.g., the browser, extension, or SDK holder that enables access to the overlay).

Table 12.1 Ecosystem rewards

<i>Recipient</i>	<i>Allocation (%)</i>	<i>Total</i>
Affiliations		10%
Meta-communities & guilds	10%	
Context Indices		10%
Topic	3%	
Category	1%	
Tags	6%	
Referrals		10%
L1	3.5%	
L2	1.5%	
L3	1%	
L4	0.75%	
L5	0.50%	
L6	0.375%	
L7	0.25%	
L8	0.125%	
Browser	1%	
Total		30%

The ecosystem distribution rewards meta-communities and guilds, referrers, and protocol enablers for their contributions, while creating a futures market for context indexes for specific tags, categories, and topics. For example, the meta-community for Belgium Brewers gets a percentage of the rewards generated by bridgers in their community. The rewards for the context indexes accrue to metabags NFTs based on the associated network activity. For example, the topic “asthma” gets 3% of the rewards generated by bridges assigned to the asthma topic.

To generate interest in the metabags, the Overweb will publish statistics on context indexes. This will also enable participants to identify the most active areas in the ecosystem and focus on them to earn rewards.

Contextual Advertising

In the future, the Overweb will offer contextual advertisements. Participants will have to opt-in to seeing advertisements and will receive value for doing so. When ad payments come in, the funds will be used to purchase OWEB (thereby hardening the protocol), which will be distributed as rewards. The ad click rewards will be split evenly with 20% each for the owner of the bridge that led to the ad clickthrough, curators of the bridge, prizes for ad watchers, the website owner, and the Overweb DAO. If a bridge was not involved, a portion of the rewards will be distributed to the link referrer.

THE WEB4 FOUNDATION

The Web4 Foundation is a non-profit organization dedicated to building the governance and infrastructure of the Overweb. The foundation’s mission is to create a decentralized and open web that is accessible to everyone, while also protecting the rights and privacy of its

members. The Web4 foundation operates as the Overweb DAO. The DAO generates operating funds from sale of meta domains, meta-bags, and reflections from OWEB transactions.

The Overweb council is responsible for guiding the DAO and the ecosystem towards progressive decentralization in order to achieve optimal efficiency, effectiveness, and inclusivity. This is accomplished by tapping into the collective wisdom and input of the community as appropriate, allowing for a more transparent and collaborative approach to decision-making. The council's ultimate goal is to ensure that the Overweb ecosystem is secure, scalable, stable, beneficial, and fair for all stakeholders.

One of the key elements of the foundation is working groups focused on different areas of the Overweb such as identity, security, privacy, data sovereignty, and rewards. These working groups are made up of experts and enthusiasts from various fields, including information technology, Web3, emerging tech, AI alignment, tokenomics, strategy, product, marketing, human resources, business development, law, policy, social sciences, and governance. They work together to develop and implement the building blocks of the Overweb, such as smart tags, and the Overweb protocol.

The foundation is committed to protecting the rights of Overweb participants through the Overweb Bill of Rights. This living document outlines the basic principles and rights that all participants on the Overweb can expect, such as privacy, security, and freedom of expression, and explains how they are protected on the Overweb.

The Web4 Foundation also supports the open-source community that is working to develop the Overweb. By collaborating with developers, researchers, and other stakeholders, the foundation aims to ensure that the Overweb is inclusive, decentralized, and accessible to everyone. The foundation also maintains the bridge registry and registries for meta-communities, meta domains, and adjacent space.

If you are interested in learning more about the Overweb and how you can get involved in the foundation's work, please visit webfour.foundation or reach out to the foundation directly. We welcome people with specific expertise and interest in the next level of the Web.

As we delve deeper into the potential of the Overweb, it's important to also consider the challenges and problems that the current web is facing. From scams and fraud to abuse and misinformation, the issues on Today's Web are not only prevalent but also complex and intractable. But what if the Overweb could offer a solution?

In the next chapter, we will explore how the decentralized and open nature of the Overweb can address these issues head-on and pave the way for a safer and more trustworthy online experience for all. From one-account-per-person, overlay apps, and smart tags, to meta-communities, we will examine the various mechanisms and structures that can tackle these issues and how they differ from today's centralized web. So get ready, as we dive into the nitty-gritty of how the Overweb is working towards a better, more secure, and reliable Internet for all.

NOTES

- 1 <https://permanent.link/to/the-metaweb/next-generation-internet-award-winners>
- 2 Available free at thenetworkstate.com. The site lists aspiring network states.
- 3 A portmanteau of digital and citizens.

Fixing the Web

Where are the so-called Internet futurists? How did they miss the Metaweb? We obviously can't fix a flat, static web of silos without a meta layer.

—Daveed Benjamin

This chapter speaks to how the Metaweb, and in particular the Overweb, addresses problems that seem intractable with Today's Web, including scams, abuse, and false information. Certainly, the Overweb will solve many other problems, but we will focus on how we can make the Web a safer place with a meta-layer. Mark our words, these problems are unsolvable without a new level of thinking. In less detail, we also explore the transition to a value-based economy that respects cognitive freedom and cultivates collective intelligence.

Please do not allow the brevity of this chapter to diminish your perception of the Metaweb's possibilities of fixing the Web. This chapter is short because we lean into two powerful aspects of the Overweb that are absent on Today's Web: accountability and metadata. Having accountability and access to metadata can play a crucial role in addressing a variety of online issues.

With accountability, people act better because they know they are responsible for the repercussions of their actions. Access to metadata helps us gain valuable insights into patterns and trends, which improves our ability to make informed decisions and solve specific problems. Through metadata analysis, we become more strategic and effective in our approach to various challenges. Together, accountability and metadata are a one-two punch that can enable us to transcend the Web's greatest challenges.

STOPPING SCAMS

The Internet is where people find information about financial services. According to the Edelman Trust Barometer, financial services are the least trusted sector. Common sense says trust and confidence in financial services are integral to the safe, stable, and efficient day-to-day workings of the financial system.

To increase trust and confidence, the adverse publicity caused by the "mischievous minority" of bad actors in financial services needs to stop. Similar to misinformation, the best response short of taking down fraudulent services (which can be difficult and time consuming) is providing context to upgrade the public's ability to judge quality and veracity. The occurrence and impact of financial malpractice, malfeasance, and misconduct would diminish if potential customers had rich, contextual information, if not explicit warnings before clicking through.

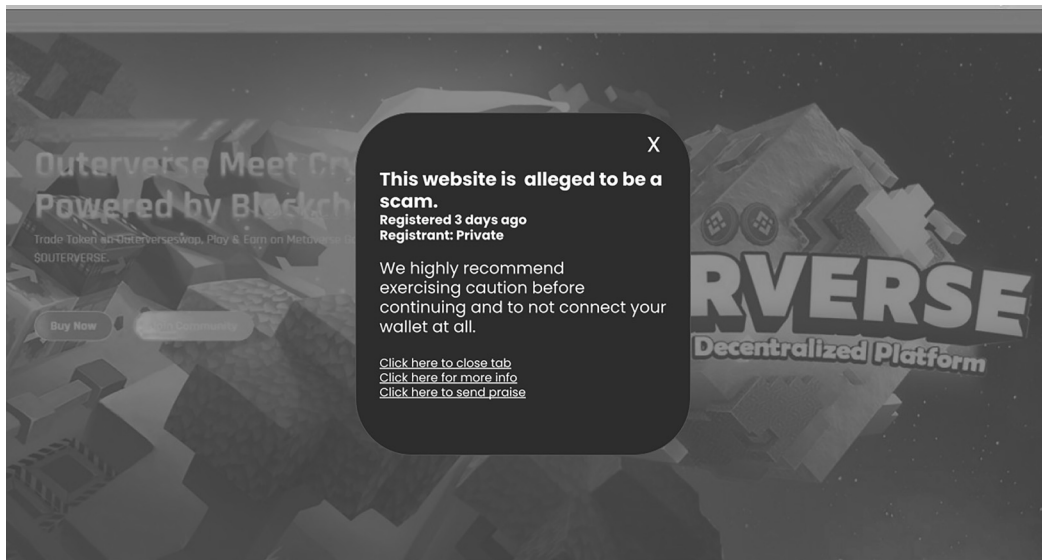


Figure 13.1 How to stop scams.

This could include identifying the “mischievous minority” and their fraudulent services, as well as actively discouraging people from enrolling in fraudulent services by providing contextual information, such as authentic comments and ratings, trust metrics, and in-depth information. Potentially damaging sites can be “marked” with bridges, notes, and other smart tags or with posts on the site’s canopies.

Figure 13.1 shows a future overlay app that prevents people from transacting with scammer sites, in this case, a fake NFT mint site. When the visitor arrives at scammer sites, a modal pop-up shows this is a previously identified scam similar to the warning when a user attempts to access a site with an invalid SSL certificate. It may provide information such as the registration date and the owner. It might disable, remove, or cover up the connect wallet button as well.

Stopping scams requires an up-to-date list of scam sites and a robust verification process. Anyone should be able to submit a potential scam. Once verified to be a scam by an approved expert, the scam goes on the public list. A fair value exchange approach would be to reward people who submit verified scam sites based on the number of people who benefited and/or donations. The overlay could also evaluate dApps and display their registration date and owner. It could further alert the visitor if anything seems suspicious and provide tips for figuring out if they are dealing with a fake.

The Overweb supports trust and veracity-related ratings for content, companies, services, products, and IP and wallet addresses. It also enables participants to tag content, financial services, and providers. For example, the crowd could tag a suspicious financial service, a scammer’s avatar, a fake NFT minting site, or a counterfeit NFT. This would alert the appropriate regulatory body and mark the site. The Overweb can also support AI in identifying fraudulent services and sites, though we would always want a human in the loop.

CURTAILING ABUSE

The web is a dangerous place, with many abusers actively seeking their next victim, including predators, trolls, catfishes, impersonators, and cyberbullies. These parties run free

because structural elements in Today's Web prevent accountability on the Web. Social accounts don't require specific information that binds the account to the person. Social platforms only require an email address or a phone number to register, and they don't prevent impersonators from spinning up duplicate profiles. Multiple phone spoofing services offer untraceable phone numbers. You don't know if an account is genuine or a predator. Parts of the Metaweb may also have this problem.

The Overweb, however, has no fake accounts, unidentified bots, or serial abusers masquerading as normal people. Anyone you encounter is in good standing. Were they not in good standing, they wouldn't be visible. Because all pseudonym accounts tie to a primary account, there is accountability across personas.

If someone's account gets deactivated, their entire account and all their personas get deactivated and they will need to go through a rehabilitation or restorative justice process to get reactivated.

NEUTRALIZING FALSE INFORMATION

Today's fact-checking apparatus is insufficient.

Hundreds of organizations, multiple alliances, and millions of people work on fact-checking across borders, languages, political perspectives, and priorities. Yet, it takes hours from claim to fact-check and few see both the false claim and the fact check. Research shows false claims are more viral on social media than authentic information. This virality of inauthentic content is driven by confirmation bias, attraction to novelty, and the abundance of fake accounts and bots on the Internet platforms.

The entire information environment from news to social media is vulnerable, as evidenced by the "infodemic" that accompanied the coronavirus:

An overabundance of information—some accurate and some not—that makes it hard for people to find trustworthy sources and reliable guidance when they need it ... In the information age, this phenomenon amplifies through social networks, spreading farther and faster like a virus.¹

Pew Research findings from their Future of Truth and Misinformation Online interviews suggest tech providing context can play a role in infodemics. A common theme was "the information environment will improve because technology will help label, filter or ban misinformation and thus upgrade the public's ability to judge the quality and veracity of content."²

Wintergreen Research agrees. WinterGreen Research, founded in 1985 by Susan Eustis, provides strategic market assessments in technology, communications, software, Internet, and advanced computer technology. They produce industry analysis, market growth predictions, and significant opportunities for new markets.

Wintergreen wrote the first Internet study in 1995, predicting 100 million users within 5 years. The telecommunications companies that bought the study invested in the Internet, laying the foundation for the prediction to come true. Early in the cell phone market, analysts predicted a maximum 3% penetration in the US and 6% in Europe. Susan predicted 95% penetration in 5 years, which was what happened.

WinterGreen recently published a report called "Overweb: Market Shares, Strategies, and Forecasts, Worldwide, 2021–2027."³ They estimated the value of services to address false

information to be \$4.5 billion by 2027. This is an annual growth rate of 27%. They suggest the Overweb provides a way to address the pervasiveness of conspiracy theories driven by social media.

Citizen Fact-Checking

Dr. Leticia Bode is a Provost Distinguished Associate Professor at Georgetown. In a WHO Infodemic Management consultation, Bode says social media can be an effective conduit for misinformation correction. She says that expert organizations are the most effective correctors. She pointed out that the platforms can also be effective correctors and that users can correct one another. Regarding user-directed corrections, she points out multiple users can achieve the same effects as platforms, but they must include a link to an expert organization. Sounds like a bridge.

We foresee an online community of citizen claim-checkers that work with AI and bots to coordinate corrective actions on social media while building a robust information ecology around matters vital to humanity. The fact checkers validate incoming threats, AI-generated response plans, and resource deployments. Participants earn rewards based on the value of their contribution. This brigade of citizen fact-checkers coordinated by AI helps:

1. Scale and improve the effectiveness of independent fact-checking.
2. Augment the fact-checking of fact-checkers, expert organizations, and social media, which are overwhelmed and/or ineffective.⁴
3. Bring fact checks to the false claim, the poster of false claims, and their networks.

Citizen claim-checkers are well-informed citizens who care about democracy and the integrity of news and are active on Twitter and/or TikTok. They're in the 64% of US adults who've seen coronavirus news or information that seemed made up. They can help the 50% of Americans who said it's difficult to discern what's true about the virus.⁵ Many are already fact-checking others on social media, correcting false (and sometimes true) claims. Their current lack of coordination, however, means there's no consistent coverage, content, approach, or results. The best practice is linking to expert pages alongside messaging that meets the errant poster where they are at. The worst is no link or an unhelpful link with an off-putting message.

CitizenFact—a would-be meta-community—provides an organizing structure for citizens who care about democracy to rally around. CitizenFact is a non-hierarchical, purpose-aligned community that aims to build a rich and robust information ecology for the digital town square. Starting as a landing page with a waiting list, an active community grows and a DAO forms around the CitizenFact overlay app, which is active over the entire web. The app identifies misinformation and false news on TikTok, Twitter, Instagram, and YouTube, and coordinates responses of fact checkers and bots.

The human-AI collaboration mediates the spread of fake news by identifying, flagging, and responding to false claims in the news. People can see flags when viewing the content. Anyone can flag false information, and, once verified, a false claim flag will appear next to the post. Because the network helps find falsehoods, the speed of identification increases with network size. The tool also makes fact checks easier to find and digest by presenting them next to false claims. More people will see the fact checks.

As news breaks and posts on social media sites, the app uses AI to analyze new stories, detect false claims, and compare them with fact-check databases. Depending on the threat, it

alerts the media, fact-checkers, and citizen fact-checkers. The AI also reviews breaking content for false claims and proposes bridges and fact checks that connect evidence to claims.

Citizens get rewarded for being first to flag misinformation, imposters, bots, fake accounts, scams, and outdated recommendations. Or, the AI flags the story or specific content as needing a fact check. The potential threats go to the verification queue. Questionable claims trigger alerts to fact-checking organizations and media companies. Citizens can also earn rewards for helping craft threat response recommendations. Response time to false news could be reduced from 13 hours to under an hour.

The overlay app enables citizen fact-checkers, journalists, media companies, fact-checkers, and NGOs to coordinate on fact-checking and steering people toward trustworthy information. Citizens create bridges, notes, and fact checks that connect evidence to claims. They also correct false claims in replies and posts on social platforms. Their activity generates an up-to-date layer of contextual information that enables participants to see fact checks next to claims in real time. They earn rewards for the value of their contributions.

People with access to the Overweb can see flags when viewing the content. As the participants browse social media, the overlay tracks their attention. If related information exists near the focus of their attention, the content displays a badge. Hovering over the badge displays an overview of the fact-checks and bridges available for that piece of content. When participants click on a fact-check or bridge, the containing page opens with the content highlighted.

Underlying the enterprise is a false news and fact-checking knowledge graph that connects the evidence to claims. This is part of the aforementioned Universal Content Graph.

Such a graph would require an unprecedented level of sharing, integration, and synergizing among the news and fact-checking communities. The Overweb could provide an open, universal, API-accessible, claim-checking graph that includes false claims, fact checks, and context. Anyone could contribute. All submissions would undergo an open validation process, and anyone could access the data.

Many types of commercial entities could use such information, including media companies, Internet platforms, fact-checking organizations, and consultants. Assuming API access fees for commercial use, the knowledge graph could pay for itself. And reward contributors based on their contribution to value.

A claim-checking knowledge graph would:

- Connect online false claims and fact-checking research.
- Provide early warnings and notifications of false claims and misinformation.
- Improve building, discovery, and access to fact-checking knowledge.
- Enable tools for professional and independent correctors.
- Encourage evidence-based dialogues among diverse thinkers.

BUILDING TRUST IN POLITICS THROUGH CONTEXT

Imagine bringing the transformative power of maps to policies and political issues, which are interdependent yet often treated separately. Everyone has different and limited views of the relevant information ecology, which inhibits collective insight, collaboration, and learning.

A map for political issues would reveal relevant connections and patterns in the information ecology and provide a shared context for communication and collaboration. The

map would self-generate from the activities of researchers, analysts, activists, advocates, and their allies using smart tags that attach to content on webpages.

Such a map would provide access to deep layers of context for political ideas, which rarely have corresponding evidence. News consumers could cross bridges that connect online information about issues, navigating from idea to idea, and enabling a trans-contextual experience of the information ecology. This would increase the discoverability of cohering information, which could lead to unforeseen breakthroughs.

When we map our research, back up assertions, fight misinformation, and defend the truth—which takes seconds to minutes using an overlay—we create a daily habit of civic engagement. We would expect an increase in civic participation to carry over to the ballot boxes and catalyze a new wave of civilian leadership around critical public issues.

By increasing knowledge and transparency in media, access to political information ecologies could help citizens make better decisions about public law and governance, thus aligning the government and the people. The overlay would also enable journalists and citizen fact-checkers to set the record straight wherever needed.

This would be especially important in election cycles. When we provide evidence in political conversations, we support grounded perspectives around critical societal issues. By helping us to better understand our collective problems, the Overweb can empower citizens to identify the best paths to progress and provide a shared context that could help make those solutions an electoral reality.

ATTENTION ECONOMY TO VALUE ECONOMY

With Web3, the transition to a value economy is underway. Web3 aims to reward creators for the value they create in systems. Yet, most people online, around 4.5 billion people, are still engaging in the Attention economy via Facebook, Twitter, Instagram, and TikTok.

The Overweb provides a parallel system—above the webpage—where participants earn the value of their contributions to the ecosystem. They own, control, and monetize their activity data and creations (e.g., bridges that many people cross and upvote). This could also evaluate and refine the state of knowledge about a specific subject. Because they are operating within meta-communities and have just one primary account, inappropriate behavior will be minimal.

Like Web3 protocols, there will be many ways to create value. We foresee overlay apps that compensate participants for applying smart tags that label content on webpages for machine learning and filtering.

In the section above, we mentioned the WinterGreen Research report that predicted a \$4.5 billion market for services that address misinformation and disinformation by 2027. This is a huge market for just one business area of the Overweb. When we think about the entire ecosystem, we see the possibility of a multi-trillion dollar market for metadata, without data-based advertising.

COGNITIVE FREEDOM

Several aspects of the Overweb support cognitive freedom. First, there are no unidentified bots or fake accounts creating false virality. Second, instead of just seeing what the page author of the algorithm wants us to see, we have access to 360° context for any idea on the web. Third, participants have access to transparent and tunable smart filters, which limit what they see as they choose.

By controlling what we see, we wrest control from those who would manipulate us. As our awareness increases, we can take back control over our lives. We're interested to see how the Overweb connects with the Cognitive Security and Education ecosystem⁶ in the future.

COLLECTIVE COGNITIVE CAPABILITIES

Bridges and smart tags such as notes, conversations, and polls that attach to content on webpages make thinking visible to anyone with access to the overlay. Together, the bridges, the other smart tags, and the Universal Content Graph into which they self-assemble create a shared contextual view. This enables people to think, learn, and build knowledge together at scale, which is foundational for developing collective cognitive capabilities. Rather than the particular perspective of the webpage author, everyone has access to all the information.

To help people make sense of web content, there will be overlay apps and smart tags for argument mapping. Extending existing argument mapping tools, they will connect relevant analyses to online articles, inquiries, and claims. This could include mapping the key inquiries, perspectives, claims, evidence, and data sources represented by online conversations. This helps us examine assertions and arguments made online to see if they hold water. Several organizations, including Society Library, Public Editor, and Canonical Debate Lab, are developing argument mapping platforms.

In robust information ecologies, conversations consider contextual information, including evidence attached to claims. This leads to evidence-based conversations. Argument mapping can further develop our collective intelligence and collective cognitive capabilities, such as sensemaking, meaning-making, and choice-making.

To reiterate, all experts seem to agree that Today's Web has zero solutions for online scams, abuse, and false information. The bad guys are always a step ahead of the police, the fact checkers, and the platforms. The former innovates and the latter three play catch up. People rarely get caught or do time. Yet, as this concise chapter corroborates, we could mitigate if not solve these seemingly intractable problems were we to widen our aperture to include a meta-layer over the web.

By making thinking visible above webpages and rewarding for value creation, we can create a next-level economy that makes scams, abuse, and false information a problem of the past. The antecedent for this approach is the "strong browser" construct (discussed in Chapter 8) that one of the world's top VCs attempted in the mid-nineties. But Microsoft's monopolistic offensive thwarted Andreessen in the mid-1990s. When meta-capable browsers finally become widely distributed, it could not only put our wicked web problems behind us but also generate context and knowledge, shifting us away from full immersion in the Attention Economy, while supporting cognitive freedom and developing our collective cognitive capabilities.

But the potential of the Overweb extends far beyond just solving the problems of Today's Web. As we step into the next chapter, we will discover how the Metaweb extends far beyond the Web. Not only an online tool, it bridges the online world, virtual worlds, the physical world, and even the conceptual realm. It has the ability to act as an overlay, seamlessly connecting everything together and revolutionizing the way we interact with the digital world, and the world around us.

The next chapter promises to be an exciting journey, where we will uncover the true extent of the Metaweb's capabilities, and how it can connect and enhance the online world, virtual worlds, the physical world, and the conceptual realm. Join us as we explore the boundless potential of the Metaweb and experience how it can shape our future.

NOTES

- 1 <https://permanent.link/to/the-metaweb/infodemic>
- 2 <https://permanent.link/to/the-metaweb/pew-research-misinformation>
- 3 <https://permanent.link/to/the-metaweb/overweb-market-shares>
- 4 <https://permanent.link/to/the-metaweb/washington-post-coronavirus-misinformation-snopes>
- 5 <https://permanent.link/to/the-metaweb/risk-details-coronavirus-fake-news>
- 6 <https://permanent.link/to/the-metaweb/cogsec-collab> and cogsec-collab.org

Into the Looking Glass

We cannot change the external circumstances, but we can change how we relate to them.

—Pema Chodron

Into the Looking Glass, where everything is not as it seems. Where a mere shift in perspective can reveal new dimensions and endless possibilities. This is the realm of the Metaweb, a world that beckons us to explore beyond the veil of Today's Web.

Pema Chodron's wise statement rings especially true for the Metaweb. We finally have the power to shape our online experience and make it reflect our values and beliefs.

Think of the Metaweb as a looking glass, a lens through which we can view the online world. It transforms unstructured online content into a construct that can be addressed and interacted with. It gives us access to a wealth of contextual information and interactions, all at our fingertips. What we see depends on our filter settings, shaped by our tribal affiliations and personal preferences.

So let us step into this world, and see what wonders and discoveries await us. Let us embrace the Metaweb and experience the hyper-dimensional web.

Recall the earlier discussion about the web cake. The bottom level is Today's Web, and the top is the Metaweb. The Metaweb is your personal lens for the online world. At the Metaweb level, you can see artifacts attached to content on webpages.

Remember how we angled the web cake like a laptop screen? Think about the bottom level of the cake—the webpage content. As your attention moves among the content, you have on-demand access to 360° of contextual information and interactions.

Welcome to the hyper-dimensional web. The Metaweb turns unstructured online content representing a claim or an idea into a construct that is addressable as a location in the Metaweb. Anyone can attach information and interactions. What someone sees depends on their tribal affiliations and personal filter settings, but the data are there—everything related to the claim or idea.

REALITY SWITCHING

We're about to throw you a curveball. Some of you may have already been expecting it.

Now imagine grabbing the sides of the Metaweb level, and ripping it away from the content level of the web cake. Hold it up so that you can see through it and scan your environment. Imagine now that, through the cake, you can see 360° of contextual information and interactions associated with items you focus on in your environment. Focus your

attention on something in your environment, and imagine everything that anyone might want to attach to it. The bottom level of the cake is now your physical reality.

Living with Context by Daveed Benjamin

I'm in my living room. I focus my attention out the window. The overlay tells me it is 27 degrees centigrade with winds of 8–16 kilometers per hour and the time is 17:09. I toggle the display to 80.6 degrees Fahrenheit, winds of 8–10 miles per hour, and the time being 5:09p. I can also request historical temperatures or the forecast for the upcoming week from various sources. My sandals are near the door. I got them last year before a trip to Amazonia, and they have served me well. The overlay gives me options to see the 3D model; info about its components, the supply chain, and the manufacturer; and the list of shops that have my size available and their distance from me, ordered by proximity.

I focus on the mango on the plate on the table. The overlay shows a 3D cross-section model of the mango and its nutritional information. I can choose options to display health benefits, an article about mangos improving sleep, song references, recipes, and cooking shows featuring mangos. Also, NFTs with mangos, a list of nearby places to get mangos, a one-click order button, a poll on favorite levels of ripeness, upcoming meetings about mangoes, and more. Everywhere I focus, I see possibilities of information and interactions.

Now strap the cake onto your head like a big pair of glasses. The bottom level of our cake is now virtual reality. A DAO is performing an animated reenactment of a historical event in the Abyssinia metaverse, which takes place in the Tigray region of Ethiopia.

As you walk the town of Adwa, a young soldier hands you an invitation from 19th-century Emperor Menelik II and Empress Taytu Betul. Activating the more-info icon next to Menelik displays metadata, including his battles, wives, children, and songs about him. Registrations are happening on the Overweb for a dinner conversation next week about Menelik and the Age of Princes. When you arrive at the emperor's castle, you see that metadata exists for almost everything in your visual field, even the injera on the table. The injera bridges to info from the Web about its ingredients and health benefits. It also bridges to other places in this and other multiverses. The farm that produced the teff. The processing "factory" where it became flour. The kitchen that made the injera. You can jump into experiences for the farm, the factory, and the kitchen to learn about the traditional process of making injera from start to finish. You can also focus on other objects in the environment to see what metadata and experiences they hold.

Flashback to the physical world. The bottom level of our cake becomes your physical reality. Through the Metaweb lens, you can access information and interactions associated with places, objects, and visuals in physical reality. In the grocery, for products on the shelf, you have access to ingredients, nutritional information, personalized warnings, and dangerous combinations. In the streets, you can access metadata about murals, signs, and geolocations. Perhaps, an artist's statement, a time-lapse of the artist's painting, or augmented reality enhancements. This could also happen in an art gallery.

Switch realities to a hyper-dimensional graph of the connections between semantic objects that represent concepts, including inquiries, perspectives, claims, evidence, and data. The bottom level of our cake is now the conceptual space. If you like, you can decompose the

semantic objects into their constituent words and phrases. For each semantic object, you can follow bridges outward to different realities. To related content on the web. To objects and places in the metaverse. And to related objects and places in a digital twin model of physical reality. Or to another semantic object.

EXTENDING TO OTHER REALITIES

Think of a hyper-dimensional knowledge graph that connects between and among semantic webs of concepts, real and virtual objects, and online content. As shown in Figure 14.1, the graph anchors the conceptual realm with bridges to the online, virtual, and physical realities.

This expansive understanding of the Metaweb suggests the “web” in “Metaweb” isn’t the World Wide Web. It’s the spider-like web that connects realities. The four realities that affect our collective cognitive capabilities are physical, online, virtual, and conceptual. Were the Metaweb to ultimately support full expression that adequately represented our collective memories and dreams, this knowledge graph would be a virtual Akashic records.

It could enable us to transcend human limitations and regenerate ourselves, our communities, and this garden planet we inherited from our grandchildren. But it could also help machines achieve artificial general intelligence (AGI), and soon thereafter, artificial super intelligence (ASI), enabling machines to transcend humans in intelligence by orders of magnitude. Humanity needs to discuss this possibility and build mechanisms that respect and protect human life.

As we discuss the two next possibilities, some may notice an impulse to get involved and help build the connective tissue of our external realities, the Metaweb. We trust some will

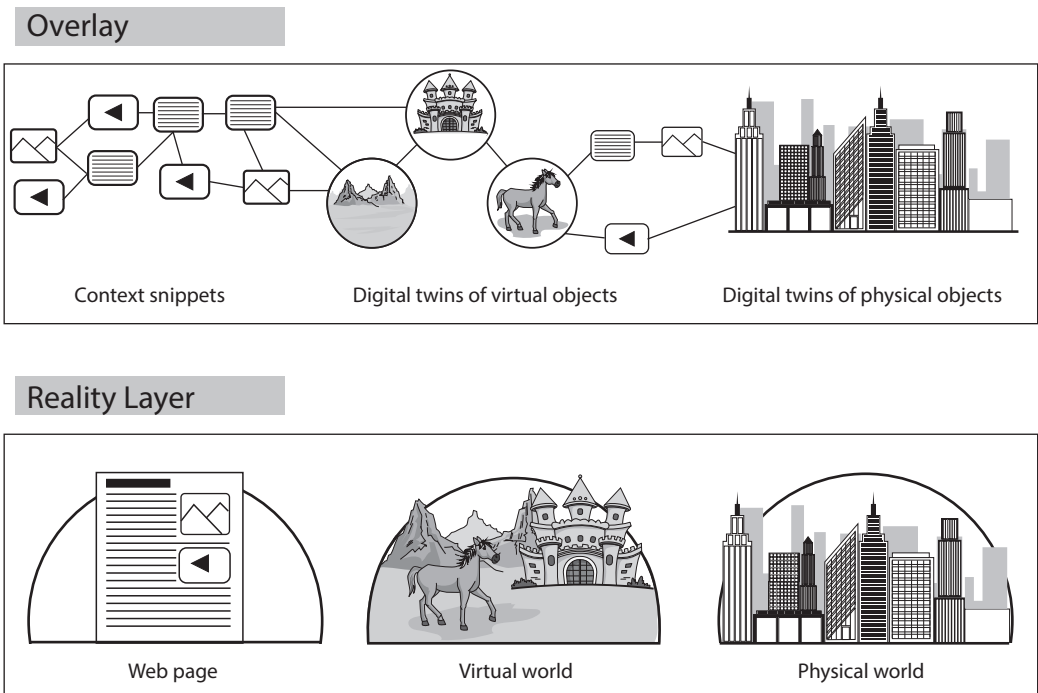


Figure 14.1 Connecting the realities.

focus on ethical analysis and preventative remedies for potential unintended consequences, and hope that all projects will consider AI safety. Once again, the integral accident.

UBIQUITOUS CONTEXT¹

One could say we understand something in how it connects to our active internal “knowledge map.” This comprises what we perceive and what we understand and recall about the focus of our attention and how it connects to our perceptual map of the world. This is our basis for sensemaking. In the future, AI will augment sensemaking, helping us make sense of our world individually and as a collective.

Engelbart protégé, David Smith, CEO of Croquet, speaks of the Augmented Conversation as a discussion between participants that is “extraordinarily enhanced” with computer tools and capabilities. The computer AI is a full participant in the conversation. It allows us to discuss and explore complex systems, datasets, and simulations as easily as we talk about the weather. A guarantee of shared truth is necessary. What I see, you must see. Doing something that affects the shared simulation must change the simulation for everyone.

Otherwise, the communication channel is not trustworthy. The shared system must enable modification and extension of the system while running for all participants. Thus, we can use the system itself to extend the system, improve it, and add new capabilities. Smith suggests a new type of operating system, built from scratch, focused on shared simulation and deep security.

We think the Overweb can accommodate the Augmented Conversation for online ecosystems and leverage existing online content, rather than starting from scratch.

In a future with ubiquitous context, wherever you are—online, in a virtual world, or IRL—you have access to deep layers of context. Every addressable location, object, and idea enables access to contextual information and interactions through its connections to a universal content graph. The accessible context is the most relevant section of the graph—the information connected to the focus of your attention.

You use your attention (line of sight, touch, cursor), gestures, and voice to navigate and interact with a rich, interactive digital overlay. Your overlay is a composite view of the relevant portion of the Universal Content Graph filtered by your digital assistant based on your preferences, needs, and activities. Your assistant bot manages your smart filter and performs actions within the overlay on your behalf.

By democratizing linking with proper incentives, the Universal Content Graph self-assembles from bridges connecting online content snippets with relationships. This graph underlies the meta-layer above the Web. As the metaverse and spatial computing grow, the graph will bridge the virtual and physical realities with the conceptual realm creating the Metaweb, the connective tissue of our realities.

Graphs are incredibly efficient in retrieving information. One can find information on a knowledge graph easily and quickly. It’s the only viable structure to connect the world’s information. No matter the size of the graph, directly connected elements are instantly retrievable. Surely, the proverbial Akashic records reflect a similar pattern. Our brain certainly does.

Thus, wherever one is—online, multiverse, or in the physical world—in the future, you’ll have access to deep layers of context within and between realities and the conceptual realm. We call this ubiquitous context. Context is available everywhere, you just need to focus your attention.

Ubiquitous context is reminiscent of how we access information IRL. With ubiquitous context, we can access more information and context by focusing our attention. That’s how



Original Photo by Snow White from Pexels

Figure 14.2 Ubiquitous context in an art gallery (courtesy of Future Text Publishing).

perception works in real life. When riding bikes, we scan the environment. If we see something to avoid, we focus our attention on changing our route. If we see a mural of a jaguar,² we may stop and admire it or take a photo. We focus on seeing details and context. With photos, we can zoom in to reveal even more details.

But ubiquitous context is much deeper. You go as far down the rabbit hole as you desire. Or go wide with context across and within realities. Keep zooming for more context, and context for the context. The graph has no beginning or end.

Figure 14.2 shows what ubiquitous context could look like in an art gallery.

Imagine a future visit to an art gallery. Since you follow the genre or artist, or activated the museum tour, a digital badge hovers next to the painting you are viewing. Moving your attention to the badge—whir—a context menu appears, providing a 360° overview of the information, interactions, transactions, and experiences connected to the painting. Yes, you can jump into the painting or at least the AI-generated 3D virtual world if you like.

Or you can follow bridges to video of the artist, work-in-process photos of the painting, and reviews by art historians and local art critics. Access is available to the notes from the artist, historians, and critics about specific parts of the painting. Depending on your filter, you may also see text and video notes of other museum-goers. You can engage in conversations and polls related to the painting.³

You can add notes, polls, conversations, and bridges, and comment on other pieces of contextual information. Options exist to purchase a print from the museum store or to have one shipped to your home. You can navigate to paintings with similar subjects and styles. Or with the same artist or era. If you like, you can quickly retrieve definitions and sentence examples for unfamiliar terms used by the local art snobs.

With ubiquitous context, wherever you focus, your attention becomes a launchpad for insight, debate, and discovery. Real and virtual objects, geolocations, and visual, aural, and

textual patterns become anchors for intertwined digital overlays. These unlock massive waves of innovation, creativity, and collaboration, as well as increase humanity's capacity for shared knowledge and collective intelligence.

The Augmented Conversation: Collaborating with AI for a Shared Reality

Doug Engelbart, a pioneer in the field of human-computer interaction, envisioned a future where computers augment human intellect and enable collaboration on a scale never before possible. With the development of Croquet, a physics engine that perfectly replicates interactive physics simulations, we are a step closer to realizing this vision.

Human communication mediated by AI and Augmented Reality devices will enable us to dynamically express, share and explore new ideas with each other via live simulations as easily as we talk about the weather. This collaboration provides a “shared truth”—what you see is exactly what I see. I see you perform an action as you do it, and we both see exactly the same dynamic transformation of this shared information space. When you express an idea, the computer AI, a full participant in this conversation, instantly makes it real for both of us, enabling us to critique and negotiate its meaning. This shared virtual world will be as live, dynamic, pervasive, and visceral as the physical.

The future of collaboration and computing is not just about adding a new layer of technology over our current reality, it's about creating a seamless, multi-user reality where the physical and virtual worlds are indistinguishable. With the development of interoperable, scaling, and development-friendly platforms, we can create a digital world that is accessible to all, regardless of device or operating system, and enable every participant to share and collaborate with every other participant in this enhanced digital world.

As we move towards this augmented reality, the virtual world will become as real as the physical world, and we will live together inside a shared information space that co-exists with and amplifies the physical plane of existence. Communication, far more than anything else, defines the true value of Augmented Reality. AR displays are transparent not just so we can see the real world with a digital overlay, but because we need to look another person in the eyes as we engage with them in this extended digital space.

The emerging AR Cloud expands the scope of how we collaborate, create and share ideas within it. This next generation of computing capability will allow us to extend and annotate the real world, but more importantly, it will allow us to easily and instantly create and explore new universes that we will build from scratch as part of our everyday discussions.

CONNECTING WITHIN AND TO THE SPATIAL WEB

The Metaweb will connect among and into the digital twins of objects in physical reality. Smart tags will attach to anything, such as a sound, a physical object, a geolocation, a sequence of words, a product's packaging, etc. The overlay could recognize people using their facial structure. To protect people's privacy, in the Overweb, we cannot attach smart tags to a person without their expressed consent.

Physical objects such as books, paintings, cafés, grocery stores, products, and more can link to relevant information ecologies on the Web. Wherever you look, you can access relevant information and interactions. In the physical reality, a virtual world, the conceptual

space, or online. For example, a physical book can connect with geolocations and objects in the book. It can connect to book reviews, book clubs looking for members, and drop-in conversations happening on the pages of the book. Even polls about controversial passages, AMAs with the book's author, and much more.

Focusing on content related to physical objects such as books or mangos can display manipulable 3D models of the object. It can also display geolocations in a digital rendering of the physical reality where the object is available or prevalent.

CONNECTING WITHIN AND TO THE METAVERSE

Every discrete object, NFT, and location in the metaverse is bridgeable and interactive between and within multiverses. Geolocations, multiverse locations, and objects can have easter eggs, polls, and meetings. Similarly, bridges can connect objects and locations between and with multiverses. For example, a Sandbox land NFT could bridge to an object or NFT in a different Sandbox location. It could also bridge to an object in Horizons World or any other environment with addressable objects.

As we move from a 2D to a 3D display, we see that attention triggering is natural for a virtual experience. The primary headset control is the user's line of sight, and they already use their focus to navigate the extended realities. In the future, activating one's focus on an object in a virtual world could bring up relevant objects, information, and interactions from the Universal Content Graph.

Imagine a refrigerator for a cooking experience in the metaverse that connects to your home refrigerator. Rather than a digital twin of your refrigerator, this is a refrigerator experience that works with refrigerators that implement its protocols. You interact by focusing your attention via line of sight and via voice or gesture commands. Opening the refrigerator displays a 3D model of your refrigerator with its doors open and visual representations of the foods inside. You can tilt the display or switch the order of products to see items that aren't visible.

But the real fun happens when you convert the 3D model into a virtual walk-in refrigerator with the contents neatly organized on shelves. You can adjust the grouping of items depending on what you have a taste for, food categories, or even proximity to expiration. You can also ask the AI to generate recipes based on what's available but not spoken for in your refrigerator at home. Activating an item enables access to nutritional information, recipes with nutrition calculations, and ideas for combining with other items in the refrigerator into a meal. It also provides access to the supply chain history of the products, remaining amounts, and edibility likelihoods. You can display a floating array of the product's ingredients, 360° videos from the manufacturer, related virtual experiences, relevant cooking shows, purchasing options, and AmazonFresh delivery times (or whatever delivery services exist at that time).

Focusing on one of the ingredients displays its history, nutritional facts, purchasing options, and virtual experiences, for example, related to its production and environmental impact. At any point in the exploration process, relevant choices are present. You see what you choose to explore, nothing more. Mind you, the creator of the cooking experience doesn't curate the cornucopia of experience options. Rather, the options are the product of human-AI collaboration in the Metaweb ecosystem for which participants receive rewards based on the value of their contributions. Importantly, smart filters apply, and, of course, you only activate the Metaweb when you need it. This entire experience could be done with a headset or from a VR-enabled next-generation home theater or an office.

The Metaweb has the potential to become the predominant discovery mechanism in the metaverse. Focus-driven navigation is natural given that line-of-sight with head-mounted displays is the primary control in virtual experience systems. Navigating with bridges doesn't require typing, which is awkward with hand controls and head gesturing. Voice and gestures are useful as secondary controls. The Metaweb doesn't rely on keywords to describe multi-dimensional objects and can handle the long tail, which could be more important in virtual experience than online.

Without a doubt, the Metaweb community will develop plug-ins for the Unity 3D and Unreal engines, the two dominant virtual reality development platforms. Thus, enabling virtual experience developers to access the Metaweb functionality within their development environments. The community will also develop a virtual experience navigation SDK for integration into spatial browsers associated with headsets/glasses (e.g., Oculus Rift, Magic Leap, Verses). This will enable people to explore the online and extended realities as well as attach smart tags and build bridges between and among these domains.

We see quantum search or focus-based navigation of the Universal Content Graph eventually becoming the predominant way to search in the metaverse, and across multiverses and realities. The participant specifies which realities are relevant. For instance, you're in the metaverse. You activate the overlay for an object of interest and filter the relevant information from the Web based on the number and strength of supporting bridges. Maybe you filter for digital twins of the object in the physical reality. Or perhaps your smart filter is using a custom ranking algorithm from a developer in the ecosystem, who gets rewarded when you use it.

We think the accountability provided by Overweb's one-person-per-account policy will raise vibes well above today's no-accountability web that's beset with inauthenticity and theft. Were there consequences, they'd think twice. Certainly, crime would be less, and mainly in environments, unlike the Overweb, with throwaway accounts. Having people take responsibility for those they introduce makes people think twice about who they bring into the fold and could ease the removal of bad actor accounts.

Absent proof of humanity, the metaverse is dangerous and more so as it gets hyper-realistic. Imagine placing your worst IRL screaming incident or argument into a virtual world. Then multiply it 100x. We must not cede the metaverse to trolls.

CONNECTING WITHIN AND TO THE CONCEPTUAL REALM

Words, thoughts, and concepts make up the conceptual realm. The conceptual realm includes a theoretical level with theories and models and an ideal level of idealized entities. The universal content graph can map the conceptual realm with words, inquiries, perspectives, claims, evidence, and data sources as nodes. This enables argument mapping, dictionaries, thesauruses, translation tools, and more.

Conceptual argument maps could connect to online content that support, contradict, or cite and to digital twins, geolocations, and virtual experiences. This would enable machines to process human knowledge at a higher level of performance and intelligence.

A word of caution. Such a knowledge graph would be useful for AGI. Therefore, its conception should address this possibility and minimize the possibility of such a graph being used against humanity. To be clear, someone will build the graph, because it's valuable. We can, however, take a proactive approach by instilling elements that protect humanity into its foundation and licenses.

As we embark on the next chapter, we find ourselves standing on the edge of a technological revolution. The Metaweb, and in particular its full instantiation, the Overweb, has the potential to unlock doors that were previously closed to us. The Overweb, with its foundation on the principles of accountability, presence, interconnection is far beyond the capabilities of Today's Web, and it will change the way we interact and engage with the digital world.

Get ready to journey with us as we explore how the Overweb can revolutionize industries, disrupt old systems and create new opportunities, increase transparency and trust, and allow for true decentralized ownership and control of data. The Overweb is the next level of the web, and its limitless potential is waiting to be unlocked. With its potential to change the way we live, work and play, it's a must see for those who are looking to shape the future.

NOTES

- 1 This section is adapted from an essay in Benjamin, D. (2020). Ubiquitous Context. In F. Hegland (Ed), *The Future of Text* (p. 118). London, UK: Future Text Publishing. <https://permanent.link/to/the-metaweb/future-of-text>
- 2 We have never been to a place that celebrates an animal as Tulum does with Jaguar. Balam, the jaguar, heralds murals, paintings, and the signs of businesses and organizations, both visually and in name. In Mayan cosmology, the jaguar represents the power to face one's fears and to confront one's enemies, and is associated with vision, the ability to see during the night and to look into the dark parts of the human heart. <https://permanent.link/to/the-metaweb/balam-jaguar>
- 3 A similar system exists at the Museum of Old and New Art (MONA) in Hobart, Tasmania, Australia, which offers a GPS-enabled handheld device instead of a guidebook. The "O" keeps track of what exhibits are nearby and offers a set of audio choices related to each piece including a summary description of the piece; the curator's notes; the museum owner's thoughts on the piece; and maybe some review of the piece from a journal or newspaper or the artist's telling of why and how they made it.

What Becomes Possible

I am not afraid of storms, for I am learning to sail my ship.

—Louisa May Alcott

Alcott's seminal quote speaks to the idea that we must embrace challenges and navigate through them in order to grow and evolve. This is especially true when it comes to the Internet.

For years, the Web has been an open sandbox, a place where anyone with access to a server could publish whatever they wanted with minimal friction. And while this approach enabled the rapid growth of web content, it also led to a number of problems. Misinformation, disinformation, and a lack of trust abound. The Web is now awash in identical and nearly identical information, much of it created for search engines to consume, not people to read.

But despite these challenges, the Web remains a metaphorical child, still growing and exploring. Everything that has happened so far has been necessary for its journey, but it is time for the Web to evolve and change direction if it is to support the shifts that humanity needs to undergo. And as we continue to navigate these waters, we must remain fearless, like the sailor in Alcott's quote, always learning and adapting in order to reach our destination.

A NEW SOCIAL CONTRACT

As a DAO, we see the opportunity to add a governance and ethics layer to the Web. The Overweb proposes a new social contract as a valuable alternative to Today's Web by providing a more decentralized, self-governing, and ethical environment for online interactions. This social contract enables communities to govern themselves and create their own norms and rules, rather than rely on third parties like Facebook or LinkedIn. This approach creates a sovereign layer above the web, where communities can meet and interact in a safe and trusted environment.

As mentioned in Chapter 12, the Overweb as a digital environment provides participants with a set of rights that do not exist on Today's Web. The Overweb also enables purpose-aligned groups to self-govern with their own norms and rules within meta-communities, which enables decentralization and self-determination. Because the meta-communities exist and operate within a network of networks model, they implicitly support subsidiarity, a principle of social organization that holds that matters should be handled by the most local or least centralized competent authority. In other words, subsidiarity devolves decision-making to the

lowest possible level of governance, so that individuals and communities can exercise greater control over their own affairs.

The Overweb's social contract is realized through the agreements around the Overweb bill of rights, their implementation, meta-communities, and the accountability that comes with the one-account-per-participant model. The one-account-per-participant model ensures that everyone you encounter on the Overweb is a real person in good standing, which enables confidence in online interactions impossible on Today's Web. The Overweb also promotes transparency and accountability for online actions, which aligns with ethical behavior.

The Overweb's social contract also supports non-violent free speech, and it holds participants accountable for their online actions. This ethical approach ensures that the platform is safe and secure for everyone, while still promoting the freedom of expression.

While a new social contract offers many benefits, many aspects of the Overweb's new social contract need to be worked out. For instance, it may not be clear how the principles of subsidiarity and self-governance will be applied in practice, and how conflicts or disputes will be resolved within and between meta-communities. Moreover, the Overweb's social contract may have broader political, social, and economic implications that need to be considered, such as its relationship to existing legal frameworks, power dynamics, and inequalities. The Overweb as a collective will work through these issues in a participatory process.

CONNECTING PEOPLE WHO NEED TO MEET

Finding IRL people with similar interests to engage can be difficult. People must be there and open to connect. This became next to impossible during the Covid lockdowns, when people were working from home, staying home, and attending virtual gatherings. During the pandemic, people met others on social media, apps, and video meetings. Post-pandemic, meeting like-minded people IRL is back, yet social media, apps, and video meetings are as strong as ever.

Unfortunately, Facebook, Twitter, and Instagram are rife with fake accounts and/or bots. It's impossible to know who's real and who's not. Catfishing is prevalent, with many people posing as fictional people or fake identities to manipulate individuals both emotionally and financially. Trolls and bots are ubiquitous, especially on Twitter. You can't meet people online as you might in a café. Online, you can't look someone in the eyes. You can't touch them.

In the Metaweb, it's easy to meet like-minded people with common interests. Imagine you're at an online art gallery, staring at a painting you find inspiring. Hundreds are looking at the image, but you can't see them. Yet, they are undoubtedly staring at the same painting with similar focus and reverence. Wouldn't it be interesting to get their perspectives?

Were you there, you might turn around and say hi. But you can't on a webpage. Unless you have access to the Overweb. The Overweb augments the painting's image and enables you to see others who were visible. It could also make the painting's image augmentable across the Web as well as the painting itself. You can take part in conversations with the painting's online and IRL admirers before you come, during your visit, and when you come back.

SHIFT JOURNALISM

The Overweb connects evidence to claims. This is interesting because people often post claims online without links, much less evidence. Yet evidence might exist in many places.

The detective's role is to find supporting or contradicting evidence for a claim. Detectives use crime boards, into which they pin photos and newspaper clippings, connected by string. Crime boards are a type of IRL bridging. The photos are content nodes, and the string is bridges. There is no expression of the relationship of a bridge, although one could express this as different colored strings.

What if we can provide tools that turn the politically curious into citizen fact-checkers? People often wonder about the vulnerability of the system to disinformation agents: bad actors who spread false information to confuse, advance political agendas, and change behavior. Methinks thou doth worry too much. As explained in Chapter 11, if they understood bridges and their governance, they might not worry so much about this issue. At least on the Overweb.

In our experience, people underestimate the extent to which we could diminish the negative impact of disinformation operations by eliminating fake accounts. As mentioned earlier, Facebook is more like Fakebook. They admit up to 5% of their active accounts are fake. It is likely that the larger disinformation campaigns are using hundreds of thousands, if not millions, of fake accounts. Without fake accounts, most disinformation operations would be dead in the water.

The bridge's structure also hardens the system to both attacks and swarming. Bridges have a relationship identifier, which distinguishes it from the hyperlink. The relationship is an objective appraisal of how two content snippets relate. Either something relates or not. If the former, a bridge is possible. If the latter, a bridge isn't possible. In the Overweb, approved validators confirm that the relationship is correct prior to posting to the public ledger.

Regardless of the relationship, a bridge does not say whether either side is true. If the relationship is supportive, we know that the content agrees. We don't know whether either side is true or false. Consider a supporting bridge between two statements: a) "Tom is a firefighter" and b) "the California Fire Authority employs Tom."

There are four scenarios to consider.

One, Tom is a firefighter with the California Fire Authority. Two, Tom isn't a firefighter or with the California Fire Authority. Three, Tom is with the California Fire Authority, but is not a firefighter. Four, Tom is a firefighter, but not with the California Fire Authority.

Contradictory relationships tell us the content disagrees and we know one is false or both are false. We can review the bridges to each statement to investigate its veracity. Consider these contradictory statements: a) "Tom is a firefighter" and b) "Tom isn't a firefighter." Semantics aside, he is a firefighter or not.

Consider these statements: a) "Tom is a firefighter" and b) "Tom is a police officer." Before jumping into analysis, we consider whether the statements are contradictory. It depends on the intention of "is" in each statement and perhaps whether jobs exist that are both firefighter and police officer.

Let's consider these additional scenarios. One, Tom is a firefighter and not a police officer. Two, Tom's a police officer, but not a firefighter. Three, Tom is neither a firefighter nor a police officer. Four, Tom is both a firefighter and a police officer. Five, Tom was a firefighter and a police officer when the statements happened.

What's the bottom line? We have to follow the bridges, examining their context and implicit assumptions. E.g., when did the statements occur? Once a firefighter, always a firefighter? What is a firefighter? Only through careful analysis of data can we arrive at the truth. Bridges provide the data.

Rather than showing whether an isolated claim on a siloed website is true or not, the Overweb shows a 360° contextual view of everything verified to relate to the claim. This could include contradicting and supporting content and analysis as represented by bridges,

notes, conversations, and other smart tags created by members of the viewer's meta-communities.

While contextual information doesn't unequivocally show whether a claim is true, it provides access to contextually relevant information for making informed decisions. Turning news consumers into curators, if not researchers, of news around specific subjects, with access to all relevant contextual information from various sources.

In this sense, the Overweb crowdsources fact-checking so that anyone can curate the relevant information ecology.

Bridger is a competition platform that gamifies bridging. Hosts set up a challenge to create bridges on a specific topic or theme or to answer a question or solve a mystery. People with access to the Overweb can join the challenge as a bridger (one who builds bridges) or a curator (one who upvotes bridges they think are insightful). For example, a teacher of a media literacy course could establish a competition where students can fact-check a breaking news story. Or a corporation could set up a challenge to connect the content of relevant research papers.

The Overweb also enables fact-checking sites to get more exposure for their work. Over 200 organizations in the international fact-checking network do fact-checks every day for online news. But very few people see their fact-checks because they live on the fact checker's destination site, a silo completely separate from the claim they are fact-checking. Imagine the improvement when non-partisan fact checks are available next to every claim on the web.

SHIFT ONLINE EDUCATION

The transition to virtual education during the COVID-19 pandemic uncovered many problems, including:

- Teachers, students, and parents struggling with virtual education, communication, and learning platforms.
- Distracted, disengaged, or absent students.
- Inability for teachers to track engagement, nonverbal cues, or to check for understanding as they would in class.

Cobbling together the available virtual solutions (e.g., ZOOM, Google Classroom, MOOCs) proved to be insufficient. It didn't provide the real-time, contextual learning experience of being in the classroom.

The education system helps shape the course of an individual's life, society, and, therefore, our collective future. Our current reality involves several existential threats. To respond to these threats, our education system needs to account for our interconnected world.

We hope humanity can solve global problems democratically. Democracy requires an informed populace. It's essential that people can discern false information (and get context) to advance their thinking and cast votes based on their values. People need to think and learn together, analyze Internet content for validity, and collaboratively construct meaning across sources, space, and time. Existing solutions don't make thinking visible or connect evidence to claims, which is foundational for critical thinking. As a result, people don't know what to believe and remain vulnerable to others trying to influence their behavior and thinking.

In 2016, futurist Thomas Frey predicted that the largest online company in 2030 will be an online school.¹

The Overweb enables on-page learning, a new educational modality, which turns the Web into a classroom and makes thinking visible.

We have conceptualized an overlay application called Skōōl that facilitates teaching and learning in a digital environment. Skōōl enables educators, administrators, learners, and parents to build knowledge together on relevant subjects and contexts. The application supports the basic smart tags, including the bridge, note, conversation, poll, and quiz, as well as educationally focused tags such as a lesson, assignment, and breakout groups.

Educators can create learning experiences from scratch on relevant subject content and share those experiences with other teachers and their students. A curriculum anchored by a constellation of bridges can grow as students learn. Students and teachers can converse and interact over webpages as if they are in a classroom, while building knowledge together.

Educators can see how students engage with online content, ask and answer questions, and even poll and quiz students on the webpage to facilitate learning and integration of content. Skōōl enables participants to share their bridges, tag and notate content on any webpage, and interact while on the same webpage.

Customers are independent and charter schools, international schools, online schools, school systems, and homeschooling parents.

On-Page Learning

In the summer of 2020, we did 15 virtual salons about on-page learning, which attracted over 150 participants, including key experts in the field from 17 countries. In each, we presented the project and a 15-slide mockup of the teaching experience. We also encouraged teachers to share how their districts were dealing with Covid and to provide feedback on the presentation. The feedback and questions were overwhelmingly positive.

The teachers understood that one-page learning was a game-changer and would enable them to engage their students at an entirely different level, more like being in a classroom, but online. They were also excited about leveraging Skōōl as a tool for formative assessment, collaboration in learning, critical thinking, creativity, and more. One student said he could learn over twice as fast if he could access the bridges other students were building.

Feedback also included an emphasis on how on-page learning aligns with learning sciences research. Specifically, on-page learning supports metacognition, cognitive flexibility, executive functioning skills, social-emotional learning, and critical thinking, and can help mitigate the effects of adverse experiences.

Other feedback was the lack of tools enabling collaborative, global interactions focused on identifying claims, evidence, and reasoning. Such tools could effectively help people develop critical life skills such as perspective-taking, recognizing misconceptions, and undergoing conceptual change needed to analyze information across the internet for validity and authenticity.

We can create a safe and engaging digital space above the webpage for learning. We envision future incarnations of Skōōl as an AI/Blockchain-based online school platform for K-12, university, corporate, and lifelong learning. Learners navigate a global marketplace for facilitated and self-directed learning experiences with the help of a personal AI that optimizes their learning trajectory and development of 21st-century skills. Learners earn microcredits towards certifications, credentials, and degrees. Skōōl introduces a new

category called On-Page Learning, which turns the Web into a classroom and makes people and thinking visible on webpages. By providing context and enabling learners to see the clear connections between different perspectives, Skōōl helps meta-cognition and changes how learners reflect on things at a deep level. Primary revenue streams are subscriptions and transaction fees.

During class time, teachers can see the students and how they interact with content. In the event of a school closure or to provide additional support, teachers can use the video conferencing option to conduct class remotely. The platform allows for teachers to check for understanding through observation and assessment, as well as to communicate directly with students and form breakouts.

DEFRAGGING THE WEB BY CONNECTING LANGUAGES

Languages fragment the Internet. There is an English web, a Chinese web, a Spanish web, etc., none of which are semantically connected. When an English-speaking person searches on Google, no recommendations will be for Spanish pages. There may be highly relevant webpages in Spanish, but English speakers won't ever see them because Google separates the results based on languages.

Now, since Google already has a translation option that allows a person to translate entire pages and search results, they could display relevant search results translated from other languages. However, Google Search has a huge amount of duplicative and irrelevant content. But adding more languages to a cesspool of duplication would worsen the Web. Besides, their focus is on increasing click-throughs on search ads, and they don't really care what is semantically relevant in other languages.

Consider the possibility that two articles, one in English and one in Chinese, speak about a specific cross-border conflict between the US and China. On Today's Web, US news consumers see US perspectives in English articles. They, however, don't understand the Chinese perspective.

The English Web and the Chinese Web are completely separate by design. Unless you speak both languages and search for news in both languages, you only get one side of the story.

Bridging, however, enables bilingual bridgers to connect ideas in one language to corresponding ideas in another language. We can make bilingual bridges between articles in English, Chinese, Japanese, Spanish, and any other language on the web. Bilingual bridges then create demand and a market for translations of the different ideas that are connected across languages.

Suppose a bridge connects two articles—Spanish and English—concerning an incident in Mexico. The connection may inspire bilingual bridgers (assisted by AI) to translate the English article into Spanish and the Spanish article into English, or at least the content of the bridge. With bridge translations, English and Spanish speakers can access perspectives in both languages.

Given the proper incentives, the Metaweb can stitch together the splinternet. And it can also create a mechanism to translate languages with little presence on the Web and for which AI is unable to translate.²

According to Carl Rubino, a program manager at IARPA (the research arm of US intelligence services), "The more interested an individual is in understanding the world, the more one must be able to access data that are not in English. Many challenges we face today, such as economic and political instability, the Covid-19 pandemic, and climate change, transcend our planet—and, thus, are multilingual in nature."

A PAN AFRICAN META-COMMUNITY

Meta-communities enable groups of people to define and enforce codes of conduct and content for their specific community. Anyone can start a meta-community focused on some combination of geography, language, identity, and/or interest. Meta-communities provide filtration services for their communities and create a safe digital space.

While affinity and identity groups and geographies can be the basis for meta-communities, the most robust meta-communities will also have a common purpose. For example, an Original Peoples meta-community might open space to allies and have a shared purpose to unify the world. Similarly, a Pan-African meta-community might have a shared purpose of creating opportunities for the people of Africa to build generational wealth or to uplift African people.

There is a strong case that if the Pan-African movement were to have success, it would likely begin digitally. As PLO Lumumba said in *the Past, Present and Future of Pan-Africanism*,³ the path to Pan-Africanism is not an easy one. It will take time and effort. But this is the path to freedom. Yet the thought of one passport, one test, etc. for Africa's 54 countries is daunting for even Pan-Africa's most ardent supporters.

Pan-Africanism, however, in the digital realm could be relatively straightforward and quick. While challenging decisions persist, such as the extent of decentralization, the technology exists in the Metaweb. The Presence browser project was born on the notion of a Black Browser for the Diaspora and Africans. A first-of-its-kind digital infrastructure that would foster new channels for economic mobility and sovereignty.

Pan-Africanism can advance by occupying the space above the webpage. We see a Pan-African meta-community for continental Africans and the diaspora, and possibly including allies as well. This meta-community would have its own sovereign layer above the entire web that is controlled by the Pan-African meta-community. Colonization from external forces will be difficult. And with effective governance structures, internal threats will lessen as well.

This meta-community would enable all Africans with a mobile phone to meet, interact, coordinate, collaborate, and build knowledge in a safe digital public space that exists over the webpage. Africans can now leapfrog would-be-colonizers regarding digital public space. This opportunity, however, will probably have a narrow window given that the code will ultimately be open source and therefore available to anyone who wants to use it. If Africans want freedom on the Web (and ultimately in the physical realm), they need to act quickly and decisively.

A meta-community will speed up the physical unification of Africa. It will enable Africans from everywhere to meet, interact and communicate about their common interests and needs. It will enable robust discussions about unification and provide a forum for all Africans with web access to take part in the governance of a digital and/or physical Pan-Africa. Such a meta-community enables Africans to express themselves and receive value for their online contributions.

We get chills thinking about the possibilities of supporting marginalized groups. Besides a Pan-African meta-community, there may be a black meta-community with the diaspora, and meta-communities for individual African states as well.

Similarly, meta-communities will emerge for Latin America, Mexico, other countries in Latin America, Spanish-speaking countries, and for their regions, states, cities, towns, even neighborhoods. Meta-communities can nest within one another.

There will be a meta-community for original peoples to stand together in solidarity. It could catalyze and coordinate a worldwide unification movement, while preserving sacred sites and sharing knowledge of living harmoniously with nature.

As a meta DAO with child DAOs, we see networks within networks as the future. Meta-communities will connect and collaborate to the extent it is mutually beneficial. And most participants will be members of multiple meta-communities. For example, we might be part of an ethnic meta-community because of our ancestry. One for our hometown. We could be in ones for the country and city we live in. And perhaps ones for the languages we speak. And these are just our ethnic, geographic, and language affinities.

What meta-communities could you join or start?

Take a moment to reflect on the different communities you are a part of - where you come from, where you live, your interests and hobbies, and think about how you could connect with like-minded individuals on the Overweb through meta-communities. With the ability to self-govern and create our own norms and rules within these communities, the possibilities are endless.

A METABOOK

The metabook is a guided and curated pathway through the relevant universe of multimedia, interactions, and experiences to express a POV. Access comes through an overlay displaying videos, information, interactions, and mixed reality experiences associated with snippets of text and images in a physical or digital book.

People will initially access the overlay with a mobile app, but in the future through mixed reality interfaces. The metabook concept supports personalization, content creation, dealing with multiple triggers, social interactions, and connecting content snippets across books. Connecting the knowledge in books on an idea-by-idea basis, book clubs, and asynchronous conversations in a digital marginalia overlay will be transformative.

In sum, the metabook is the hyper-dimensional digital twin to a physical or digital book that grows over time with the individual reader and the reader community. The metabook can host book clubs, AMAs,⁴ notes, reviews, polls, and conversations related to pieces of text and images on specific pages of the book. Thus, in the Metaweb future, the physical book becomes the contextual footprint for intertwined worlds of people, information, and interactions associated with content on the pages of the book.⁵

CONTEXTUAL ADVERTISING

Unlike typical display advertising, which seeks to capture attention based on micro-targeting with activity data, the Metaweb serves ads that relate to what people are focusing on. We call this contextual advertising or content-driven advertising. The idea is to only provide the participant with information that is contextually relevant.

The Overweb tracks the participant's attention on the screen and progressively reveals meta-level information related to the focus of their attention. Thus, when participants have contextual ads turned on, they see ads (and smart tags) related to what they are focusing on.

Notice what happens here. The Attention Economy targets based on activity data. This involves getting a lot of information about the person and their activity on the Web and on the mobile phone and displaying information and advertisements intending to change behavior.

With contextual advertising, there is no targeting based on the participant's activity history or personal information. Ads relate to the focus of attention. Interest in the subject

triggers the ads. Rather than capture attention, contextual ads support the attention by providing information to purchase at exactly the right moment. Ads, as with all content on the Overweb, are subject to the participant's smart filter. If a former alcoholic has no tolerance for content that promotes alcohol consumption, related ads and smart tag content won't make it through the filter.

We emphasize—in the Overweb—ads are off by default. The participant must explicitly enable advertisements. Ad viewers receive a reward or chance to win tokens for their attention.

When advertisers wish to display offers associated with ideas, they can buy advertising clicks with tokens and ultimately through demand-side systems that connect with the Overweb. The payer's account gets a click credit for the token value. The value of the click credit doesn't fluctuate over time, as fluctuations in token value don't affect past purchases.

With Overweb ads, the website owner gets 40% if they are registered. The remaining 60% is split among the owner of the bridge that led to the ad, a curation pool, and an ad-watching pool. If the website isn't taking part in the Overweb, their 40% goes to the other actors proportionally.

As mentioned earlier, the long tail notion also extends to advertising. This will unlock many advertising opportunities involving pockets of information that are unrelated to page metadata, yet could be highly effective for targeting customers by interest.

MAINTAINING SAFETY, ACCOUNTABILITY, AND PRIVACY

The Overweb places a strong emphasis on maintaining safety, accountability, and privacy for its participants. To achieve this, the platform implements several measures to ensure the authenticity of users and their actions.

First, the Overweb uses decentralized identifiers (DIDs) to create unique accounts for each person. This makes it difficult for impostors, burner accounts, or bots to take advantage of the system. Additionally, the account holder's private key protects their identity information and grants access to it, making it nearly impossible for the Overweb to doxx users to the authorities.

The Overweb also allows users to create multiple personas that are either visibly or invisibly tied to their main account. This enables users to maintain their privacy and remain pseudonymous if they choose to do so. However, if a persona engages in behavior that breaches the social contract of the meta-community, their main account and other personas may be affected.

Furthermore, the Overweb does not record user activity without consent. If a user wants their activity tracked, the Overweb encrypts it on-chain and gives the user control over who can access their data. This ensures that the privacy of the user is respected while still maintaining safety and accountability.

The Overweb takes the protection of its participants' safety, accountability, and privacy very seriously. Through the use of decentralized identifiers, multiple personas, and controlled data tracking, the Overweb ensures that its users can trust each other and engage in meaningful interactions with peace of mind.

PERSONALIZED HEALTHCARE AND WELLNESS

Nikki is a single mom of two kids living in a three-bedroom home in the suburbs. She has a "natural" diet and seems healthy based on outside appearance, but she suffers from two

chronic illnesses: sickle cell anemia and diabetes. Additionally, her six-year-old was born with high-functioning autism that she manages through a diet that's slightly different from that of her two-year-old daughter who has allergies.

Sickle cell and diabetes are difficult to navigate nutritionally. Especially since Nikki, like most people, doesn't have the personal healthy literacy needed to navigate the onslaught of advertisements and misinformation that's influencing her food purchases.

It's not Nikki's fault; the amount of time needed to learn to navigate the food choices, doctor visits, medications, lifestyle, body goals, and mental well-being is staggering. Nikki needs instant access to impactful data curated by contextually relevant communities, which includes the researchers who connect the dots, health and wellness professionals who diagnose which dots apply to which clients, and the brands who manufacture and sell products and services to the individuals who create the demand for actionable information in real-time.

DPHR is building a personalized healthcare and wellness app that can help people like Nikki navigate the food shopping experience, as shown in Figure 15.1. Here's the new experience of what's possible. When Nikki approaches the grocery store on her weekly visit, she gets anxious thinking about the 3 diets and 63 meals she needs to shop for. This feeling often intensifies as she surveys shelves with dozens of options for every packaged product that she needs.

With a feeling of relief, she remembers DPHR, an overlay app that is used in the diabetes meta-community she recently joined. With DPHR's augmented reality, she points her phone at the product shelves. For a number of the products, she can see color-coded overlays. The red overlay is for products that include ingredients which would be detrimental to her health. The yellow overlay is for products that include ingredients which may be detrimental but more information is needed. The green overlay is for products whose ingredients are safe. She can also shift the color-coding to account for her children individually or the entire family at once.

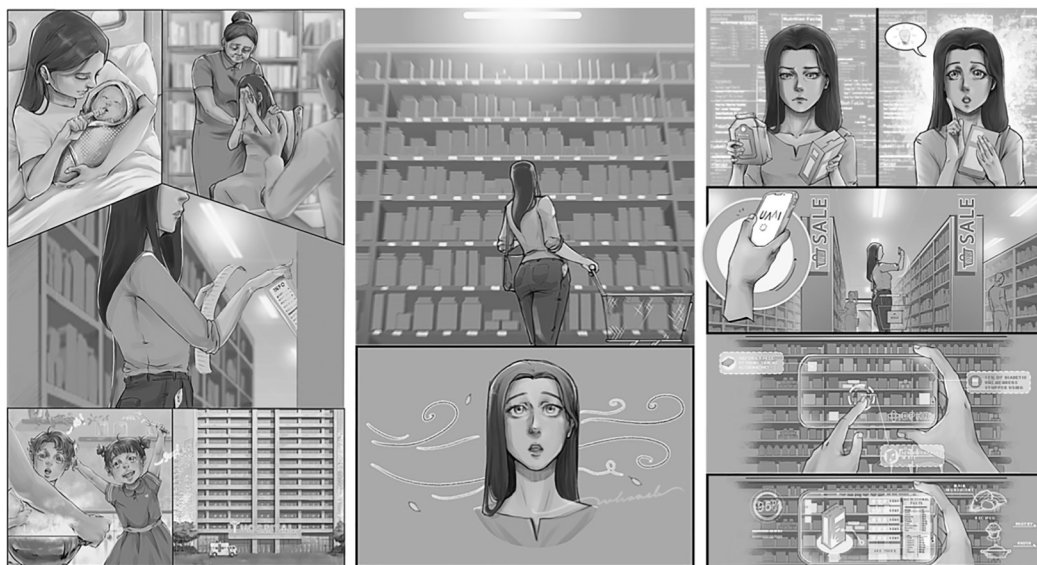


Figure 15.1 Navigating complex food choices with DPHR.

The DPHR app works on datasets cultivated in partnership with health-related meta-communities, which build and share knowledge on the Overweb. With its data, DPHR is able to support the diabetic, sickle cell, autistic, and allergic meta-communities with actionable regenerative information in real-time across various mediums including online, virtual, and phygital worlds.

COLLECTIVE INTELLIGENCE

Collective intelligence is arguably the aim of science. But the efficacy of science has faced many diminishing factors. Notably, the increasing specialization and subdivision of the sciences has researchers so concentrated on their specialty that they ignore other potentially relevant information outside of their walled gardens.

Not to mention bias, the challenges of synthesis, the difficulties of dissemination, and the limitations of the extent to which non-scientists can contribute meaningfully. Science is seeking greater collective intelligence through more open peer review processes, making tax-funded research freely available, and opening access to datasets for published works. But the process is slow and limited in effect.

Yet collective intelligence is an inherent capability for groups of humans. We have ways to do this in small groups such as group facilitation, wisdom councils, citizens' juries, citizens' deliberative councils, and the International Bateson Institute's Warm Data Labs.⁶ These in-person activities, however, are small by design. We don't have effective ways to communicate and collaborate online with large groups of people.

With the Metaweb, we increase the likelihood of collective intelligence by:

- Connecting people around information and their common interests.
- Enabling people to leave stigmergic traces of metadata for web content.
- Connecting ideas with relationships.
- Building topical information ecologies.
- Rewarding creations based on their value to collective intelligence.
- Creating better datasets for AI to build collective intelligence.
- Augmenting human conversations with AI.
- Measuring collective knowledge and wisdom.

The next three sections speak to different aspects of collective intelligence.

MAPPING THE COLLECTIVE MIND

The noosphere is the sphere of human consciousness and mental activity, especially regarding its influence on the biosphere and in relation to evolution. It began forming with early human thinking and has sped up with modern technologies that enable communication and interconnection. The noosphere itself integrates the immense variety of human beings and communities that continue to create it. A healthy noosphere respects the biosphere from which it emerged and continues to rely upon.

A large-scale neural network model of the noosphere or collective mind could be based on the Overweb's Universal Content Graph. As mentioned earlier, the closest metaphor for the graph is the human brain, with ideas being neurons and bridges being the synapses that

connect neurons. Because the graph comprises bridges built by all participants and all the meta-level information attached to ideas, it reflects the collective conscious mind. The bridges are externalizations of thoughts that connect ideas in our mind that were deemed important enough to memorialize.

The research process has two phases: gather relevant data and perform analysis. The Universal Content Graph—especially bridges, labels, and other structured annotations—dramatically shortens the time and computational power necessary in the data-gathering phase of online research. It also underlies tools that can improve and speed up the analysis phase. Bridgers get rewards when researchers cross and upvote their bridges. Data providers receive compensation when their data is used to train models, calibrate marketing campaigns, and filter content.

As participants upvote, cross bridges, and interact with ideas, they are strengthening the connections, which are the synapses of a global brain. These interactions can feed back into the AI training process. The Universal Content Graph, including curation and interaction data, can help us understand ourselves. This includes how people think about topics, how ideas connect, what paths people traverse through ideas, and what drives human opinion in specific topic areas. A global brain can also help us understand how we are collectively thinking, our why, and identify opportunities where more and better access to information can help.

The concept of the collective mind has potential applications in the following fields.

- **Complexity Science:** The concept of the collective mind can be understood through the lens of complexity science, which explores how simple components can combine to form complex systems and patterns.
- **Collective Intelligence:** The concept of collective intelligence refers to the shared and coordinated mental processes of a group of people, and how these processes can produce intelligence that is greater than the sum of its parts.
- **Network Science:** Network science studies the structures, dynamics, and properties of complex networks, and can be applied to the collective mind as a network of ideas, connections, and interactions.
- **Social Networks:** Social networks, such as online communities, can be seen as a manifestation of the collective mind and studied to understand how people interact, collaborate, and form collective opinions and beliefs.
- **Collective Awareness:** The concept of collective awareness refers to the shared understanding and awareness of a group of people regarding a particular issue or phenomenon, and how this awareness can influence behavior and decision-making.
- **Artificial General Intelligence (AGI):** The concept of AGI, or artificial intelligence that is capable of general cognitive abilities and performing tasks that typically require human intelligence, is relevant to the mapping of the collective mind as it suggests that a machine could eventually mimic or surpass human intelligence.
- **Epistemology:** Epistemology is the branch of philosophy that studies knowledge and belief, and can be applied to the study of the collective mind to understand how individuals and communities form, share, and evolve their knowledge and beliefs.
- **Human-Computer Interaction (HCI):** The field of HCI explores the interaction between humans and computers and how technology can be designed to enhance and augment human cognitive processes. This is relevant to the mapping of the collective mind as it suggests that technology can be used to enhance and augment collective intelligence.

SELF-KNOWLEDGE

The development of the Universal Content Graph has the potential to revolutionize how people understand and interact with information. Beyond providing a vast and interconnected platform for knowledge, the Universal Content Graph could play a crucial role in helping individuals better understand their own thinking patterns and biases.

Cognitive biases are a common phenomenon that lead to perceptual distortion and inaccurate judgment. Confirmation bias, for example, is the tendency to seek out and give weight to information that confirms one's existing beliefs. This can create a subjective reality that is far from the actual truth, and dictate an individual's behavior in harmful ways.

By tracking relevant cognitive biases, the Overweb aims to provide participants with interventions to mitigate their effects, making it possible to identify, understand, and overcome these biases. The dashboard will provide specific interventions, which will help individuals better understand how they think and help them to create their "subjective reality" free of perceptual distortion, inaccurate judgment, illogical interpretation, or irrationality.

As individuals gain more self-knowledge through this process, they become more intelligent. They can then share their lessons and insights with others, helping to grow collective intelligence.

Furthermore, the dashboard could support the building of one's active purpose, which contributes to both individual and collective intelligence. This is particularly relevant in peer groups, such as masterminds, where individuals can collaborate and learn from each other.

This development is significant, as it has the potential to promote a more informed, self-aware, and intelligent society. By giving people the tools to understand their own thinking patterns, the Universal Content Graph could help individuals overcome the limitations imposed by their cognitive and societal biases, leading to better decision-making and improved outcomes. The ability to understand and mitigate cognitive biases has far-reaching implications for fields such as politics, business, and technology, where clear and rational thinking is crucial.

TRAINING AI

As the world becomes more reliant on artificial intelligence, the need for high-quality training data becomes increasingly crucial. While the Internet has made a vast amount of information available, the minimal connectedness and low signal-to-noise ratio make it difficult for AI systems to effectively process and make use of unstructured data. This has led to the rise of data labeling companies such as Data Labelers, which employs over 1,000 individuals to manually label data for moderation purposes.

The Overweb, however, is set to revolutionize this process. The Universal Content Graph will enable the transformation of webpages into structured data that is connected and easily accessible. The Overweb will introduce a "smart tag" labeling system, in which participants can label content with relevant metadata, and be rewarded for doing so. This metadata can be added through public or private labeling schemas, which provide information on data format, eligibility, and rewards.

For example, participants can use a public labeling schema to tag content that is violent, sexual, or offensive, and receive rewards based on the use of this data by machine learning algorithms and smart filters. This creates a massive, high signal-to-noise dataset, which can be used to train AI systems to understand and process information in a more effective manner.

The Overweb will also support the creation of data cooperatives, allowing participants to license their activity data for machine training purposes, in return for compensation. This will ensure that individuals are fairly compensated for the use of their data, and incentivize the creation of high-quality training data.

In sum, the Universal Content Graph and the Overweb have the potential to transform the way AI is trained, providing AI systems with vast amounts of high-quality data, and supporting the development of more advanced and effective AI systems. The smart tag system and data cooperatives will ensure that individuals are fairly compensated for their contributions to the AI training process, and will encourage the growth of a more diverse and inclusive AI ecosystem.

TRANSFORMING TWITTER

On April 25, 2022, Twitter's buyout announcement quoted Musk:

"Free speech is the bedrock of a functioning democracy, and Twitter is the digital town square where matters vital to the future of humanity are debated."

Six months later, on the eve of the deal's completion, Musk doubled down in a tweet to Twitter advertisers, saying civilization needs a common digital town square, where people can debate a wide range of issues without violence. He continued:

"There is currently great danger that social media will splinter into far right wing and far left wing echo chambers that generate more hate and divide our society. In the relentless pursuit of clicks, traditional media has fueled and catered to those polarized extremes, as they believe that is what brings in the money, but in doing so, the opportunity for dialogue is lost."

Characterizing Twitter as a "digital town square" makes sense. Twitter users debate everything, including "matters vital to the future of humanity." Statista contends Twitter is more like a digital debate club because its user base is much smaller than Facebook and Instagram. And, only 30% of social media users in the US regularly use Twitter and even fewer in major European markets such as France and Germany.

We disagree. While we wonder how many Twitter users are actually bots,⁷ we also have a strong sense that real people debate real ideas on Twitter. We also recognize that most of a town may not have shown up, much less spoken in the traditional town square. Apparently, in former times, the largest crowds would show up for hangings and beheadings. The packed square could only hold a fraction of the town population. In debates, most people were watchers rather than speakers. Still, the effects of town square debates rippled into the collective consciousness and town life.

While Twitter isn't a functional digital town square everywhere, it's influential for the English-speaking world, which has shaped global socio-economics in the last century. Twitter is where serious online conversations are happening in the world's largest economy in the de facto world language.

Thus, we think the digital town square is a fair characterization, an important aspiration, and a fantastic opportunity.

But here's the rub. A digital town square is not enough. In fact, as another silo, it often degenerates into people arguing for their confirmation bias. In ephemeral arguments, facts take a backseat to emotions, unsupported opinions, and logical fallacies. People choose sides, and battles begin. People get canceled, shadow banned, and de-platformed. Worst, people polarize into fixed mindsets. Gridlock continues. Collective intelligence remains a fleeting thought.

Such is today's Twitter. Musk plans to make Twitter a better town square. Authenticating humans, eliminating spam bots, opening algorithms, and making advertising contextually relevant are important steps. But they're not enough.

A digital town square may be essential to humanity's future, but it's not enough on its own. Platforms like Twitter allow individuals to connect with people who share similar views, gain attention, express poorly supported opinions, and harass those with opposing viewpoints. However, this type of engagement does not foster much growth in perspective. This may still be the case after Musk has implemented his plans. Musk's town square could illuminate what people think and what algorithms do. It could also surface suppressed voices and narratives. But it won't build knowledge and may not lead to better outcomes.

Operating by itself, the digital town square is a firehose of disparate voices with no sense of a collective or continuing conversation. Rather, it's a continual stream of unstructured information, impossible for humans to make sense of without computers. And the insight generated is ephemeral. Most of the insight that ripples beyond the square is untraceable, occurring in people's minds, and resurfacing in conversations and tweets, often without reference to origin or substantiation. Hence, it's difficult to measure Twitter's impact on humanity's wellbeing or intelligence.

A functional town square is not a silo. It operates in a context, as part of a greater system; the town. A digital town square should also operate as part of a greater system; the metaphorical digital town that is the online world. One of the most important aspects of any town is its library.

The Digital Town Library

The library's purpose has always been to collect knowledge, learn from it, and use it to improve life. Throughout history, libraries have disseminated important advances in agriculture, architecture, medicine, art, manufacturing, war, and more via vast collections of books and printed materials.

With over 17,000 libraries and 2.5 billion materials circulating annually, libraries are an important part of the US landscape. As libraries modernize, however, they face a harsh budget environment, and technological disruption in media, scholarship, and education. Declines in usage of traditional library services at universities over the past 25 years suggest students are looking elsewhere⁸ for information. Perhaps their laptops. While the public still values access to printed books, today's libraries offer a wide range of digital, educational, social, and entrepreneurial tools, which is attracting significant public investment. Libraries have also been at the forefront of organizing public makerspaces and workspaces to promote collaboration and entrepreneurship.

We think the digital town square needs a digital town library to realize its value to humanity. A digital town library provides a shared contextual view so humanity can have an informed conversation, increasing the likelihood of outcomes that address the needs and circumstances of people and the planet. The main activity of the library is the assembly, arrangement, and connecting of information into knowledge.

We see the possibility of a virtuous cycle. The town square generates conversation. The library archives, analyzes, and turns it into knowledge and information, which influences decision-making and cycles back into the town square. Town decisions and conversations become grounded in reality, yet continue to reflect human values and concerns.

The digital town library connects online information into an information ecology that provides a shared context for communication, sensemaking, and coordination. The information ecology grows as conversations continue in the digital town square and the library

assembles it into knowledge. Robust information ecologies connect information across the webpage silos endemic to Today's Web, creating possibilities for breakthroughs, collective cognition, and new perceptions, choices, and outcomes.

A Collective Knowledge System

Together, the digital town square and library comprise a collective knowledge system, a term coined by Tom Gruber in *Collective Knowledge Systems: Where the Social Web meets the Semantic Web*.⁹ Collective knowledge systems are human-computer systems that collect and harvest large amounts of human-generated knowledge.

Collective knowledge systems have social and semantic aspects. Twitter is a social web with 238 million daily active users tweeting their opinions, creating searchable and sharable information. Gruber calls this "collected intelligence." The digital library is a semantic web, which classifies and connects social content and uses computational technology, including machine learning and text mining techniques, to find structures and patterns.

Web inventor Tim Berners-Lee said:

The Semantic Web is not a separate Web but an extension of the current one, in which information is given well-defined meaning, better enabling computers and people to work in cooperation.

According to Gruber, the challenge for next-generation knowledge systems is to match what's put online (e.g., tweets) with methods for doing useful reasoning with the data. True collective intelligence can emerge if the data collected can aggregate and recombine into new knowledge and ways of learning that individuals can't do by themselves.

The traditional Semantic Web approach would be to embed semantic tags into the webpage code. But the least friction implementation is to add semantics in a "library layer" above the webpage. In the Metaweb. This implementation abstracts the semantic content. It also extends Twitter's reach to every relevant webpage. This meta-layer could connect tweet content to the relevant content anywhere online, providing context and building an information ecology that generates value. Twitter already has internal capabilities to develop such a meta-layer with its fact checking annotation tool Birdwatch, though we think the library layer is best as an open-source project.

In the Metaweb, "information ecologists" would earn compensation for cultivating the information ecology and creating value. They would gather information to "feed" the information ecology by creating new entries, verifying submissions, and updating existing information. They could work alone or in small teams, delving into tweet archives and external sources to find substantiating information about important tweets, through scientific publications, news sites, relevant experts, research institutes, and so on.

Thus, the collective knowledge system would not only create new value streams for Twitter but also new categories of meaningful remote work and a shared contextual view of matters vital to humanity's future.

The Elusive Boundary Infrastructure

This marriage of the digital town square and digital town library also creates boundary infrastructure, the holy grail of knowledge production. Boundary infrastructures create an ecosystem where groups use and generate different knowledge primitives as part of their particular mission-driven interactions with the system. But boundary infrastructures are

rare. Most existing infrastructures that build knowledge focus on a singular information object—e.g., Twitter focuses on tweets of 280 characters or fewer—with limited use cases and often within a specific firm or group.

A study from two Parisian universities about rare disease research defined the term “boundary infrastructure.” Boundary infrastructures produce knowledge as one output of their work and articulate it with other kinds of activities and matters of concern. Boundary infrastructures serve multiple communities of practice, within a single organization or distributed across multiple organizations. The benefit of the boundary infrastructure concept over traditional unitary vision infrastructures is support for differing information objects within diverse communities of practice that share the infrastructure.

According to a seminal study in *Information Infrastructure Studies*:¹⁰

To negotiate the political, moral and epistemic dimensions of the boundary infrastructure they contribute to and rely on, actors resort to infrastructural inversion: they discuss explicitly the infrastructure itself, and strive to represent its inner workings, shortcomings and desirable evolutions. Infrastructural inversion, therefore, is not only a methodological lens for the analyst to capture how things like databases and classification systems are embedded in the many practices of collectives It is also constitutive of the practices of these collectives themselves.

Thus, participants in boundary infrastructures are in an ongoing process of reflection on the system and its processes. This reminds us of Doug Engelbart’s Bootstrap Principle, where a human-machine system simultaneously harvests collected knowledge and evolves the technology for collective learning. In these systems, both humans and machines actively build knowledge, each doing what they do best.

With Twitter, different actors could come together in a boundary infrastructure for matters vital to humanity. Creators, including video shows, journalists, authors, and graphic artists, would create meaningful tweets. Companies, NGOs, and influencers would reply and retweet the content. Content consumers, including researchers, read and interact with the content. Information ecologists would connect and label content and conversations, verify and fix submissions, and monitor the system. AI would suggest classifications and connections, look for patterns, and filter content. This ecosystem of actors could build knowledge from social interactions that are already occurring with minimal coordination.

In sum, Twitter as a digital town square needs a digital town library. The union of the town square and library could create a boundary infrastructure for matters vital to humanity’s future. This boundary infrastructure would in fact be a collective knowledge system where Twitter provides the social content and information ecologists build semantic content in a meta-layer above the webpage. The collective knowledge system would build collective intelligence that enables humanity to better address its global challenges.

Free the Bird!

Here’s an exercise to complete before moving on to the next chapter. Reflect on the following questions and how a hyper-dimensional (HD) Twitter might use the Overweb pattern and building blocks from Chapter 12:

1. How could Twitter benefit from applying the Overweb pattern to improve the digital town square and manifest the digital town library?

2. What Overweb tools would be helpful in creating a boundary infrastructure for social interaction that manifests as a collective knowledge system?
3. How might a hyper-dimensional Twitter deploy bridges, other smart tags, overlay apps, canopis, meta-communities, and one-account-per-participant?
4. What becomes possible for Twitter in embracing the next level of the Internet?

Keep in mind that Twitter, as an example, could create a custom overlay library that loads with their site, thus providing meta-layer functionality without the need for installing a browser extension. Alternatively, they could also embed the meta-layer information directly into their webpages.

However it plays out, the Bridgit DAO team would be eager to see your ideas. This exercise can be applied to any internet platform or destination site or even an aggregation of related sites, and could even inspire the creation of “Meta-tech” enterprises with a powerful advantage over the original. We would be particularly interested to see your ideas for platforms such as TikTok, Google, ChatGPT, YouTube, Wikipedia, Amazon, Bumble, Reddit, LinkedIn, Circle, Etsy, Mighty Networks, MidJourney, Substack, Instagram, Facebook, Trip Advisor, and Fandom to name a few.

As you’ve seen, if we expand our conception from a flat, static web to the hyper-dimensional Metaweb, the opportunities are mind-boggling. Any project can grow and expand its impact. We will enjoy new degrees of freedom. Today’s Web does not go away. The webpage remains the bedrock, becoming the contextual footprint for intertwined worlds of information, interactions, transactions, and experiences that emanate from pieces of content on the page.

As we progress to the next chapter, we find ourselves in an urgent situation, metaphorically trapped in an escape room as humanity, and the clock is ticking. The challenges facing our planet are numerous and pressing, and we need to solve them before the doomsday clock strikes midnight. But we are not alone in this endeavor. Each one of us has the potential to be a part of the solution.

The next chapter is all about how you can get involved in shaping the next level of the Internet and humanity. The Metaweb, and in particular the Overweb, is a decentralized system that relies on the participation and collaboration of its participants. You too can become a part of this movement and take on one of the many roles available in the system, from becoming a Bridger, to contributing to the open-source code, and much more.

Let’s grow!

NOTES

- 1 <https://permanent.link/to/the-metaweb/futurist-online-school>
- 2 <https://permanent.link/to/the-metaweb/languages-defy-auto-translate>
- 3 <https://permanent.link/to/the-metaweb/plo-lumumba-pan-africanism-video>
- 4 AMA is short for “ask me anything” and is a type of interactive post in which someone answers questions, usually in real time.
- 5 Our intention is for this to be the first metabook. Stay tuned at metawebbook.com

- 6 <https://permanent.link/to/the-metaweb/warm-data-labs>
- 7 This Twitter thread from former Twitter CEO speaks to the difficulty of identifying bots and why:
<https://permanent.link/to/the-metaweb/parag-bot-stream-bots>
- 8 <https://permanent.link/to/the-metaweb/platform-manipulation>
- 9 <https://permanent.link/to/the-metaweb/collective-knowledge-system>
- 10 <https://permanent.link/to/the-metaweb/information-infrastructure>

Your Move

The best time to unleash the Overweb was 20 years ago. The second best time is now.

—Bridgit DAO

Their egos were on the line as the giant countdown clock loomed overhead. They were close to solving the final puzzle. With it, they'd make their escape and enjoy the glory of success and bragging rights. But they had to untie 4 giant knots and discover the final door's passcode. With just two minutes left!

This was their first experience of escape rooms.¹ They didn't know what to expect. They soon realized, to escape, they would need to solve a series of puzzles before time ran out. The escape room challenged their ability to collaborate on a shared goal. "It's like team-building on steroids." They involve a theme or story and have varying degrees of difficulty. Escape rooms can bring out the best and worst in teammates as the clock ticks on and stress sets in.

We think humanity's in an escape room. Humanity needs to escape from its self-terminating path, which has us barreling towards existential threats. Time is short. According to the UN, we have until 2050 to make a massive shift. To escape, we must cultivate collective intelligence. We doubt humanity can pass this initiatory test without getting the best out of our primary tools. We're not even close with the Web.

To date, humanity's tools haven't supported the level of connection, communication, and collaboration needed to address our global challenges. While the Web has plenty of tools for individuals, it doesn't provide tools for humanity to work as a collective, such as a shared contextual view of the information ecology. This absence limits our capacity to sense-make and work together, much less to develop collective intelligence. Further, most people are disinterested; unaware of our global challenges, unwilling to prioritize them, and/or lacking confidence in their ability to provide value. We are seeing the worst in some of our teammates.

Time to Move Up the Doomsday Clock

The symbolic Doomsday Clock began in 1947 to warn humanity of the dangers of nuclear war (Figure 16.1). It shows how close our planet is to complete annihilation. It started at 7 minutes to midnight. In January 2020, the Bulletin of the Atomic Scientists moved the clock up from 2 minutes in the previous year to only 100 seconds away from midnight. They made the change because of nuclear proliferation, failure to tackle climate change, and "cyber-based disinformation."

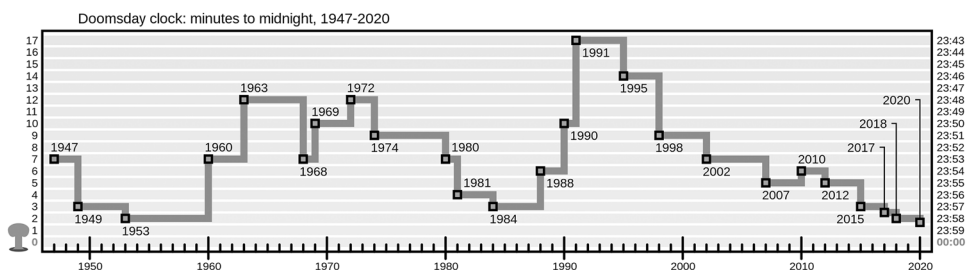


Figure 16.1 The Doomsday clock.

In January 2022, they maintained the same level in their annual statement. But in January 2023, they moved up the Doomsday Clock to 90 seconds, closer than ever to midnight, largely because of the war in Ukraine and the risk of nuclear escalation. The new clock time was also influenced by the continuing threat of climate change, and “the breakdown of global norms and institutions needed to mitigate risks associated with advancing technologies and biological threats.”

In their annual statement, Rachel Bronson, PhD, President and CEO of the Bulletin of the Atomic Scientists, said: “We are living in a time of unprecedented danger, and the Doomsday Clock time reflects that reality. 90 seconds to midnight is the closest the Clock has ever been set to midnight, and it’s a decision our experts do not take lightly. The US government, its NATO allies, and Ukraine have a multitude of channels for dialogue; we urge leaders to explore all of them to their fullest ability to turn back the Clock.”

The Metaweb can help turn back the clock by providing powerful new tools for collaborating on global challenges, such as climate change, biological threats, and AI alignment. For example, forming a massive online research collaborative or MORC would enable researchers, practitioners, policymakers, and other stakeholders worldwide to collaborate without direct coordination. It can also connect the Splinternet into a shared contextual view that could potentially be helpful for supporting diplomatic responses to regional conflicts. By working together, we can harness the power of technology to build a safer and more sustainable future for all.

Considering our existential quandary as an escape room, we need massive amounts of freedom, collaboration, expression, and creativity to solve its puzzles. But that’s what Today’s Web and the Attention Economy suppress if not prevent. Today’s Web is neither safe for nor supportive of creative collective responses to humanity’s situation. It is unsafe because of scammers, abusers, and false information. In optimizing for engagement, the Attention Economy amplifies herd behavior, meme slinging, and clout chasing. Further, we don’t have cognitive freedom or adequate online tools for working together as a collective.

In one of his videos, Boyce Watkins, an American author, social commentator, and investor, said his #1 job was to encourage his viewers to be free. Boyce says:²

Let’s lay the ideas on the table. Freedom of expression, thought, ability to brainstorm, and look for the best ideas wherever they lie. You can’t solve an equation when you have too many constraints. The more constraints, the harder to solve the problem, the narrower the set of potential solutions.

This chapter is an open invitation to those called to shape the future of the Internet and humanity. It will guide you on how to get started, what roles are available, and how you can contribute to this decentralized system. Join us in this exciting journey, as we work together to solve planetary challenges and build a better future.

Together, we can ensure that the doomsday clock stops in the nick of time and that humanity has a solid foundation to thrive on and offline. So come, be a part of this movement, and let's build the future we want to live in.

FINDING YOUR PLACE

The Metaweb has opportunities galore. Like webpages, the space above a single webpage is unlimited. The emergent action above the webpage is in the Overweb the Mask Network (mask.io), and a Web3 platform built on the NEAR Protocol called Dapplets. With any ecosystem, it's about finding your place, and how to create value. We'll focus on the Overweb and we encourage you to check out our friends at Dapplets as well. In the next section, we explore the bridger role and other ways to contribute and reap rewards.

Dapplets

On Today's Web, only website owners can create and earn money from business processes that run on their websites. This is because ownership means full control over the users' communications.

The Dapplets Project disrupts this paradigm by creating technology that enables new kinds of applications—dapplets—that democratize the web experience. Dapplets run on top of existing websites, in the website context, but out of the website owner's control. This new paradigm enables brand-new applications and business opportunities.

Dapplets technology also has an important social effect. It allows censorship-resistant user flows on websites, even for banned content and banned users. Dapplets cancel cancellations.

Dapplets is a true Web3 project: unstoppable, permissionless, anonymous, open-source, and decentralized, not only on the backend but on the UX level too. Find out more at dapplets.org.

ARE YOU A BRIDGER?

Bridgers organize, share, and monetize their research by building bridges.

Bridgers are the 15% of people who search multiple times daily. They can't help re-searching their topics of interest on the Web, seeking new information and perspectives that help them understand the Why. Bridgers think it's all about "connecting the dots," and the bridge is emblematic of what humanity needs. Let's make bridge the word-of-the-decade.

For bridgers, most research begins with online search. They seek to fill gaps in their understanding and the relevant information ecology. Bridgers combine a drive to learn more about a topic with knowledge of keywords, search terms, advanced search expressions, and which search engines are best for specific use cases. They may spend hours searching on the Internet.

Bridgers discover and learn the most amazing things. They find needles in digital haystacks. They have frequent breakthroughs in understanding.

But until now, they haven't had effective options to organize, share, or monetize what they learn.

They could write a post or make a video about it. But that's time-consuming with uncertain benefits. Unless they have a following, the potential exposure (and therefore impact) is fleeting and limited. Hence, they often relegate their amazing discovery to a preferred information silo. This could include pasting into Google Docs or Evernote, texting or emailing themselves or friends, bookmarking, or posting to a Telegram group or their Facebook Wall.

But with Metaweb, they can make bridges, monetizing their research. Hence, we consider them information ecologists.

In sum, bridgers are avid knowledge creators and consumers who share some commonalities. They:

- Do informational and transactional searches.
- Are interested in understanding *the Why* and applying knowledge.
- Want to learn and advance knowledge within their communities.
- Want to supplement income or donate efforts to support a cause.

OUR COMMUNITY

The Overweb is an inclusive community that provides income opportunities for people of all backgrounds and major web languages. We all have our own unique combination of awareness and blind spots. By holding space for different perspectives and bridging different languages, the Overweb creates the possibility of people learning from one another if not together.

The ecosystem has many roles for individuals, including

- Bridge builders aka Bridgers
- Ad watchers (a compensated activity)
- Content curators
- Translators
- Transcribers of video and audio
- Content validators
- Bad actor detectives
- Community managers
- Bridgework service providers
- Community voters

BROAD APPLICABILITY

The Overweb is building an ecosystem which includes bridgers, creators, freethinkers, as well as websites, advertisers, web marketing companies, and organizations that want to engage with their audience around information and content.

We expect sports, food, politics, health, celebrities, hip-hop, pets, and blockchain/crypto to be among the first information ecologies to harden. We're excited to see what kinds of bridges and topics create the most value. It could be elections, football or fútbol, Kardashians, or cats. Or perhaps more serious issues like the Ukraine war, boosters, crypto, or regeneration. Your guess is as good as ours. We're excited to see what bridges become popular and how they can elevate humanity and help identify false information.

The Metaweb can organize information ecologies and provide decision context for many domains, including those that involve many data and content sources, such as but not limited to:

- Upleveling knowledge and wisdom
- False info analysis and fact checking
- Legal texts and records
- Community decision making
- Learning subjects
- Researching subjects
- Literature reviews
- Proofs
- Analysis of thinking and biases
- Decision analysis
- Risk mitigation
- Product development
- Corporate decision making
- Developing marketing plans
- Story design and analysis
- Sports draft choices
- Investment
- Reviewing performance videos
- Analysis of differences in beliefs
- Analysis of groupthink
- Education
- Psychological analysis
- Sociological analysis
- Political analysis
- Biological analysis
- Chemical analysis
- Engineering analysis
- Design processes
- Book/film/music reviews and analysis
- Criminal investigations
- System design
- Business model canvas
- Project management and evaluation
- DYOR (Do your own research)

The Metaweb supports several types of enterprise participants as well:

- Advertisers who want to reach viewers focusing on topics related to their offerings.
- Partners with aligned products and services who want to reach consumers focused on topics related to their offerings.
- Online sites with high-quality content who want traffic and engagement. Examples include YouTube shows, podcasts, radio and television shows; news sites; organic/sustainable products; knowledge economy sites; teams; nonprofits; and blogs. The Overweb can provide them with an analysis of attention on their site so they can understand which ideas garner attention, keep people on their site, and generate advertisement click-throughs.
- Organizations with teams working on projects that involve organizing and processing information from the Internet. These include journalism, detective work, research and development, advocacy, and hundreds of other use cases. This is especially useful for large corporations and organizations that may have teams in multiple divisions and/or geographies and network organizations and other types of alliances and associations of multiple organizations that need to share best practices and/or otherwise engage in collaborative knowledge building.
- Knowledge ecology companies who create and trade bridges for the asset value, create content (e.g., state-of-the-research summaries), and provide services to the Overweb ecosystem (e.g., building genesis constellations of bridges on a topic). We see companies in less developed countries with a history of having been sites for outsourcing—such as India, the Philippines, and Africa—forming businesses or adding divisions to pursue the Overweb business opportunity.
- Social networks and other sites who want to extend their presence to overlays that operate anywhere on the web.
- DAOs, online communities, and companies who want to extend their reach to any page on the web.

- Journalists, independent researchers, activists, research institutions, universities, and others who collaborate and build upon each other's research.
- They all can receive compensation for the value their contributions create in the network. Together, bridgers will break knowledge silos, question the status quo and mainstream narratives, propose alternative perspectives, and connect disciplines.

Bridging surfaces the perspectives of diverse communities as they become relevant. Often, marginalized peoples have skills and world views that enable them to survive if not thrive in difficult circumstances. They have knowledge to share. We see enterprising individuals and groups in lower-income communities and developing countries creating businesses around building bridges and/or offering bridgework and other Overweb services to the developed world.

As the Overweb ecosystem grows, bridging could become a digital land grab similar to NFTs and domain registration, but for knowledge. Bridging rewards creativity, insight, and breakthroughs while creating rich data for supra-information interactions, next-generation apps, chatbots, and search engines.

Bridges self-assemble into the Internet of Ideas, which enables rapid content discovery, discernment of false information, learning breakthroughs, and unparalleled creative expression. Thus, by decentralizing links, the Overweb resolves many of our earlier concerns about the Web. It also builds a shared context and collective intelligence that can help fulfill the Internet's original promise. Let's go, bridgers!

METAWEB MINDSET FOR CREATORS

Now that you understand what the Metaweb is, we suggest trying on "the Metaweb mindset," a four-step process for opening up value creation possibilities in your field(s) of interest. This process provides guidance for engaging with the Metaweb.

First, identify the most pressing and intriguing issues and topics. Bring them to the forefront of your awareness. Note questions for which you need answers. Review and update your to-do list. Look at the news to see any developments which affect your priorities. Identify your priority issues or topics for the moment. Note any keywords or hashtags that can help find and distinguish your issues and topics.

Second, think about the types of people who might be helpful as you move forward with your priority issues or topics. List the webpages or types they are likely to visit. Also, keep a list of thought leaders, pundits, and/or influencers with whom you have common interests. If you don't know relevant webpages or people, search for some. You may also want to join relevant meta-communities.

Third, get active in the Metaweb on the pages you identified in the previous step. Install the Presence extension. Visit the pages and filter the canopis with keywords to surface relevant conversations and people. You can reply to the conversations or send the person a direct message or even a friend request. Next, look at the visible list. Filter to find people with your interests. You can send them a direct message or a friend request.

The fourth step requires a higher level of awareness. Pay attention to your thoughts and emotions when you are scanning webpages. Notice what captures your attention or triggers thoughts and emotions as you scan the page. Note the specific content or claims to which you are reacting.

1. Focus on interesting pieces of content or claims.
2. If information or interactions exist, explore them.
3. Ask yourself what additional information or interactions would be helpful.

To fill this void, consider making bridges, notes, or using other smart tags for the content or claim. If you have relevant thoughts, make a public or private note. If inclined, search for related information and make bridges.

The following information could produce useful bridges:

- Studies related to the subject
- The source of the information
- More general information related to the subject
- Historical information
- Supporting evidence
- Contradicting evidence
- Models
- Opinions
- Judgments

METAWEB MINDSET FOR BUILDERS

As a builder, the Metaweb is your oyster. Your main tools are smart tags and overlay applications. In the future, you will extend meta-communities, create on-page interactions in the Presence sidebar, and build open-source algorithms. The section discusses the mindset for building on the Metaweb.

First, identify the problem you are addressing and what's not working on Today's Web. For example, SaaS sites may want to extend their reach to the entire web and add new on-page functionality. Or, perhaps, a community you are working with needs more information to make optimal decisions, or would benefit from new types of interactions on relevant webpages.

Second, identify use cases that would resolve the problem. With the existing web service, we may want to consider how smart tags can attach information and interactions from our application to content snippets. We can also consider how an overlay application can surface relevant information and interactions from our application on relevant webpages without someone having to attach a smart tag.

Consider what information would improve decisions, how to build and access this information, and how to deliver it. If we wanted community members to disseminate information, we would consider a smart tag implementation. If it wants to be automatic, it's better suited for an overlay application.

Third, design and develop the user experience and interface within the open-source frameworks for smart tags and overlay applications. Register your creations with the Overweb application registry.

LET'S HEAD IN THIS DIRECTION

The Web is one of the most influential technologies in history. In three decades, over 4 billion people are online. The Web promised to change how we interact and gain knowledge. It was to raise all boats by making knowledge accessible to everyone. Though it has fulfilled some of its promise, the Web hasn't made humanity more collectively intelligent. We, however, need collective intelligence now. Humankind must change how it operates to address our global challenges. The Web was a good start.

We need a next-generation web that connects information and people and gives them a presence above the webpage. No longer trapped in social handles with limited rights and

freedom of movement, people can meet on webpages of their common interest. As you read these lines, know that you can meet your people online now. On a canopi.

Canopis side-step filter bubbles, connecting real people who share a common interest. Jump on one of our canopis or other sites where people congregate in canopis. Children's book lovers might visit the Pachaverse canopi. The Metaweb-curious and serious might visit the canopis for Bridgit DAO, Presence Browser, and the Overweb.

We also need to get information out of silos. This is done with bridges, the connective tissue of the Metaweb. Bridgers connect two pieces of content with a relationship. If bridges have the correct relationship, they post to the public ledger and self-assemble into a universal knowledge graph that connects the world's information on an idea-by-idea basis.

We need to shift from an Attention Economy that harms people to a knowledge economy that builds a thriving ecosystem and creates a cornucopia of value, builds knowledge, and inspires real action in the world by leveraging our collective wisdom. To do this, we must harness the Web to build collective intelligence and the key collective cognitive capabilities: sensemaking, meaning-making, and choice-making.

Imagine a web where platforms no longer exploit user data or think of people as users. This is about how we choose to spend our time and energy. If you weren't already clear, the transition is now. If we're heading to the New Earth, everything needs to change. We need tools that bring out our best. Humanity needs a next-level Web for real people, not serial abusers, fake accounts, or unidentified bots. Communities need safe digital spaces.

The Metaweb can host this technological shift.

Decentralized public space above the webpage enables us to exercise choice and regain control over what we see online. We use evidence from the Metaweb for sense-making and a shared context to build knowledge, learn, and think together.

The Metaweb defrags the Web and transforms it into an integral information ecology with deep layers of context for online ideas, building a Web worthy of our trust. A web where algorithms and online business models create value for real people and respect digital rights, including the right to inspect the code. Providing a shared context that supports discernment of what to believe online opens the cone of possibilities to an engaged and informed populace and a future that is accountable, free, and democratic.

Last but not least, we now need to leverage the potential power of the Web to support the unprecedented levels of connection, coordination, and collaboration that are needed to address our existential threats. As Elon Musk says, the future is bright, but we cannot be complacent. We now understand what is missing. If you want to take part, the future hyper-dimensional web is your oyster.

Without a doubt, the Web is the best way to share information. The Web can reach anyone on the planet. With the Metaweb, it'll be even better. If humanity is to become wise, it will only happen with a meta-layer above the web. Your move is to join us and help build a future to be proud of.

A BICYCLE FOR THE MIND

At the close of the last decade, with—for example—the Cambridge Analytica election hacking and the exploitative data practices of Internet Platforms, we saw in retrospect what can go wrong when the platforms adhere to a “Move fast & break things” ethos that ignores effects on humans and society. As we consider how to repair trust and reimagine relationships between Internet platforms, people, and data in the coming decade, a view into how computers began provides valuable context for focusing our attention and resources.

In a 1989 documentary, Steve Jobs said the computer is our most remarkable tool. “The equivalent of a bicycle for our minds.” Riding a bicycle is the most efficient means of locomotion. Next best are salmon, horses, and jet transport.³

In an earlier presentation about Apple, Jobs said the bicycle shows humans, as toolmakers, can make tools that amplify our inherent abilities. The bicycle improved humans’ ability to move efficiently. Building computers improves our ability to think efficiently.⁴

All due respect to Mr. Jobs, but thinking more efficiently has proven insufficient. Despite billions of computers in use, far surpassing the human population, humanity faces mounting existential risks (e.g., nuclear wars, artificial intelligence, and ecological crises caused by unsustainable human activity). A democracy seeking to mitigate such risks cultivate an informed populace and develop tools that build humanity’s collective intelligence.

BICYCLES FOR THE COLLECTIVE MIND

This era reflects great division among people, massive degradation of the planet’s life support systems, and unprecedented distraction. All exacerbated by technology. Most tech extends systems that create or contribute to the problems. Most impact-focused innovations address symptoms rather than the problem’s roots. Without addressing the roots, solutions are likely to be temporary fixes, with problems manifesting through alternate pathways.

Our premise is that one root of all the problems is the lack of collective intelligence and wisdom, often reflecting a corresponding absence of shared knowledge. The logic goes: if we had more shared knowledge, collective intelligence and wisdom, we’d be smarter and wiser, and therefore make better choices, diminishing if not resolving the threats. We may be smart as individuals, but as the human collective, we need to uplevel. Especially as machines barrel towards AGI.

Although collective intelligence is the aim of science, to date, humanity has experienced little progress. Perhaps it’s the lack of a technological category or meta-category focused on collective intelligence or wisdom. As new systems emerge, a “named” category can provide a container to connect, support, and illuminate projects.

As toolmakers, we can make tools that amplify our shared knowledge and collective intelligence. Let’s make bicycles for the collective mind.

We propose WisdomTech—a new genre of computing that focuses on improving thinking, accelerating learning, and building collective intelligence in service to a democratic, regenerative future and our growing human capacities and consciousness. WisdomTech—anchored by emerging technologies such as AI, extended reality, big data, and distributed ledger technologies—can, for example, expose misinformation and help bring about a more useful virtual experience, while protecting personal data.

WisdomTech could address different aspects of the information value spectrum from data (unitary facts and figures) to knowledge (data in context), sense-making, and wise choice-making, including

- Improving dissemination and access to knowledge.
- Building knowledge while keeping a reference to its provenance.
- Improving pattern discovery, critical thinking, and decision-making.
- Convening evidence-based dialogues among diverse thinkers.
- Exposing fake news and misinformation

—all the while protecting data ownership, free speech, and human rights.

Acknowledging that exponential technology and, in particular, AI will far outpace advances in human intellectual capacities, WisdomTech pioneers collective approaches to data, sense-making, and choice-making that prioritize safety, resilience, regeneration, inclusion, and trust. WisdomTech enables the creation and improvement of technologies, products, companies, and markets that strive for positive social and environmental impact, including protecting and nourishing our potential and human rights. Aligned with the European Union's flagship Next Generation Internet program, WisdomTech supports an Internet that is sustainable, resilient, inclusive, and trustworthy.

WisdomTech would create bicycles for the collective mind.

Bicycles for the collective mind are tools that build shared knowledge and collective intelligence. By definition, shared knowledge and collective intelligence require harnessing the “wisdom of the crowd.” As noted by James Surowiecki in the book “The Wisdom of Crowds,” crowds often outperform experts when there is a diversity of opinion, independent thinking, local knowledge, and an aggregation mechanism.

The Overweb provides such an aggregation mechanism in the Universal Content Graph, which enables ecosystem participants to build online knowledge maps of ideas and the relationships between them, creating a shared context for collaboration and communication across intellectual boundaries. Shared context is what the International Bateson Institute calls warm data,⁵ but for online information. The Overweb aims to connect all the world's information, enabling 360° context for any idea—whether on the web, in a digital overlay on physical reality, or the metaverse.

Again, we call this Ubiquitous Context. Imagine the possibilities for collective intelligence if everyone—no matter where they are, in a digital world, online, or in physical reality—had access to deep layers of context.

Mapping online knowledge is but one approach. Humanity must build our capacity for collective intelligence in this upcoming decade, leveraging crowd wisdom across disciplines and worldviews in service to a democratic and sustainable future.

A NEW WINNING NARRATIVE

A winning narrative explains how a system can meet the needs of all stakeholders. It emphasizes our best and guards against our worst. It's the story we want to tell.

This new winning narrative aims for honesty, pragmatism, and frankness in charting a new course for humanity. The new narrative addresses the paradoxical reality that—at the same time—many key economic and wellness metrics (e.g., GDP and life expectancy) are at all-time highs, yet other metrics concerning, for example, resource depletion, topsoil levels, and species extinction portend a less than bright picture. Not to mention the existential threats that humanity faces, such as exponential weaponry, climate disruption, artificial intelligence, robotics, and nanotechnology.

Consider that age-old maxim, Knowledge is Power. From the beginning, humans have leveraged knowledge to consolidate power. For most of human existence, knowledge was sequestered, secured, and rationed in silos—libraries, academia, companies, institutions, and mystery schools. Gaining knowledge was often expensive in time, money, social capital, and/or lives.

But we are living in revelatory times. Unprecedented amounts of information are at our fingertips through mobile devices and chatbots. We're unable to ingest and process this information now. Once we develop such capabilities, we'll better understand our world. As we shine a light in the shadows, patterns will reveal themselves. More truth and

understanding will rise to the surface of human consciousness. Collective cognitive breakthroughs will reveal solutions for our intractable problems.

The WisdomTech Emerging narrative recognizes that humanity has made huge strides, co-evolving with tools, so much that not-so-ancient societies would see modern humans as gods. Yet humanity's existence has been a momentary blip if we consider all time.

An apt metaphor for humanity is a teenager. Young people operate with underdeveloped prefrontal cortexes until their early-to-mid twenties and are thus prone to rash decisions. The Internet is a child. It's natural that the Internet—prioritizing potential for impact—is subject to reinvention. Hence, the conversations around the Next Generation Internet, led by Europe, as well as the Web3 movement, the AI containment movement, and this book.

Meanwhile, thought leaders and social enterprises are taking a stand for technology that builds knowledge, raises consciousness, and grows collective wisdom. They believe that increasing our collective intelligence is the most important thing that humanity can do. We can start by reinventing humanity's open knowledge base, the World Wide Web.

Regarding wisdom and knowledge building, today's Internet is dysfunctional.

We think the Internet is far from reaching its potential for building humanity's knowledge base, our understanding of what it all means, and our ability to make smart choices on an individual and collective basis. This not only affects our emotional and mental well-being and our capacity for self-actualization and self-transcendence, but also cripples our capacity for collective intelligence and other collective cognitive capabilities.

The problem is the Internet proliferates content rather than building knowledge, understanding, and wisdom. Social accounts pump out content to gain followers. Companies produce a lot of content for machines, bots, and web spiders. New content is often uninformed by previous content and the links are based on what the page author knows and who they or advertising networks want to promote.

At its essence, the Web is transient—we discuss new content for an instant. Research from Google Trends found that the average 'big' news story lasts about seven days before the public moves on to the next crisis. Although they remain available, online conversations have short lifespans unless they go viral. False information is more likely to go viral than the real thing. Conversations are haphazard because of the structure of the containers that hold them—think about how challenging it is to glean different perspectives, logic, and sources from a TikTok feed, a thread of tweets, and Facebook comments.

Even Reddit, which seems more oriented towards knowledge sharing, shuts down conversations after six months by "archiving" them. Reddit—like the comments on Facebook and forums—is unstructured, so it is challenging to get a full sense of the community's perspectives and virtues. Reddit encourages anonymity, which diminishes transparency regarding conflicts of interest that may exist for contributors and makes it easy for people to keep their online comments separate from their actual identity. Hence the rampant troll problem on the platform.

WisdomTech would help gather relevant information for a decision. It would help us discern false claims in the news, as well as resolve conflicting information. It would also disincentivize if not prevent the creation of duplicate content. WisdomTech would help us with sensemaking, identifying patterns, and understanding *the Why*. WisdomTech would help us make wise decisions based on our knowledge and understanding.

It would keep content and conversations alive by creating spaces where people can have safe, structured ongoing conversations about vital matters. WisdomTech would enable smart choices by individuals and communities. It would enable people to observe their own thinking (e.g., their confirmation biases) and how the community thinks about issues.

Improving critical thinking, knowledge discovery, and collaboration, as well as conflict resolution, could enable us to transcend circumstances that separate us; come together in dialogue around claims, perspectives, and needs; and make better choices.

We must live the story we want to tell. In 2030, what do we want to say about now? An unacceptable story is a beleaguered and traumatized humanity looking back from the future and reflecting that we never tried to use technology to make ourselves smarter, despite our deep understanding of the Internet's shortcomings. Despite five plus decades of warnings from Doug Engelbart and his progeny. Despite this book.

Fortuitously, a cadre of next-level web companies backed by forward-thinking investors are emerging in the WisdomTech arena, and hold great promise for raising consciousness and collective wisdom. An example is Betaworks' Tools for Thinking accelerator out of New York City. We expect little from Silicon Valley.

If we are up to the challenge, Wisdom Tech changes the game for:

- Emotional and mental health.
- Human knowledge, understanding, and wisdom.
- The prospect of humanity making smart choices about its future.

THE INTERNET'S SAVAGE DAUGHTER

At the end of 2020, inspired by the song "Savage Daughter," sung by a trio of shamanesses in a sound healing ceremony in the tropical dry forest along the coast of San Juan del Sur in Nicaragua, a plant told us a story about a wild feminine spirit long suppressed in most aspects of human experience that is now emerging alongside the coronavirus.

The refrain of Mistress Windreth Berginsdottir's song "Savage Daughter" is:⁶

"I am my mother's savage daughter,
The one who runs barefoot, cursing sharp stones.
I am my mother's savage daughter,
I will not cut my hair, I will not lower my voice."

Below we recount our recollection of the plant's story.

From time immemorial, the plant told us, this wild feminine spirit has been associated with truth. In ancient Egypt, the deity of truth, Ma'at, was a goddess. Leonardo da Vinci thought of truth as female, "Truth was the only daughter of Time." Truth and women both align with beauty and purity.

Although suppression of this wild feminine spirit has been uneven, the plant noted, "Nowhere has her suppression been deeper and more complete than in the invisible mechanisms and machinations of the Internet." But not how you think.

The plant described a future Internet—a veritable truth machine. A deeply connected, multi-dimensional Internet that layers knowledge on top of webpages. The Internet of Ideas. Imagine a more feminine, truthful, and authentically wild Internet.

An Internet connecting information and people. Information is no longer in silos. People are no longer in bubbles. Imagine an online future with fair value exchanges. Platforms no longer exploit user data. Users are now participants. Anyone you see is real and in good standing. No unidentified bots, fake accounts, or serial abusers.

This technological savage daughter liberates our minds. She enables us to regain control of our attention and exercise choice in what we see online and what we don't. The savage

daughter gives us evidence-based information and knowledge for sensemaking. She provides a shared context for collaboration that enables humanity to think, learn, and build knowledge together for the first time.

Search and filter bubbles become the lazy choice. We value, nurture, and measure trust. Online business models and algorithms create value for real people and respect their digital rights.

The Web's de-fragmentation into an integral information ecology provides access to deep layers of context for online ideas. People have the tools to determine for themselves what to believe online. Informed and engaged citizens take part and vote on policy in this wild democratic future.

As we listened to the plant's story, we became excited about this technologically savage daughter and her potential to heal our deeply fragmented and polarized digital world. We had the sense that such a shift in our vital technologies could also heal our inner worlds. But our pragmatic side realized not everyone wants a truth machine.

Sensing our need for reassurance, the plant riffed off celebrated author and political activist Arundhati Roy and Mistress Windreth⁷, "Another Internet is not only possible, she is on her way. On a quiet day, I can hear her breathing ... One of her names is Bridgit."⁸ She will not cut her hair. She will not lower her voice."

We need the Savage Daughter that demands to connect everything, for us all to be heard, for us to integrate our shadow. She enables the higher-order masculine, which runs through us, the deep desire that many of us feel to make the major shifts that right our relationships with ourselves, one another, and the planet.

We need to conquer ourselves as the human collective and become a higher-order humanity compatible with life and freedom. But without embracing the Savage Daughter, we cannot give the higher-order masculine what it needs for high-order perception, agency, and meaningful problem-solving.

Perhaps we need a high-order androgynous web that balances the feminine and masculine and brings out the best in our human collective.

From one angle, we see high-order feminine energy manifesting in deep authentic connections between people and information. From another angle, high-order masculine energy layers knowledge on knowledge, in an ever-evolving quest for truth. Steel sharpens steel.

This high-order androgynous web can be the masculine extreme, deactivating the accounts of serial abusers—a digital "off with their heads!" And it can also be the feminine extreme, creating spaces for the most intimate of digital connections. Hence, the Web can be the androgynous, peaceful yet sword-wielding Athena!

We need a future web that balances masculine and feminine energies, enabling us to surf interconnected information flows. One that enables us to grow, maximize agency, coordinate, and collaborate as a collective of collectives that steers humanity towards healthy relationships with itself and the planet. Were this the next level of the Internet, it would shine a light on our collective shadow, enabling us to move forward in balance and alignment, so that humanity ascends to the next level of consciousness. Anything short of this risks utter disaster.

NOTES

1 <https://permanent.link/to/the-metaweb/first-escape-room-experience>

2 <https://permanent.link/to/the-metaweb/boyce-watkins-video>

3 <https://permanent.link/to/the-metaweb/steve-jobs-documentary>

- 4 <https://permanent.link/to/the-metaweb/apple-presentation-video>
- 5 <https://permanent.link/to/the-metaweb/warm-data>
- 6 <https://permanent.link/to/the-metaweb/savage-daughter-blog-video>
- 7 Of the Shire Shattered Oak, Principality of Northshield in the Middle Kingdom.
- 8 Brigitte, Brigid, Brigid, or Bridgit is both the name of the Celtic deity for all things high including intelligence and wisdom and the name of the DAO—Bridgit DAO—that we started in 2018 to bring bridging to the world.

Epilogue

United We Compute or Divided We Fall

The key thing about all the world's big problems is that they have to be dealt with collectively. If we don't get collectively smarter, we're doomed.

—Douglas Engelbart

Friends, we are all participants in a grand experiment in consciousness and wisdom. The Internet, a marvel of technology that promised to bring us closer together and make us collectively smarter, has not lived up to its promise. The problems facing our world today are complex and overwhelming, from poverty and inequality to misinformation and cyberattacks. Hence, the term “Polycrisis”¹ is emerging alongside an international organization agenda² “to master the future.”³

And yet, the words of Douglas Engelbart ring truer than ever. Engelbart's bottom-up approach was to empower individuals with technology to increase their collective intelligence and ability to collaborate, rather than relying on top-down solutions imposed by central authorities. Bottom-up is made practical by the idea of subsidiarity, which holds that social and economic issues should be dealt with at the most local level possible, and that higher-level authorities should only intervene in issues that cannot be effectively addressed locally.

This epilogue is a reflection on this bottom-up journey towards collective computing, and a call for future-conscious humans to come together, reshape the Internet, and create a better future for all. For, as we modify the old saying, “united we compute or divided we fall.”

LET'S GO AGAIN SALLY⁴

At first, we thought the Metaweb would receive a hearty welcome in Silicon Valley. After all, it solved their problems. Or so we naively thought. But as we soon realized, we were swimming upstream. We were told, “People won't install an extension,” “Google will crush you,” and “Many have tried this. What makes you think you can be successful?” Yet, we never heard, “Why are you different?”

Supposedly “open” organizations like the W3C (the World Wide Web Consortium) and Mozilla proved impenetrable. Incubator programs focused on “Fixing the Internet,” like Mozilla Builders Program and Betaworks, were disinterested. The Tools for Thinking accelerator didn't think much of us. Foundations committed to stopping “fake news” like the Craigslist Foundation and the Omidyar Group gave us a stiff arm. Pharrell's Black Ambition wasn't happy with our application. We almost got a grant from the NEAR foundation. Even American's Seed Fund didn't support this sprout.

We harbor no resentment. Everything happens for a reason. Perhaps we weren't far enough along.

Like any groundbreaking technology, the Metaweb in its various nascent forms has been told "NO" countless times. The Metaweb (as we have represented it) has been denied entry to prestigious accelerators, financing, grants, speaking slots, mentorship, and other types of access in Silicon Valley. Our ideas of a hyper-dimensional web were apparently antithetical to the prevailing yet faltering wisdom that is lurching us into a future that most of us don't want. Even some leaders in the annotation space have been reluctant to throw unequivocal support behind the Metaweb.

The preponderance of evidence suggested that the Metaweb would not have its breakthrough in Silicon Valley. As George Bernard Shaw so aptly reminds us in *Man and Superman*, "The reasonable man adapts himself to the world: the unreasonable one persists in trying to adapt the world to himself. Therefore, all progress depends on the unreasonable man." So we left ...

THE PROBLEM IS BIGGER THAN WE IMAGINE

Even after learning about the Metaweb, it is not uncommon for people to question its purpose or financial viability. A novel analysis, however, using a modified Metcalfe's law reveals that an immense amount of value is untapped. The Metaweb presents an unprecedented opportunity to capture this value for the collective good while addressing the most pressing issues of our time.

Yet the legacy web has significant inertia, and people are right to ask, "what's a big enough problem that's going to inspire enough people and companies to adopt a monumental shift in thinking around the Web?" The answer lies in understanding humanity's relationship with computers. We have evolved from having hard drives the size of commercial refrigerators seven decades ago, to now having slim handheld devices that enable us to instantly retrieve information and communicate wherever we are.

The Metaweb is the next step in this evolution, enabling us to break free from the cognitive cages of the legacy web and unlock indirect collaboration via digital stigmergy. It's a monumental yet necessary shift for us to collectively capture the value of collaboration and democratically address the most pressing issues of our time.

For clarity, Metaweb theory solves this problem:

Our personal computing paradigm traps us in cognitive cages that obfuscate our incarceration and prevent us from accurately conceiving our position, much less operating as a planetary collective. Hence, we leave tremendous value on the table and remain unable to effectively address our planetary challenges and existential threats.

The impact of the problem is clear. It is widely acknowledged that humanity is facing significant challenges, whether it be in the news, economic indicators, case numbers, or unintended consequences of emergent technologies. It is also evident that traditional solutions such as political and regulatory approaches and global cooperation are insufficient. We can utilize network mathematics to evaluate the opportunities that are being missed. While these methods are not perfect, they still provide valuable insights.

Rather than jump directly into the analysis, let's recall the story of humanity's relationship with computers, which is one of constant evolution and adaptation. We used early computers for calculations and data processing. They were large, bulky, and required specialized

knowledge to operate. As technology advanced, computers became smaller and more accessible to the general public. With the advent of the Internet, computers became humanity's primary tool for communication and connection and knowledge base.

Along the way, we realized that the potential of computers was far greater than we had imagined. We saw the possibility of using computers to enhance our ability to learn, work, and play. We saw the potential for computers to improve our productivity, connect us with people from all over the world, and make our lives easier.

However, as we became more reliant on computers, we also began to see the negative effects. The Internet, which was once a tool for connection, became a source of fragmentation, isolation, and exploitation. We found ourselves trapped in filter bubbles, surrounded by clickbait, propaganda, and misinformation. Our attention was constantly being pulled in different directions, and our ability to focus and think critically was diminished. Our computers had captured our imagination and constrained us in cognitive cages with invisible bars.

The integral accident of the personal computer is that it indeed is personal, allowing a singularly focused direct connection to itself. Its purpose is to provide content, interactions, and ads for us to engage with. Other people show up as content authors and through messages. The integral accident is reinforced by the structural constraints of Today's Web—being flat, static, and centralized—which naturally led to fragmentation through competition for mindshare. There are multiple sites, apps, and communities for every perspective on every topic, often promoted with superlatives, such that we don't need to think. And with the coming flood of generative AI content, it's only going to get more muddled.

The specter of deplatforming, shadow banning, and other demonetizing controls exacerbate the web's fragmentation. For example, many alternative voices on YouTube have backup channels in case their main channel gets community guidelines strikes. To make up for income lost to demonetization, many creators offer more provocative content through Patreon channels. Common practice is now for creators to share scrubbed content on YouTube, and uncensored content on multiple smaller outlets that support free speech such as Rumble, Minds, Parler, Gettr, Truth Social, and Locals.

In this informational quagmire, rather than think critically to form our own opinions, we shift our attention among the ever-ready streams of perspectives that feed our confirmation biases, and engage when so inclined. In the ever-growing sea of content, sites, and apps, attention becomes the scarcest of commodities. Everyone competes for eyeballs. Differentiation and separation is the result. Algorithms and auctions for your attention are the mechanism.

And because computers were beholden to the individual rather than a network, the user was made to be accountable for everything that occurred to them. No one even considered making systems that held people accountable for harm to others because it wasn't an obvious derivative from personal computing or social media business models. Spammers, scammers, catfishes, impersonators, cyberbullies, trolls, and sexual predators proliferated on the Web. Caveat emptor was the unspoken rule, embedded in the terms and conditions that almost no one read; use at your own risk.

The effect was to isolate us in a dangerous new digital world. We have multiple devices, all providing direct access to the Web, yet none offer us a real-time digital presence or community, although this is envisioned in the metaverse. Yet still without accountability or safety. When we are on the web, we are alone in a digital den of invisible thieves and extractors.

Messaging may reduce the feeling of isolation. But it can be distracting when off-topic, often requiring us to code-switch. It is also mediated by the machine and often unencrypted, which has privacy ramifications. Again, use at your own risk.

We don't have safe, durable spaces for sharing relevant, contextual information so when we find things, we put them in silos, usually on our devices or on the cloud. And more of us are now retreating from the major platforms to smaller more authentic spaces that are not infested by trolls and fake accounts. This furthers the fragmentation.

Personal computing is about our direct, one-on-one relationship with the computer, and by extension with the virtual machine that comprises the wires, networking equipment, servers, databases, and organization across the planet that seek to capture our attention. Four billion of us are hooked up to the machine, many having few or fleeting meaningful connections with other humans online or IRL. In many Western countries, most people don't even know their neighbors. When you watch YouTube, when you read a Wikipedia article, when you look at your feed, when you read your email, you're completely alone.

Being isolated opens us up to exploitation by algorithms and platform business models. The funders of these systems are quick to point out the systems' celebrated network effects, yet their platforms do not provide you network effects in getting your problems solved, your questions answered, or your life improved. You don't get to meet people who can supply the missing pieces to your puzzles or connect with people that you uniquely can benefit from. Apparently, what's good for the goose is not good for the gander. #FowlPlay

The machine wants your attention, money, and time. It doesn't care how you feel except to the extent that it can use that knowledge to get more from you. The machine is an extractor; it wants to suck whatever energy it can get from you. It wants you to think that being online is enriching your life so you keep coming back for more. But it doesn't care if it is actually good for you. It wants more and more of our time, attention, and money.

Many entities are part of the machine—the manufacturers who make the machine, the electricity producers and providers, the network operators, the access providers, and the platform operators. The machine includes anyone who thinks of you as a stakeholder, consumer, or victim. They may not know who you are but they have you profiled. And they compete to get access to you, through auctions and through your choices.

They bid to get in front of you; to get the chance to extract from you. They use everything you give them—your behavior, your preferences, your creations, and your data—to figure out how to best extract from you. And again, they don't care how you feel as long as you keep clicking. In fact, they benefit from dysfunctional, mal-intentioned, trolling interactions, and inauthentic activity just as much from beneficial activity. The machine is indifferent. Any virality is good virality because it pays.

Together, fragmentation, isolation, and exploitation limit our imagination, not only for what we imagine is possible for the Web but also for our online and IRL lives, putting us in a cognitive cage. Many don't believe it's there because they can't imagine the contours of the bars, which gives the machine plausible deniability. We don't see the possibility of anything different because we can't see the web changing—literally or figuratively—and we intuit that there is no real solution. Yet we know we're in the cognitive cage when we realize we aren't able to align our individual and collective interests or even act in our own best interests online.

We know there are major problems in the world, but we have no agency as singular isolated beings, floating alone in an ocean of trending content. We're just one person. Even with people, governments, and nonprofits supposedly working on fixing the Web, the problems don't seem to go away. In fact, the machine shows us they are getting worse. We're stuck in our cognitive cage, isolated online, and unable to imagine a better world. So we shut down parts of ourselves that would do something if it could. The Wicked Web of the West is now in control.

IN THE FOOTSTEPS OF ANTS

All great inventions emerge from a long sequence of small sparks; the first idea often isn't all that good, but thanks to collaboration it later sparks another idea, or it's re-interpreted in an unexpected way. Collaboration brings small sparks together to generate breakthrough innovation.

—Scott Belsky, founder, product, investor

This reminds us that the way out is to go in ... together. It's much too late to pull the plug on the machine. The cat is out of the bag, and together we've made a mess out of the house. We need computers to help us organize the clean up. But rather than 4 billion separate conversations with the machine, we need the machine to see us as a coordinated network of cleaner uppers so that we can do our respective parts, ideally with zero duplication, minimal coordination, and leveraging our strengths, interests, and desires to the extent possible.

We often hear the phrase “standing on the shoulders of giants” when it comes to great success, for example, in reference to the founding fathers of the US or in award recipient speeches. We should, however, also take inspiration from another tiny but mighty creature—ants. Ant colonies are incredibly organized and efficient, all without the need for direct communication. It's time to learn from these masters of collaboration and organization.

We have to change our thinking about the Web such that it supports a shift from a personal computing paradigm that disempowers and extracts our energies to one that embraces and enhances the collective good. A paradigm where we use computers together as a network of networks, as a collective of collectives, for the greater good, following in the footsteps of ants.

What could that look like? Whatever problems you are dealing with or working on, you would be connected to other people in the world who are working on the same thing and who are able and willing to help with these types of problems, and who are available when you need them. Or who have left traces on the Web that can lead us to resolution. The hard part is finding one other and connecting, which is facilitated by on-page presence. Just go to the relevant webpage and see who's there, or who has been there recently. Or what they wrote in the canopi. Besides on-page presence, AI will also be helpful in helping us find one another.

Collaborations can start with a basic agreement on principles of coordination, with the views of the problem and solution expected to converge in the course of the coordination. Related bridgework may provide a shared contextual view of the problem and solution spaces that evolves over time. That way, when we are talking about something, and looking online in the same direction, we see the same thing.

Here are some examples:

- A team of molecular biologists is working on a delivery system for an enzyme that breaks down viral pathogens. They're having some issues with getting the enzyme to the most affected cells and tissues. One of them opens a recent study about relevant enzyme delivery to the proximate areas of the body. Through on-page presence, she reviews the visible list to find researchers who have visited the page recently. She reviews the page chat to see who may be working on relevant enzyme delivery issues.
- A parent and their teenage son are having a hard time communicating about the son's porn addiction. The parent visits a webpage about porn addiction. Their AI assistant also takes them to relevant sites for authoritative parenting, parental monitoring, educative parenting, and emotion coaching. Through on-page presence, the parent

finds a network of parents, counselors, former porn addicts, and therapists that are able to support them separately and together, when they need them.

- Rural transplants have purchased their dream country home in the backroads of Tennessee but in the late Fall, they discover that the wood-fired stove is not producing enough heat. This is their first time operating a wood-fired stove. If they don't remedy the situation, it's going to be a long winter. The first thing he thinks of is on-page presence. He goes to a wood-fired stove site but no one's there, so he heads to a site about rural living where he speaks to someone in Germany's Black Forest, who suspects it could be a blocked airflow and suggests that visit his local community page. At the community page, he finds a neighbor willing to help. He knows the person is a real person in good standing because of their digital footprint. The neighbor comes by and helps him determine that, thank goodness, it is blocked flue rather than a poorly placed air intake or a lack of proper ventilation. On the way out, the neighbor also suggests that they load less wood into the stove at a time.
- A utilitarian cyclist's old bike lock is not turning with the key. She goes to the product page of the lock on the manufacturer's site thinking that they would have troubleshooting information, but the lock is too old. Using on-page presence, she reviews the comments from others on the page chat, and finds that the lock can be opened with a hairpin. After unlocking her bike, she comes back to get some recommendations for her next lock. Better safe than sorry.

Whatever and whenever we need, the collective is there, whether through content, online, or in person.

As we continue to evolve our relationship with computers, we are again seeing that its potential is far greater than we had imagined, as will be demonstrated by the analysis below. We see massive potential for value creation through on-page presence, knowledge building, and collaboration. We realize, by working together, we can tap into the full potential of the collective and its innate intelligence. We see the potential for computers to bring us together, and to make us smarter both individually and as a collective, and we can imagine the feedback loops between the two.

Hence, the future of our relationship with computers is one of collective computing, where we use computers together as a collective of collectives to address our individual needs as well as our planetary challenges.

Adopting a collective computing approach would help address the United Nations Sustainable Development Goals (SDGs) in a comprehensive and unified manner. Economic interests may prioritize certain SDGs that align with their goals, but connecting researchers, practitioners, and initiatives within and between SDG goals, building a bridge-based information ecology for the SDGs, and reconfiguring SDG incentives, or providing additional incentives, could encourage stakeholders to consider SDGs equally and holistically.

The shift towards collective computing would enable new forms of collaboration, and help develop humanity's collective cognitive capabilities, which are also essential for addressing global challenges. Furthermore, it would allow for the creation of a collaborative online knowledge-based economy, where the value of information is determined by the quality, relevance, and accuracy of the data, rather than the attention it generates.

This is problematic, as the attention economy not only exacerbates our global challenges but hampers us from addressing them. On the other hand, a knowledge-building economy could be foundational in solving them.

By connecting and collaborating in ways that are impossible with Today's Web, we can tap into the full potential of the collective and create something greater than the sum of its

parts. The difference in value between 4 billion isolated users on the Web compared to 4 billion collaborators is truly staggering.

Using Metcalfe's law, we can measure the difference in value mathematically. Metcalfe's law states that the value of a network is proportional to the square of the number of its participants:

$$V = n * (n - 1)/2$$

where V is the network value and n is the population.

The more participants connected in a network, the greater the network value. In the context of the web participants, the value increases exponentially with the connections between participants. The increase in connections between participants can lead to various benefits such as an increase in productivity, connection, communication, collaboration, sensemaking, and collective intelligence. It creates diversity in perspectives, talents, and interests that can be leveraged for deeper engagement and better decision-making.

If we think of the web as an orchestra, each individual user is a solo musician with their own talents and abilities. The orchestra can only work if the solo musicians collaborate. Imagine an orchestra of solo musicians, each playing with headphones on, only hearing their own contributions. The result would be a disjointed, chaotic, and cacophonous performance.

On the other hand, when these solo musicians are able to work together with the help of a conductor, the result is a beautiful and powerful symphony. In the same way, when 4 billion isolated web users are able to collaborate and work together with the help of a collaboration machine, the result could be a more coordinated and productive experience for all. By connecting and collaborating, we can tap into the full potential of the collective and create something greater than the sum of its parts.

Applying Metcalfe's Law to the scenario of 4 billion isolated web users, the network value is 4 billion. When, however, we apply Metcalfe's Law to the scenario of 4 billion people who are able to collaborate on webpages, the network value increases astronomically. The value of the network is equal to:

$$V = n * (n - 1)/2 = 4 \text{ billion} * (4 \text{ billion} - 1)/2 = 8 * 10^{18}$$

where V is the network value and n is the population.

This is $2 * 10^9$ times greater than the value of the network when 4 billion people are isolated. This staggering difference in value theoretically could occur if 4 billion people were able to collaborate and connect on the Web. But this obviously is not a reasonable assumption given human constraints such as time, attention, and access.

Applying Metcalfe's Law to 4 billion humans collaborating on the Internet is problematic for several reasons. First, human relationships and connections are not as simple or straightforward as connections between devices in a network. Humans have limited time and capacity for close, stable relationships, which can impact the formation and maintenance of connections on the Internet.

Second, the value of a connection between humans is determined by the quality and nature of those connections. In other words, not all connections are created equal and the value of a connection can vary greatly depending on the context and purpose of the connection.

Third, humans are not purely rational actors and their behavior cannot be easily predicted or modeled, which makes it difficult to accurately measure the value of connections between humans on the Internet.

That said, the stigmergic nature of the Overweb creates an environment where the value of a connection is not solely determined by the people we know. The Overweb allows us to indirectly collaborate with all those who share our interest in a topic, by leaving traces in the form of smart tags, canopi chats, and visibility lists that layer knowledge on webpages. This expands the number of people that we can collaborate with well beyond the number of stable relationships or even the number we can directly communicate with. It also allows for larger group sizes and a more diverse set of perspectives. In other words, the Overweb's stigmergic nature creates a more flexible, inclusive, and diverse collaboration environment.

To make allowances for our humanness though, we conservatively assume that even with powerful stigmergic collaboration tools, people on average will choose to collaborate within small networks. We can modify Metcalfe's law to calculate the network value assuming that people collaborate in groups, as follows:

$$V_s = n/s * s * (s - 1)/2 = n * (s - 1)/2$$

where V_s is the network value for group size s , s is the group size, and n is the population.

A good starting point for group size is the Dunbar number, a concept introduced by British anthropologist Robin Dunbar in the 1990s. It refers to the maximum number of stable social relationships that a human can maintain at any one time. He found that the size of the neocortex, the part of the brain responsible for complex thinking and social cognition, is proportional to the size of social groups in primates. He proposed that this relationship also applies to humans and that the average human can maintain stable social relationships with around 150 individuals.

The Dunbar number is often used as a benchmark for the size of social groups in human societies, such as the size of a tribe or a village. It is also commonly used in the context of social networks and online communities, where it is believed that people have difficulty maintaining more than 150 stable relationships.

To calculate the network value of 4 billion people collaborating in groups of 150, the Dunbar number, the value of the network is equal to:

$$V_{150} = 4 * 10^9 * (150 - 1)/2 = 298,000,000,000 = 2.98 * 10^{11}$$

So, according to Metcalfe's Law, the network value of 4 billion people collaborating in groups of 150 is 74 times larger than that of an isolated network like Today's Web.

Figure 17.1 shows how the increase in the network value varies according to collaborative group size. The network value of 4 billion people in groups of 10, 50, and 100 collaborators is 350%, 2350%, and 4850% larger, respectively, than that of 4 billion isolated users.

Besides the value of collaboration based on person-to-person connections, we can also use Metcalfe's Law to estimate the contextual value of networks, specifically from connecting information with bridges. The bridges generate the Universal Content Graph, which serves as the foundation of the Overweb. As discussed in the synaptic web section above, context enables and enhances sensemaking, collective intelligence, search effectiveness, usability and navigation of the web, and productivity.

First, we need to estimate the number of bridgeable pieces of online content. Competing factors are at play. There are 1.7 billion websites, each of which can have many pages, and each of which can have multiple bridgeable pieces of content (e.g., Wikipedia, an average video or podcast). We can leave out the 96% of the Internet that is the deep web and darknet. We can also assume that a substantial fraction of surface web content is duplicate or worthless SEO pages. Taking this all into account, the value of the network is equal to:

Value Increase by Collaborative Group Size

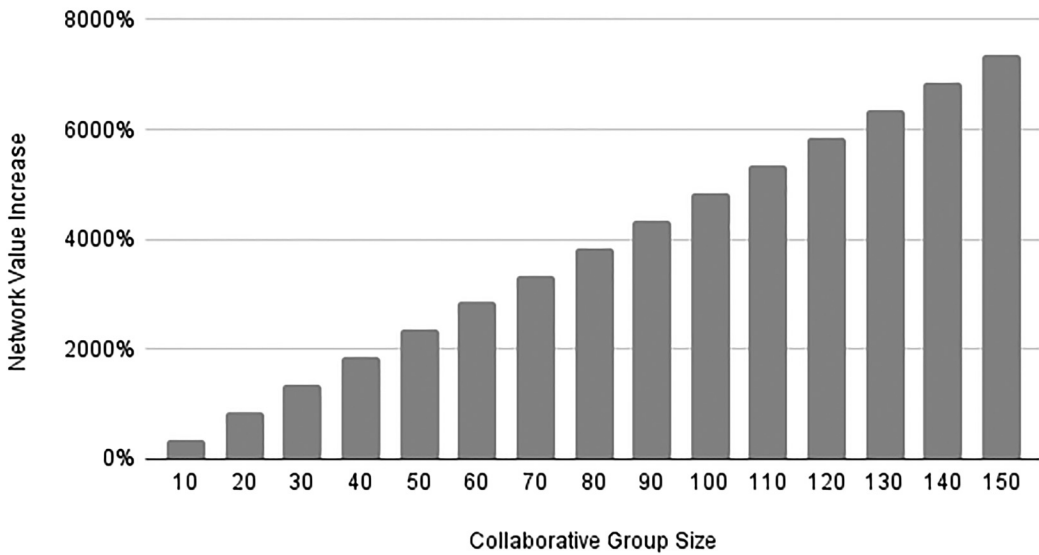


Figure 17.1 Untapped network value increases exponentially with the size of collaborative groups.

$$C = S * sw * p * c * d$$

where C is total pieces of bridgeable content, S is total sites or 1.7 billion, sw is the fraction of the surface web or 0.04, p is pages per site—estimated at 10, c is pieces of content per page—estimated at 10, and d is the fraction of useful content on the surface web—estimated at 0.8. This resolves to:

$$C = 1.7 * 10^9 * 0.04 * 10 * 10 * 0.8 = 5.44 * 10^9 = 5.44 \text{ billion}$$

The relatively modest bridgeable content estimate of 5.44 billion unique pieces is fewer than the world population. More so than humans, most pieces of bridgeable content are not connectable. Content only bridges to content to which it directly relates. To account for this, we need to limit the average number of bridges per piece of content. Let's assume that each piece of content has 100 bridge connections.

Using the modified Metcalfe's Law for groups from above to calculate the network value of 5.44 bridgeable pieces of content with 100 bridge connections, the contextual value of the network is equal to:

$$V_{100} = 5.44 * 10^9 * (100 - 1)/2 = 269,280,000,000 = 2.69 * 10^{11} = 269 \text{ billion}$$

We can also calculate the contextual value of Today's Web. First, we need to estimate the number of linked pages. In this context, we are only interested in outbound non-promotional links. As before, we need to screen out the deep web and darknet as well as duplicative and SEO content.

$$P = S * sw * p * l * d$$

where P is total linked pages, S is total sites or 1.7 billion, sw is the fraction of the surface web or 0.04, p is pages per site—estimated at 10, l is outbound non-promotional links per page—estimated at 2 (which may generous), and d is the fraction of useful content on the surface web—estimated at 0.8. This resolves to:

$$P = 1.7 * 10^9 * 0.04 * 10 * 2 * 0.8 = 1.088 * 10^9 = 1.088 \text{ billion}$$

Using the modified Metcalfe's Law for groups from above to calculate the network value of 1.088 linked pages with 2 links each, the value of the network is equal to:

$$V_2 = 1.088 * 10^9 * (2 - 1)/2 = 544,000,000 = 5.44 * 10^6 = 544 \text{ million}$$

But we're not done. We must also consider, whereas bridges are bi-directional, links are unidirectional. Therefore, the contextual value of a link is half that of a bridge, which puts the contextual value of the web at 272 million. This analysis suggests that bridging could increase the contextual value of the Web by several orders of magnitude:

$$\text{Increase in value from bridges} = 269 \text{ billion}/272 \text{ million} = 988$$

This huge increase in value isn't shocking given the paucity of links on Today's Web and the relative flexibility and power of bridging. But how does the value manifest? The value comes from context-based services that enable or enhance sensemaking, collective intelligence, search effectiveness, usability and navigation of the web, and productivity. These all depend on the Universal Content Graph, which is the aggregation of bridges.

But how do we really know the value is there? We know that people value search the most out of all Internet services. A 2017 study by MIT and the University of Groningen in the Netherlands found that people would have to be paid \$17,530 per year to stop using search engines.⁵ Assuming this value scales with inflation, it would be over \$21,000 in 2023. With the release of ChatGPT in late 2022, we are already seeing less dependence on Google; it is not clear how much lower the stop-using-search figure would be or how high the stop-using-ChatGPT value would be, given that ChatGPT cannot at this time replace search engines.

With bridges aggregating into the Universal Content Graph, we will see an improvement in content discovery as well as usability and navigation of the web, which could also improve search engines that incorporate the data. Imagine having the ability to focus on a specific topic and easily access verified, relevant information related to it, without having to search for it. Not only information but interactions such as conversations, meetings, polls, and more; and you only see what makes it through your smart filter, which you control. This not only saves time and effort, but also creates a highly efficient and valuable hyper-dimensional experience.

Coupling the Universal Content Graph and AI enables a global representation of human consciousness that could anticipate what you need before you ask for it. Imagine the value of the data for discovery, sentiment, intent, and polling as well as training and tuning AI and even more so if capabilities such as A/B testing are native. People could potentially earn a basic income from allowing the data to be captured and used in this decentralized global brain.

Beyond content discovery, we know that people value truth. The context created by the Universal Content Graph enables sensemaking, fact checking, and collective intelligence. There is also huge value in helping people avoid fake accounts, scams, cyberbullies, trolls, abuse, catfishes, and false virality. The AI will enhance all of these as well.

And vice versa. The Universal Content Graph enhances the AI, providing it training data, helping it find relevant information, and enabling it to generate in context. There is much largely unrecognized value in generating information and content where you are, freeing you from having to copy, paste, and code switch over to a chatbot silo. Imagine seeing a meme. Long click to select the image, prompt to generate 7 original memes based on the meme template that relate to your project, select and make any needed edits to those you want to use, schedule them, and move on to the next task.

That all said, it may take years to reach the assumed level of 10 bridges per piece of bridgeable content. Hence, we will continue to the next step, building upon the previous analysis of the network value of collaborative groups, which is to estimate the actual value we are leaving on the table by fixating on the personal computing paradigm.

While there are no definitive estimates for the value of the Internet, a 2019 report valued the U.S. Internet Consumer Surplus at almost \$8 trillion. Consumer surplus is an economic concept that refers to the difference between the amount that consumers are willing to pay for a good or service and the amount that they actually pay for it.

In the case of the Internet, consumer surplus can be thought of as the value that consumers derive from using the Internet beyond what they pay for it. This can include things like the ability to access information, communicate with others, and participate in online communities and marketplaces. Assuming the US is a quarter of the world economy, the world Internet consumer surplus would be about \$32 trillion.

Assuming this is the entire value of the Internet, enabling collaborative groups of 10, 50, and 100 would value the collaborative Internet at \$112 trillion, \$752 trillion, and \$1.5 quadrillion, respectively.

That's no small potatoes. Obviously, there are many degrees of imprecision and assumptions embedded in those estimates. Nonetheless, they are directionally correct as we are missing out on the huge gains available through collective computing.

But you may still be wondering how massive value can emerge from within the current system? History shows us a pattern of parallel systems rising to dominance. These systems can emerge within an existing system by offering advantages and benefits that existing systems cannot match. Over time, they gain acceptance and become the dominant system.

The tapestry of history is woven with systems superseding one another, and in most cases, the next one creates much more value than the previous, hence the transition. For example, the rise of Pythagoras' spherical Earth, the Roman Republic, Christianity, the Industrial Revolution, and the Internet all exhibited this pattern, which can be explained in part by the Two Loop model discussed in Chapter 7.

The first instantiation of the Metaweb—the Overweb—is an emergent parallel computing system that supports collective computing and even has a complementary currency system through the use of tokens. With the Overweb, we go from a flat and static Web to a hyper-dimensional space with unlimited parallel layers, each with a meta-view of Today's Web, unlocking a world of new possibilities and potential. It's exciting to think about the impact this technology could have on our society and the way we live our lives.

TODAY'S WEB IS FAR FROM DECENTRALIZED EVEN WITH WEB3

If humanity is a teenager and the Web is a child, the Metaweb is in the birth canal. While the Metaweb incorporates elements of Web 2.0 and Web3 and builds upon—and over—Web 1.0, it differs from them. The existing iterations of the Web—including Web3—do not decentralize the Web experience, which is perhaps the most important aspect of the Web.

While Web 1.0 decentralizes information, Web 2.0 decentralizes publishing, and Web3 decentralizes ownership and transactions, even in their most idealistic and elusive incarnations, the webpage remains under the control of the page author.

In Web 1.0, Web 2.0, and Web3, the content on a page is sacrosanct. None of these support their users in contemplating, questioning, or discussing information. They allow the page to stand on its own, so that users can consume, digest, and share it, regardless of value, accuracy, and relevance. Anyone can publish on the Web and social media. But it's difficult to get anything taken down.

The Metaweb takes a novel approach to the web experience. While it doesn't alter webpage content, it creates decentralized public space above every webpage. Communities can create worlds of information, interactions, transactions, and experiences above any webpage.

LEAPING FROM WEB3 TO WEB5

On June 10, 2022, Jack Dorsey, co-founder of Twitter and CEO of Block, said his team's most important contribution will be Web5, an "extra decentralized web platform." Web5 brings decentralized identity and data storage to applications. Developers can focus on creating delightful user experiences while returning ownership of data and identity to users.

The logic is that $\text{Web5} = \text{Web 2.0} + \text{Web 3}$, combining the captivating experiences of social media with decentralized ownership of data and identity, in this case, built on the Bitcoin network. While Web 2.0 and Web3 operate in silos, and a marriage of the approaches would be a giant step forward, it leaves Web 1.0 out of the picture.

The Web has over 1.7 billion websites, with hundreds of thousands being added every day.⁶ The overwhelming majority of these are Web 1.0, as Web 2.0 is less than 300 notable social media networks⁷ and Web3 has less than 5,000 active dApps.⁸ Thus, Jack's Web5 (which differs from the Web5 proclaimed by web founder Tim Berners-Lee in 2009⁹) may only affect thousands if not hundreds of sites in the near term.

This begs the question, does yet another web technology silo merit having its own web generational number? Web 2.0 and Web3 are web silos. Should Web5 be another web silo that combines two silos? Time will tell.

Jack's proclamation created a void that the Internet noticed: what happened to Web4?

Before Jack's Web5 and more since claims to the Web4 mantle emerged from startups and pundits. A semantic web driven by AI and machine learning. An invisible yet active web with virtual assistants that know your habits and tastes. A seamless and minimal friction combination of blockchain, machine learning, AI, VR, AR, robotics, 5G, and IoT. Mobility and voice interaction between the user and robots. A web connecting mobile devices in real-time in the physical and virtual worlds. While these are likely on the way (or have already started on Web 3.0), none seems worthy of the Web4 distinction, given their timelines. Some may ultimately become part of Web 4.0 with the Metaweb and a fully immersive metaverse, but this remains to be seen.

ENTER THE METAWEB

The Metaweb is here, and it brings together elements of Web 1.0 and Web3, providing a wallet, annotations, and interactions over billions of webpages. It can also work over Web 2.0 social media sites and traditional dApps. Hence, the Metaweb is a safe, social layer with

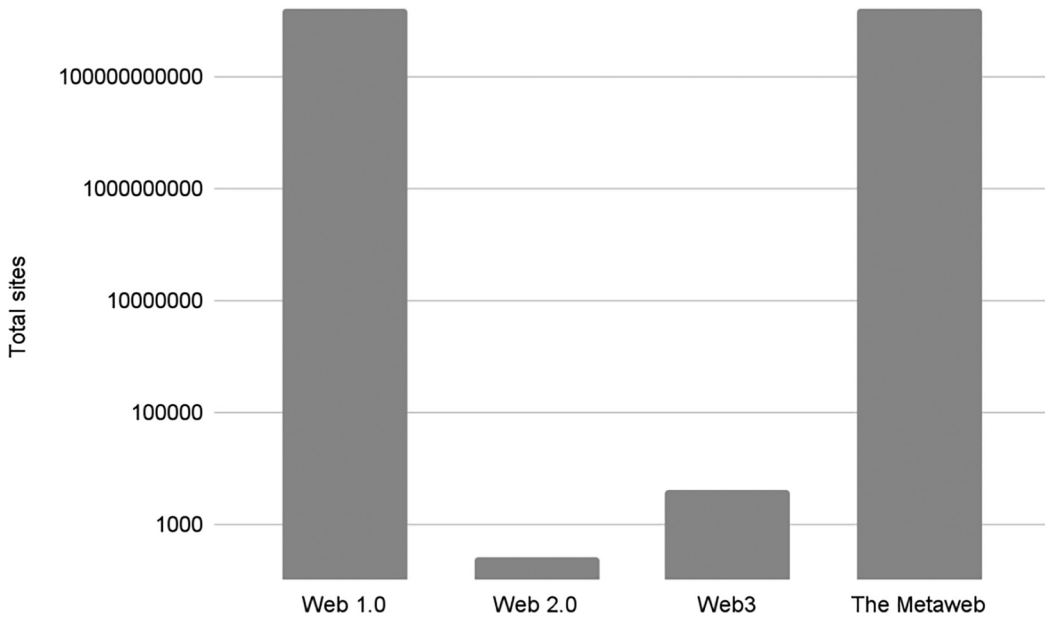


Figure 17.2 The number of sites involved in each generation of the Web.

ownership and accountability above the entire web. It's a Web above the Web. Or more precisely, a hyper-dimensional Web over the Web. An Internet over the Internet.

Thus, the Metaweb is Web4.

As shown in Figure 17.2, the number of sites addressed by the Metaweb dwarfs the number of sites involved in both Web 2.0 and Web3.

Whereas the other Web4 claimants are neutral regarding democracy, the Metaweb aligns with a democratic future. It is inconceivable that extrapolating Today's Web into the metaverse or any AI, virtual assistant, 5 G mobile-first world without unprecedented safeguards will portend the fair and just democratic future that is our birthright.

The Web is our most important communication and collaboration tool, and it anchors modern life. Our global challenges need to be addressed collectively with unprecedented levels of connection, communication, and collaboration. Given our extensive global challenges, we must leverage our most important tools—and especially the Web—to their fullest collaborative expressions. Humanity is already exploring the Web's potential for negative and harmful expressions. Now it's time to discover what's possible in the positive polarity. If we continue to rely on a web rife with scams, abusers, and false information, we can't imagine humanity being able to transcend its existential threats.

The Metaweb unlocks this possibility.

TOWARDS COLLECTIVE INTELLIGENCE

From the 1960s to the 1980s, a shy engineer named Douglas Engelbart sought to bring collective computing to the Web. He wanted to equip teams working on global challenges with much better tools. Although best known for the Mother of All Demos in 1968, where he unveiled the computer mouse, Doug's most significant contributions were envisioning tools that would increase humanity's collective intelligence.¹⁰

As the rate and scale of change increased worldwide, Engelbart foresaw the exponentially increasing complexity and urgency of problems and opportunities we are now facing. Thus, he surmised, our collective capability for pursuing these challenges would need to increase, if not surpass, this rate of change for humanity to thrive on Earth. One of his two primary focuses was boosting human intellect, which he later called Collective IQ. It was a measure of how a team or organization could address a complex, urgent problem or opportunity. The other was boosting humanity's ability to apply Collective IQ to improve its Collective IQ; getting smarter at getting smarter.

Doug designed systems to enable people to work together on important challenges by drawing people in, making connections between them, and advancing their ideas towards solutions. He designed tools, processes, and organizational structures to develop solutions for important problems by leveraging collective cognitive capacities into applicable knowledge. These capabilities include collective perception, memory, insight, vision, planning, reasoning, foresight, and experience. The systems would evolve. Both the people and information could change the process, enabling the system and its various elements to improve over time.

Doug wanted humanity to think, learn, and build knowledge together, to address our then-forthcoming but now-pressing global challenges.

But the computer revolution took hold around him. Opposing paradigms such as office automation, artificial intelligence, WYSIWYG, and personal computers gained popularity, closing off Engelbart's funding sources and the organizational support for his work. Thus, the world veered away from Doug's vision of collective computing that boosts human intellect towards the personal computing-based hot mess that we have today.

We, at Bridgit DAO, are proud to follow in the giant footsteps of Douglas Engelbart, Marc Andreessen, Bernadette Farias López, Vannevar Bush, Ted Nelson, Ada Lovelace, Tim Berners-Lee, Grace Hopper, Vint Cerf, Wendy Hall, Dr. Thomas Mensah, Dan Whaley, Ruben Brave, and other pioneers who brought collective computing to this point. We're exploring new capabilities for boosting collective intelligence, including new tools and ways of interacting with knowledge and one another. These involve novel methodologies, symbols, organizational roles, structures, and languaging.

We are excited about the co-evolution of the Web, the spatial web, the conceptual realm, and the Metaweb, as well as knowledge-building processes and decentralized organizations. Together, emergent systems can enable the continuous building and application of knowledge, supporting a growing collective intelligence that enables humanity to transcend its current and future challenges.

Yes, our challenges are enormous, including the need to break free of our cognitive cages so we can overcome the limitations of flat web thinking. Furthermore, the coming Dead Web of generative content, misalignments with both the Attention Economy and with AI, and the United Nations' Sustainable Development Goals all speak to interlinked planetary problems that require planetary solutions. By transcending our mental blocks and working collaboratively, we can create a better future for all. Let's seize this opportunity and build a brighter, more democratic, and free world.

Now is the time to come together as a united force and follow in the footsteps of ants! United we compute or divided we fall. Let's collaborate indirectly, in real time, and in person to create the future we all want to live in. May we rise to the challenge and build a better, more prosperous, regenerative, and fair world for all beings.

May the future be bright, democratic, and free. Imbued with balance, deep connection, and reverence. Anchored by love, light, and truth All above, on, and beyond the webpage.

NOTES

- 1 <https://permanent.link/to/the-metaweb/polycrisis>
- 2 <https://permanent.link/to/the-metaweb/world-economic-forum-global-risks>
- 3 <https://permanent.link/to/the-metaweb/world-economic-forum-master-the-future>
- 4 Kudos to the first to independently decipher this puzzling section title and inscribe it as a digital artifact on Bitcoin
- 5 <https://permanent.link/to/the-metaweb/internet-consumer-surplus>
- 6 <https://permanent.link/to/the-metaweb/how-many-websites> accessed July 28, 2022
- 7 <https://permanent.link/to/the-metaweb/wikipedia-social-networking-services> accessed July 28, 2022
- 8 dApps are distributed applications. <https://permanent.link/to/the-metaweb/how-many-dapps> reported 4073 when accessed on July 28, 2022
- 9 <https://permanent.link/to/the-metaweb/berners-lee-web5>
- 10 <https://permanent.link/to/the-metaweb/engelbart-tools-collective-intelligence>

Glossary

Explanations in italic are homegrown in the Overweb

Adjacent Space *The adjacent space is the virtual space around a website that can contain web content and virtual reality experiences. When on the Overweb, one can access a website's adjacent spaces.*

AI-DAO *David Shapiro coined the term AI-DAO as a DAO (decentralized autonomous organization) with both AI and human members. it is a way to keep humans in the loop with autonomous AI systems. AI-DAOs may require autonomous agents seeking membership to have a specific cognitive architecture (e.g., constitutional AI) or implement heuristic imperatives. An AI-DAO itself may also implement heuristic imperatives.*

Annotation A meta datum (e.g., comment, explanation, question) attached to a location on a webpage. Annotations are a layer on top of the page.

API A set of functions and procedures that allow applications to access the features or data of an operating system, application, or another service.

Artificial General Intelligence (AGI) AGI is a type of AI system that can learn and perform any task a human can. Unlike AI, AGI systems can think and reason at a level that is comparable to human intelligence.

Artificial Intelligence (AI) AI is a field of computer science that focuses on creating machines that can think, learn, and act in a manner similar to humans. AI systems are used to solve complex problems in a wide variety of domains.

Artificial Narrow Intelligence (ANI) ANI is a type of AI system that performs a specific task or solves a specific problem. These systems are highly efficient, but lack the general intelligence of AGI systems.

Artificial Super Intelligence (ASI) ASI is a type of AI system that is much more powerful than any human. ASI systems can process large amounts of data and make decisions that are far beyond what a human could do.

Automated DAO Factory *An application that progressively forms a DAO and walks a community through building its initial governance based on engagement KPIs.*

Automated Market Maker An Automated Market Maker (AMM) is a type of decentralized exchange (DEX) algorithm that uses a mathematical formula to determine the prices of assets and facilitate trades on the blockchain. It allows for the buying and selling of assets without the need for an order book and liquidity providers.

Blockchain Blockchain is a distributed ledger technology that stores records of transactions between two or more parties in an immutable, secure, and verifiable manner.

Bitcoin Bitcoin is the first decentralized digital currency that can be sent between users peer-to-peer without intermediaries, like banks.

Bridge A bi-directional conceptual link between two content snippets (e.g., a piece of text, a part of an image, a segment of a video). Bridges have a semantic relationship (e.g., supporting, contradicting). The bridge can encode and track additional information as needed. We can attach multiple bridges to a single idea.

Bridge Constellations A set of bridges associated with a subject intended to provide scaffolding for the participant to learn about the subject.

Bridge Validation The process that checks whether bridges are structurally sound. Are content elements valid? Is the relationship correct? Are metadata correct?

Bridger Someone who builds bridges on Bridgit.

Bridgit DAO A meta-DAO that catalyzes, supports, and launches social DAOs focusing on regeneration, cognitive freedom, and collective intelligence. The initiator of the Overweb. The author of the book, *"The Metaweb: The Next Level of the Internet."*

Catfishing Catfishing is when a person creates and impersonates a fictional persona or fake identity on a social networking service. They usually target a specific victim, often romantically and/or as a scam.

Canopi Canopi is an overlay app and communication tool that enables people to meet and interact on webpages. People can see each other. They can take part in the page chat. Canopi is available through the Presence extension and Canopi Enterprise.

Canopi Enterprise Canopi Enterprise is a SDK that activates Canopi on a website. The website must install three lines of code. People can see each other. They can take part in the page chat. People can also become friends and take part in one-on-one and group chats.

Constitutional AI Constitutional AI refers to AI systems that are designed and governed in a manner that aligns with the principles and values embodied in constitutions. This can involve incorporating legal and ethical principles in AI systems to ensure that they respect human rights, promote fairness and impartiality, and protect privacy and freedom of expression. The goal is to ensure that AI systems operate within a legal and ethical framework that serves the greater good and advances the public interest.

Contextual Advertising Serving ads based on the focus of the participant's attention, as tracked by a cursor, touch, or line of sight, depending on the viewing device. Also known as focus-driven ads.

Cryptography Cryptography is the practice of secure communication by converting plain text into an unreadable format, called ciphertext, and back to plain text again. It is used to protect sensitive information from unauthorized access or tampering.

Curation Reviewing and judging content with an eye on whether it merits inclusion or if aspects need to be fixed.

DAO A decentralized autonomous organization (DAO) is an organization that is run through rules encoded in computer programs called smart contracts directed by the voting of members.

dApp Decentralized applications that are run on a peer-to-peer network, such as a blockchain.

Dead Internet Theory The Dead Internet Theory is a conspiracy theory that suggests the Internet has died and that much of the content we see online is now artificially generated by AI to manipulate the world population. The theory raises concerns about the impact of AI on propaganda, art, and journalism.

DeFi Decentralized finance (DeFi) is an emerging financial technology based on secure distributed ledgers to remove third parties in financial transactions.

DEX A decentralized exchange that enables peer-to-peer cryptocurrency transactions.

Meta-community A meta-community is a purpose-aligned community with its own dedicated layer above the Web. A safe community above the webpage where people can meet and interact directly and through content. Often managed as a DAO.

- Ethereum** Ethereum is a decentralized, open-source layer one blockchain with smart contract functionality. Ethereum's native cryptocurrency is Ether (ETH).
- Focus-Driven Ads** Ads that are served based on the participant's focus, as tracked by a cursor, touch, or line of sight, depending on the viewing device. Also known as contextual advertising.
- Genesis Constellation** The first set of bridges associated with a specific site, intended to provide scaffolding for the participant to learn and interact with the site's content.
- Generative Tech** Generative technology is a type of artificial intelligence that uses machine learning to produce new and unique outputs from existing data. It is used to create content such as images, text, audio, and video, and can create new ideas and insights in various fields.
- Highlight** A selection of text or an image that appears as an overlay.
- IRL** In real life.
- Internet of Ideas** A new layer of smart links or "bridges" that connect information on an idea-by-idea basis. Collectively, the bridges make up the Internet of Ideas (IOI).
- Intertwingularity** A term coined by Ted Nelson to express the complexity of interrelations in human knowledge. Nelson wrote, "EVERYTHING IS DEEPLY INTERTWINGLED. In an important sense, there are no 'subjects' at all; there is only all knowledge, since the cross-connections among the myriad topics of this world simply cannot be divided up neatly ... Hierarchical and sequential structures, especially popular since Gutenberg, are usually forced and artificial. Intertwingularity is not generally acknowledged—people keep pretending they can make things hierarchical, categorizable and sequential when they can't."
- Ledger** A collection of transactions.
- Liquidity** Liquidity is a pool of funds that token developers need to create to enable their investors to buy and sell instantly. Absent the pool, investors must wait for someone to match their buy or sell order, but there is no guarantee of completion.
- Liquidity Pools** Pooling creates liquidity in a new token and one that's established, like ETH or BNB, in DEXs like Uniswap or PancakeSwap. This pool of funds gets deposited in the exchange and the liquidity provider receives liquidity pool (LP) tokens in return, which are redeemable for pool funds.
- Locked Liquidity** If liquidity is unlocked, then the token developers can do a "rug pull." We lock liquidity by renouncing the ownership of liquidity pool (LP) tokens for a fixed time period by sending them to a time-lock smart contract. Without ownership of LP tokens, developers cannot get liquidity pool funds back.
- Machine Learning** A subset of artificial intelligence that uses statistical techniques to enable computers to progressively improve task performance through data, without being explicitly programmed.
- Metadata** Descriptive metadata describes a resource for discovery and identification. It includes elements such as title, abstract, author, and keywords.
- Metaverse** The metaverse is a network of 3D worlds and a rich virtual space where people can work, play, shop, and socialize that mixes augmented and virtual reality.
- Metabags** Topical NFTs that 1) contain other related NFTs and 2) receive an Overweb royalty stream when the topic appears as category, category, or tag in a bridge.
- Meta Domains** NFTs that convey ownership of the space above and adjacent to the website.
- Metaweb** The Metaweb is the space above the webpage. A meta-layer over the Web that enables people, interactions, and computations above the webpage that is accessible through a browser overlay. The Metaweb will also overlay virtual worlds, the physical world, and the conceptual world of words and ideas.

(To the) Moon In crypto terms, “mooning” means a cryptocurrency’s valuation is “going to the moon,” i.e., experiencing a significant spike in price and volume.

Network State Coined by Balaji Srinivasan, a network state is a highly aligned online community with a capacity for collective action that crowdfunds territory around the world and eventually gains diplomatic recognition from pre-existing states.

Network Union Coined by Balaji Srinivasan, the network union is the antecedent of the network state: a social network with a blockchain, a leader, and a purpose. <https://thenetworkstate.com/network-union>.

New Earth The New Earth is an expression from the Bible—the Book of Isaiah (65:17 & 66:22), 2 Peter (3:13), and the Book of Revelation (21:1)—to describe the final state of redeemed humanity. The regeneration and regenerative communities adopted the New Earth as the moniker for the new world they want. A world where all beings on Earth thrive in harmony with one another and the planet, and live to their full potential.

Nodes Nodes are computers running a blockchain protocol and validating blocks of transactions that are broadcast to the network. They are an essential part of the blockchain network, helping to secure the network and enabling communication between participants in the network.

Note A web annotation that has a title, body, tags, category, and atype.

On-Page Interactions On-page interaction gives computation—as interactions, transactions, and experiences—a real-time presence on webpages, similar to how you can tag your aunt in a photo on Facebook.

On-Page Presence On-page presence gives people a real-time presence on webpages similar to Google Docs. People can see each other. They can take part in the page’s chat.

Overlay Content that is displayed on top of other content.

Overlay App A dApp enabling participants to view and/or interact in the Overweb.

Overweb Safe decentralized public space above the webpage; the first full instantiation of the Metaweb. A hyper-dimensional web above the Web. A safe meta-layer above the webpage where people can meet and interact directly and through content.

NFTs NFTs, or Non-Fungible Tokens, are digital assets that are unique and indivisible, allowing for the tokenization of digital assets such as art, music, and gaming items.

Presence DAO A decentralized autonomous organization that is building the Presence Browser and catalyzing the Overweb.

Presence Browser Overlay The Presence browser overlay provides access to the Overweb, a safe digital space in which real people and information have presence and interactions over the webpage. The overlay is accessible through browser extensions, an SDK, and a forthcoming mobile app. Browsers can support it by adopting the protocol.

Presence Extension A Chrome browser extension that provides access to the Overweb. People can see others that are visible, take part in a page-level chat, and join Canopi events. They can also become friends and take part in one-on-one and group chats. People can build bridges and make notes.

Probably nothing Web3 and NFT speak, often used sarcastically, to say it’s probably something.

ReFi The Regenerative Finance (ReFi) movement uses blockchain and Web3 to address climate change, support conservation and biodiversity, and create a more fair and sustainable financial system.

Reflection Token A type of cryptocurrency token that provides holders with a share of the revenues of profits generated by a specific project or ecosystem, typically in the form of dividends or buybacks.

- Regenaissance** Regenaissance is a portmanteau of the words “regeneration” and “renaissance,” and it refers to a period of renewed growth, creativity, and innovation. The Regenaissance is a movement to regenerate the Earth’s ecosystems and create the New Earth.
- Regeneration** Regeneration in ecological and social systems is the process of restoring and revitalizing ecosystems and communities in order to maintain and sustain the long-term health of these systems. It involves activities such as restoring natural habitats and resources, as well as initiatives to promote social inclusion, economic development, and environmental protection.
- Ransomware** Ransomware is a malware that threatens to publish the victim’s personal data or permanently block access to it unless a ransom is paid.
- Safe Digital Space** Safe digital space is a one-account-per-participant gated space above the webpage.
- Search Engine Optimization (SEO)** The process using metadata and webpage content to affect the visibility of a website or a webpage in a search engine’s unpaid results.
- Sensemaking** A cognitive process that helps people make sense of the information they receive and navigate complex and uncertain situations. Sensemaking is an ongoing process that involves interpreting, analyzing, and making connections between events and data to form mental models and understand the context in which they are embedded. Sensemaking is an important skill for decision-making and problem-solving, particularly in dynamic and rapidly changing environments.
- Shadow banning** A practice on social networks that restricts or limits a user’s visibility without notifying them by blocking their content from appearing in public view or search results. Common uses are to prevent spam or other unwanted content from being seen by other users.
- Smart Contracts** A computer protocol intended to facilitate, verify, or enforce the negotiation or performance of a contract. Smart contracts allow the performance of credible transactions without third parties.
- Smart Filter** An adaptive, transparent, and tunable filter for content on the Overweb.
- Smart Tags** Smart tags are computation code that attach information and interactions to content on webpages.
- Startup Society** Coined by Balaji Srinivasan, a startup society is a new community built Internet-first, usually to solve a specific social problem in an opt-in way.
- Stigmergy** Stigmergy is a method of communication and coordination between individuals or groups in which the environment is used as a means of information sharing and organization. It is a way of using the traces left by one’s actions to communicate with others, rather than direct communication.
- Universal Basic Income** A government program that gives every adult citizen a set amount of money regularly. The goals are to reduce poverty and replace other need-based social programs that require greater bureaucratic involvement.
- Virtual Experience** Another term for the possibilities with virtual reality, augmented reality, and mixed reality.
- Visibility** Participants can “go visible” on specific webpages, and see other visible participants on the webpage.

